

Installation Guide | PUBLIC
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Installation and Maintenance Guide

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THE BEST RUN SAP

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1 Document History

Version	Date	Description
1.0	May 2021	First published version

2 Before You Start

About This Document [page 7]

Target Audience [page 9]

What's New in This Document [page 9]

Additional Information [page 10]

Document Abbreviations [page 11]

Document Definitions [page 12]

2.1 About This Document

> → Remember
>
> The commands used in this guide have been tested in the different scenarios with the dedicated tools. However, using these tools is not mandatory: other tools are available according to your preferences. In this case, the commands may differ.

Installing SAP Convergent Charging

- Installing SAP Convergent Charging [page 21]

The **Installation and Maintenance Guide** is an *SAP Product Document* that explains how to install an SAP Convergent Charging 2020 system landscape, including the necessary components such as:

- Server systems
- User interfaces (desktop applications, Web applications)
- Additional components

This guide provides installation guidance for setting up system landscapes from development to production, grouped among 2 different installation scenarios:

- Installing a Monohost System in the Landscape [page 28], which corresponds to landscapes hosting the minimum required software components on a single Microsoft Windows host
- Installing a Multihost System in the Landscape [page 29], which corresponds to landscapes hosting a more realistic set of software components on multiple Linux hosts that are configured and sized according to the business requirements

These 2 installation scenarios are described afterwards in this guide, and represent specific sets of installations steps that correspond to procedures available in the Procedures [page 88] section. You can adapt each installation scenario to fit specific needs by using different or additional procedures, and thus create specific scenario such as:

- A monohost landscape running on a Linux host
- A multihost landscape running on Microsoft Windows hosts and using an Oracle Core Database
- And so on

Once your SAP CC system landscape is installed, some post-installation operations [page 31] may be necessary to administrate and secure your landscape or integrate it with other systems.

> ⚠ Caution
>
> This guide does not provide any guidance for installing the following elements:
>
> - HTTP Client and Message Client user interfaces
> - IBM DB2 pureScale databases

Uninstalling SAP Convergent Charging

- Uninstalling SAP Convergent Charging [page 33]

This SAP product document describes the uninstallation of SAP Convergent Charging 2020.

Updating and Maintaining SAP Convergent Charging

- Updating and Maintaining SAP Convergent Charging [page 35]

This SAP product document describes the maintenance of SAP Convergent Charging 2020 server systems and how to install a patch [page 38], a feature package stack (FPS), or a service package (SP) [page 36].

Upgrading SAP Convergent Charging

- Upgrading SAP Convergent Charging [page 40]

At last, this SAP product document also explains how to upgrade from the replaced SAP CC release 5.0 to the newer release 2020 (FPS 0 minimum). The target version is the latest feature packacge stack (FPS) or support package (SP) with the latest patch level delivered by SAP.

> → Recommendation
>
> SAP recommends that you upgrade to the newest maintenance version of the SAP CC 2020 software. Determine the target version and plan your upgrade project to benefit from the features provided by the release 2020 of SAP Convergent Charging and by its latest maintenance support packages (SPs).

The main new features in the upgrade process in release 2020 are:

- The SAP installer user interface has been modified to support the upgrade process from former SAP CC release 5.0 to SAP CC release 2020. For further information, refer to the Upgrade Path [page 49] section
- The JDBC[1] drivers are changed to support the latest releases of RDBMSs[2].
- As of SP 1, you must prepare the database by creating 2 new tablespaces or filegroups in your Core Database: MONITORING_DATA and MONITORING_INDX. For further information, refer to the Preparing the Back-End Database Systems [page 61] section
- As of SP 5, a new version of SAPinst is available as part of SAP Software Provisioning Manager 1.0 and associated to a renewed user interface SL Common GUI (aka SAP Software Logistics Client UI).

[1] Java Database Connectivity
[2] Relational Database Management System

2.2 Target Audience

This guide is intended for the following audience:

- Solution and Technology Consultants
- Application and System Administrators

2.3 What's New in This Document

What's New in SAP CC 2020 FPS 2?

As of FPS 2, SAP Convergent Charging 2020 provides you with the following modifications of this SAP product documentation:

- Enhancement of the Securing Communications with the Core Server System [page 221] procedure, considering the communications with an OAuth 2.0 authorization server (AS) system

What's New in SAP CC 2020 (in FPS 0)?

As of FPS 0, SAP Convergent Charging 2020 provides you with the following modifications of this documentation:

- Modification of the Downloading the Oracle JDBC Driver [page 97] procedure
- Modification of the Installing and Launching Core Tool [page 192] procedure
- Modification of the Installing and Launching BART Tool [page 195] procedure
- Modification of the Installing and Launching CAT Tool [page 198] procedure
- Modification of the Installing and Launching Cockpit [page 201] procedure
- Modification of the Installing and Launching Admin+ [page 207] procedure
- Modification of the Installing and Launching BART+ [page 209] procedure
- Modification of the Installing and Launching Setup Tool [page 211] procedure
- Modification of the Installing and Launching BART Setup Tool [page 212] procedure
- Modification of the Installing and Launching Config Tool [page 214] procedure
- Migration of the "update" part of the former SAP CC 5.0 Operation Guide in this SAP document. Updating and Maintaining SAP Convergent Charging [page 35].

2020 is a new main release of SAP Convergent Charging (SAP CC).

See Also

- What's New in This Document in the SAP CC 2020 Security Guide
- What's New in This Document in the SAP CC 2020 Tuning Guide

Related Information

What's New in Feature Package Stack 2

2.4 Additional Information

For more information about specific topics, see the quick links as shown in the table below:

Content	Quick Link
SAP CC documentation	help.sap.com/cc2020 **Note** SAP CC is part of the SAP solution SAP Billing based on SAP Business Suite or SAP S/4 HANA. Consult the central solution information at the following address: https://cx.sap.com/fr/products/billing

Content	Quick Link
Related SAP notes	support.sap.com/notes Important SAP Notes for the Upgrade

SAP Note	Description
2772452	Delivery and Information Note for SAP CC 5.0 Support Package 5
2778886	Technical Implementation of SAP CC 5.0 SP 5
2315313	Delivery and Information Note of SAP CC 4.1 Support Package 2
2332433	Technical Implementation of SAP CC 4.1 Support Package 2
197623	Temporary license keys
94998	Requesting license keys and deleting systems
611361	Hostnames of SAP servers
1453216	CA Introscope (IS) for SAP Convergent Charging
397175	SAP Cryptographic Software - Export control
1036194	Processor Driver for Windows Server 2003 must be installed
1367498	SAP JVM installation prerequisites
1375494	SAP system does not start after applying SAP kernel patch

Related release notes	https://support.sap.com/en/my-support/knowledge-base.html
SAP Solution Manager	http://support.sap.com/solutionmanager

2.5 Document Abbreviations

Abbreviation	Meaning
AAA	Authentication, Authorization, and Accounting
BART	Batch Acquisition and Rating Toolset
CAT	Connector Administration Tool
CI	Chargeable Item
CIF	Charging output Integration Framework
CMA	Customer Management Area
DBA	DataBase Administrator
GUI	Graphical User Interface
HA	High Availability

Abbreviation	Meaning
HCI	HTTP Communication Interface
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secured
IEC	Import/Export Connector
JDBC	Java Database Connectivity
JRE	Java Runtime Environment
JVM	Java Virtual Machine
MDC	Multiple Data Container
NFS	Network File system
PAM	SAP Product Availability Matrix
PKCS	Public-Key Cryptography Standards
PSE	Personal Security Environment
RAC	Real Application Cluster
RDBMS	Relational Database Management System
SAPSID	SAP System Identifier
RDBMS	Relational Database Management System
SAR	SAP Archive
RDBMS	Relational Database Management System
SID	SAP System Identifier
SNC	Secure Network Communications
TIF	Transaction Integration Framework
URL	Uniform Resource Locator
VAT	Value Added Tax
XML	eXtended Markup Language

2.6 Document Definitions

2.6.1 SAP Global Host

The SAP Global Host is the first physical host on which an instance of an SAP Convergent Charging system is installed.

According to your installation scenario, your SAP CC system landscape may thus contain multiple SAP Global Hosts: one for each installed SAP CC system.

> **Example**
>
> Considering an SAP system landscape containing 3 physical hosts:
>
> - Host H1 is dedicated to the installation of a dispatcher instance and an updater instance of the Core Server system.
> - Host H2 is dedicated to the installation of additional instances of the Core Server system.
> - Host H3 is dedicated to the installation of the BART Server system.
>
> Then:
>
> - H1 is considered as the SAP Global Host for the Core Server system. This system is distributed over two hosts and is composed of multiple instances.
> - H3 is considered as the SAP Global Host for the BART Server system.

2.6.2 SAP Central Repository

The *SAP Central Repository* is a concept related to SAP systems that are made up of instances distributed over multiple physical hosts, and that need to have a central location containing the executable programs and profiles of all these instances.

- **In a high availability (HA) setup, this repository is located on a network storage solution** such as a NAS, a SAN, or another network storage solution. For sharing and access optimizations, redundancy and failover mechanisms, the use of network storage technologies is recommended for SAP Convergent Charging (SAP CC).
  When you implement real-time charging services (online charging scenarios) based on SAP CC, high levels of availability are required.
  The storage can be a single point of failure (SPoF) in these typical business scenarios.
- **In a non-HA setup, this repository is located on the** *SAP Global Host* **of each SAP CC system** in your SAP CC system landscape and is shared to the different hosts that make up this SAP CC system.
  When you implement batch charging services (offline charging scenarios) based on SAP CC, no or low levels of redundancy are required.

> **Caution**
>
> Depending on your requirements for your production SAP CC system landscape, you adapt the levels of the redundancy and failover mechanisms. For your test (TST) or development (DEV) SAP CC system landscapes, you can adopt other levels.

On Microsoft Windows operating systems, the SAP installer:

- Automatically creates the SAP Central Repository using the `<DRIVE>:\usr\sap\<SID>\SYS` directory.
- Shares the `<DRIVE>:\usr\sap` directory using the `sapmnt` and `saploc` share names.

On Linux operating systems:

- The SAP Central Repository must be manually created using the `/sapmnt` directory.

- The SAP Central Repository must be manually shared to the other hosts using the `sapmnt` share name, and mounted on each host different than the SAP Global Host.
- The SAP installer automatically creates the `/usr/sap/<SID>` directory and some associated symbolic links.

> **Note**
>
> As far as SAP Convergent Charging is concerned, only the Core Server system can be distributed over multiple hosts. The concept of SAP Central Repository is thus only relevant for the SAP CC Core Server system.

The following table contains information about the content of the SAP Central Repository:

Subfolder	Description	Location	
		Global Host	Host
`exe`	The `exe` folder is a physical folder that is located on each host of your landscape but contains subfolders mapped onto the Global Host of your system. Considering `<OS>` as the operating system of the host, the `uc/<OS>` subfolder contains global executable programs and SAP CC programs that are located into different subfolders according to your SAP CC system:	■	■

System	Folder Structure
Core Server	• `exe/uc/<OS>/CC_CORE_SERVER/` • `exe/uc/<OS>/CC_CORE_SERVER/config/` • `exe/uc/<OS>/CC_CORE_SERVER/core_sql/` • `exe/uc/<OS>/CC_CORE_SERVER/session_sql/` • `exe/uc/<OS>/CC_CORE_SERVER/jars/` • `exe/uc/<OS>/CC_CORE_SERVER/licenses/` • `exe/uc/<OS>/CC_CORE_SERVER/tax/`
BART Server	• `exe/uc/<OS>/CC_BART_SERVER/` • `exe/uc/<OS>/CC_BART_SERVER/bart_sql/` • `exe/uc/<OS>/CC_BART_SERVER/config/` • `exe/uc/<OS>/CC_BART_SERVER/examples/` • `exe/uc/<OS>/CC_BART_SERVER/jars/`
Diameter Server	• `exe/uc/<OS>/CC_DIAMETER_SERVER/` • `exe/uc/<OS>/CC_DIAMETER_SERVER/config/` • `exe/uc/<OS>/CC_DIAMETER_SERVER/jars/`

Subfolder	Description	Location	
		Global Host	Host
`profile`	Also called **SAP Profile Folder**, the `profile` directory contains the configuration files related to boot and startup operations of all the instances (`jstart.config`, `boot.config`), available in the following subfolders: • `profile/boot`, within which the `boot.config` file is present • `profile/jstart`, which contains a subfolder for each installed instance, each subfolder containing the adequate `jstart.config` file	■	□

(■: Physical folder, □: Shared folder)

2.6.3 SAP Installer

As of SP 5 for release 5.0, SAP Convergent Charging is delivered with a new SAP installer that is based on a newer installation toolset named SAP Software Provisioning Manager 1.0. This installer is a web application.

The SAP installer is customized for SAP Convergent Charging, and is made up with:

- SAP Software Logistics UI Client (aka *SL Common GUI*) as a front-end application
- SAPinst Framework 749 as a back-end application

For further information, refer to SAP Notes 1680045, 2480194, and 2393060.

See here:

- Launching the SAP installer [page 190]
- SL Common GUI for SAP CC

2.6.4 System Landscape Directory (SLD)

System Landscape Directory (SLD) of SAP NetWeaver is the central information repository for your system landscape. It contains information about technical systems, landscapes, business systems, products, and software components in your system landscape. You have to configure manually SAP CC systems to connect to SLD and send data periodically.

All the server systems of SAP CC 2020 can send information about installed hardware and software components in your system landscapes to facilitate the central configuration of the landscape (applications, machines):

- Core Server
- BART Server
- Diameter Server

The SAP CC systems regularly update the System Landscape Directory repository. Installation information is available in SAP Solution Manager [page 16].

For further information, refer to the dedicated documentation on SAP Help Portal.

2.6.5 SAP Solution Manager for SAP CC

SAP Solution Manager is a platform that provides integrated content, tools, and methodologies that you need to implement, support, operate, and monitor your enterprise´s solutions from SAP.

SAP Solution Manager enables you to accelerate and optimize the implementation process and access full support and continuous improvement offerings from SAP.

With SAP Solution Manager, you can:

- Minimize risk and increase the reliability of your IT solutions
- Reduce total cost of ownership (TCO) throughout the solution life cycle
- Manage your core business processes and link business processes to the underlying IT infrastructure
- Get more on your return-on-investments (ROI), because it supports both SAP and non-SAP software

> ⓘ Note
>
> To benefit from some important features of SAP Solution Manager, you need to install the infrastructure for CA Introscope third party. Refer to the CA Introscope for SAP CC [page 17] section in this document.

> → Recommendation
>
> Install and use CA Introscope and SAP Solution Manager for your SAP CC 2020 system landscapes. Configure SAP Solution Manager with the agent framework and CA Introscope infrastructures.
>
> CA Introscope is delivered as a component of SAP Solution Manager, you shall consider to install the correct version regarding to your SAP CC release and support package number.

Application Operations

Application Operations in SAP Solution Manager provides all capabilities for central monitoring, message alerting, analysis, and administration of SAP CC and SAP solutions. It helps you to reduce TCO by predefined content and centralized management tools for all aspects of operations in SAP Solution Manager. For further information, refer to the *Application Operations Guide* documentation on the Help Portal for SAP Solution Manager - Use.

Work Centers Used for SAP CC

SAP Solution Manager provides the necessary central work environments in the form of specialized work centers. These work centers are the central access to functions and methods for monitoring your SAP solutions and accelerating problem solving. This improves the availability of your solutions. For further

information, refer to the *Application Operations Guide* documentation on the Help Portal for SAP Solution Manager - Use.

Related Information

- Monitoring and Alerting: https://help.sap.com/viewer/c413647f87a54db59d18cb074ce3dafd/7.2.10/en-US/d45d8afe9f104fd98bb4fbb4b8cb4c2f.html
- Root Cause Analysis: https://help.sap.com/viewer/82f6dd44db4e4518aad4dfce00116fcf/7.2.10/en-US/92c65043f2004b098fda16b8e56ef702.html

2.6.6 CA Introscope for SAP CC

SAP Convergent Charging comes with a packaged solution that provides you with application performance management tools. This solution supports Java-based applications and, although it is based on the CA Introscope 9.x software, it is entirely compatible with further versions of CA Introscope.

CA Introscope (IS) allows the technical monitoring of SAP CC in live production environments.

- It provides a central server system that collects performance data pushed from agents located in the monitored SAP CC systems and makes metric data available to the monitoring consoles and to SAP Solution Manager
- User interfaces provide specialized dashboards that enable to monitor manually the system health, performance, and stability. This is the starting point for detecting issues and investigating a root cause analysis

> **Note**
>
> - CA IS is a component of SAP Solution Manager as an additional infrastructure
> - SAP provides you with the necessary CA IS instrumentation and console dashboards dedicated to the support of SAP Convergent Charging 2020

Features

CA Introscope for SAP provides you with the:

- Necessary components customized for SAP CC and for SAP Solution Manager
- Instrumentation agents and probes specialized for SAP CC and Java applications
- Monitored metric definitions (throughput, latency, other performance data)
- Console dashboards including graphs and alert indicators

> **Note**
>
> The dashboards and console alerts are available in an additional installation package that includes all the Introscope Management Modules developed by SAP. Make sure that you install the

> Management Module that includes the monitoring dashboards specialized for SAP Convergent Charging: SAP_SRM_CRM (Products for "SRM and CRM"). Refer to the SAP Note 1579474.

With CA IS, you can:

- Operate the technical monitoring
- Determine the source of problems quickly
- Conduct incident triage by identifying the source of the issues (front-end systems, internal components, database, host resources, file system)
- Perform root cause analysis by isolating the problems and determining where to investigate

For further information, refer to:

- Help Portal for SAP Solution Manager - Implement > *CA Introscope - Setup Guides*
- Help Portal for SAP Solution Manager - Use

2.6.7 SAP Management Consoles for SAP CC

SAP provides two applications for central system management that are installed with SAP Convergent Charging:

- SAP Management Console (SAP MC)
- SAP Microsoft Management Console (SAP MMC)

Depending on the operating system of the host for the SAP CC system to install, these applications are deployed or not. See: Starting and Stopping the Server Systems [page 144]

Both applications provide a common framework for centralized system management.

SAP Management Console (SAP MC)

The SAP Management Console (SAP MC) application provides a common framework for centralized system management. It allows you to monitor and perform basic administration tasks on the SAP systems centrally, thus simplifying system administration.

The SAP MC is a Java applet that can be run from any web browser supporting Java. Thus, you can administer remote systems without needing a local installation.

Using SAP MC you can manage your SAP CC system landscapes:

- Monitor and control (start, stop, or restart) the SAP systems, ABAP, and Java instances with a single tool.
- Display SAP system log and trace files, start profiles, instance parameters, the system environment, SAP environment, and so on.
- Display and control Java and ABAP processes.
- Monitor system alerts.
- Save the current console configuration in a file to reuse it later or to forward it to other users.
- Start third-party tools (such as Telnet), if available, to manage an application server.

For more information, consult: SAP Management Console

SAP Microsoft Management Console (SAP MC)

The SAP Microsoft Management Console (SAP MMC) application provides a common framework for system management. It integrates several tools in a common user interface for centralized system management.

Using SAP MMC you can manage your SAP CC system landscapes:

- Monitor and control any number of SAP systems, application servers and ITS instances with a single tool if they are running on Microsoft Windows.
- Once you have configured the console, you can save the configuration in a file and easily forward it to other users.
- The start service is available on all platforms and offers a Web service interface, called SAPControl with better control and monitor capabilities. The snap-in requires the new Web service interface. You can access the Web service interface in your own applications from a Web service-enabled programming environment like C, Java, ABAP or .NET. Alternatively, you can use the "sapcontrol" command line client to access the Webservice interface.
- Display all SAP trace files, start profile, instance parameters, the system environment and SAP environment.
- Display the most serious alerts in the alert tree.
- Display the current status tree.
- Display and acknowledge current alerts.
- Display the SAP syslog for a functioning, offline, or malfunctioning SAP system.
- Display the ABAP work process overview.
- Display the ABAP and ICM queue statistics (dpmon).
- Display Application Server Java (AS Java) processes, caches, threads, web sessions, EJB sessions, remote objects, application aliases, and shared memory tables.
- Display Java virtual machine garbage collection and heap memory information of the Application Server Java.
- Display ICM threads, connections, caches and proxy connections.
- Display Enqueue locks and statistics.
- Start or stop one or all application servers of a system with a mouse-click.
- Restart the service without having to restart the application server.
- Log on to an application server.
- Start analysis tools in the SAP system for nodes in the alert tree.
- Control Java or ABAP processes.

SAP Management Console (SAP MC)	SAP Microsoft Management Console (SAP MC)
	• Modify dynamic profile parameters. • Remove enqueue locks manually. • Create compressed instance or system snapshots for offline problem analysis and problem reporting. • Start the following third-party tools, if available, to manage an application server or ITS instance: • Telnet • SSH client • Windows Terminal Server Client • pcANYWHERE • Computer Management snap-in For more information, consult: Standalone SAP Microsoft Management Console

3 Installing SAP Convergent Charging

Plan and perform an initial installation and deployment of SAP Convergent Charging 2020.

SAP CC Software Components [page 21]

Sizing the SAP Convergent Charging Landscape [page 22]

Installing a Monohost System in the Landscape [page 28]

Installing a Multihost System in the Landscape [page 29]

Adding Elements to Your System Landscape [page 30]

Post-Installation Operations [page 31]

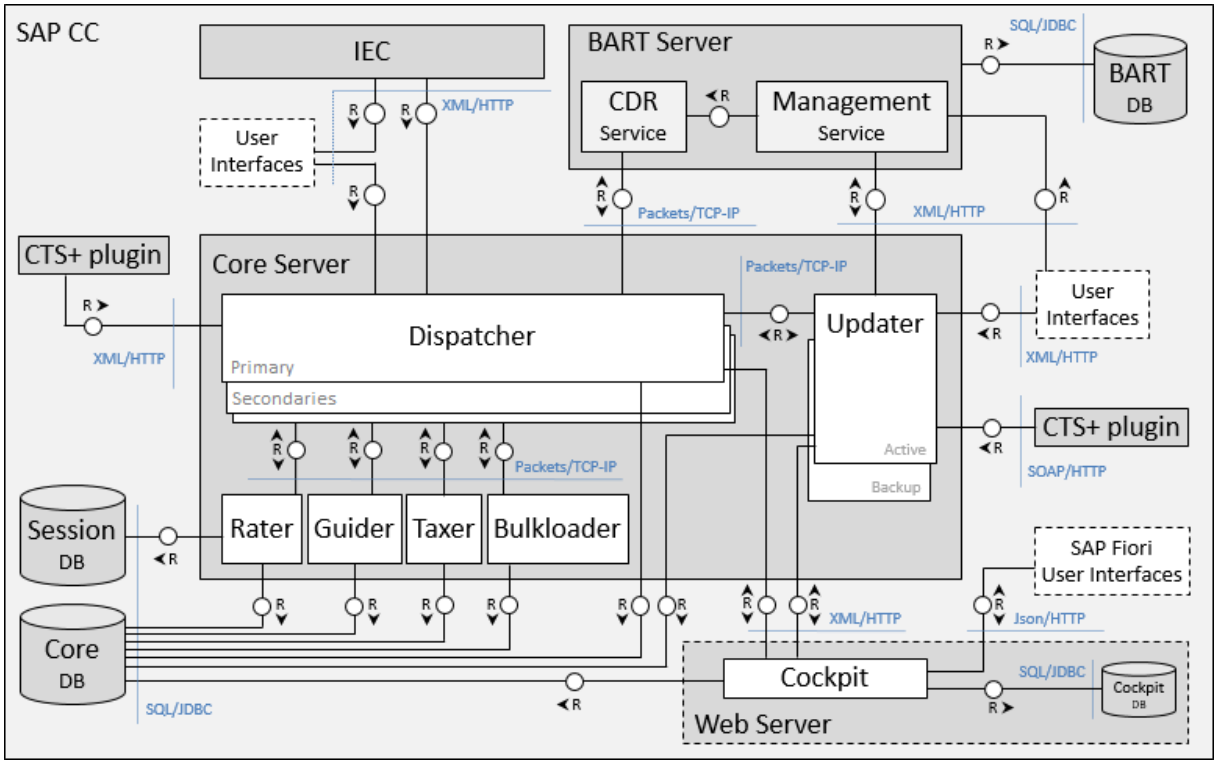
3.1 SAP CC Software Components

SAP Convergent Charging 2020 is made up with the following set of Java-based software components that you can install to perform your business operations. For further information about the different software components, refer to the PAM [3] available at the following location: support.sap.com/pam, or to the *Architecture* section of the SAP CC 2020 Application Help documentation.

Server System	User Interface	Database (and RDBMS)
• Core Server	• Core Tool • Cockpit • Admin+ • Setup Tool • Config Tool • HTTP Client, Message Client	• Core Database • Session Database • Cockpit Database
Variant: Offline charging (batch mode rating and charging) • BART Server • Import/Export Connector	• BART Tool, CAT Tool • BART+ • BART Setup Tool	• BART Database • IEC Database

The following schema provides you with an overview of the interactions between the different software components of SAP CC:

[3] SAP Product Availability Matrix



Each software element is available in the installation materials of SAP Convergent Charging, that you can download using the Downloading the Installation Materials [page 92] procedure available in this guide.

3.2 Sizing the SAP Convergent Charging Landscape

SAP Convergent Charging provides a rating and charging solution for high-volume processing in service industries, that delivers pricing design capabilities, high performance rating and convergent balance management. Capable of operating in real-time and in batch mode, it includes hybrid prepaid/postpaid charging capabilities, unrivaled marketing agility for service pricing, carrier-grade scalable performance and high availability.

→ Recommendation

The installation of SAP Convergent Charging requires an **initial sizing** with the vision of the SAP system landscape and business requirements (data volume, rating/charging operations throughput). This expert sizing analyzes future customer-specific data in detail. The main objective is to determine the resource consumption of storage, resources, systems, and databases using comprehensive measurements.

The sizing covers the development, test, quality assurance, preproduction, and production SAP CC system landscapes.

During the implementation phase, you verify and validate the initial sizing and take into accounts some evolutions of your business (traffic, data volume, service level objectives).

To ensure these performance and high availability (HA) capabilities, it is necessary to respect the following recommendations when deploying an SAP Convergent Charging (SAP CC) system landscape:

- Landscape Recommendations [page 23]
- Network Recommendations [page 24]
- OS Recommendations [page 24]
- Disk Recommendations [page 26]
- JVM Recommendations [page 26]

Landscape Recommendations [page 23]

Network Recommendations [page 24]

OS Recommendations [page 24]

Disk Recommendations [page 26]

JVM Recommendations [page 26]

Minimum Hardware Requirements for an SAP Convergent Charging Development Environment [page 28]

3.2.1 Landscape Recommendations

Concerning your SAP Convergent Charging landscape, SAP SE recommends that you implement the following instructions in terms of hosts sizing:

- **dispatcher** and **guider** instances should be co-located on the same physical host. All deployed dispatchers are running, but one of them is considered as the primary dispatcher. The other dispatchers are considered as slaves
- **rater** instances should be deployed on dedicated physical hosts. The more raters are installed, the less number of partitions are assigned to each rater, which reduces impacts in case of rater failure. As a consequence, a higher number of small-size raters is preferable to a smaller number of large-size raters
- **updater** instances can be deployed on dedicated physical hosts, or co-located on the same physical host as another instance. All deployed updaters are started, but only one of them is nominated by the primary dispatcher as active (ACTIVE state). The other updaters are considered as backups (WAITING state)
- **bulkloader** instances can be deployed on dedicated physical hosts, or co-located on the same physical host as the rater they are working with (due to constraints related to shared file system)

Considering the rules above, SAP SE recommends that you deploy the following instances in order to meet HA[4] requirements:

- 3 dispatcher+guider couples should be deployed, each couple being deployed on a dedicated physical host
- 3 raters should be deployed, each rater being deployed on a dedicated physical host
- 2 updaters should be deployed

> → Remember
>
> Do not deploy all instances of a same type on a common host (for example, all the dispatchers on the same machine).

[4] High Availability

> **Note**
>
> SAP Convergent Charging is designed for horizontal scalability, which means that it is possible to deploy additional hosts in order to increase the global performance of your landscape. For these additional hosts, the above recommendations must be followed.

In addition to this deployment strategy, SAP SE recommends that you implement the following requirements for each host in terms of hardware:

- **Hyper-threading technology**: To improve parallelization of computing operations, hyper-threading technology should be activated. Each host should have up to 64 hyper-threaded cores (or 32 non-hyper-threaded cores) installed and split among the available sockets, for 2 or 3 deployed raters. In case more raters are necessary, the horizontal scalability of SAP CC gives the possibility to add other hosts. In case bigger machines (with more sockets) are used, these additional installed raters shall be bound to dedicated resources using the NUMA[5] technology to optimize memory accesses
- CPU[6]/Core **frequency**: To improve the processing time of the events handled by SAP CC, the fastest available CPUs shall be used, such as CPUs with a minimum frequency of 3.2 Ghz

3.2.2 Network Recommendations

The different components of your SAP Convergent Charging system landscape communicate over the network, which is very solicited and implies to implement the following instructions to ensure the lowest possible latency:

- **Technologies**: To ensure a maximum bandwith, the fastest network technologies shall be used.
- **Shared folders**: In case of host failure, folders shared by bulkloaders and raters must be accessible from other running hosts to ensure the continuation of the data files processing operations.

3.2.3 OS Recommendations

Every host of your landscape takes part of the overall performance of SAP Convergent Charging. It is thus necessary to implement the following instructions to handle a maximum of incoming events :

- **Virtualization** technology: Using virtualization to split resources of bigger machines and install multiple instances of SAP CC on similar physical hardware is not recommended
- **Power-saving** mode: Modern CPUs[7] can run in power-saving mode, which usually corresponds to the default option. As SAP CC requires the maximum CPU available speed, double-check and possibly adjust the parameters to ensure that the CPU is running in “performance” mode, which means using highest frequency
- (Linux only) **nproc** setting: The “nproc” linux setting gives the possibility to limit the number of processes that a user can execute simultaneously. Considering that SAP CC starts a high number of java threads, it is thus necessary to increase the “nproc” setting in order to avoid OutOfMemory exceptions (that could

[5] Non-Uniform Memory Access
[6] Central Processing Unit
[7] Central Processing Unit

occur in case the JVM[8] could not start a givent thread). To increase this limit, open the `limits.conf` file located in the `/etc/security/` folder and add or update the following lines accordingly:
`<sapcc_user> soft nproc <tuned_value>`
`<sapcc_user> hard nproc <tuned_value>`
Where:
  - `<sapcc_user>` corresponds to the operating system user related to SAP CC
  - `<tuned_value>` should be greater than the sum of all java threads started by the SAP CC instances, including CA Introscope agent threads, JVM Garbage Collection Threads, and so on... In case of any doubt, you can set this value to the maximum value allowed by the concrete operating system
- (Linux only) **nofile** setting: The "nofile" linux setting gives the possibility to limit the number of file descriptors that can be simultaneously opened. Considering that SAP CC handles a high number of files, it is thus necessary to increase the "nofile" setting in order to avoid failures during files opening operations. To increase this limit, open the `limits.conf` file located in the `/etc/security/` folder and add or update the following lines accordingly:
`<sapcc_user> soft nofile <tuned_value>`
`<sapcc_user> hard nofile<tuned_value>`
Where:
  - `<sapcc_user>` corresponds to the operating system user related to SAP CC
  - `<tuned_value>` should be large enough to ensure that SAP CC can open as many files as necessary. In case of any doubt, you can set this value to the maximum value allowed by the concrete operating system
- (Linux only) **huge pages** memory: Memory is managed in blocks that are called "pages" and whose default size is set on Linux operating systems to 4096 bytes. CPUs have a built-in management unit that contains a list of these pages, each page being referenced through a page table entry whose browsing is CPU consuming. Considering that SAP CC manages large amounts of memory, it is thus necessary to enable the concept of "huge pages", that consists in blocks of memory coming in 2MB, which divides the number of page references by 512 (that is more suitable for managing multiple gigabytes of memory. To enable huge pages, open the `/etc/sysctl.conf` file that contains the configuration of the Linux kernel and add or update the following lines accordingly:
`vm.hugetlb_shm_group = <group_id>`
`vm.nr_hugepages = <nb_pages>`
`shmmax = <tuned_value>`
Where:
  - `<group_id>` corresponds to the group identifier of the SAP CC "`<SYSTEM_ID>`adm" user that will reserve huge pages
  - `<nb_pages>` corresponds to the multiplier of the huge page size to compute reserved memory, which should include JVM memory and SAP CC cached structures
  - `<tuned_value>` should be greather than the reservable memory

And execute the following command to apply the modifications:

```
sysctl -p
```

Then, allow the user to use and lock memory by opening the `/etc/security/limits.conf` file and adding or updating the following lines accordingly:

```
@<sapcc_user> soft memlock unlimited
```

```
@<sapcc_user> hard memlock unlimited
```

[8] Java Virtual Machine

Where `<sapcc_user>` corresponds to the operating system user that will reserve memory.

Finally, update the JVM settings by adding the `-XX:UseLargePages` option within the `jstart.config` files of each instance, and restart these instances.

3.2.4 Disk Recommendations

The processing time of the events handled by SAP Convergent Charging also depends on the low latency of disk I/O operations. As a consequence, SAP SE recommends that you implement the following instructions to ensure performance and reliability of your SAP CC system landscape:

- **Network** storage: For sharing and access optimizations, the use of network storage technologies such as NAS or SAN shall be preferred to typical solutions based on hosts' folders sharing.
- **SSD** technology: To minimize latency during writing operations, SSDs are recommended to store data files generated by SAP CC. In addition, due to folders sharing requirements, the use of CFS[9] shall be preferred.

3.2.5 JVM Recommendations

SAP Convergent Charging is made up of a set of Java-based applications whose execution is performed by JVM[10] instances that can be configured to fit specific requirements. For further information about the available options, refer to the Java Virtual Machine dedicated section afterwards.

Every Java object created by the SAP CC instances is handled in 2 different Java memories:

- The Heap memory, which is divided into the following portions known as "generations", whose content is reorganized as time goes by for optimization and performance purposes:
  - **Young** generation, that represents the most active portion of the heap and that is used by SAP CC to store the most-used objects whose lifecycle rapidly evolves
  - **Old / Tenured** generation, that represents the less active portion of the heap, used by SAP CC to store longer-lived objects such as the objects belonging to catalogs
  - **Permanent / Metaspace** generation, that contains metadata required by the JVM

  Each generation is continuously reorganized to ensure performance regarding the access to the contained objects. This reorganization of the heap is made during "garbage collection" (GC) operations that may impact the global performance of your landscape.
- The Direct memory, that provides access to native I/O operations, and that is used by the SAP CC instances to store the cached structures

To ensure performance and stability of your landscape, SAP SE recommends that you implement the following instructions to set up the JVM options described in the Java Virtual Machine dedicated section:

[9] Clustered File System
[10] Java Virtual Machine

JVM Option	Recommendation
`-Xmn<SIZE>`	The size of the young generation of the heap memory depends on your business. **→ Recommendation** SAP SE recommends you to consult the SAP Note 2006073 documentation, that you can consider as an starting point to set a relevant size for the young generation used by your SAP CC instances
`-Xms<SIZE>`	Considering a default 4 GB size used to store the catalog data within the tenured generation of the heap memory, SAP SE recommends you to set the following value: `-Xmn + 4 GB`
`-Xmx<SIZE>`	Same value as -Xms
`-Xss<SIZE>`	512k (864k for HP-UX servers)
`-XX:CompileThreshold=<NUMBER>`	10000
`-XX:SurvivorRatio=<RATIO>`	16
`-XX:TargetSurvivorRatio=<NUMBER>`	90
`-XX:ConcGCThreads=<NUMBER>`	Set a value that corresponds to: `<Total nb of cores on the host>/<nb of instances on the host>` **Note** This number should not exceed the number of cores allocated to the instance (for example in case of NUMA[11] binding)
`-XX:ParallelGCThreads=<NUMBER>`	Same value as -XX:ConcGCThreads
`-XX:PromotedPadding=<NUMBER>`	5 **for raters only**
`-XX:+UseParallelOldGC`	Enable the option
`-XX:+UseSplitVerifier`	Enable the option
`-XX:+ScavengeBeforeFullGC`	Enable the option
`-XX:+UseAdaptiveSizePolicy`	Enable the option for all instances **except for the raters to be set to** `-UseAdaptiveSizePolicy`

[11] Non-Uniform Memory Access

JVM Option	Recommendation
`-XX:+TieredCompilation`	Enable the option

3.2.6 Minimum Hardware Requirements for an SAP Convergent Charging Development Environment

A development environment (DEV) for SAP Convergent Charging (SAP CC) is dedicated to the creation of charge plans and rate plans, price tables, and the test of pricing configurations, including unit and integration test throughout the BRIM solution.

An SAP Convergent Charging development environment is used for functional testing only, it is not intended for performance or volume testing.

> → Recommendation
>
> According to your needs, refer the minimal sizing in SAP Note 3568851.

Related Information

Installing a Monohost System in the Landscape [page 28]

3.3 Installing a Monohost System in the Landscape

This installation and deployment scenario corresponds to landscapes containing the minimum set of software components, all installed on the same physical host (acting as the SAP Global Host [page 12]) using a temporary license. This scenario consists in deploying on a host 4 instances of the Core Server system (1 dispatcher, 1 guider, 1 rater, and 1 updater):

- Working in conjunction with a pre-installed database
- Without any encryption of the communication channels

The first step of this installation scenario consists in retrieving the different materials that are necessary to deploy the expected SAP CC elements. Execute the following common procedures, taking into account the associated recommendations:

- Defining Your System Landscape [page 90]
- Choosing System IDs [page 91]
- Downloading the Installation Materials [page 92]

The second step of this installation scenario consists in using the Setting Up a New Host for a Landscape [page 141] procedure to set up the system of the SAP Global Host before deploying the 4 instances of the Core Server system.

The third step of this installation scenario consists in preparing the database using the dedicated Preparing the Core Database [page 106] procedure.

The last step of this installation scenario consists in deploying the expected SAP CC elements using the Installing a Core Server on a Monohost Landscape [page 123] procedure that you can consider as an entry point for installing:

- A stand-alone landscape, that is not integrated with any third-party system and does not give the possibility to work with billable items or consumption items
- A landscape that is integrated with SAP Convergent Invoicing and gives the possibility to work with billable items and consumption items

When these 3 steps have been successfully executed, you can use the following procedures:

- Starting and Stopping the Server Systems [page 144], to manage your installed instances
- Testing a Landscape Basically [page 139], to ensure that your landscape is properly working

Even if your landscape only concerns development or test operations, you might need to tune it to get better performances. For further information about the tuning possibilities, refer to the SAP CC 2020 Tuning Guide documentation.

3.4 Installing a Multihost System in the Landscape

This installation scenario corresponds to SAP system landscapes containing a more realistic set of software components used for production purpose. This scenario consists in deploying on different physical hosts 4 instances of the Core Server system (1 dispatcher, 1 guider, 1 rater, and 1 updater):

- Working in conjunction with a database pre-installed on a dedicated host
- Using a temporary license
- Without any encryption of the communication channels
- Every physical host running under a Linux operating system

> **Note**
>
> For scalability purpose, you can extend your system landscape by installing additional instances, respecting the following basic rules:
>
> - For performance reasons, every rater/bulkloader couple shall be deployed on a dedicated host, and every dispatcher/guider couple shall be associated to 2 rater/bulkloader couples
> - 5x9 high availability for charging operations is available when the landscape contains either:
>   - 3 hosts each containing a dispatcher/guider couple + 3 hosts each containing a rater/bulkloader couple
>   - Or 3 hosts each containing 1 dispatcher, 1 guider, 1 rater and 1 bulkloader instances
> - Every updater instance shall be deployed on a dedicated host, to avoid latencies during charging or refilling operations. 5x9 high availability for provisioning operations is available when 2 updaters are

deployed within your landscape. Deploying more than 2 updaters is possible but not necessary to ensure high availability

The first step of this installation scenario consists in retrieving the different materials that are necessary to deploy the expected SAP CC elements. Execute the following common procedures, taking into account the associated recommendations:

- Defining Your System Landscape [page 90] considering:
  - 1 host (the SAP Global Host [page 12]) containing 1 dispatcher and 1 guider
  - 1 host containing 1 rater
  - 1 host containing 1 updater
  - 1 host containing the Core Database
- Choosing System IDs [page 91]
- Downloading the Installation Materials [page 92]

The second step of this installation scenario consists in using the Setting Up a New Host for a Landscape [page 141] procedure to set up the system of the following physical hosts before deploying the expected SAP CC elements :

- The SAP Global Host, that will contain the installation materials and the following 2 instances of the Core Server:
  - 1 dispatcher
  - 1 guider
- The host dedicated to provisioning operations, containing 1 updater
- The host dedicated to operations such as charging, refilling, and so on, containing 1 rater

The third step of this installation scenario consists in preparing the database using the dedicated Preparing the Core Database [page 106] procedure.

The last step of this installation scenario consists in deploying the expected SAP CC elements using:

- The Installing a Core Server on a Multihost Landscape [page 131] procedure, that you can consider as an entry point for managing the SAP Global Host and installing:
  - A stand-alone landscape, that is not integrated with any third-party system and does not give the possibility to work with billable items or consumption items
  - A landscape that is integrated with SAP Convergent Invoicing and gives the possibility to work with billable items and consumption items
- The Adding Core Server Instances in a Multi-Hosts Landscape [page 142] procedure to manage the other hosts of the landscape

When these 4 steps have been successfully executed, you can use the following procedures:

- Starting and Stopping the Server Systems [page 144], to manage your installed instances
- Testing a Landscape Basically [page 139], to ensure that your landscape is properly working

3.5 Adding Elements to Your System Landscape

After the installation of the Core Server system, you can extend your SAP system landscape by:

- Installing the following SAP CC 2020 systems to fulfill your business requirements:
  - BART Server, using the dedicated Installing a BART Server System in an Existing Landscape [page 167] procedure
  - Import/Export Connector, using the dedicated Installing an IEC (Import/Export Connector) in an Existing Landscape [page 187] procedure
  - Diameter Server, using the dedicated Installing a Diameter Server in an Existing Landscape [page 173] procedure
- Deploying the following user interfaces to interact with the installed systems:
  - Convergent Charging Cockpit, using the dedicated Installing and Launching Cockpit [page 201] procedure
  - Core Tool, using the dedicated Installing and Launching Core Tool [page 192] procedure
  - BART Tool, using the dedicated Installing and Launching BART Tool [page 195] procedure
  - CAT Tool, using the dedicated Installing and Launching CAT Tool [page 198] procedure
  - Admin+, using the dedicated Installing and Launching Admin+ [page 207] procedure
  - BART+, using the dedicated Installing and Launching BART+ [page 209] procedure
  - Setup Tool, using the dedicated Installing and Launching Setup Tool [page 211] procedure
  - BART Setup Tool, using the dedicated Installing and Launching BART Setup Tool [page 212] procedure
  - Config Tool, using the dedicated Installing and Launching Config Tool [page 214] procedure

3.6 Post-Installation Operations

Once your SAP CC system landscape is installed, you may need to perform post-installation operations such as:

- Administrating your landscape (permanent license installation, instances management, and so on)
- Configuring elements to fit your business requirements (VAT[12] rules and rates, public holidays, and so on)
- Integrating your SAP CC landscape with other systems available within a global infrastructure, such as SAP System Landscape Directory, SAP Solution Manager, SAP S/4HANA, SAP ERP/FI-CA, SAP CRM, and so on
- Copying a Core Server system from a host to another

You can use the following list to perform post-installation operations:

- Administrating your landscape:
  - Installing a Permanent License [page 261]
  - Starting and Stopping the Server Systems [page 144], to manage the instances deployed in your system landscape
  - Securing an SAP CC System Landscape [page 216]
- Integrating your landscape with other systems:
  - Integrating SAP CC with a SCIM-Based System [page 263]
  - Integrating SAP CC with CA Application Performance Management [page 264]
  - Integrating SAP CC with SAP CTS+ [page 267]

[12] Value Added Tax

- Integrating SAP CC with SAP Convergent Invoicing [page 271]
- Integrating SAP CC with SAP S/4HANA Cloud Public Edition [page 275]

- Managing your landscape:
  - Copying an SAP CC Core Server [page 279]

4 Uninstalling SAP Convergent Charging

For any reason, you can modify your SAP CC 2020 system landscape by uninstalling some components:

- One or multiple instances in an installed Core Server system
- A complete SAP CC system
- A back-end database system
- A user interface

Uninstalling an Instance of a Given Host [page 33]

Uninstalling a Back-End Database System [page 33]

Uninstalling a User Interface [page 34]

4.1 Uninstalling an Instance of a Given Host

The SAP installer used to install instances within your SAP Convergent Charging system also gives the possibility to uninstall instances. To remove one or several instances of a given host within an existing system, refer to the Removing System(s) or Instance(s) from a Given Host [page 262] dedicated procedure.

> ⚠ Caution
>
> If your SAP CC system is a multihost system, you must first uninstall instances deployed on hosts different than the SAP Global Host [page 12]. The SAP Global Host of every installed SAP CC system must be the last host you uninstall elements from.

4.2 Uninstalling a Back-End Database System

SAP Convergent Charging 2020 does not provide any mechanisms to uninstall a back-end database system from an SAP system landscape.

> → Recommendation
>
> Refer to your DBA[13] to get the relevant procedures.

[13] DataBase Administrator

4.3 Uninstalling a User Interface

SAP Convergent Charging 2020 provides a set of user interfaces that are either automatically deployed by the SAP installer during the installation process, or manually deployed when necessary.

To uninstall a given user interface, perform the following procedure:

1. Open the procedure dedicated to the launching of the concerned user interface.
2. Open the folder that contains the script used to launch the user interface.
3. Remove the launch script from this folder to avoid any further use.

> ⓘ Note
>
> The uninstallation of an SAP CC system removes the user interfaces installed in the system.

5 Updating and Maintaining SAP Convergent Charging

Implement a maintenance version (feature package stack, patch level) in an already-installed SAP CC 2020 system landscape.

Manage the software changes in your SAP system landscapes.

> → Remember
>
> If you have installed the replaced release 5.0, you must consider the Upgrading SAP Convergent Charging [page 40] section that is dedicated to the planning, preparation, and processing of this typical and specific project: the upgrade to the release 2020.

Updates

- Implementation of a Feature Package Stack (FPS) [page 36]
- Implementation of a Patch Level [page 38]

System maintenance relating to the installation and deployment

- Adding Core Server Instances in a Multi-Hosts Landscape [page 142]
- Removing System(s) or Instance(s) from a Given Host [page 262]

Implementation Plan [page 35]

Implementation of a Feature Package Stack (FPS) [page 36]

Implementation of a Patch Level [page 38]

5.1 Implementation Plan

Before you begin the actual update, you must first plan it carefully. This includes requesting all the SAP Notes you need. Careful planning is a prerequisite for a successful SAP system update.

The update sequence includes the following main steps:

- A: Update of the Core Server systems of SAP Convergent Charging
- B: Manual restart of the Core Server instances of SAP Convergent Charging
- C: Update and restart of the other server systems of SAP Convergent Charging
  1. Diameter Server
  2. BART Server
- D: Update of the client applications and systems that use the Java Library of SAP Convergent Charging such as SAP Convergent Mediation by Digital Route
- E: Update of the graphical user interfaces and the IEC[14] application of SAP Convergent Charging
  - Core Tool

- Cockpit
- BART Tool
- CAT Tool
- Import/Export Connector

→ Recommendation

SAP recommends that you first update all the server systems of SAP Convergent Charging. Then you update all the client applications and systems connected to these SAP CC server systems:

- External applications or other systems that use the Java Libraries of the SAP Convergent Charging
- Graphical user interfaces (GUIs) of SAP Convergent Charging
- SAP CC IEC

When updating an application, the configuration of data may change: it may not be compliant any longer with non-updated applications.

Do not maintain a system landscape with different versions of the SAP CC server systems, different versions of the SAP CC GUIs and different versions of the SAP CC Java Libraries installed in the client applications or systems connected to a server system of SAP Convergent ChargingSAP Convergent Charging.

⚠ Caution

Refer to the relevant SAP Notes before implementing a feature package stack (or a patch) to update the SAP CC 2020 server systems in your SAP system landscapes. Consult the update procedures and prerequisites in these documents.

5.2 Implementation of a Feature Package Stack (FPS)

A feature package stack (FPS) is a complete maintenance version of the SAP CC 2020 software product. You can install it as initial version or you can update an existing SAP system landscape with the latest version. It relates to all the server systems, user interfaces, and other applications of SAP Convergent Charging (SAP CC).

If you plan an initial installation based on this FPS, then refer to the section Installing SAP Convergent Charging [page 21].

If you want to update an existing SAP CC 2020 system landscape, then use the dedicated procedures [page 286] in this documentation.

ⓘ Note

Before implementing and installing a feature package stack (FPS), check:

- If a patch of SAP CC is available for this level of version. Find delivery notes in the knowledge base (KBA) or items to download in the software center.
- If a higher FPS is available for the release 2020.

[14] Import/Export Connector

→ Remember

SAP Convergent Charging 2020 provides you with all the delivered feature package stacks (FPSs) in the *Support Packages and Patches* subarea of the SAP Software Download Center (SWDC).

You can download it on SAP Support Portal to maintain an SAP CC product component that is already installed in your SAP system landscape.

Note that it does not include the SAP installer specialized for the SAP CC software product.

For an initial installation of SAP Convergent Charging 2020, use the *Installations and Upgrades* subarea on SAP Support Portal.

You update the following components of SAP Convergent Charging in your SAP system landscape:

SAP CC Component	Software Unit: Server System	Software Unit: User Interface	Software Unit: Library	Software Unit: Plugin
Core Server and Tool *Online and offline charging system:* • Server system • User interfaces • Java libraries	"CC Core Server" • Server system • Admin. utilities	• "CC Core Tool" • Core Tool • Utilities • "CC Cockpit" • Web Graphical User Interface (GUI or UX)	"CC Core SDK" • Java libraries • Application Programming Interface (APIs) specifications *For integration with mediation systems*	"CC CTS+ Plugin" • Plugin for SAP CTS [page 267]
BART Server and Tool *Offline charging system (batch mode, past service consumptions)*	"CC BART Server" • Server system • Admin. utilities	"CC BART Tool" • BART Tool • Utilities	"CC BART SDK" • Java libraries • API specifications *For integration with mediation systems*	
Diameter Server ⚠ Caution From SAP Convergent Charging 5.0 SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189. *Online charging based on diameter communications*	"CC Diameter Server" • Server system			
Import/Export Connector (IEC) and Toolset	"CC IEC Application"	"CC CAT Tool"		

Legend: IEC: Import/Export Connector

To find the latest version of SAP CC 2020, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Click the *Support Packages & Patches* tab to access this subarea.
3. Expand the *By Alphabetical Index (A-Z)* section.
4. Click the *C* letter.
5. Click the *SAP Convergent Charging* item of the list.
6. Click the **SAP Convergent Charging 2020** item to open the information page dedicated to SAP CC 2020.

> **Note**
>
> Previous versions of SAP CC are also available but this procedure only concerns the 2020 release.

7. Click the *Feature Package Stacks and Patches* link available in the upper part of the screen in order to open the *Downloads* section.
8. Click the component concerned by the update to open the list of available downloads.
9. Click on each element of the table to download it.

> **Note**
>
> A feature package stack (FPS) is also included in a multispanning archive that is distributed to several files available for downloads. Download all the files in the same local repository and unpack the archive. Refer to SAP Note 886535 for more information.
>
> You can use this deliverable for your maintenance operations. See Downloading the Installation Materials [page 92].
>
> Once unpacked, the local repository includes a copy of an installation media. The name of the root directory is the SAP material number of the installation media. It includes several files and the `DATA_UNITS` directory.

5.3 Implementation of a Patch Level

A patch is a partial maintenance version of SAP CC 2020. It relates to one or several software units of SAP CC 2020.

To facilitate the technical implementation of a patch level, consult its delivery note and the following SAP Note 2977857.

Patches may exist for the following software units and categories:

Software Unit Category (or Technical Name)	Software Unit	Product Component	Product Instance in PAM[15]
BART SDK 2020	CC BART SDK	BART Server and Tool	BART Tool / Frontend

[15] SAP Product Availability Matrix

Software Unit Category (or Technical Name)	Software Unit	Product Component	Product Instance in PAM[15]
BART SERVER 2020	CC BART Server	BART Server and Tool	BART Server
BART TOOL 2020	CC BART Tool	BART Server and Tool	BART Tool / Frontend
CAT TOOL 2020	CC CAT Tool	Import/Export Connector (IEC) and Toolset	IEC Tool / Frontend
CC COCKPIT 2020	CC Cockpit	Cockpit	CC Cockpit / Frontend (Web-Based)
CORE SDK 2020	CC Core SDK	Core Server and Tool	Core Tool / Frontend
CORE SERVER 2020	CC Core Server	Core Server and Tool	Core Server
CORE TOOL 2020	CC Core Tool	Core Server and Tool	Core Tool / Frontend
CTS+ PLUGIN 2020	CC CTS+ Plugin	Core Server and Tool	
DIAMETER 2020 **⚠ Caution** From SAP Convergent Charging 5.0 SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.	CC Diameter Server	Diameter Server	Diameter Server
IEC 2020	CC IEC Application	Import/Export Connector (IEC) and Toolset	IEC / Frontend

To find the latest patches for SAP CC 2020, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Click the *Support Packages & Patches* tab to access this subarea.
3. Expand the *By Alphabetical Index (A-Z)* section.
4. Click the *C* letter.
5. Click the *SAP Convergent Charging* item of the list.
6. Click the **SAP Convergent Charging 2020** item to open the information page dedicated to SAP CC 2020.

   ⓘ Note

   Previous versions of SAP CC are also available but this procedure only concerns the 2020 release.
7. Click the *Support Packages and Patches* link available in the upper part of the screen in order to open the *Downloads* section.
8. Click the component concerned by the update to open the list of available downloads.
9. Click on each element of the table to download it.

Once download is finished, refer to the Update and Maintenance [page 286] procedure.

[15] SAP Product Availability Matrix

6 Upgrading SAP Convergent Charging

Plan and perform a complete upgrade project to replace the former release 5.0 of SAP Convergent Charging with the current release 2020.

This section explains how to upgrade an SAP system landscape as a productive edition: a production environment that is live and delivers all the business and technical services.

> → Remember
>
> If you have an older release, consider the successive upgrades, release level by release level. You must follow the upgrade path [page 49]. Contact your SAP Support Team.

Upgrade Plan [page 40]

Planning the Upgrade Project [page 40]

Preparations for the Upgrade [page 54]

Upgrading [page 66]

6.1 Upgrade Plan

Before you begin the actual upgrade, you must first plan it carefully. This includes requesting all the SAP Notes you need. Careful planning is a prerequisite for a successful SAP system upgrade.



6.2 Planning the Upgrade Project

This section of the SAP CC product documentation and user assistance contains information about planning your upgrade project. The following information helps you planning your upgrade so that downtime is reduced to a minimum, and the upgrade runs as efficiently as possible.

Upgrade in an SAP System Group [page 41]

Technical Overview of Installed SAP CC Components [page 43]

Planning the Upgrade Schedule [page 44]
Steps of the Upgrade Phase [page 45]
Detailed Upgrade Sequence [page 46]
Upgrade in an SAP Solution Scenario with SAP S/4HANA, SAP ERP, or SAP CRM [page 48]
Upgrade Path [page 49]
Upgrade Stages [page 50]
Technical Details of the Upgrade Process [page 51]
Database Specific Aspects [page 52]
Upgrade of Distributed and High-Availability Systems [page 52]
Upgrade of CA IntroScope (IS) for SAP [page 53]
Upgrade of SAP Solution Manager [page 53]

6.2.1 Upgrade in an SAP System Group

If you have more than one SAP system, for example, a production system, a quality assurance system, and a development system, you must consider the entire system group.

Upgrade the SAP systems of your group with the global sequence:

1. Development and test system
2. Quality assurance and validation system
3. Preproduction system
4. Production system

Validate always an upgrade stage before performing the next upgrade.

Global Process Sequence

If your system group consists of several SAP systems and includes a quality assurance system that has been set up between the development and production system, upgrade in the following order:

1	Development system or landscape	This system landscape should include all the software units of SAP Convergent Charging that are running on the production system. The external systems that are connected with a server system of SAP CC are not always present in this development landscape. If an interfaced system or application uses the Java libraries (see Software Development Kit) provided by SAP Convergent Charging, it must be part of the development landscape to test the upgrade of these libraries. Note that for test purposes with realistic production data, set up a quality assurance system or pre-production system between the development system and production system. This system can be a copy of the production system.
2	Test system or landscape	Depending on the testing scopes and levels, the landscape includes all the systems interfaced with a server system of SAP Convergent Charging: CRM, mediation, billing. Such an interfaced system can be simulated to facilitate the testing. Certain systems are not necessary for the tests.
3	Quality assurance system or landscape	The landscape must include all the systems interfaced with a server system of SAP Convergent Charging: CRM, mediation, billing. The hardware and landscape architecture differs from the production system or landscape.
4	Validation system or landscape	A validation system is a simulation of the production system and data. It is a light copy of the production system and data. The hardware and landscape architecture simulates the production system or landscape. Use this system (or landscape) to validate the upgrade process when you have no preproduction system (or landscape).
5	Preproduction system or landscape	A preproduction system is a copy of the production system and data. The hardware and landscape architecture is similar. The preproduction system or landscape is used to: • Validate the hardware sizing of the production landscape • Setup performance tests • Train the operations team • Test and validate any software maintenance operation Use this system or landscape to validate the upgrade process.
6	Production system or landscape	This SAP system landscape includes the production systems of SAP Convergent Charging and all other The production system landscape is isolated from the other landscapes.

For test purposes with realistic production data, setup a quality assurance system between the development system and production system. This system can be a copy of the production system.

For validation purposes, setup a preproduction system between the quality assurance system and the production system.

6.2.2 Technical Overview of Installed SAP CC Components

An SAP system landscape for SAP Convergent Charging includes several server systems installed on different hosts:

- SAP CC Core Server (required)
- SAP CC BART Server
- SAP CC Diameter Server

The server systems to upgrade depend on your implementation and integration project.

After the upgrade of the server systems, some other applications must be upgraded.

- The applications that use Java Libraries provided in the Software Development Kits (SDKs) for SAP Convergent Charging:
  - SAP CC Core SDK
  - SAP CC BART SDK (* required if SAP CC BART Server is installed)

  For example, Java-based client applications or systems connect to SAP CC server systems by using the Java APIs of SAP Convergent Charging. See also: Upgrading the Client Applications [page 80]
- The Graphical User Interfaces (GUIs) of SAP Convergent Charging are part of the landscape and are installed on different hosts or laptops:
  - SAP Convergent Charging Cockpit
  - SAP CC Core Tool
  - SAP CC BART Tool (* required if SAP CC BART Server is installed)

  See also: Upgrading the SAP CC GUIs [page 81] and Upgrading the Cockpit Web Applications [page 82]

> ### Example
>
> Several applications can be integrated with SAP Convergent Charging by using the SAP CC Java Libraries:
>
> - A real-time mediation system that sends charging operations in high volume
> - A connector application between an offline mediation system and an SAP CC system
> - A provisioning system that massively creates and maintains master data related to end customers to be configured in SAP CC
>
> Consult the *SAP CC Application Help* for more details about the Technical Interfaces provided by SAP Convergent Charging.

This chapter covers the upgrades of all the elements in the following table:

System/Application of SAP CC 2020	Java-Based Client Application	Frontend GUI	Third Parties	RDBMS
SAP CC Cockpit	-	Cockpit front-end application	• Java Web Server • JVM	SAP CC Cockpit Database
SAP CC Core Server	SAP CC Core SDK	SAP CC Core Tool	• JVM • Special (US taxes for telecommunications): Avalara AvaTax (formerly known as Billsoft EZTax) for the SAP CC Core Server systems	SAP CC Core Database
SAP CC BART Server	SAP CC BART SDK	SAP CC BART Tool	JVM	SAP CC BART Database
SAP CC Diameter Server	-	-	JVM	-
SAP CC Import/Export Connector (IEC)	SAP CC IEC SDK	SAP CC CAT Tool	JVM	• (Customer Database) • (SAP CC IEC Database)
See also: Upgrading the SAP CC Server Systems [page 67]	See also: Upgrading the Client Applications [page 80]	See also: Upgrading the SAP CC GUIs [page 81]	See also: Upgrading the Cockpit Web Applications [page 82]	

6.2.3 Planning the Upgrade Schedule

Create an exact schedule for the upgrade of your SAP system landscapes.

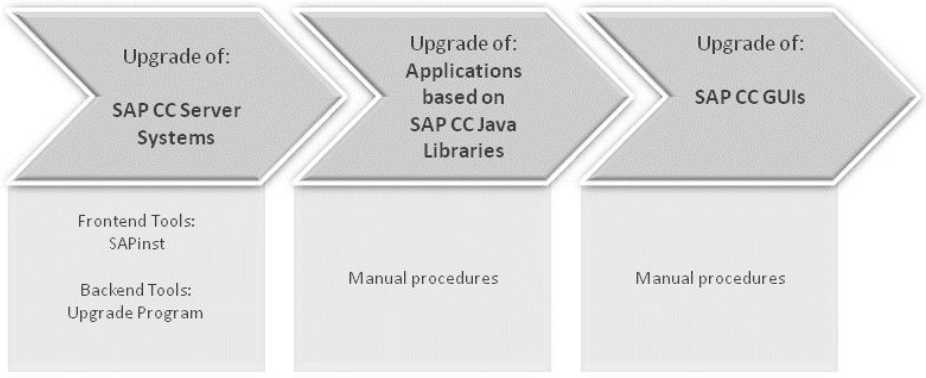
Step	Description
1: You decide when to begin preparing for the upgrade	When you start the upgrade, preparations depend not only on the length of the preparations but also on the length of the upgrade itself. To determine the expected total length of the upgrade of your production system, perform a test upgrade of an SAP system with a similar set of data (a system copy of your production system is ideal).

Step	Description
2: You decide when to start the upgrade	If you have special requirements regarding the availability of your SAP system, adjust the process and the length of the upgrade to fit the expected amount of downtime or restricted services. Avoid peak traffic periods. In a high availability (HA) environment, consider periods with limited business traffic. In a non-HA environment, you stop the SAP systems. During the upgrade, various actions are performed in the system. The runtime of these actions depends on various factors and is therefore specific to each SAP system.
3: You decide when a full backup is performed after the upgrade	The full database backup is a prerequisite for resuming production operation of your system. It must be made after you have performed all necessary follow-up activities. This has the advantage that all database changes resulting from follow-up activities are included in the full backup.
4: You decide when production operation resumes	If possible, resume production operation only after you have performed all required follow-up activities.

6.2.4 Steps of the Upgrade Phase

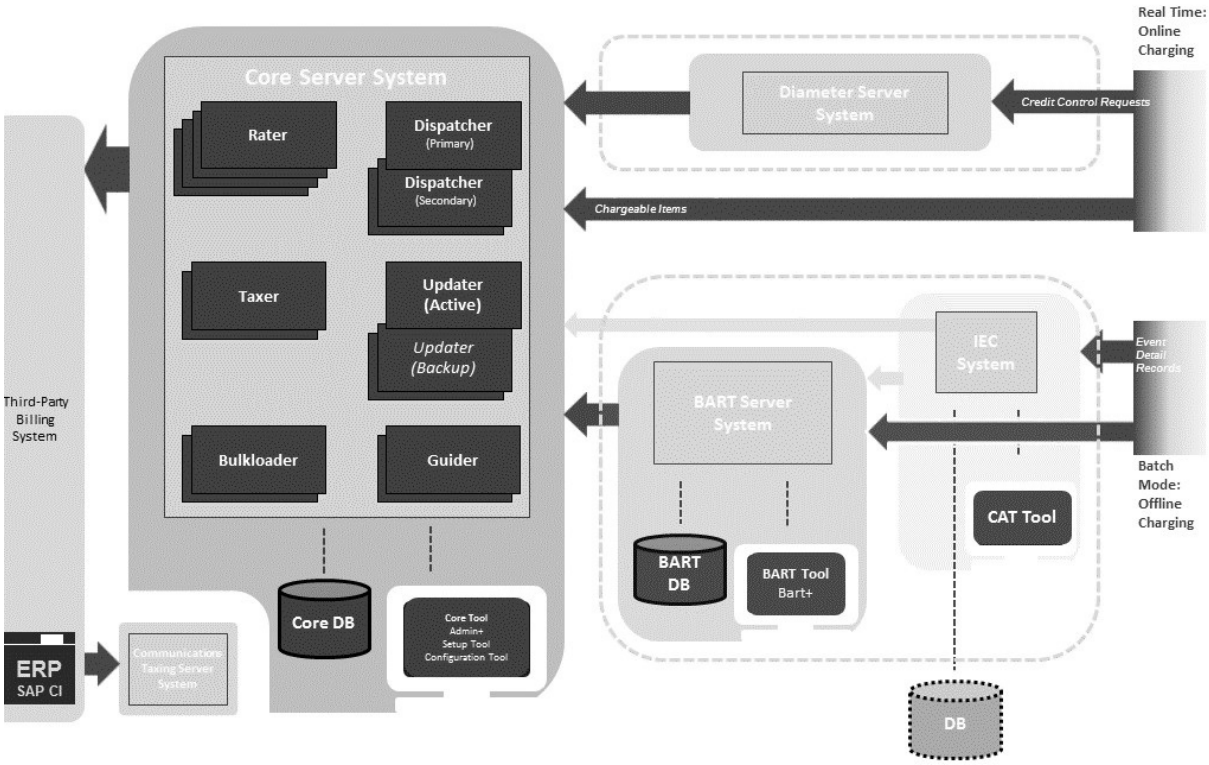
The upgrade phase of the upgrade plan can be divided in a sequence that includes the following main steps:

- A: Upgrade of the server systems of SAP Convergent Charging
- B: Upgrade of the systems that use the Java Library of SAP Convergent Charging
- C: Upgrade of the graphical user interfaces (GUIs) of SAP Convergent Charging



6.2.5 Detailed Upgrade Sequence

The upgrade sequence is split into detailed steps by using the software units present in the landscape:

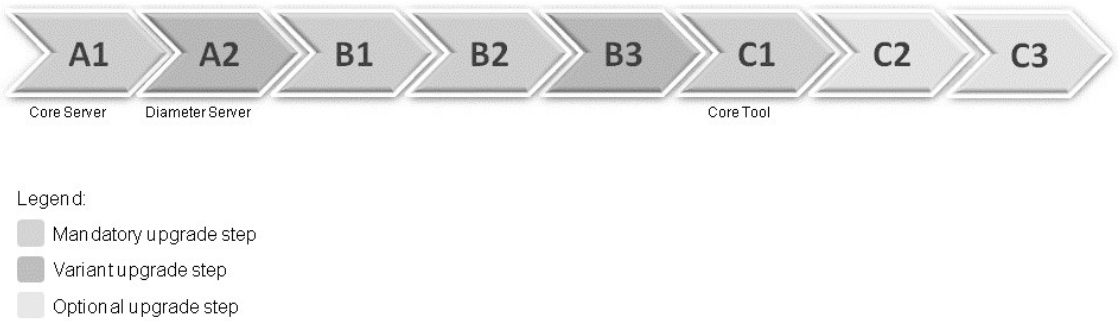


Online Scenario [page 46]

Offline Scenario [page 47]

6.2.5.1 Online Scenario

In a scenario for online charging, the Core Server system is the central systems of your landscape.



Step	SAP CC Software Unit
A1	Core Server
B1	Mediation System
B2	Provisioning System
C1	Core Tool

Options

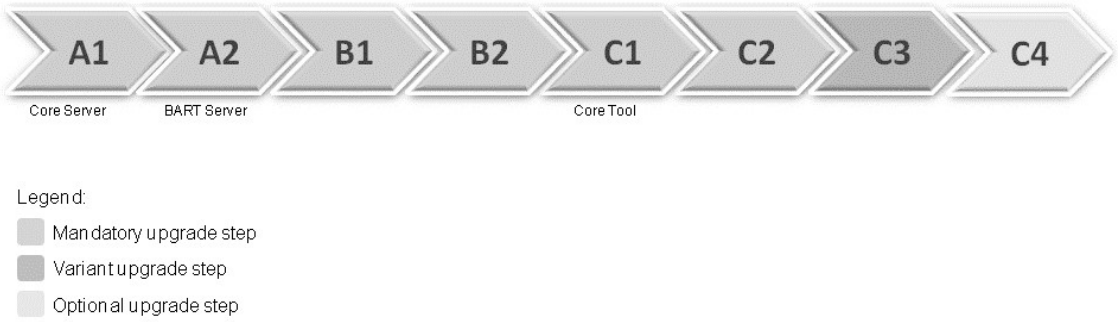
Step	SAP CC Software Unit
C2	CAT Tool and Import/Export Connector

Variant

Step	SAP CC Software Unit
A2	Diameter Server
B3	Network Monitoring System

6.2.5.2 Offline Scenario

In a scenario for offline charging, both the Core Server system and the BART Server system are the central systems of your landscape.



Step	SAP CC Software Unit
A1	Core Server

Step	SAP CC Software Unit
A2	BART Server
B1	Mediation System
B2	Provisioning System
C1	Core Tool
C2	BART Tool

Variant

Step	SAP CC Software Unit
C2	CAT Tool and Import/Export Connector

6.2.6 Upgrade in an SAP Solution Scenario with SAP S/4HANA, SAP ERP, or SAP CRM

SAP Convergent Charging (SAP CC) 2020 can be part of SAP system landscapes integrated with:

- SAP ERP of SAP Business Suite 7i2016 (incl. SAP CRM)
- SAP S/4HANA (and SAP CRM or not)
- SAP S/4HANA Cloud Public Edition (as of SAP CC 5.0 SP 5)

Use the relevant integration guides to verify the compatibility matrix and upgrade prerequisites. See Integration Guides for SAP BRIM.

System Landscape Based on SAP S/4HANA [page 48]

System Landscape Based on SAP Business Suite 2016 [page 49]

System Landscape Integrated with SAP S/4HANA Cloud Public Edition [page 49]

6.2.6.1 System Landscape Based on SAP S/4HANA

If your SAP CC production system is part of an SAP system landscape that is based on the SAP S/4HANA product (also known as SAP Business Suite 4 SAP HANA), plan the upgrade for already installed SAP Convergent Charging in the overall upgrade schedule for the global landscape.

SAP CC 2020 is compliant with multiple on-premise editions of SAP S/4HANA.

Refer to the implementation documentation:

- Integration Guide for SAP S/4HANA BRIM Using SAP Subscription Order Management

- help.sap.com/brim

6.2.6.2 System Landscape Based on SAP Business Suite 2016

If your SAP CC production system is connected to an SAP CRM or SAP ERP system, plan the upgrade for SAP Convergent Charging in the overall upgrade schedule for the SAP Business Suite landscape or for one or both SAP components (SAP CRM or SAP ERP). Refer to the implementation documentation:

- Integration Guide for Offer-to-Cash Based on BRIM Using SAP ERP

Related Information

Integration Guides for SAP BRIM

6.2.6.3 System Landscape Integrated with SAP S/4HANA Cloud Public Edition

If your SAP CC system is integrated with SAP S/4HANA Cloud Public Edition, no specific operation relates to SAP CC in the overall upgrade of the global landscape.

6.2.7 Upgrade Path

The upgrade to SAP CC 2020 FPS 2 (or a later version) must be performed in an SAP system landscape using SAP CC 5.0 SP 6 or a later support package (SP). This upgrade path may change during the maintenance cycles of both software main releases.

To facilitate the upgrade of an old SAP CC 5.0 system, first update your SAP system landscapes by implementing both the latest support package (SP) and the latest patch level that are available for this replaced release of the SAP Convergent Charging software product.

Plan this software maintenance with the current delivered SAP product documentation, available SP delivery notes, and existing patch notes.

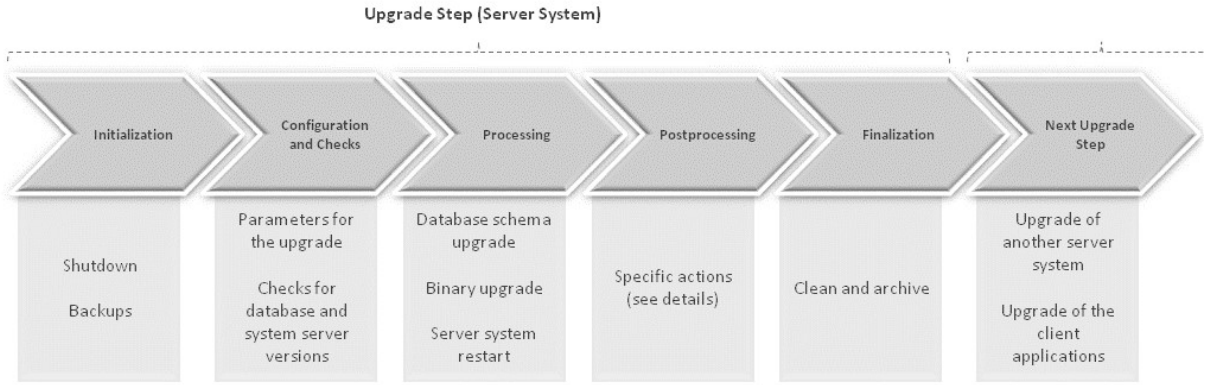
Then prepare and plan the upgrade to the target available version of SAP CC 2020.

> **Note**
>
> Make sure that there is a valid upgrade path between the last delivered version of the SAP CC 5.0 software release and the last version of the SAP CC 2020 software release. Consult the production SAP Notes that relate to each delivery version.

If the latest SAP CC 5.0 support package or the patch level is more recent than the current SAP CC 2020 deliverable, contact your SAP Support Team to know if there is a valid upgrade path between this SAP CC 5.0 version and the current SAP CC 2020 version.

6.2.8 Upgrade Stages

For an upgrade of an SAP CC server system, the upgrade step is based on the following stages:



The table summarizes the stages for the upgrade process of a server system. For more information, see the Technical Details of the Upgrade Process [page 51] section.

Stage	Level	Description
Initialization	Shutdown	
	Database Backup	
	Software Backup	
Configuration & Checks	SAP installer: • Choose Option • Define Parameters • Summary	Check the current version of the software and database of the SAP CC system to check that the upgrade is possible
Pre-processing	SAP installer: • Execute • Completed	• Update the database of the system (if applicable for the system being upgraded) by executing SQL[16] upgrade scripts • Update the content of the archive folder in the central repository of the system
	System Administrator	The administrator checks the new launch scripts of the SAP CC server system.

[16] Structured Query Language

Stage	Level	Description
Processing	System Administrator	The administrator restarts the shutdown SAP CC server system. The system upgrades automatically its directories. HA[17]: the SAP installer has already restarted the running instances.
Post-processing	System Administrator	The administrator monitors and checks the restarted SAP CC server system.
Finalization	System Administrator	The administrator archives the logs generated by the SAP installer that are no longer required and completes the upgrade.

6.2.9 Technical Details of the Upgrade Process

SQL Upgrade Scripts [page 51]

Locations of the SQL Upgrade Scripts [page 51]

6.2.9.1 SQL Upgrade Scripts

The SAP installer tool executes a sequence of SQL[18] scripts to upgrade the database schemas from the former SAP CC5.0 release to the last SAP CC 2020 version.

You can consult the relevant scripts to know the detailed changes in the database systems.

The executed SQL scripts depend on the technology of database and on whether database-partitioning function is used.

The SAP installer tool displays the list of SQL scripts.

6.2.9.2 Locations of the SQL Upgrade Scripts

For SAP CC 2020 Core Server and SAP CC 2020 BART Server, the upgrade process includes modifications to the database schema of both database systems.

The necessary SQL[19] upgrade scripts are located in an archive on the installation media.

For Core Server, the archive is named `CC_CORE_SERVER.SAR`, and includes the scripts in a directory named `core_sql`.

For BART Server, the archive is named `CC_BART_SERVER.SAR` and includes the scripts in a directory named `bart_sql`.

17 High Availability
18 Structured Query Language
19 Structured Query Language

6.2.10 Database Specific Aspects

Before upgrading, be sure to back up the existing data to be able to roll back your upgrade when required.

- You have multiple options:
  - Cold backup of the whole database
  - Hot backup of the whole database
  - Backup the SAP CC old schema
  - Duplicate the SAP CC old schema and upgrade the duplicated schema

  See your database documentation to find the best way of achieving these tasks.

  > → Recommendation
  >
  > The prefered option is the duplication.
  >
  > With this option, the SAP CC server system is completely unmodified and the old service can restart at any time. But this assumes that the database can handle a duplicated schema in terms of data size.
- Refer to the section Backup and Restore for SAP Convergent Charging and third parties.

6.2.11 Upgrade of Distributed and High-Availability Systems

The upgrade of high-availability systems differs from the upgrade of normal systems.

> ⚠ Caution
>
> Make sure you apply the procedures dedicated to high-availability (HA) systems if your SAP CC system landscape manages real-time charging services.

The high-availability concept of SAP Convergent Charging 2020 includes the real-time charging services.

> ⓘ Note
>
> The high-availability concept of SAP Convergent Charging 2020 does not include the data provisioning services (master data, business data, and customer master data). It is not possible to create or maintain any master data in the SAP CC system during the upgrade processing:
>
> - Master data: pricing catalogs of service providers
> - Customer master data: subscriber accounts, contracts, or business data tables

Core Server

The SAP CC Core Server system can be distributed in your system landscape to ensure high service levels (SLA) for real-time charging services.

High-availability systems represent a key constraint for the upgrade process. When a primary system instance fails (for several reasons), the other running instances get and share the business traffic until the failed instance is up again.

In addition, the SAP CC Core Database instance is replicated across multiple hosts in a cluster. When a primary node fails, proprietary switchover software automatically switches the failed software unit to another hardware node in the cluster (see Oracle RAC[20] technology or IBM DB2 pureScale).

Diameter Server

The SAP CC Diameter Server system can be distributed in your system landscape to ensure high service levels (SLA) for real-time charging services based on the Diameter protocol.

High-availability (HA) systems use switchover clusters to guarantee the availability of the system. SAP CC Diameter Server is across multiple hosts in a cluster. When a primary node fails, proprietary switchover software automatically switches the failed software unit to another hardware node in the cluster.

6.2.12 Upgrade of CA IntroScope (IS) for SAP

When CA IntroScope (IS) for SAP is the monitoring application for your SAP CC systems, you plan the upgrade of the running CA IS Enterprise Manager system with the release 9.7.

Refer to the SAP Note 1565954 for more information.

> **Note**
>
> SAP recommends that you also implement SAP Solution Manager release 7.2 in your SAP system landscape to support SAP Convergent Charging 2020 and CA Introscope 9.x:
>
> - When you operate with SAP Solution Manager release 7.0 or 7.1, plan the upgrade of your technical landscapes.
> - When you operate with SAP Solution Manager release 7.2, plan an update to implement the latest support package and patch level. SPS 6 is the minimal recommended version.
>
> See also: Upgrade of SAP Solution Manager [page 53]

For more information, refer to the CA Introscope for SAP CC [page 17] section of this documentation.

6.2.13 Upgrade of SAP Solution Manager

When you use SAP Solution Manager as central monitoring system (CEN), upgrade your productive edition of SAP Solution Manager with the release 7.2 and implement the latest support package stack (SPS) and patch level in your SAP system landscape. SPS 6 is the minimal recommended version.

[20] Real Application Cluster

For more information, refer to the SAP Solution Manager for SAP CC [page 16] section of this documentation.

6.3 Preparations for the Upgrade

This section of the SAP CC product documentation and user assistance contains important information about preparatory tasks that you need to complete before starting your update or upgrade process. Consider the steps of the preparation phase.

1. Identifying the Hosts for the Preparation [page 54]
2. Checking the Version of the SAP System to Upgrade [page 55]
3. Identifying the Updater Instances for the Core Server System (HA Only) [page 56]
4. Checking Free Space [page 58]
5. JVM Option Settings [page 59]
6. System Encoding [page 59]
7. Making a Backup Copy of the Launch Scripts for the Server Systems [page 60]
8. Making a Backup Copy of the Software Configurations for the Introscope Agents [page 60]
   Back up the configurations of the installed Introscope Agents that run within the Java process of the monitored SAP CC systems.
9. Preparing the Back-End Database Systems [page 61]
10. Preparing the Upgrade of Client Applications [page 63]
11. Preparing the Upgrade of the Cockpit Applications [page 63]
    You need to prepare the upgrade of the Cockpit web applications that are deployed in your SAP system landscapes.
12. Preparing the Installation/Upgrade Folders [page 63]
13. Checking the State of the SAP Start Services [page 64]
14. Checking the Enablement of the Event Date Shifting in Core Server (Integrated SAP Solution with SAP CI) [page 65]
15. Checking the Status of the Network Data Transfer (NDT) Mechanism [page 66]
    As a system administrator you want to check if the network data transfer (NDT) mechanism is implemented in your SAP system landscapes.

6.3.1 Identifying the Hosts for the Preparation

To upgrade the SAP CC server systems, you will have to launch the SAP installer tool:

- On the global host for the Core Server system
- On the machine hosting the:
  - BART Server system
  - Diameter Server system

You must identify all these hosts before starting the upgrade.

> **Note**
>
> To find the global host for the SAP CC Core Server system:
>
> - Go to the following directory on any machine hosting instances of the Core Server:
>   - `<DRIVE>:\usr\sap\<SID>\SYS\profile\ (Microsoft Windows operating system)`
>   - `/sapmnt/<SID>/profile/ or /usr/sap/<SID>/profile/ (Linux and UNIX operating systems or other operating systems)`
> - Open the profile file `DEFAULT.PFL`
> - Find the value of the variable `SAPGLOBALHOST`

Parent topic: Preparations for the Upgrade [page 54]

Next: Checking the Version of the SAP System to Upgrade [page 55]

6.3.2 Checking the Version of the SAP System to Upgrade

You need to check if the version of the SAP CC software allows the upgrade to SAP CC 2020.

Refer to the Upgrade Path [page 49] section in this document for more information.

Core Server System

- Use the SAP CC Admin+ [page 207] user interface to retrieve the technical version number of the instances of the SAP CC Core Server system to upgrade. Use the command: `get VERSION`
- Check that the version of the former SAP CC software is at least '4.6.2.0' (the forth digit may be greater)

Other SAP CC Systems

- Open the log file of any system instance of the SAP CC server systems to upgrade
- Locate the line containing the instruction `LAUNCHER_STARTED`. It contains the version number of the SAP CC software installed (on 4 digits).
- Check that the version of the former SAP CC software is at least '4.6.2.0' (the forth digit may be greater).

If the version of the software of the SAP CC system does not satisfy this condition, update the SAP CC system to the latest support package (SP) of SAP CC 4.1 before attempting to upgrade to SAP CC 2020.

SAP CC Database

For an SAP CC Core Server system, you need to check if the version of the SAP CC Core Database allows the upgrade to SAP CC 2020:

- Execute the following query on the database of the system:

```
SELECT VERSION_ID
FROM VERSION
WHERE INSTALL_DATE = (SELECT MAX(INSTALL_DATE) FROM VERSION);
```

- Check that the result of the query is at least: `4.6.2`

If the version of the database of the SAP CC system does not satisfy this condition, update the SAP CC system to the latest support package (SP) of SAP CC 4.1 before attempting to upgrade to the SAP CC 2020 newest release.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Identifying the Hosts for the Preparation [page 54]

Next: Identifying the Updater Instances for the Core Server System (HA Only) [page 56]

6.3.2.1 Checking the Version of the SAP CC Cockpit to Upgrade

Verify that the running Cockpit application version is "5.0.0.2.PL", where PL is the patch level already implemented in your SAP system landscape. Use the *About* page of the Convergent Charging Cockpit user interface or use the logs/traces of the web application.

6.3.3 Identifying the Updater Instances for the Core Server System (HA Only)

To upgrade the SAP CC Core Server system in high-availability (HA) environment, all the updater instances must be stopped before.

During the upgrade processing the data provisioning of master data, business data, or customer master data is not possible.

The SAP CC users and the external systems or application cannot configure this data during the upgrade.

Procedure

- You must identify the updater instances of your SAP CC system
- Write down the instance names for each hostname in the SAP CC system landscape

> **Note**
>
> To find the updater instances of the SAP CC Core Server system:
>
> Go to the following directory on any machine hosting instances of the Core Server system:
>
> - `<DRIVE>:\usr\sap\<SID>\SYS\profile\ (Microsoft Windows operating system)`
> - `/sapmnt/<SID>/profile/ or /usr/sap/<SID>/profile/ (Linux and UNIX operating systems or other operating systems)`
>
> Locate the files containing the instance profile: they are named with the format `<SID>_<INSTANCE_NAME>_<HOSTNAME>`.
>
> Where:
>
> - `<SID>`: The SAP system identifier (3 characters)
> - `<INSTANCE_NAME>`: The name of the SAP CC system instance
>
> The files with an instance name starting with the string `CCU` correspond to the updater instances of the SAP CC system.
>
> Get the host name from the appropriate profile file

> **Example**
>
> An SAP CC system landscape includes two updater instances: CCU23 and CCU34.
>
> The first updater instance is installed on a host machine HSTA. Several instances of several SAP systems are installed on this host machine. CCU23 is the 23rd instance of an SAP system.
>
> The second updater instance of an SAP CC system is installed on another host machine (HSTB) belonging the same landscape. CCU34 is the 34th instance of an SAP system.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Checking the Version of the SAP System to Upgrade [page 55]

Next: Checking Free Space [page 58]

6.3.4 Checking Free Space

Checking Free Space for Updated Files

During the upgrade process, the SAP installer copies new versions of the software units (SAP CC, SAP JVM, CA Introscope Agent) to the central repository. When the instances of the SAP CC system are restarted after the upgrade, these archives are copied to the local directories of each instance of anSAP CC system.

You need to check that there is enough free space available in these directories for these files:

Operating System	Free Space in the Central Repository	Free Space in each Instance Directory
Microsoft Windows 64-bit	1,100 MB	600 MB
AIX	650 MB	650 MB
HP-UX	1,050 MB	1,050 MB
Linux and UNIX operating systems	650 MB	650 MB
Solaris	550 MB	550 MB

> ⓘ Note
>
> The free space required depends on the operating system (OS) on which the SAP CC system is installed.

Checking Free Space for Temporary Files (Linux and UNIX like operating systems only)

For all Linux and UNIX like operating systems, the `/tmp` directory is used by the SAP installer tool to create temporary files. Each time you launch the installer, you need to check that you have 400MB free space for storing these temporary files.

> ⓘ Note
>
> The installer creates the following directories:
>
> - `/tmp/sapinst_dir`
> - `/tmp/sapinst_exe*`
>
> You can delete the `/tmp/sapinst_exe*` content once the installation with SL Common GUI is completed to get free space. On the contrary, it is recommended to keep `/tmp/sapinst_dir` because it contains the installation log files, which could be useful in case an issue arises during later installations.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Identifying the Updater Instances for the Core Server System (HA Only) [page 56]

Next: JVM Option Settings [page 59]

6.3.5 JVM Option Settings

If your production system was installed prior to SAP CC 4.1 SP 2, we recommend that you verify the startup configuration files (`jstart.config`) of the deployed SAP CC Core Server system.

This recommendation applies to all the preproduction, quality assurance, and performance test systems of SAP Convergent Charging in your system landscapes.

- Verify if the jstart.config files contain parameters dedicated to the Concurrent Mark Sweep (CMS) garbage collector such as:
  - `-XX:+UseConcMarkSweepGC`
  - `-XX:+CMSIncrementalMode`
  - `-XX:+CMSIncrementalPacing`
  - `-XX:CMSInitiatingOccupancyFraction`
  - `-XX:CMSIncrementalDutyCycleMin`
  - `-XX:CMSIncrementalDutyCycle`
- If yes, schedule an additional maintenance plan before or after your current upgrade plan. Refer to the SAP Note 2006073 to determine the best settings for your systems
- Consult your support specialists for more information

Parent topic: Preparations for the Upgrade [page 54]

Previous: Checking Free Space [page 58]

Next: System Encoding [page 59]

6.3.6 System Encoding

Refer to Setting Up the System Encoding [page 100] section.

Parent topic: Preparations for the Upgrade [page 54]

Previous: JVM Option Settings [page 59]

Next: Making a Backup Copy of the Launch Scripts for the Server Systems [page 60]

6.3.7 Making a Backup Copy of the Launch Scripts for the Server Systems

The upgrade process will modify automatically the launch scripts dedicated to the instances of your SAP CC server system. These files are located in the central repository of the global host. You need to back up these files and identify the files different from the default files provided with SAP Convergent Charging.

Do a backup copy of the following launch scripts of the server systems:

- The files named `jstart.config` located in the directory:
  - `<DRIVE>`:\usr\sap\`<SID>`\SYS\profile\jstart\ (MS Windows operating system)
  - /sapmnt/`<SID>`/profile/jstart/ (UNIX and Linux or other operating systems)

Parent topic: Preparations for the Upgrade [page 54]

Previous: System Encoding [page 59]

Next: Making a Backup Copy of the Software Configurations for the Introscope Agents [page 60]

6.3.8 Making a Backup Copy of the Software Configurations for the Introscope Agents

Back up the configurations of the installed Introscope Agents that run within the Java process of the monitored SAP CC systems.

Check if the CA APM Introscope Agent is installed with your running SAP CC systems. This bytecode agent for Java is used by the CA IS Enterprise Manager system to collect performance and health metrics in order to monitor your SAP CC systems. SAP Solution Manager also gets these metrics.

In this situation, check if your installations contain some custom configurations such as:

- Agent unlocking
- Specific probes activation

Agent profiles are stored in files named `IntroscopeAgent.profile` and colocated with the SAP CC system installation directories.

Back up the corresponding software configurations in order to be able to restore these specialized settings in case of a failure of the upgrade process:

- The file named `IntroscopeAgent.profile` is located in the directory:
  - `<DRIVE>:\usr\sap\<SID>\<INSTANCE_ID>\config\` (MS Windows operating system)
  - `/usr/sap/<SID>/<INSTANCE_ID>/config/` (UNIX and Linux or other operating systems)

> → Tip
>
> For more information about this Introscope bytecode agent for Java, see the SAP Note 1453216.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Making a Backup Copy of the Launch Scripts for the Server Systems [page 60]

Next: Preparing the Back-End Database Systems [page 61]

Related Information

Backup and Restore
Detailed Backup and Restore

6.3.9 Preparing the Back-End Database Systems

Checking the Backup Strategy for the Database Schemas [page 61]

Creating New Tablespaces or Filegroups in the Core Database [page 61]

Creating Tables and Views in the Session Database [page 62]

Parent topic: Preparations for the Upgrade [page 54]

Previous: Making a Backup Copy of the Software Configurations for the Introscope Agents [page 60]

Next: Preparing the Upgrade of Client Applications [page 63]

6.3.9.1 Checking the Backup Strategy for the Database Schemas

Before upgrading, be sure to back up the existing data to be able to roll back your upgrade.

Refer to the section Backup and Restore in the SAP CC 2020 Administration Guide documentation.

6.3.9.2 Creating New Tablespaces or Filegroups in the Core Database

> → Remember
>
> The following procedure is not relevant for SAP HANA database.

According to your database technology, you must determine and create the following tablespaces or filegroups, that are mandatory in release 2020:

- `MONITORING_DATA`
- `MONITORING_INDX`

These tablespaces or filegroups include data and database indexes related to the new technical feature in SAP CC 5.0 SP 1 that allows you to store metadata information about the data files generated by the Core Server system during business operations (charging, refilling, and activation). These data files are also known as "item files" in the SAP documentation.

After the upgrade, your Core Database includes such data when you activate the feature. For further information, refer to:

- The `FILE_METADATA_RECORDING_ENABLED` system parameter
- The database preparation procedure that relates to your RDBMS in the Installation Procedures [page 88] procedures of this documentation

6.3.9.3 Creating Tables and Views in the Session Database

If your business implements the “Dual Database” feature, your SAP Convergent Charging landscape contains the Session Database. As of SAP CC 5.0 SP 3, a new Table Rollover for Rating Sessions feature is available to avoid the fragmentation of the `RATING_SESSION` table in case of a huge volume of stored rating sessions.

> **Note**
>
> For further information about the Dual Database feature, refer to the dedicated SAP Note 2189251.

To use this new feature, it is necessary to execute the following procedure:

- Grant the `CREATE VIEW` privilege to the SAP CC Session Database administrator using the following commands:

```
sqlplus / as sysdba
grant CREATE VIEW TO <sessiondb_user>;
```

- Get the `patch4512to503_oracle_enterprise_sessiondb.sql` script located into the `CC_CORE_SERVER/session_sql/` directory
- Update the content of the Session Database using the following command:

```
sqlplus <sessiondb_user>/<sessiondb_pwd> <
patch4512to503_oracle_enterprise_sessiondb.sql
```

After the upgrade, your SAP CC Session Database includes the relevant elements that are mandatory to use the Table Rollover for Rating Sessions feature.

6.3.10 Preparing the Upgrade of Client Applications

You need to prepare the upgrade of software applications or systems that are Java based and that use the Java Libraries provided by SAP Convergent Charging (ex: mediation application to communicate with).

Determine if some Java based applications use the Java Libraries of SAP Convergent Charging.

Determine the upgrade sequence and procedures for each application.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Preparing the Back-End Database Systems [page 61]

Next: Preparing the Upgrade of the Cockpit Applications [page 63]

6.3.11 Preparing the Upgrade of the Cockpit Applications

You need to prepare the upgrade of the Cockpit web applications that are deployed in your SAP system landscapes.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Preparing the Upgrade of Client Applications [page 63]

Next: Preparing the Installation/Upgrade Folders [page 63]

6.3.11.1 Checking the Tomcat Settings

In the Apache Tomcat Server system that runs a Cockpit web application, determine the current installation/deployment settings:

- Basic installation and automatic deployment in a Tomcat Server: Verify the app name and path in the Java Web Server system. Check the `<TOMCAT_HOME>/webapps/` web app base directory.
- Advanced installation based on a Tomcat Context: Verify the location and usage of the WAR file that was installed previously. Check the location of the Cockpit configuration file.

6.3.12 Preparing the Installation/Upgrade Folders

You copy the installation material to each machine hosting an SAP CC server system.

Prepare the Installation/Upgrade Folders for the Cockpit Upgrades [page 64]
You copy the installation/upgrade materials to each machine hosting an Apache Tomcat Server system that runs a Cockpit web application to maintain.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Preparing the Upgrade of the Cockpit Applications [page 63]

Next: Checking the State of the SAP Start Services [page 64]

6.3.12.1 Prepare the Installation/Upgrade Folders for the Cockpit Upgrades

You copy the installation/upgrade materials to each machine hosting an Apache Tomcat Server system that runs a Cockpit web application to maintain.

1. Get the ZIP file of the version 2020 FPS 2 (or a later FPS). It is available on the installation media in the directory `CC20202022_TOOLS_CONTENT_UC_OSIND`.

   > **Note**
   >
   > See the Downloading the Installation Materials [page 92] section for more information about the media and the ZIP files to use.

2. Uncompress the delivered ZIP file.
3. Log on to the Tomcat host by using the operating system (OS) user running the Tomcat Server system. See Installing and Launching Cockpit [page 201].
4. On this Tomcat host, in the HOME directory, create a *working directory* and copy the uncompressed files and subfolders.
5. Check that you have extracted and copied the following items:
   - A WAR file `cccockpit.war`
   - A configuration template file `sap_cc_cockpit_[WEB_APP_NAME].properties.sk`

6.3.13 Checking the State of the SAP Start Services

You have to check that the SAP Start Services run for each system instance on every host of the SAP CC server systems. It is implemented as a service on Microsoft Windows, and as a daemon on Linux and UNIX operating systems.

Perform one of the following procedures on each host of the SAP CC server system, depending on the operating system:

- Microsoft Windows:
  - Open the *Computer Management* window

    - Open the view *Services and Applications* > *Services*
    - The services related to the system are named SAP<SID>_<INSTANCE_NB> (for example SAPSCC_00). Check that all the services of the SAP CC system are running. If they are not, start them (starting the service does not start the related server instance).
- Linux and UNIX operating systems:
  Execute the script `/usr/sap/sapservices` as the root user. It starts all the services of the host that are not already started.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Preparing the Installation/Upgrade Folders [page 63]

Next: Checking the Enablement of the Event Date Shifting in Core Server (Integrated SAP Solution with SAP CI) [page 65]

6.3.14 Checking the Enablement of the Event Date Shifting in Core Server (Integrated SAP Solution with SAP CI)

In an integrated SAP Solution with SAP Convergent Invoicing (SAP CI) in SAP ERP/FI-CA, you must enable the Event Date Shifting function in the rater instances of your running SAP CC Core Server system. There is no basic system parameter to view but there is an advanced configuration record to check.

In the SAP CC Setup Tool [page 211] user interface, export the CIF[21] configuration settings and check that the `eventDateShifter` XML nodes are present in the configuration file.

To verify the enablement of the Event Date Shifting function:

- Open a terminal in the `/usr/sap/<SID>/CCD<INST_NB>/script/` directory on the SAP Global Host
- Determine if the SAP CC system is configured to send both billable items and consumption items to SAP CI in SAP ERP/FI-CA:
    - With the Admin+ [page 207] user interface, check the value of the EIF_EXTERNAL_SYSTEM_CODE_CHARGEABLE_ITEM system parameter. The value must be `CI`
- Export the CIF configuration for the rater instances of the SAP CC system. As a system administrator, use the Setup Tool [page 211] user interface with the `cif export` command:

  ```
  ./setup.sh cif export rater <TO_FILE_PATH>
  ```
- View the exported file with a text editor
- Check the content. An `eventDateShifter` XML element is required either:
    - Around each `sequence` XML element which surrounds a `billableItemValidator` XML element
    - Around each `billableItemValidator` XML element which is not surrounded by a `sequence` XML element
- If the function is disabled, refer to Enabling the Event Date Shifting Mechanism for Data Files Generated by the Rater Instances section of the SAP CC 2020 Tuning Guide to enable it

[21] Charging output Integration Framework

Parent topic: Preparations for the Upgrade [page 54]

Previous: Checking the State of the SAP Start Services [page 64]

Next: Checking the Status of the Network Data Transfer (NDT) Mechanism [page 66]

6.3.15 Checking the Status of the Network Data Transfer (NDT) Mechanism

As a system administrator you want to check if the network data transfer (NDT) mechanism is implemented in your SAP system landscapes.

In the running Core Server systems, check the values of the RATER_PARTITION_SWITCH_CACHE_TRANSFER_ENABLED and GUIDER_PARTITION_SWITCH_CACHE_TRANSFER_ENABLED system parameters.

If the values are set to `true`, contact the team members of the implementation and integration project or contact your SAP Support Team for some expert guidance and to verify if this mechanism is relevant for your production SAP system landscapes.

Parent topic: Preparations for the Upgrade [page 54]

Previous: Checking the Enablement of the Event Date Shifting in Core Server (Integrated SAP Solution with SAP CI) [page 65]

6.4 Upgrading

> → Remember
>
> From SAP Convergent Charging 5.0 SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

This section of the SAP CC product documentation and user assistance provides you with information to upgrade all the SAP CC server systems and other components in your SAP system landscapes.

Depending on your upgrade plan, you perform the following main steps:

- Upgrading the SAP CC Server Systems [page 67]
- Upgrading the Client Applications [page 80]
- Upgrading the SAP CC GUIs [page 81]
- Upgrading the Cockpit Web Applications [page 82]

> ⚠ Caution
>
> In a high-availability (HA) system landscape, make sure that you conform to the following conditions:
>
> 1. Upgrade of the SAP CC Core Server systems:
>   - If your SAP CC system is installed in a high-availability environment, make sure that only the mediation system that requests real-time charging services is connected to the SAP CC Core Server system. The other systems (CRM, ERP, provisioning, invoicing, and billing) must not connect to the Core Server system during the upgrade process
>   - The business traffic must be limited or off-peak. Avoid peak traffic periods.
>   - The data provisioning services are stopped during the upgrade processing:
>   - It is not possible to configure some objects in the pricing catalogs with the SAP CC Core Tool user interface.
>   - External systems cannot create or maintain master data or customer master data in your SAP CC system.
> 2. Upgrade of the SAP CC Diameter Server systems:
>   - If your SAP CC system is installed in a high-availability switch-over environment, make sure that the failover capabilities of the cluster switch-over software are disabled during the downtime. This is to prevent unwanted failover actions during stop actions of the upgrade procedure.

> ⚠ Caution
>
> In a non high-availability system landscape, the upgrade process requires you to stop all the SAP CC systems to maintain. The business traffic must be limited.

Upgrading the SAP CC Server Systems [page 67]

Using the Accessibility Mode [page 78]

Restarting the SAP installer (Upgrade GUI) [page 79]

Resetting the Upgrade of a Server System [page 79]

Upgrading the Client Applications [page 80]

Upgrading the SAP CC GUIs [page 81]
As part of the upgrade project from release 5.0 to 2020, you must upgrade all the already-deployed user interfaces in your SAP system landscapes.

Upgrading the Cockpit Web Applications [page 82]
Use this procedure when you want to upgrade your Cockpit applications from release 5.0 to 2020.

Follow-Up Activities [page 83]

6.4.1 Upgrading the SAP CC Server Systems

1. Initialization Stage [page 68]
2. Processing Stage [page 70]
3. Post-Processing Stage [page 76]

Next Steps

1. Upgrading the Client Applications [page 80]
2. Upgrading the SAP CC GUIs [page 81]
3. Upgrading the Cockpit Web Applications [page 82]
4. Follow-Up Activities [page 83]

Initialization Stage [page 68]

Processing Stage [page 70]

Post-Processing Stage [page 76]

6.4.1.1 Initialization Stage

1. Shutting Down the Server System [page 68]
2. Backing Up the Software [page 69]
3. Backing Up the Database [page 70]

6.4.1.1.1 Shutting Down the Server System

To start the upgrade procedure for a normal system, you first have to shut down the SAP CC server system you want to upgrade to version 2020. This upgrade cannot be done online.

If your system is installed in a high-availability (HA) environment, you have only to shut down all the updater instances of the Core Server system.

> ⚠ Caution
>
> For high-availability services (real-time charging), do not stop all the instances of the SAP CC Core Server system but only the updater instances.

HA Setup

To shut down the updater instances of the SAP CC Core Server system:

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to upgrade
- Select the system

- Select the updater instance (CCUxx) and right-click
- In the context menu, choose *Stop...*
  - You may have to enter the logon and password of the SAP system user (<SID>adm)
- Wait until this instance is fully stopped
- Apply the same procedure with the other updater instances in the running SAP CC system

Non HA Setup

To shut down an SAP CC server system (Core Server, BART Server):

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to stop.
- Select the system and then, from the context menu, choose *Stop...*
  - You may have to enter the logon and password of the OS[22] user for the SAP System Administrator (named <SID>adm)
- Wait until all the instances in the SAP CC system are fully stopped

> ⓘ Note
>
> For further information about the Start and Stop procedures, refer to the Starting and Stopping the Server Systems [page 144] section of this documentation.

Parent topic: Initialization Stage [page 68]

Next: Backing Up the Software [page 69]

6.4.1.1.2 Backing Up the Software

Perform the procedures decided during the preparation phase.

Parent topic: Initialization Stage [page 68]

Previous: Shutting Down the Server System [page 68]

Next: Backing Up the Database [page 70]

[22] Operating System

6.4.1.1.3 Backing Up the Database

Perform the procedures decided during the preparation phase.

Back up the data in the Core Database system and in the BART Database system.

Parent topic: Initialization Stage [page 68]

Previous: Backing Up the Software [page 69]

6.4.1.2 Processing Stage

1. Starting the Upgrade GUI [page 70]
2. Making Entries for the SAP installer [page 71]
3. Processing by SAP installer [page 73]
4. Restoring the Configuration of the SAP CC Introscope Agent [page 73]
5. Restarting the Upgraded Server System Manually [page 74]

6.4.1.2.1 Starting the Upgrade GUI

The upgrade of the SAP CC server systems in a system landscape is performed by using the upgrade tool that is the SAP installer. Its configurations are similar than during the implementation of a support package.

> ⓘ Note
>
> The instances which have been shut down in preparation of the upgrade will not be restarted automatically at the end of the upgrade operations performed by the installer. Only the instances running at the time of the upgrade will be restarted automatically by the installer.
>
> In HA[23] setup, a manual restart is recommended.

- Go to the `CC50_IM_<OS>` installation material where `<OS>` is your OS[24]
- Start the SAP installer by running the `sapinst.exe` program for Microsoft Windows operating systems or the `sapinst` script for Linux and UNIX operating systems
- Select the option *SAP Convergent Charging* > *Software Life-Cycle Options* > *Update* > *Update - SAP Systems*
- Make entries for the upgrade preprocessing done by the SAP installer

[23] High Availability
[24] Operating System

> **Note**
>
> - You can use the accessibility mode: refer to the Using the Accessibility Mode [page 78] section of this documentation
> - You can have to restart the upgrade program: refer to the Using the Accessibility Mode [page 78] section of this documentation
> - If you have to reset the upgrade process for a server system, refer to the Resetting the Upgrade of a Server System [page 79]

Parent topic: Processing Stage [page 70]

Next: Making Entries for the SAP installer [page 71]

6.4.1.2.2 Making Entries for the SAP installer

Enter the following information or confirm the existing values when prompted by the SAP installer:

1. **Choose Option** screen:
   (Installation Mode)
   - *Typical*: the upgrade is performed with default settings. You only have to respond to a small selection of prompts
   - *Custom*: you are prompted for all the parameters

   > **Note**
   >
   > If you do not choose the *Custom* mode, you can still change later any parameter on the *Parameter Summary* screen.
2. **Define Parameters** screen:
   - *SAP System Identification*: specify the Profile Directory of the system to upgrade
   - *SAP System Administrator Password*: enter the password of the SAP System Administrator (Microsoft Windows with remote hosts only)
   - *Update Oracle JDBC Driver*: use this screen to select a new JAR archive file containing the relevant Oracle JDBC[25] driver to use
   - *Database Update*: select the SQL[26] scripts to execute.

     > → Recommendation
     >
     > By default, the SQL scripts necessary for the upgrade are selected. It is recommended not to modify the default selection.
   - *Update Archives*: specify the archives to be used for the upgrade

[25] Java Database Connectivity
[26] Structured Query Language

> **Note**
>
> - By default, only the archive corresponding to the SAP CC system being upgraded (`CC_<SYSTEM>_SERVER.SAR`) is selected
> - Select the archive for CA Introscope for SAP (`ISAGENT.SAR`) only if you want to update the version of the CA Introscope Agent that is installed with SAP CC
>
> > **→ Recommendation**
> >
> > CA Introscope 9.7 must be available in the system landscape. Upgrade first your CA IS Enterprise Manager system. SAP recommends that you also update the archive for the SAP JVM (SAPJVM8.SAR).

- *Restart Instances*: choose if you want to automatically restart the running SAP CC instances at the end fo the upgrade process.
  At the end of the upgrade process, each SAP CC instance needs a restart to make the upgrade effective. As long as an instance is not restarted, it keeps on running in its previous version.

  > **→ Recommendation**
  >
  > The *Restart SAP Convergent Charging instances after installation* option is ticked by default, meaning that SAP CC instances will be automatically restarted by SAP installer at the end of the upgrade process.
  >
  > - In production (or preproduction) landscapes, SAP SE highly recommends to untick this option. This will allow you to manually manage the SAP CC instances restart sequence and thus keep control of the upgrade.
  > - In HA[27] environments, SAP SE also recommends to untick this option. This will allow you to check that all the updater instances are stopped.

  > **Note**
  >
  > In development or test landscapes, you can leave this option ticked

3. **Summary** screen: the installer displays the summary of the actions scheduled. You can modify any of the selected parameters by clicking on the appropriate checkboxes and then clicking on the *Revise* button.

The installer displays a final summary for the operations it will perform. When you are ready, click *Next* to start the next stage.

Parent topic: Processing Stage [page 70]

Previous: Starting the Upgrade GUI [page 70]

Next: Processing by SAP installer [page 73]

[27] High Availability

6.4.1.2.3 Processing by SAP installer

The installer tool performs mostly the Processing stage of the upgrade process but the system administrator has to complete the processing.

The tool performs the following actions during this stage.

1. Backup unpacked archives in the central repository to allow you to restore previous software versions
2. Unpack the selected archives containing the new versions of the software units to upgrade and store them in the central repository. The archive for the SAP CC software is systematically updated. The archives for CA Introscope and the SAP JVM[28] may also be updated during this stage
3. Upgrade the back-end database of the SAP CC server system (if applicable for the system being upgraded) by executing the appropriate SQL[29] upgrade scripts
4. Update the launch configuration of all the server instances to adapt the Java classpath of the server instances to take into account the new **JAR** files introduced in the new main version of SAP CC software
5. In a high-availability (HA) environment, sequentially restart all the instances except the updater instances (Core Server only)

Parent topic: Processing Stage [page 70]

Previous: Making Entries for the SAP installer [page 71]

Next: Restoring the Configuration of the SAP CC Introscope Agent [page 73]

6.4.1.2.4 Restoring the Configuration of the SAP CC Introscope Agent

As of SAP CC 4.1 SP 2, the version of the SAP CC Introscope Agent is updated from the 9.1.5 version to the 9.7.1 version. As this new version is automatically deployed by the SAP installer during the upgrade process, you must restore all the modification (such as agent unlocking or specific probes activation) you performed on your Introscope Agent before restarting the SAP CC systems.

Parent topic: Processing Stage [page 70]

Previous: Processing by SAP installer [page 73]

Next: Restarting the Upgraded Server System Manually [page 74]

[28] Java Virtual Machine
[29] Structured Query Language

6.4.1.2.5 Restarting the Upgraded Server System Manually

Once the SAP installer ended the upgrade operations, you must manually restart the instances of the Core Server system that were not running during the upgrade process (or because you unticked the *Restart SAP Convergent Charging instances after installation* option). This restarting operation really performs the upgrade of the software from the central repository to the repository of each instance of an SAP CC server system. In case of issue, check the logs to get detailed information about the encountered errors. For further information about instances restarting procedures, refer to the Starting and Stopping the Server Systems [page 144] section of this documentation.

> → Recommendation
>
> When upgrading an SAP CC Core Server system, SAP SE recommends you to:
>
> - Sequentially restart all the dispatcher instances first
> - Restart the updater instances
> - Restart the other instances **sequentially** (waiting until an instance has restarted before restarting another one), respecting the following order: guiders, raters, bulkloaders, and taxers.

> ⚠ Caution
>
> In high-availability landscapes, SAP SE recommends that you determine the number of instances (for each type of instance) that must run simultaneously to handle the expected business traffic.

Core Server in HA Landscape

To start again the updater instances manually:

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to manage
- Select the updater instances in the system and then, from the context menu, choose *Start...*

  > ⓘ Note
  >
  > You may have to enter the logon and password of the OS[30] user for the SAP System Administrator (named `<SID>`adm)

- Wait until all the updater instances of the SAP CC system are fully started

To restart an instance (of the SAP CC Core Server system) manually:

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to manage
- Select the instance in the system and then, from the context menu, choose *Restart...*

[30] Operating System

> **Note**
>
> You may have to enter the logon and password of the OS user for the SAP System Administrator (named <SID>adm)

- Wait until the instance of the SAP CC system is fully started and has finished its warm-up

Core Server in non-HA Landscape

To start the SAP CC Core Server system manually:

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to start again
- Select the system and then, from the context menu, choose *Start...*

> **Note**
>
> You may have to enter the logon and password of the OS user for the SAP System Administrator (named <SID>adm)

- Wait until all the instances of the SAP CC system are fully started

BART Server

To restart an SAP CC server system (Core Server, BART Server) manually:

- Start the SAP Management Console
- In the navigation pane, open the tree structure and navigate to the node of the SAP CC system that you want to start
- Select the system and then, from the context menu, choose *Start...*

> **Note**
>
> You may have to enter the logon and password of the OS user for the SAP System Administrator (named <SID>adm)

> **→ Tip**
>
> For further information about restarting servers, refer to the Starting and Stopping the Server Systems [page 144] section of this documentation.

Parent topic: Processing Stage [page 70]

Previous: Restoring the Configuration of the SAP CC Introscope Agent [page 73]

6.4.1.3 Post-Processing Stage

1. Checking the Restart of the Server System [page 76]
2. Resetting or Verifying the Tracing Domains in the Core Server System [page 76]
3. Cleaning the Platform [page 77]
4. Changing the Database Users of the Core Database and that Are Dedicated to the SAP CC Cockpit Web Applications [page 77]

6.4.1.3.1 Checking the Restart of the Server System

You check the status of the SAP CC server system and control its logs. The log format has changed since release 3.0 of SAP Convergent Charging. Refer to the SAP CC 2020 Application Help documentation for more information about the formats.

The startup process has changed since release 3.0 of SAP Convergent Charging: a system instance tries first to connect to the back-end database system. The UDP[31] discovery function is now disabled.

- Check the error and warning entries recorded in the log files
- Determine the resolution and the necessary corrective or preventive actions

> ⓘ Note
>
> Refer to the SAP CC 2020 Log Message Reference and Error Troubleshooting documentation for more information about the logs and the corrective or preventive action plans to setup.

Parent topic: Post-Processing Stage [page 76]

Next: Resetting or Verifying the Tracing Domains in the Core Server System [page 76]

6.4.1.3.2 Resetting or Verifying the Tracing Domains in the Core Server System

As of SP 1, SAP recommends that you verify the values of the LS_TRC_DOMAIN system parameter in your running SAP CC Core Server systems. New tracing domains are available: DATA, PERF, and PPI. In a production SAP CC system, set up the system parameter with all the existing domains.

You can use the `reset` command of the Admin+ [page 207] user interface

Parent topic: Post-Processing Stage [page 76]

[31] User Datagram Protocol

Previous: Checking the Restart of the Server System [page 76]

Next: Cleaning the Platform [page 77]

6.4.1.3.3 Cleaning the Platform

You archive the logs of the installer and complete the upgrade:

- As a system administrator, you archive the logs generated by the installer tool. These logs are no longer required but can be needed by SAP Support Team or your technical support (local organization) for error analysis. Refer to the Upgrade Troubleshooting [page 299] section of this documentation. The files are saved in the following directory:
  - Microsoft Windows operating systems: `<DRIVE>:\Program Files\sapinst_instdir\CC2020\SLM\UPDATE\`
  - Linux and UNIX operating systems and other operating systems: `/tmp/sapinst_instdir/CC2020/SLM/UPDATE/`
- You remove the temporary files and directories that are no longer required.

Parent topic: Post-Processing Stage [page 76]

Previous: Resetting or Verifying the Tracing Domains in the Core Server System [page 76]

Next: Changing the Database Users of the Core Database and that Are Dedicated to the SAP CC Cockpit Web Applications [page 77]

6.4.1.3.4 Changing the Database Users of the Core Database and that Are Dedicated to the SAP CC Cockpit Web Applications

If you upgrade to SAP CC 2020 FPS 2 and if you have deployed SAP CC Cockpit web applications, then you need to change the configuration of the database users (or database accounts) for the SAP CC Core Database systems in your SAP system landscapes. **You need to maintain every database user (as a service user) that is dedicated to the communication between an SAP CC Cockpit back-end application and an SAP CC Core Database.** These service users need to be granted a read access to the database views pointing on the CERT_STORE database table and some other tables.

The complete list of SAP CC database tables is:

- **CERT_STORE (as of SAP CC 2020 FPS 2)**
- FILE_METADATA (as of version 5.0 SP 1)
- FILE_METADATA_PATH (as of version 5.0 SP 1)

- METRIC_RECORD (as of version 5.0 SP 2)
- OBJECT_CHANGE_LOG (as of version 5.0 SP 2)
- OBJECT_SNAPSHOT (as of version 5.0 SP 2)
- USER_OPERATION (as of version 5.0 SP 2)

Consult the Database Users/Accounts section about the recommended access rights for this particular database user of an SAP CC Core Database.

SAP CC Database: **Core Database**

Database User/Account: service users dedicated to the SAP CC Cockpit web applications

> → Remember
>
> Maintain these database users or database accounts before the upgrade of your already-installed SAP CC Cockpit web applications.

See Also

- Database Users/Accounts in SAP CC 2020 Security Guide
- Configuring Cockpit in SAP CC 2020 Administration Guide

Parent topic: Post-Processing Stage [page 76]

Previous: Cleaning the Platform [page 77]

6.4.2 Using the Accessibility Mode

You can run the upgrade program in accessibility mode. This offers the following features:

- Keyboard access
- High-contrast color
- Custom font setting

The high-contrast color and custom font setting features are derived from the Windows display properties, which you have to adjust initially.

Activating and Adjusting Accessibility Settings on Windows

Activate and adjust the relevant settings for the font size and color schemes as follows:

- Right click on your Windows desktop and choose *Personalize*
- Select *Adjust font size (DPI)* and choose *Larger scale (120 DPI)* (to define other font size schemes, choose *Custom DPI*)

- In the right-hand pane, select *Window Color and Appearance*. Select a color scheme from the *Color scheme* drop-down box. To define your own color schemes, choose *Advanced*

Switching On the Accessibility Mode

To switch on the accessibility mode, run the SAP installer with the launch option SAPINST_GUI_ACCESSIBLE set to true.

6.4.3 Restarting the SAP installer (Upgrade GUI)

When you restart the installer (as your upgrade GUI[32]), it continues at the point where it had stopped. You may need to restart the SAP installer in the following cases:

- If an error occurred and an upgrade phase terminated
- If a system failure occurred

If an error occurred, you first have to correct the error. For more information about finding and correcting errors, see the Upgrade Troubleshooting [page 299] section of this documentation. If you are not sure how to proceed, contact your SAP Support Team.

Before you restart the upgrade after a system failure, you must make sure that you meet the requirements for restarting the system successfully.

You can restart theSAP installer by executing the same startup script you used to initiate the upgrade process on the right host machine.

6.4.4 Resetting the Upgrade of a Server System

You can reset the upgrade, that is, undo all changes that have been made in the systems (SAP CC, back-end database) by the SAP installer.

For example, you may want to reset the upgrade in the following cases:

- You want to repeat one of the upgrade step to change an entry.
- You ran the upgrade stage for information purposes only and do not want to upgrade the system now, but at a later point in time.
- You cannot complete the upgrade in time and therefore need to reset the system to enable production operation again.

> ⓘ Note
>
> The procedures for resetting the upgrade depend on the stage and the level you have reached.

[32] Graphical User Interface

- You can reset a stage by pressing the *Back* button of the SAP installer during the initialization stage
- Once the summary report is displayed do not use the *Back* button anymore: use the *Revise* button
- The procedures depend on the level you have reached during the processing stage:

Failed Level	Reset Procedure
Database SQL patching	In case an SQL upgrade script fails during the Upgrade Database Schema level, you must restore the latest database backup: • Restore the back-end database (Core Database, BART Database, IEC Database) • Fix the cause of the failure • Restart the upgrade from the beginning **⚠ Caution** Do not retry to launch the SAP installer before the complete restoration of the database.
Other levels	Start the upgrade from the beginning

ⓘ Note

- Before you overwrite or delete the upgrade directory and the `/usr/sap/<SID>` directory, make a backup of these directories. The SAP Support Team or your technical support may need this backup to analyze the error.
- If you want to reset the upgrade after you have upgraded the operating system or database system, you might need to downgrade these software components. For downgrading options, see the vendor´s specification. The safest method is to reload a full backup of the whole system, including the operating system.

6.4.5 Upgrading the Client Applications

If you have Java client applications that use the Java Libraries (APIs) of SAP Convergent Charging 5.0, consult this section to upgrade to the release 2020.

The JAR files from the previous and replaced releases of SAP Convergent Charging used by your client applications or systems, must be replaced by the JAR files from the Software Development Kit of (SDK) of SAP CC 2020.

This procedure is different for each application installed in the SAP CC system landscapes.

ⓘ Note

You may need to change the technical configurations of your client applications that use the SAP CC 2020 Java Libraries in order to take into account the new JAR files.

Perform the procedure identified during the preparation phase (from Identifying the Hosts for the Preparation [page 54] section to Checking the Enablement of the Event Date Shifting in Core Server (Integrated SAP Solution with SAP CI) [page 65] section).

Next Steps

1. Upgrading the SAP CC GUIs [page 81]
2. Upgrading the Cockpit Web Applications [page 82]
3. Follow-Up Activities [page 83]

6.4.6 Upgrading the SAP CC GUIs

As part of the upgrade project from release 5.0 to 2020, you must upgrade all the already-deployed user interfaces in your SAP system landscapes.

Upgrade the Already-Deployed SAP CC User Interfaces (UIs)

To upgrade the graphical user interfaces (GUIs) of SAP Convergent Charging, you need to:

- Backup the previous configuration files of each concerned tool, available in the config directory of each tool.
- Update your JVM to the SAP JVM 8.1 version (as of SAP CC 4.1 SP 2).
- Install the last version of each concerned tool.
- If necessary, restore the previous configuration, by overwriting the configuration file with the backed up one.

> **Note**
>
> If your SAP system landscape contains an import/export connector (IEC) working in conjunction with the BART Server system, copy the new `bart_iec.jar` files from your BART Server installation to the `/jars/` directory of the CAT Tool user interface. An old version of this file should be already present.

For more information about the installation of the last version of the SAP JVM and the SAP CC graphical user interfaces, refer to the dedicated procedures available in the Downloading the SAP JVM [page 94] section of this documentation.

Next Steps

1. Upgrading the Cockpit Web Applications [page 82]
2. Follow-Up Activities [page 83]

6.4.7 Upgrading the Cockpit Web Applications

Use this procedure when you want to upgrade your Cockpit applications from release 5.0 to 2020.

> ⚠ Caution
>
> Before upgrading a replaced release, SAP recommends that you first implement all the support packages and patches for this former release. Please identify these up-to-date versions and maintain your SAP system landscapes. See the relevant SAP Notes.
>
> If you upgrade from **SAP CC 5.0 SP 0**, SP 1, or SP 2, or SP 3, perform the following procedure at https://help.sap.com/viewer/d1d04c0d65964a9b91589ae7afc1bd45/5.0.latest/en-US/f524ea6e1463448cb643583cf893913f.html because of technology changes.

In order to upgrade from 5.0 to 2020:

1. You have implemented the version SAP CC 5.0 FPS 2 in your SAP system landscapes.
2. Your dba or sys dba has modified the database users/accounts that are used for the communications between the SAP CC Cockpit web applications and the SAP CC Core Databases. See Changing the Database Users of the Core Database and that Are Dedicated to the SAP CC Cockpit Web Applications [page 77].

Upgrade a Cockpit Installation to Release 2020

By default, you must upgrade all the installed Cockpit user interfaces of SAP CC with the latest delivered version as a patch, FPS, or support package (SP).

Before to start this software upgrade operation, consult the Installing SAP Convergent Charging [page 21] section of this documentation for more information about the installation [page 88] procedures and recommendations.

To upgrade the Web applications in your Tomcat Server systems, you perform the following procedure:

1. Verify the last-minute information in the latest SAP Notes delivered with your SAP Convergent Charging version (patch notes, SP notes, technical implementation notes).

   See SAP Note 3023817 as a starting point.
2. Make sure there is no supplementary procedure that relates to the technical implementation of the SAP CC software.
3. Determine the current installation settings:
   - Basic installation and deployment in a Tomcat Server: Verify the app name and path in the Java Web server.
   - Advanced installation based on "Tomcat Contexts": Verify the location and usage of the `WAR` file that was installed previously.
4. Plan the upgrade processing and inform all the SAP CC users about the software upgrade project.
5. Get the `ZIP` file. It is available on the installation media in the directory `CC2020_TOOLS_CONTENT_UC_OSIND`. See the Downloading the Installation Materials [page 92] section for more information about the media and the `ZIP` files to use.

6. Unzip the file in the installation directory to extract the `WAR` file.
7. Verify there is no user connected to the Web application.
8. Depending on your installation, deploy, rename and deploy, or manage the Tomcat Contexts and the `WAR` file.
9. Save copies of the old configuration files (`*.properties`).
10. Create the necessary configuration files by copying the newly delivered templates and customize the configuration of the corresponding Cockpit user interfaces.
11. Close and relaunch the browser.
12. Relaunch each updgraded Web application to verify that they can run and display the sign-in window. See Launching Cockpit in a Web Browser [page 206].

6.4.8 Follow-Up Activities

1. Monitoring the System Landscape [page 83]
2. Implementing New Features of SAP Convergent Charging (SAP CC) [page 84]
3. Optimizing your System [page 85]
4. Upgrading the SAP Online Documentation [page 86]
5. Implementing Support Packages After the Upgrade [page 86]
6. Backing Up [page 86]
7. After Resuming Production Operations [page 87]

6.4.8.1 Monitoring the System Landscape

Monitor the performances of the business services before resuming production operation:

- Real-time charging and refilling operations (Online Charging Services)
- Batch charging operations, rerating operations (Offline Charging Services)
- Configuration operations of customer master data (initial provisioning, data maintenance)

Parent topic: Follow-Up Activities [page 83]

Next: Implementing New Features of SAP Convergent Charging (SAP CC) [page 84]

6.4.8.2 Implementing New Features of SAP Convergent Charging (SAP CC)

After the upgrade process is completed, you can configure the upgraded systems to use the new or enhanced features provided by SAP Convergent Charging release 2020 before resuming production operation.

Implementing New Features of SAP CC 2020 [page 84]

Parent topic: Follow-Up Activities [page 83]

Previous: Monitoring the System Landscape [page 83]

Next: Optimizing your System [page 85]

6.4.8.2.1 Implementing New Features of SAP CC 2020

After the upgrade process is completed, you can configure the upgraded systems to use the new or enhanced features provided by SAP Convergent Charging release 2020 before resuming production operation.

The table below suggests some changes available in the new main release of SAP Convergent Charging and that you can benefit in your SAP system landscape after the complete upgrade.

First refer to the SAP CC 2020 Release Notes [page 10]/What's New documents for a complete overview of the enhancements and changes.

In SAP CC release 2020, consider the following possible features:

Area	Feature	Condition/Procedures
Security	Authentication	As of 2020 FPS 2: The OAuth 2.0 authentication protocol is supported by individual users of the Cockpit user interface (web application).
System Administration	Cockpit	As of release 5.0, you can implement the web application in your system landscape. Refer to: • Installing and Launching Cockpit [page 201] section in this documentation • Configuring the Web Applications section of the SAP CC 2020 Tuning Guide • SAP CC 2020 Security Guide • SAP CC 2020 Application Help
Back-End Database	Data Store	As of 2020 FPS 1: • Microsoft SQL Server 2017

Area	Feature	Condition/Procedures
Charging Services	Outputs	As of 2020 FPS 2: • Immediate loading of billable items (BITs) into SAP Convergent Invoicing
Troubleshooting	Thread Dump Files	As of 2020 FPS 2: The SAP CC system instances generate a new XML element named rfc in thread dump files.

→ Tip

Consider the features provided with the previous release of SAP CC: 5.0.

Related Information

What's New in Feature Package Stack 2

6.4.8.3 Optimizing your System

→ Remember

We recommend that you schedule an additional maintenance plan to optimize the performances of your running SAP CC systems by fine configuring the JVM[33] options.

To perform this advanced configuration, refer to the SAP Note 2006073 to determine the best settings for your systems (Core Server, BART Server).

Consult your support specialists for more information.

Refer to the SAP CC 2020 Tuning Guide for more information.

Parent topic: Follow-Up Activities [page 83]

Previous: Implementing New Features of SAP Convergent Charging (SAP CC) [page 84]

Next: Upgrading the SAP Online Documentation [page 86]

[33] Java Virtual Machine

6.4.8.4 Upgrading the SAP Online Documentation

You must perform the following actions before you can display the online documentation in your SAP system landscape:

- Install the SAP online documentation in your SAP system as described in the `README.TXT` file contained in the root directory of the online documentation material, which is delivered as part of the upgrade package.
- Configure the upgraded graphical user interfaces (Core Tool, BART Tool, CAT Tool)

Refer to the SAP CC 2020 Tuning Guide for more information.

Parent topic: Follow-Up Activities [page 83]

Previous: Optimizing your System [page 85]

Next: Implementing Support Packages After the Upgrade [page 86]

6.4.8.5 Implementing Support Packages After the Upgrade

To make sure that your SAP Convergent Charging software is running smoothly, SAP recommends that you regularly implement the new Support Packages (SP) in your system as soon as they become available. This also applies after the upgrade, to avoid potential problems in your system before they occur.

If you have not included all the necessary support packages in the upgrade, you can still import them after the upgrade.

Refer to the Implementation of a Feature Package Stack (FPS) [page 36] section of this documentation for more information.

Parent topic: Follow-Up Activities [page 83]

Previous: Upgrading the SAP Online Documentation [page 86]

Next: Backing Up [page 86]

6.4.8.6 Backing Up

After an upgrade, SAP SE recommends you to do a full backup of your systems.

Parent topic: Follow-Up Activities [page 83]

Previous: Implementing Support Packages After the Upgrade [page 86]

Next: After Resuming Production Operations [page 87]

6.4.8.7 After Resuming Production Operations

During a limited period, daily monitor the performances to ensure that the upgrade has been well performed.

Parent topic: Follow-Up Activities [page 83]

Previous: Backing Up [page 86]

7 Procedures

As a system administrator use the procedures available in this section to complete your tasks.

Installation Procedures [page 88]

Update and Maintenance [page 286]

Upgrade Procedures [page 295]

7.1 Installation Procedures

Defining Your System Landscape [page 90]

Choosing System IDs [page 91]

Downloading the Installation Materials [page 92]

Downloading the SAP JVM [page 94]

Downloading the SAPCAR Utility [page 95]

Downloading the SAP Cryptographic Library [page 96]

Downloading the Oracle JDBC Driver [page 97]

Checking Free Space [page 98]

Creating Required Directories [page 99]

Setting Up the System Encoding [page 100]

Installing C Shell (csh) [page 101]

Setting Up the System Time Zone [page 102]

Setting Up the System Environment Variables [page 104]

Preparing the Core Database [page 106]

Installing a Core Server on a Monohost Landscape [page 123]

Installing a Core Server on a Multihost Landscape [page 131]

Testing a Landscape Basically [page 139]

Setting Up a New Host for a Landscape [page 141]

Adding Core Server Instances in a Multi-Hosts Landscape [page 142]

Starting and Stopping the Server Systems [page 144]

Creating a Database User/Account of the Core Database and Dedicated to Communications Between SAP CC Cockpit and the SAP CC Core Database [page 151]

Preparing the BART Database [page 151]

Installing a BART Server System in an Existing Landscape [page 167]

Installing a Diameter Server in an Existing Landscape [page 173]

Preparing the Oracle Session Database [page 177]

Preparing the IEC Database [page 179]

Installing an IEC (Import/Export Connector) in an Existing Landscape [page 187]

Launching the SAP installer [page 190]

Installing and Launching Core Tool [page 192]

Installing and Launching BART Tool [page 195]

Installing and Launching CAT Tool [page 198]

Installing and Launching Cockpit [page 201]

Installing and Launching Admin+ [page 207]

Installing and Launching BART+ [page 209]

Installing and Launching Setup Tool [page 211]

Installing and Launching BART Setup Tool [page 212]

Installing and Launching Config Tool [page 214]

Securing an SAP CC System Landscape [page 216]

Securing Communications with the Core Server System [page 221]

Securing Communications with the BART Server System [page 235]

Securing Communications with the Diameter Server System [page 242]

Securing Communications with Cockpit and Tomcat Server [page 246]

Securing Communications with Admin+ [page 251]

Securing Communications with Core Tool [page 256]

Installing a Permanent License [page 261]

Removing System(s) or Instance(s) from a Given Host [page 262]

Integrating SAP CC with a SCIM-Based System [page 263]

Integrating SAP CC with CA Application Performance Management [page 264]

Integrating SAP CC with SAP CTS+ [page 267]

Integrating SAP CC with SAP Convergent Invoicing [page 271]

Integrating SAP CC with SAP S/4HANA Cloud Public Edition [page 275]

Copying an SAP CC Core Server [page 279]

Configuring the Required SAP CC Users and Accounts [page 285]

As part of your post-installation operations, you create the SAP CC users who are relevant and required for your implementation scenario.

Deleting the Super Administrator and Emergency User [page 286]

7.1.1 Defining Your System Landscape

Description

During the installation of your SAP Convergent Charging (SAP CC) system landscape, you will have to perform manual operations and checks on the different software components concerned by the installation. This procedure summarizes the information relating to the different hosts that make up this SAP system landscape, such as hostnames, IP addresses, Operating Systems, software components to install, individual and service users (and associated passwords), and so on.

Prerequisites

- Before deploying your SAP Convergent Charging landscape, the sizing [page 22] has been estimated and meets your needs (especially concerning the high availability and the required SLO for each service: pricing, provisionning, and rating).
- For more information about naming rules for SAP servers and systems, refer to SAP Note 611361.

Procedure

Fill the following table with the information related to your landscape:

Useful Information (Friendly Name, O.S, ...)	Hostname + IP Address	Software Components	User + Password

7.1.2 Choosing System IDs

Description

The SAP System ID (or SID) corresponds to the identifier for an SAP system. This procedure provides you with information related to the selection of the SIDs of each SAP CC system you want to install on your landscape (Core Server or BART Server).

Prerequisites

Before choosing your SIDs, please take into account the following rules:

- The SID must be unique throughout your organization and consistent throughout your system installation landscape. If you want to install an additional instance, make sure that no gateway instance with the same SID already exists in your SAP system landscape
- The SID must consist of **exactly** three alphanumeric characters
- Only uppercase letters are allowed
- The first character must be a letter (not a digit)
- The following SIDs are reserved and cannot be used:

ADD	ALL	AMD	AND	ANY	ASC	AUX	COM	CON
DBA	END	EPS	FOR	GID	IBM	INT	KEY	LOG
LPT	MON	NIX	NOT	NUL	OFF	OMS	PRN	RAW
ROW	SAP	SET	SGA	SHG	SID	SQL	SYS	TMP
TOP	UID	USE	USR	VAR				

> **Note**
>
> The complete and up-to-date list of reserved SAP System Identifiers is available in the SAP Note 1979280

Procedure

> **Caution**
>
> Choose your SID carefully. Renaming is complicated and requires that you re-install your system.

Write down in the following table the different SIDs you have chosen for your SAP CC systems:

Software Component	System ID
Core Server	
BART Server	

7.1.3 Downloading the Installation Materials

Description

The installation materials of SAP Convergent Charging (SAP CC) 2020 contain all the SAP CC software components and installable data units relating to a given Support Package (SP) and required for either an initial installation or a maintenance update of an existing SAP CC 2020 system or application. As of release 2020, FPS are available. You can use this procedure to download these installation materials from SAP ONE Support Launchpad and possibly customize its content to fit your installation scenario.

Preliminary Notes

The installation materials of SAP CC 2020 are available as a **multispanning archive** made up with multiple files. For more information and user assistance about multispanning archives, refer to the SAP Note 886535.

Procedure

To download the SAP Convergent Charging 2020 materials, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Click the *Installations & Upgrades* tab to access this subarea.
3. Expand the *By Alphabetical Index (A-Z)* section.
4. Click the *C* letter.
5. Click the *SAP Convergent Charging* item of the list.
6. Click the **SAP Convergent Charging 2020** item to open the information page dedicated to SAP CC 2020.

   > **Note**
   >
   > Previous versions of SAP CC are also available but this procedure only concerns the 2020 release.

7. Click the *Installation* link available in the upper part of the screen in order to open the *Downloads* section.
8. Click on each element of the table to download it.

Once downloaded, refer to the SAP Note 886535 to unpack the multispanning archive and get the content of the installation materials.

Check that you get the following unpacked content in the specified destination folder:

Directory	Content Description
/	The root directory corresponds to the SAP material number of the downloaded deliverable. In addition to the directories listed afterwards, it contains the following files: • CDLABEL.ASC • CDLABEL.EBC • COPY_TM.HTM • COPY_TM.TXT • LABEL.ASC • LABEL.EBC • LABELIDX.ASC • MD5FILE.DAT • MID.XML • SHAFILE.DAT • VERSION.ASC • VERSION.EBC
/DATA_UNITS/	The DATA_UNITS directory contains a list of subdirectories that correspond to each data unit available in the installation materials.
/DATA_UNITS/ CC2020_<SU>_CONTENT_UC_OSIND	These directories correspond to the different SAP CC 2020 systems that are available in the installation materials. Available directories are: • CC2020_**CORE**_CONTENT_UC_OSIND for the Core Server system • CC2020_**BART**_CONTENT_UC_OSIND for the BART Server system • CC2020_**DIAM**_CONTENT_UC_OSIND for the Diameter Server system
/DATA_UNITS/ CC2020_IM_<OS> and /DATA_UNITS/ INST_CC2020_UC_<OS>	These directories contain the installation materials. Available directories are: • CC2020_IM_<OS> for the SAP installer • INST_CC2020_UC_<OS> for the installation framework, that contains a version of the SAP JVM[34] (available as a SAR[35] file) Where <OS> corresponds to the supported operating systems, listed in the Product Availability Matrix: support.sap.com/pam
/DATA_UNITS/ CC2020_TOOLS_CONTENT_UC_OSIND	This directory contains the different user interfaces and documentation of the development libraries provided with SAP CC 2020.

Once unpacked, you can customize the installation materials for optimization purpose. See Customizing the Installation Materials [page 93].

7.1.3.1 Customizing the Installation Materials

[34] Java Virtual Machine
[35] SAP Archive

Once unpacked, you can customize the installation materials for optimization purpose. Use the following procedure to remove the non-relevant content and only keep the `LABELIDX.ASC` file plus the different directories that are necessary to install the relevant software components.

> ⚠ Caution
>
> Be careful when removing the non-relevant content. In case you remove a content that is required for your SAP system landscape, you will have to unpack the installation materials again to recover this content.

1. Remove all data units directories that do not refer to the operating systems of your SAP system landscape.
2. Remove all data units directories that are not relevant for your landscape, according to the following table:

Data Unit	Core Server	BART Server	Diameter Server
CC2020_CORE_CONTENT_UC_OSIND	Required	-	-
CC2020_BART_CONTENT_UC_OSIND	-	Required	-
CC2020_DIAM_CONTENT_UC_OSIND	-	-	Required
CC2020_IM_<OS>	Required	Required	Required
INST_CC2020_UC_<OS>	Required	Required	Required
CC2020_TOOLS_CONTENT_UC_OSIND	Required	Required	-

> ⚠ Caution
>
> As of SAP Convergent Charging 5.0 SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

3. Remove all **files** located in the root directory, except the `LABELIDX.ASC` file that must be kept. You now have a customized version of the SAP Convergent Charging 2020 installation materials, whose content can be copied or shared in your SAP system landscape.

7.1.4 Downloading the SAP JVM

Description

SAP Convergent Charging 2020 is a set of Java-based applications that require the SAP JVM 8.1. A version of this SAP JVM is included within the installation materials as a SAR file, but you can also download the latest available version if necessary.

Preliminary Notes

For more information about the SAP JVM installation prerequisites, refer to SAP Note 1367498.

Procedure

If you want to get the SAP JVM from the installation materials, take it from the following folder: `/DATA_UNITS/INST_CC2020_UC_<OS>/DBINDEP`

If you want to download the SAP JVM, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Search for the *SAP JVM 8.1* element: in the search field named *Search in Software Downloads*, type this text and click the search icon button.
3. Click the *SAP JVM 8.1* category within the list of search results in order to open the *Downloads* section.
4. Select your operating system (OS).
5. Browse the list of available SAP archives and click the one that corresponds to the highest patch level in order to download it.

Once you retrieved the SAR file relating to the SAP JVM, you have to use the SAPCAR utility to uncompress it. For further information about the SAPCAR utility, refer to the Downloading the SAPCAR Utility [page 95] procedure. Then, execute the following procedure:

1. Copy the SAPCAR utility into the folder containing the SAP JVM SAR file you retrieved.
2. Execute the following command into a command line prompt:

```
<SAPCAR_FILENAME> -xvf <SAR_FILENAME> -R ./SAPJVM
```

The newly created `SAPJVM` folder contains the uncompressed version of the SAP JVM, that includes the different tools used throughout the overall installation procedure (such as *keytool* in particular, that can be used to secure the SAP CC landscape).

7.1.5 Downloading the SAPCAR Utility

Description

SAPCAR is a utility used by SAP to compress and/or uncompress SAP archive files named SAR files (SAR: SAP Archive). You can use this procedure to download the latest available version of SAPCAR for your operating system (OS).

Preliminary Notes

For more information about the SAPCAR utility, refer to SAP Note 212876

Procedure

To download the SAPCAR utility, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Click the *Support Packages & Patches* tab to access this subarea.

   > → Remember
   >
   > The SAPCAR you want to download is not available in the "Installations & Upgrades" subarea.

3. For example, expand the *By Alphabetical Index (A-Z)* section.
4. Click the *S* letter.
5. Click the *SAPCAR* element of the list.

   > → Tip
   >
   > If SAPCAR is not visible in the list, maybe you are in the "Installations & Upgrades" subarea. SAPCAR is available in the "Support Packages & Patches" subarea.

6. Select the highest available version of SAPCAR in order to open the *Downloads* section. For example: 7.53
7. Select your operating system (OS).
8. Click the element you want to download.

Once downloaded, execute the downloaded EXE file to use the SAPCAR utility (setting appropriate execution rights when necessary) to compress or uncompress SAR files.

7.1.6 Downloading the SAP Cryptographic Library

Description

The SAP Cryptographic Library represents the security product provided by SAP for encryption with SAP systems. In an SAP Convergent Charging landscape, this library is required in case of secured communications performed using the RFC[36] over TCP/IP[37] communication channel.

You can use this procedure to download the relevant version of the SAP Cryptographic Library for your operating system.

[36] Remote Function Call
[37] Transmission Control Protocol/Internet Protocol

Prerequisites

- SAP CC supports different versions of the SAP Cryptographic Library to secure communications performed using the RFC over TCP/IP communication channel. To determine the version of the SAP Cryptographic Library that can be used, refer to SAP Note 2471232.

Procedure

To download the SAP Cryptographic Library, complete the following procedure:

1. Go to *SAP for Me* at: https://me.sap.com/softwarecenter/
2. Click the *Support Packages and Patches* tab.

   > → Remember
   >
   > The SAP Cryptographic Library you want to download is not available in the subarea named "Installations & Upgrades".

3. On the top of the screen, select the *Downloads* element of the list, and type **`cryptolib`** in the search field.
4. Click the *Search* button.
5. Select the relevant version of the library in order to open the *Downloads* section.
6. Select your operating system (OS).
7. Click the element you want to download.

Once downloaded, refer to SAP Note 2471232 for further information about the installation of the SAP Cryptographic Library.

7.1.7 Downloading the Oracle JDBC Driver

Description

According to your business requirements, your SAP Convergent Charging landscape can contain Oracle databases. This procedure explains you how to download the relevant JDBC[38] driver that is required at installation time.

Procedure

To download the Oracle JDBC driver, execute the following procedure:

[38] Java Database Connectivity

1. Go to the following address: https://repo1.maven.org/maven2/com/oracle/database/jdbc/ojdbc8/
2. Click the link corresponding to your Oracle version to access to the download list
3. Click the *ojdbc8-<VERSION>.jar* link (where <VERSION> corresponds to the Oracle version)
4. Save this file

7.1.8 Checking Free Space

Description

Every host on which you install software components of SAP Convergent Charging 2020 must have enough free space, either for installation materials, or for running tasks (log files, temporary files, and so on). You can use this procedure to check if your hosts have enough free space.

Procedure

The following table contains the list of required free space that your host must have:

Elements	Required Space
Installation materials	5 GB
SAP installer depot (*)	1 GB
SAP CC Core Server system and its instances	1 GB per instance to deploy on the host
SAP CC BART Server system	1 GB

(*) The SAP installer depot corresponds to a directory created and used by the SAP installer during the installation process to create temporary files and log files. Each time you launch SAP installer, the following directories are automatically created:

- `<DRIVE>:\Program Files\sapinst_instdir` for *MS Windows* hosts
- `/tmp/sapinst_instdir` and `/tmp/sapinst_exe*` for *UNIX and Linux operating systems* hosts

> **Note**
>
> Once the installation with SAP installer is completed, you can delete the content of the `SAPInst_exe*` directory to get free space. On the contrary, it is recommended to keep the directory because it contains log files related to the installation process, that can be useful in case an issue occurs during later installations.

7.1.9 Creating Required Directories

Description

Some directories must be created on each host on which you install software components of SAP Convergent Charging (SAP CC), either for installation materials, or for running tasks (such as high volume data files generated during the charging operations). In case of a high availability (HA) setup, the directories are created on the network storage. You can use this procedure to create the following directories:

- `sapmnt`, which corresponds to the SAP Central Repository [page 13] of each installed SAP CC system and contains the different programs and profiles related to the different software components to install. This directory must be created:
  - **On your NAS, SAN, or another network storage solution in case of a high availability (HA) setup**.
  - **On every SAP Global Host [page 12]** running on a Linux operating system. On Microsoft Windows operating systems, the SAP installer automatically creates the SAP Central Repository and shares it using the `sapmnt` and `saploc` share names.
    If your SAP CC system is a multihost system, you must also share this directory so that the other hosts related to the same SAP CC system can access it.
- `workingDirectory`, which corresponds to a directory used by the instances of the Core Server system to store data files resulting from business operations

> ⚠ Caution
>
> When modifying the path of the Core Server Working Directory, groups of two subsequent backslashes are interpreted as only one backslash.
>
> Backslashes in the working directory path may be protected: this means that whether the folder names are separated by one or two subsequent backslashes, only one of them will be kept.
>
> To avoid misinterpretations, SAP SE recommends to systematically protect all backslashes. In particular, network paths must then start with four backslashes (i.e. two protected backslashes).

Procedure (Linux and UNIX Operating Systems)

1. In case your landscape is (or is planned to be) a multi-hosts landscape, you must create the SAP Central Repository [page 13] and share it with the adequate permissions to the other hosts (or future hosts).

   To create this folder **on the SAP Global Host [page 12]** of your SAP CC system, execute the following commands as the root user:

```
mkdir /sapmnt
/etc/init.d/nfs start
/sbin/chkconfig nfs on
chmod -R 766 /sapmnt
echo "/sapmnt *(rw,sync,no_root_squash)" >> /etc/exports
/usr/sbin/exportfs -ra
```

```
/etc/init.d/nfs restart
```

2. If your landscape is a multi-hosts landscape, you must mount the SAP Central Repository that has been created on the **SAP Global Host** on every other host related to your SAP CC system. Execute the following procedure (adapting the IP address of the SAP Global Host) as the root user **on every host different than the Global Host** of your SAP CC system in order to access the shared `sapmnt` directory:

```
mkdir /sapmnt
echo "/sapmnt *(rw,sync,no_root_squash)" mkdir /sapmnt>> /etc/exports
/usr/sbin/exportfs -ra
/etc/init.d/nfs restart
<IP_GLOBAL_HOST>
/etc/init.d/nfs start:/sapmnt   /sapmnt   nfs   defaults   0 0" >> /etc/fstab
mount /sapmnt
```

3. If your landscape is a multi-hosts landscape, you must create the SAP CC Working Directory on a host where you want to install instances of a Core Server system. To create this directory, execute the following commands as the root user **on each concerned host of your landscape**:

```
mkdir /workingDirectory
chmod 777 /workingDirectory
```

   And in case you need to share the SAP CC Working Directory, execute the following additional commands:

```
echo "/workingDirectory *(rw,sync,no_root_squash)" >> /etc/exports
/usr/sbin/exportfs -ra
/etc/init.d/nfs restart
```

Procedure (Microsoft Windows Operating System)

To create the SAP CC Working Directory on a host where you want to install instances of a Core Server system, execute the following command as a member of the Administrators group **on each concerned host of your landscape**:

```
mkdir C:\workingDirectory
```

And in case you need to share the SAP CC Working Directory, execute the following additional command:

```
net share workingDirectory=C:\workingDirectory /grant:Everyone,FULL
```

7.1.10 Setting Up the System Encoding

Description

To install a software component of SAP Convergent Charging on a *Linux or UNIX* operating system host, the SAP installer requires to use the UTF-8 encoding.

> ⚠ Caution
>
> *MS Windows* does not allow modifying the default encoding of the whole system. As a consequence, this procedure only concerns hosts running Linux and UNIX operating systems.

To set the UTF-8 encoding on your *Linux or UNIX* operating system hosts, connect to the concerned hosts as the root user, and set or modify the **current locale** to the **`en_US.UTF-8`** value.

Refer to the product documentation for your operating system about the relevant procedures.

7.1.11 Installing C Shell (csh)

Description

Every host running the SAP installer requires the presence of the C shell command language interpreter. To ensure that C shell is already installed on the host before running SAP installer, verify the existence of the `/bin/csh` script file. When not existing, use this procedure to install C shell on your Linux host.

Preliminary Notes

- This procedure only concerns hosts running *Linux or UNIX* operating systems
- For further information about the Yum installer, refer to its dedicated documentation
- For further information about the Zypper installer, refer to its dedicated documentation

Linux and UNIX Operating Systems

To set up the system encoding on a given *Linux or UNIX* operating system host, execute the following procedure as the root user:

1. Execute the following command to install the adequate packages:
   - For Red Hat distributions:

     ```
     yum install csh
     ```

   - For SuSe distributions:

     ```
     sudo zypper install csh
     ```

2. Check that the `/bin/csh` script file exists using the following command:

   ```
   ls -al /bin/csh
   ```

7.1.12 Setting Up the System Time Zone

Description

For both business and technical reasons, it is necessary to ensure that the *time zone* is correctly configured on each host containing software components of SAP Convergent Charging. Furthermore, in a multihost and multisystem landscape, this *time zone* must be similar for each host. Ideally, it is recommended to use a NTP[39] server to synchronize the clocks of your hosts.

This procedure explains you how to check the time zone of a given SAP CC host, and set it up when necessary.

> ⚠ Caution
>
> Do not modify the time zones of your hosts **after** the installation of SAP CC.

Checking the system time zone

Linux and UNIX Operating Systems

To check the current *time zone* on a given *Linux or UNIX* operating system host, execute the following command as the *root* user:

- For Red Hat distributions:

```
date
```

- For SuSe distributions:

```
date
```

- For AIX:

```
grep TZ /etc/environment
```

- For Solaris:

```
date
```

Microsoft Windows Operating System

To check the current *time zone* on a given *Microsoft Windows* host, execute the following command as a member of the *Administrators* group:

```
tzutil /g
```

[39] Network Time Protocol

Modifying the system time zone

Linux and UNIX Operating Systems

In case you need to modify the *time zone* on a given *Linux or UNIX* operating system host, execute the following procedure as the *root* user:

1. Execute the following command to open a graphical user interface dedicated to the management of time zones:
   - For Red Hat distributions:

     ```
     timedatectl (or system-config-date for releases older than RHEL7)
     ```

   - For SuSe distributions:

     ```
     yast timezone
     ```

   - For AIX:

     ```
     smit chtz
     ```

   - For Solaris:

     ```
     sysconfig configure
     ```

2. Select your time zone and validate the modification.
3. Finally, test that your modification has been taken into account by executing the dedicated Checking the system time zone [page 102] procedure above.

Microsoft Windows Operating System

In case you need to modify the *time zone* on a given *Microsoft Windows* host, use the *Date and Time* menu of the *Control Panel*, or perform the following procedure as a member of the *Administrators* group:

1. Execute the following command (or refer to *Microsoft MSDN online documentation*) to get the list of time zones (display names, zone IDs, and UTC time offsets) supported by Microsoft:

   ```
   tzutil /l
   ```

2. Execute the following command to modify the current time zone of your host, considering that `<TIME_ZONE_ID>` corresponds to the name of the time zone:

   ```
   tzutil /s <TIME_ZONE_ID>
   ```

3. Finally, test that your modification has been taken into account by executing the dedicated Checking the system time zone [page 102] procedure above.

7.1.13 Setting Up the System Environment Variables

Description

SAP Convergent Charging (SAP CC) requires to set up some environment variables, either used by the SAP installer during the installation process, or used by some user interfaces. Use this procedure to set up the following environment variables for each host in your SAP CC system landscape:

- `SAPCC_JAVA_HOME`, used by SAP CC GUIs[40], and thus only relevant if a GUI must be installed on this host, or used by an IEC[41].
- `SAP_IPv6_ACTIVE`, used for JCo[42] communications, only relevant for hosts that use *IPv6* addresses.
- `NI_USEIPv6`, used for communication with SAP MMC[43] and only relevant for hosts that use *IPv6* addresses.

Linux and UNIX Operating Systems

The `SAP_IPv6_ACTIVE` and `NI_USEIPv6` environment variables only concern hosts that use *IPv6* addresses. To check whether your host uses *IPv4* or *IPv6* addresses, execute the following procedure:

1. Connect as the *root* user to the host, and execute the following command to display information about the network configuration of your host:

   ```
   ifconfig
   ```

2. Locate the section dedicated to the Ethernet connection:
   - The `inet` string means that your host uses *IPv4* addresses.
   - The `inet6` string means that your host uses *IPv6* addresses.

To set up the environment variables, execute the following procedure as the *root* user:

1. Execute the following command to edit the configuration file containing the environment variables:

   ```
   vi /etc/environment
   ```

   .

2. Assuming that they do not already exist, add the following lines at the end of the file:
   - `SAPCC_JAVA_HOME=<SAP_JVM_PATH>`, where `<SAP_JVM_PATH>` corresponds to the path of the folder that contains the SAP JVM 8.1. This environment variable is only relevant if a GUI must be installed on this host.
   - `NI_USEIPv6=1`, if this host uses *IPv6* addresses
   - `SAP_IPv6_ACTIVE=1`, if this host uses *IPv6* addresses

[40] Graphical User Interface
[41] Import/Export Connector
[42] Java Connector
[43] SAP Microsoft Management Console

3. Save the file. These environment variables will be available when reconnecting to the host.

Microsoft Windows Operating System

The `SAP_IPv6_ACTIVE` and `NI_USEIPv6` environment variables only concern hosts that use *IPv6* addresses. To check whether your host uses *IPv4* or *IPv6* addresses, execute the following procedure:

1. Open a command line prompt and execute the following command to display information about the network configuration of your host:

   ```
   ipconfig
   ```

   .
2. Locate the section dedicated to the Ethernet connection. If an *IPv6* address is displayed, it means that your host uses *IPv6* addresses

To create the environment variables, use the *System* menu of the *Control Panel* and execute the following procedure:

1. Click on *Advanced system settings* to open the *System Properties* dialog.
2. Select the *Advanced* tab, and click the *Environment Variables* button to open the dialog dedicated to the management of user and system variables.
3. If a GUI or an IEC must be installed on this host:
   1. Click the *New...* button of the *System variables* section in order to open the *New System Variable* dialog. If you are not granted adequate rights to create system variables, click the *New...* button of the *User variables for xxx* section in order to open the *New User Variable* dialog.
   2. Enter **`SAPCC_JAVA_HOME`** as the *variable name*, and specify the path of the folder that contains the SAP JVM 8.1 as the *value*.
   3. Click *OK* to create the new variable.
4. If this host uses *IPv6* addresses:
   1. Click the *New...* button to open the *New System Variable* dialog. If you are not granted adequate rights to create system variables, click the *New...* button of the *User variables for xxx* section in order to open the*New User Variable* dialog.
   2. Enter **`NI_USEIPv6`** as the *variable name*, and **`1`** as the *value*.
   3. Click *OK* to create the new variable.
   4. Click the *New...* button to open the *New System Variable* dialog. If you are not granted adequate rights to create system variables, click the *New...* button of the *User variables for xxx* section in order to open the*New User Variable* dialog.
   5. Enter **`SAP_IPv6_ACTIVE`** as the *variable name*, and **`1`** as the *value*.
   6. Click *OK* to create the new variable.
5. Close the dialog windows.

7.1.14 Preparing the Core Database

> → Remember
>
> The Core Database is required when a Core Server system is part of your SAP Convergent Charging (SAP CC) system landscape.
>
> See Installing a Core Server on a Multihost Landscape [page 131] or Installing a Core Server on a Monohost Landscape [page 123].

This section includes the procedures for the system database administrator (sys dba) or database administrator (dba) to prepare the Core Database that is the dedicated back-end database of the Core Server system.

Preparing the Oracle Core Database [page 106]

Preparing the SQL Server Core Database [page 109]

Preparing the SAP ASE Core Database [page 113]

Preparing the SAP HANA Core Database [page 118]

Preparing the IBM DB2 Core Database (w/o pureScale Feature) [page 120]

7.1.14.1 Preparing the Oracle Core Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, Session Database, BART Database and IEC Database). This procedure explains you how to prepare your Core Database running under an Oracle RDBMS[44] (Standard or Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Oracle, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/oracle

[44] Relational Database Management System

- http://scn.sap.com/docs/DOC-7888

Prerequisites

An instance of an Oracle RDBMS must be installed and available in your landscape. This database instance must use the UTF-8 character encoding to work in conjunction with the different SAP CC software components, and the *Oracle SQL*plus* tool must be installed to execute this procedure. Refer to your system administrator to get information about these prerequisites relating to the SAP CC Core Database.

Procedure

The preparation of your SAP CC Core Database consists in the following operations:

- Setting up some initialization parameters
- Creating a user account and granting the adequate roles and privileges to this user
- Creating the different tablespaces

1. SAP Convergent Charging requires to modify some initialization parameters of the Oracle database, located in the `init.ora` file which contains the persistent parameters used during database startup. You can use the following procedure to:
   - Set the `_optim_peek_user_binds` parameter to `FALSE` in order to disable the bind peeking feature, which is not compatible with SAP CC
   - Increase the maximum number of processes and sessions by setting the processes and sessions parameters to 300 multiplied by the number of databases installed on this Oracle instance, i.e. 1 (as only the Core Database is installed on this instance)

   To correctly set up the initialization parameters of your Core Database, execute the following commands:

   ```
   sqlplus / as sysdba
   alter system set "_optim_peek_user_binds"=false scope=spfile;
   alter system set processes=300 scope=spfile;
   alter system set sessions=300 scope=spfile;
   shutdown immediate
   startup
   ```

2. An Oracle user must be created and granted the adequate roles and privileges to administrate the Core Database. Execute the following commands by replacing `<DB_USER>` by the name of the Core Database administrator and `<DB_PASSWORD>` by its password:

   ```
   sqlplus / as sysdba
   define dbuser=<DB_USER>";
   define dbpwd="<DB_PASSWORD>";
   create user &dbuser identified by &dbpwd;
   grant CONNECT, RESOURCE TO &dbuser;
   grant EXECUTE_CATALOG_ROLE to &dbuser;
   grant EXP_FULL_DATABASE, IMP_FULL_DATABASE to &dbuser;
   grant CREATE PROCEDURE to &dbuser;
   grant CREATE SESSION to &dbuser;
   grant CREATE TABLESPACE, DROP TABLESPACE to &dbuser;
   grant ALTER TABLESPACE to &dbuser;
   grant CREATE ANY DIRECTORY to &dbuser;
   grant DROP ANY DIRECTORY to &dbuser;
   ```

```
grant EXECUTE ON dbms_lock TO &dbuser;
```

3. The data storage of your SAP CC Core Database must be optimized by creating the following mandatory tablespaces, whose size depends on the edition of your RDBMS (Standard or Enterprise edition):

		Minimal Size (MB)	
Tablespace name	**Description**	**SE**	**EE**
CATALOG_DATA	Tables used to store the objects belonging to the catalog, and the operations audit.	180	2000
CATALOG_INDX	Indexes used for the objects belonging to the catalog, and the operations audit	220	220
SUBSCRIBER_DATA	Tables containing objects belonging to provider contracts and subscriptions	100	20000
SUBSCRIBER_INDX	Indexes used for the objects belonging to provider contracts and subscriptions	200	20000
ACCESS_DATA	Tables used to store chronologies of mappings between technical identifiers and corresponding contract or subscription	10	1000
ACCESS_INDX	Indexes used for chronologies of mappings between technical identifiers and corresponding contract or subscription	20	250
PRERATING_DATA	Tables used to store prerating data	10	2000
PRERATING_INDX	Indexes used for prerating data	10	3000
COUNTER_DATA	Tables used to store the counters declared in subscriptions and provider contracts	20	1000
COUNTER_INDX	Indexes used for counters	10	10
C_SNAPSHOT_DATA	Tables used to store snapshots of counter values for subscriptions and provider contracts	20	1000
C_SNAPSHOT_INDX	Indexes used for snapshots of counter values	10	10
SESSION_RATING_DATA	Tables used to store session-based charging events and charging session histories for failover purposes	1500	7000
SESSION_RATING_INDX	Indexes used for session-based charging events and charging session histories for failover purposes	500	500
OBJECT_CHANGE_DATA	Tables used to store data related to the object change log for audited operations	100	2000
OBJECT_CHANGE_INDX	Indexes used for object change log for audited operations	100	1000
ALLOWANCE_DATA	Tables used to store allowances	20	1500
ALLOWANCE_INDX	Indexes used for allowances	15	500
MONITORING_DATA	Tables used to store the objects used for monitoring purpose, such as data files metadata	10	1000
MONITORING_INDX	Indexes used for the objects used for monitoring purpose, such as data files metadata	5	500

To create the mandatory tablespaces, execute the following command by replacing:

- `<TABLESPACE_NAME>` by the name of the tablespace to create

- <FILE_LOCATION> by the full path of the file that will contain the tablespace, located in the `oradata` directory
- <SIZE> by the recommended value associated to the tablespace, listed in the above table and using the `AUTO_EXTEND` option

```
sqlplus / as sysdba
```

And then, for each tablespace listed in the above table:

```
CREATE BIGFILE TABLESPACE <TABLESPACE_NAME> DATAFILE '<FILE_LOCATION>' SIZE
<SIZE> AUTOEXTEND ON MAXSIZE UNLIMITED UNIFORM SIZE 1M LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
```

Followed by this command **for Oracle 12c databases**:

```
ALTER USER <DB_USER> QUOTA UNLIMITED ON <TABLESPACE_NAME>;
```

Example

For an Oracle 12c Standard Edition RDBMS installed on a Linux host, you can use the following statement:

- To create the "SUBSCRIBER_DATA" tablespace
- To give the "sapcc" user the possibility to access this tablespace

```
CREATE BIGFILE TABLESPACE SUBSCRIBER_DATA DATAFILE '/oradata/vol1/
SUBSCRIBER_DATA.dbf' SIZE 100M AUTOEXTEND ON MAXSIZE UNLIMITED UNIFORM
SIZE 1M LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
ALTER USER sapcc QUOTA UNLIMITED ON SUBSCRIBER_DATA;
```

7.1.14.2 Preparing the SQL Server Core Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, Session Database, BART Database and Import/Export Connector database). This procedure explains you how to prepare your Core Database running under a Microsoft SQL Server RDBMS[45] (Standard or Enterprise edition).

[45] Relational Database Management System

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/sqlserver
  - http://scn.sap.com/docs/DOC-8286

Prerequisites

An instance of a SQL Server RDBMS must be installed and available in your landscape. The *SQL Server Management Studio* application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC Core Database.

Procedure

The preparation of your SAP CC Core Database consists in the following operations:

- Creating a SQL Server user account and granting the adequate roles to this user on the adequate server
- Creating the Core database with the user as the owner
- Creating the different filegroups within this database

1. An SQL Server user must be created for your SAP CC Core Database, and granted the adequate roles and privileges. Execute the following procedure to create this SAP CC user:
   - Open the *SQL Server Management Studio* application
   - The *Connect to Server* dialog window opens. Use the following settings:
     - *Server Name*: Select *(local)*
     - *Authentication*: Select *Windows Authentication*, in order to use your operating system user
     - Click the *Connect* button
   - Open the *View* > *Object Explorer* menu to display the *Object Explorer* and expand the *(local)* server
   - Right-click the *Security* element of the tree and open the *New / Login...* menu to display the **Login** screen
   - Fill the *General* tab with the following information:
     - *Login name*: Fill with the name of the Core Database administrator
     - Select the *SQL Server authentication* option
     - *Password*: Fill with a password respecting the SAP password policy
     - *Confirm password*: Re-type the chosen password
     - *User must change password at next login*: Untick the checkbox
     - *Default database*: Select the *master* database

- In the *Server Roles* tab, select the following roles:
  - *dbcreator*
  - *diskadmin*
  - *serveradmin*
- Click the *OK* button
- Right-click the *(local)* server and click *Disconnect*

> → Tip
>
> The SQL Server authentication method can be replaced by the Windows authentication method:
>
> 1. In the *General* tab, enter the following information:
>   - *Login name*: Fill with the `<DOMAIN>`\SAPService`<SID>` login
>   - Select the *Windows authentication* option
>   - Ensure that the Default Database is the master one
> 2. Change the launch configuration:
>   - Edit the `boot.config` file of each dispatcher instance in the folder:
>
>   `<drive>:\usr\sap\<SID>\CCD<XX>\config\boot.config`
>   - Add the string property `IntegratedSecurity=true` to the value of the SQLHELPER_JDBC_URI system parameter to enable the Windows authenticatio login method
>
>   > Example
>   >
>      > ```
>      > SQLHELPER_JDBC_URI=jdbc\:sqlserver\://
>      > <dbhost>\:<dbport>;Databasename\=<dbname>;IntegratedSecurity\=true
>      > ```
>   - (Optional) Remove the values of the following parameters:
>
>   SQLHELPER_LOGIN
>
>   SQLHELPER_PASSWORD
>
>   Execute this step each time a dispatcher is added.
> 3. Change the database configuration: launch [page 207] the Admin+ user interface and identify as the administrator of the Core Server system and change the value of the SQLHELPER_JDBC_URI system parameter:
>
>    ```
>    set SQLHELPER_JDBC_URI jdbc:sqlserver://
>    <dbhost>:<dbport>;Databasename=<dbname>;IntegratedSecurity=true
>    PERSISTENT
>    ```
> 4. Restart all instances

2. Once the SQL Server user dedicated to the Core Database has been created, it is necessary to reconnect to the server using the previously created user in order to create the Core Database. Execute the following procedure:
   - Open the *File* > *Connect Object Explorer...* menu or click the *Connect Object Explorer* button to re-open the *Connect to Server* dialog window. Use the following settings:
     - *Server Name*: Specify the full name of the machine hosting the database
     - *Authentication*: Select *SQL Server authentication*
     - *Login*: Fill with the previously created user
     - *Password*: Fill with the password of the previously created user

    - Click the *Connect* button
- Right-click the *Databases* element of the tree and open the *New Database...* menu to display the **New Database** screen
- Fill the *General* tab with the following information:
    - *Database name*: Fill with the name of the Core Database
    - *Owner*: Fill with the name of the previously created user
- Fill the *Options* tab with the following information:
    - *Collation*: Collation defines the way strings are managed in SQL Server databases. The Core Database must be case-sensitive (CS) and compatible with ASCI Unicode (AS). Thus select the element that corresponds to the appropriate language, ending by the *CS_AS* string (e.g. *Latin1_General_CS_AS*)
    - *Recovery model*: Select *Full*
    - *Compatibility level*: Ensure that *SQL Server 2008 (100)* is selected
    - Click the *OK* button

3. The data storage of your SAP CC Core Database must then be optimized by creating the following mandatory filegroups:

Filegroup name	Description
CATALOG_DATA	Tables used to store the objects belonging to the catalog, and the operations audit.
CATALOG_INDX	Indexes used for the objects belonging to the catalog, and the operations audit
SUBSCRIBER_DATA	Tables containing objects belonging to provider contracts and subscriptions
SUBSCRIBER_INDX	Indexes used for the objects belonging to provider contracts and subscriptions
ACCESS_DATA	Tables used to store chronologies of mappings between technical identifiers and corresponding contract or subscription
ACCESS_INDX	Indexes used for chronologies of mappings between technical identifiers and corresponding contract or subscription
PRERATING_DATA	Tables used to store prerating data
PRERATING_INDX	Indexes used for prerating data
COUNTER_DATA	Tables used to store the counters declared in subscriptions and provider contracts
COUNTER_INDX	Indexes used for counters
C_SNAPSHOT_DATA	Tables used to store snapshots of counter values for subscriptions and provider contracts
C_SNAPSHOT_INDX	Indexes used for snapshots of counter values
SESSION_RATING_DATA	Tables used to store session-based charging events and charging session histories for failover purposes
SESSION_RATING_INDX	Indexes used for session-based charging events and charging session histories for failover purposes
OBJECT_CHANGE_DATA	Tables used to store data related to the object change log for audited operations
OBJECT_CHANGE_INDX	Indexes used for object change log for audited operations
ALLOWANCE_DATA	Tables used to store allowances
ALLOWANCE_INDX	Indexes used for allowances

Filegroup name	Description
MONITORING_DATA	Tables used to store the objects used for monitoring purpose, such as data files meta-data
MONITORING_INDX	Indexes used for the objects used for monitoring purpose, such as data files metadata

To create the mandatory filegroups, execute the following procedure by replacing <DATABASE_NAME> by the name of the Core Database, <FILEGROUP_NAME> by the name of the filegroup to create, and <FILE_LOCATION> by the full path of the file that will contain the filegroup directory:

- Right-click your server and open the *New Query* menu
- For each filegroup listed in the above table, type the following query and execute it:

```
ALTER DATABASE "<DATABASE_NAME>" ADD FILEGROUP <FILEGROUP_NAME>;
ALTER DATABASE "<DATABASE_NAME>" ADD FILE
(
  NAME = <FILEGROUP_NAME>,
  FILENAME = '<FILE_LOCATION>',
  SIZE = 1MB,
  MAXSIZE = 1000MB,
  FILEGROWTH = 5MB
)
TO FILEGROUP <FILEGROUP_NAME>;
```

Example

To create the "SUBSCRIBER_DATA" filegroup in the "SAP CC" Core Database of an SQL Server RDBMS, you can use the following query:

```
ALTER DATABASE "SAPCC" ADD FILEGROUP SUBSCRIBER_DATA;
ALTER DATABASE "SAPCC" ADD FILE
(
  NAME = SUBSCRIBER_DATA,
  FILENAME = 'c:\tmp\SUBSCRIBER_DATA.ndf',
  SIZE = 1MB,
  MAXSIZE = 1000MB,
  FILEGROWTH = 5MB
)
TO FILEGROUP SUBSCRIBER_DATA;
```

7.1.14.3 Preparing the SAP ASE Core Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, Session Database, BART Database and Import/Export Connector database). This procedure explains you how to prepare your Core Database running under an SAP Adaptive Server Enterprise RDBMS[46].

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of SAP ASE, refer to the following pages:
  - http://infocenter.sybase.com
  - https://answers.sap.com/questions/12305881/sap-ase-installation.html
  - http://scn.sap.com/docs/DOC-34995
  - http://scn.sap.com/docs/DOC-34996

Prerequisites

An instance of an SAP ASE RDBMS must be installed and available in your landscape. This instance must be:

- Correctly licensed in order to fit specific needs such as data partitioning, communication securing, system high-availability, and so on
- Installed with the following configuration:
  - **Application Type**: Mixed (OLTP/DSS)
  - **Page Size**: 8k. Note that this value cannot be modified after the installation. The modification of the value leads to a warning message informing that the newly specified page size must be similar between databases used for import purposes
  - **Default Character Set**: utf-8 : Unicode 3.1 for UTF-8 Character Set
  - **Default Sort Order**: binary (id=25) or binaryalt (id=99)

In addition, the *Interactive SQL tool* must be installed to execute the different steps of this procedure. For further information about this tool, refer afterwards. To get information about these prerequisites relating to the SAP CC Core Database, refer to your System Administrator.

> ⚠ Caution
>
> Free Developer Edition or Small Business Edition of SAP ASE are not supported by SAP CC. You must install a licensed copy of the Sybase Adaptive Enterprise Suite.

Using the Interactive SQL tool

Interactive SQL is a command-line tool provided by SAP ASE to execute SQL statements:

- Interactively, to execute individual commands
- In a batch mode, to execute script files that contain multiple commands

To use the interactive mode, you have to:

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- Launch isql using the following synopsis, providing the username and password specified at installation time:

```
isql -U <USERNAME> -P <PASSWORD>
```

- Type the command to execute, followed by the "go" statement:

```
<command to execute>
go
```

To use the batch mode, you have to:

- Create a text file containing the list of commands to execute, each command being followed by the "go" statement:

```
<command to execute>
go
<command to execute>
go
```

  and so on
- Launch isql using the following synopsis, providing the username and password specified at installation time, and specifying the previously created script file:

```
isql -U <username> -P <password> -i <script file>
```

> **Note**
>
> - Whatever the execution mode is, you can use the `quit` command to exit the isql tool.
> - For further information about the isql tool, refer to its dedicated documentation.

Procedure

The preparation of your SAP CC Core Database consists in the following operations:

- Setting up some parameters of the SAP ASE Server
- Creating the devices for data and logs, and the database relying on these devices
- Setting up some database options
- Creating a user account and granting the adequate roles and privileges to this user

1. Some parameters of the SAP ASE Server must be modified in order to fit specific needs of SAP Convergent Charging. Use the following list of commands to modify the configuration of the ASE Server accordingly:

```
use master
GO
sp_configure "number of open objects", 10000
GO
sp_configure "number of open indexes", 2000
GO
sp_configure "max network packet size", 16384
GO
sp_configure "default network packet size", 16384
GO
sp_configure "optimization goal", 0, 'allrows_mix'
```

```
GO
sp_configure "disable varbinary truncation", 1
GO
sp_configure "number of user connections", 200
GO
sp_configure "number of locks", 1000000
GO
sp_configure "deadlock checking period", 800
GO
sp_configure "lock hashtable size", 16384
GO
sp_configure "row lock promotion HWM", 214748364
GO
sp_configure "row lock promotion LWM", 214748364
GO
sp_configure "SQL batch capture", 1
GO
sp_configure "kernel resource memory", 12000
GO
sp_configure "enable monitoring", 1
GO
sp_configure "sql text pipe active", 1
GO
sp_configure "sql text pipe max messages", 2000
GO
sp_configure "plan text pipe active", 1
GO
sp_configure "plan text pipe max messages", 2000
GO
sp_configure "statement pipe active", 1
GO
sp_configure "statement pipe max messages", 2000
GO
sp_configure "errorlog pipe active", 1
GO
sp_configure "errorlog pipe max messages", 2000
GO
sp_configure "deadlock pipe active", 1
GO
sp_configure "deadlock pipe max messages", 2000
GO
sp_configure "lock timeout pipe max messages", 2000
GO
sp_configure "lock timeout pipe active", 1
GO
sp_configure "wait event timing", 1
GO
sp_configure "process wait events", 1
GO
sp_configure "object lockwait timing", 1
GO
sp_configure "statement statistics active", 1
GO
sp_configure "per object statistics active", 1
GO
sp_configure "max SQL text monitored", 512
GO
sp_configure "enable stmt cache monitoring", 1
GO
sp_configure "aux scan descriptors", 1000
GO
sp_configure "select for update", 1
GO
```

If you use the database partitioning function for your database, execute the following commands:

```
sp_configure "enable semantic partitioning", 1
```

```
GO
sp_configure "number of open partitions", 100000
GO
```

If you use secured communications with your database, execute the following command:

```
sp_configure "enable SSL", 1
GO
```

2. To create the devices for data and logs, and then the database that relies on these devices, use the following list of commands considering that:
   - <DB_NAME> corresponds to the name of your database
   - <DEVICE_PATH> corresponds to the path of the device
   - <DEVICE_SIZE> corresponds to the amount of space to allocate to the device, made up with a quantity followed by one of these supported unit specifiers:
     - k or K (kilobytes)
     - m or M (megabytes)
     - g or G (gigabytes)
     - t or T (terabytes)

```
disk init name="<DB_NAME>_data", physname="<DEVICE_PATH>\<DB_NAME>_data",
size=<DEVICE_SIZE>, dsync = true
GO
disk init name="<DB_NAME>_log", physname="<DEVICE_PATH>\<DB_NAME>_log",
size=<DEVICE_SIZE>, dsync = true
GO
create database <DB_NAME> on <DB_NAME>_data=2000000 log on
<DB_NAME>_log=2000000
GO
```

3. Some options of the previously created database must be modified. Use the following list of commands, considering that <DB_NAME> corresponds to the name of your database:

```
sp_dboption <DB_NAME>,"ddl in tran", true
GO
sp_dboption <DB_NAME>, "select into", true
GO
```

4. A dedicated user must be created to administrate the Core Database, and must be declared as the owner of the Core Database. Use the following list of commands considering that:
   - <DB_USER> corresponds to the name of the Core Database administrator
   - <DB_PASSWORD> corresponds to the password of the Core Database administrator, that respects the SAP password policy
   - <DB_NAME> corresponds to the name of the Core Database

```
create login <DB_USER> with password <DB_PASSWORD>
GO
use <DB_NAME>
GO
sp_changedbowner <DB_USER>
GO
```

7.1.14.4 Preparing the SAP HANA Core Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, the Session Database, the BART Database and the IEC Database). This procedure explains you how to prepare your Core Database running under an SAP HANA in-memory database system using a single container or a multiple containers mode.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of SAP HANA, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/developer-center/hana
  - http://scn.sap.com/docs/DOC-53955

For further information about data types, operators, functions, statements, and so on, refer to the SAP HANA SQL and System Views Reference documentation available on SAP Help Portal:https://help.sap.com.

Prerequisites

An SAP HANA database system must be installed and available in your landscape. The *SAP HANA Studio* application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC Core Database.

Procedure

The preparation of your SAP CC Core Database consists in the following operations:

- Declaring your system within the SAP HANA Administration Console of the *SAP HANA Studio* application
- Creating a user account and granting the adequate roles and privileges to this user on this system
- Disabling the expiration of the user's password

1. An SAP HANA system must be created to administrate your SAP CC Core Database. Execute the following procedure to create this system:

- Open the *SAP HANA Studio* application
- Open the *Window* > *Perspective* > *Open Perspective* > *SAP HANA Administration Console* menu to display the perspective dedicated to the management of SAP HANA systems
- Open the *Window* > *Show View* > *Systems* menu to display the view containing the list of SAP HANA systems
- Click the *Add System* icon to open the *System* dialog window
- Use the following settings to fill the *Specify System* screen:
  - *Host Name*: Fill with the master host name
  - *Instance Number*: Fill with the instance number of your SAP HANA system
  - *Mode*: Select *Single container* or *Multiple containers*, according to your needs. In case of **multiple containers**, select then *Tenant database* and provide its name in the *Name* field below
  - *Description*: Fill with a text that describes your system
  - Click the *Next* button
- The *Connection Properties* screen opens. Select the *Authentication by database user* option and use the following settings:
  - *User Name*: Fill with the name of the SAP HANA system administrator
  - *Password*: Fill with a password respecting the SAP password policy
  - Click the *Finish* button to log on the newly declared SAP HANA system

2. Once you are connected to the relevant SAP HANA system, it is necessary to create a user account and grant the adequate roles and privileges to this user in order to administrate the Core Database. Execute the following procedure to create this user:
   - Expand the *Security* folder, and right-click the *Users* element to display the context menu corresponds to the name of your database
   - Right-click the *Users* element of the tree and open the *New User* menu to display the screen dedicated to the creation of new users
   - Fill the *User* tab with the following information:
     - *User Name*: Fill with the name of the Core Database administrator
     - Select the *Password* option in the *Authentication* fieldset
     - *Password*: Fill with a password respecting the SAP password policy
     - *Confirm password*: Re-type the chosen password
   - In the *Granted Roles* subtab, click the + button to add the following roles to the user:
     - *CONTENT_ADMIN*
     - *MODELING*
   - In the *System Privileges* subtab, click the + button to add the following privileges to the user:
     - *EXPORT*
     - *IMPORT*
   - Click the *Deploy* button to finalize the creation of the user
3. Once the user has been created and deployed on your SAP HANA system, it is necessary to disable the expiration of its password. Execute the following procedure:
   - Right-click the previously created SAP HANA system and open the *Open SQL Console* menu to display the screen dedicated to the execution of SQL statements

- Type the following SQL statement within the editor, considering that <USERNAME> corresponds to the name of your previously created user. For further information, refer to the documentation of the ALTER USER command available in the SAP HANA documentation:

```
ALTER USER <USERNAME> DISABLE PASSWORD LIFETIME;
```

- Click the *Execute* button to execute the SQL statement

7.1.14.5 Preparing the IBM DB2 Core Database (w/o pureScale Feature)

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, Session Database, BART Database and Import/Export Connector database). This procedure explains you how to prepare your Core Database running under an IBM DB2 RDBMS[47] without the pureScale feature.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of IBM DB2, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/db2-for-linux-unix-windows
  - http://scn.sap.com/docs/DOC-8211
- This procedure uses the following variables:
  - <DB_GROUP>, which corresponds to the name of the group containing the DB2 administrators, specified at DB2 installation time
  - <DB_NAME>, which corresponds to the name of the Core Database
  - <DB_USER>, which corresponds to the name of the Core Database administrator, which corresponds to a user granted adequate roles and privileges
  - <DB_PASSWORD,> which corresponds to the password of the Core Database administrator, respecting the SAP password policy but without any expiration date
  - <TABLESPACE_NAME>, which corresponds to the name of the tablespace to create
  - <FILE_LOCATION>, which corresponds to the full path of the directory that will contain the tablespaces

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Prerequisites

An instance of an IBM DB2 database must be installed and available in your landscape. The *IBM DB2 db2cmd tool* must be installed to execute this procedure. Refer to your Application and System Administrators to get information about these prerequisites relating to the SAP CC Core Database.

Procedure

The preparation of your SAP CC Core Database consists in the following operations:

- Creating a user account and adding this user to the relevant group
- Creating the Core Database and setting up some configuration parameters
- Granting the newly created user the adequate roles and privileges
- Creating a user account and granting the adequate roles and privileges to this user
- Creating the different tablespaces within this database

1. SAP Convergent Charging requires to create a dedicated user for your SAP CC Core Database, and add this user to the group containing the DB2 administrators. Execute the following procedure:
   - For a Microsoft Windows operating system, execute the following commands into a command lineprompt:

     ```
     net user <DB_USER> "<DB_PASSWORD>" /ADD /PASSWORDCHG:NO /EXPIRES:never
     ```

     ```
     wmic UserAccount where Name="<DB_USER>" set PasswordExpires=False
     ```

     ```
     net localgroup <DB_GROUP> <DB_USER> /add
     ```
   - For Linux and UNIX operating systems, execute the following command:

     ```
     usermod -a -G <DB_GROUP> <DB_USER>
     ```
2. SAP Convergent Charging requires to modify some configuration parameters when creating the DB2 database. You can use the following procedure to:
   - Create the database
   - Set the default page size of the database to 16K
   - Specify an amount of storage for recovery log files by pre-allocating a number of 50 primary log files
   - Set the amount of the database heap memory to use as a buffer for recording logs before writing these records to disk to 1024 4KB pages
   - Set the size of each primary and secondary log file to 16K 4KB pages. This size limits the number of records that can be written in these files before they become full and require the creation of new ones
   - Specify the percentage of changed pages at which the asynchronous page cleaners will be started (if they are not currently active)
   - Activate the dynamically distribution of the available memory resources between the memory consumers
   - Activate the automatic sampling rate determination, based on table size
   - Set the default prefetch size of tablespaces to 960 pages

- Set an automatic percentage for the lock list held by an application that must be filled before performing lock escalation
- Set an automatic amount of 4K pages to store the lock list

To create the SAP CC Core Database, execute the following procedure:

```
db2cmd
db2
create database <DB_NAME> pagesize 16384
update database configuration for <DB_NAME> using LOGPRIMARY 50
update database configuration for <DB_NAME> using DECFLT_ROUNDING
ROUND_CEILING
update database configuration for <DB_NAME> using LOGBUFSZ 1024
update database configuration for <DB_NAME> using LOGFILSIZ 16384
update database configuration for <DB_NAME> using CHNGPGS_THRES 40
update database configuration for <DB_NAME> using SELF_TUNING_MEM ON
update database configuration for <DB_NAME> using AUTO_SAMPLING ON
update database configuration for <DB_NAME> using DFT_PREFETCH_SZ 960
update database configuration for <DB_NAME> using MAXLOCKS AUTOMATIC
update database configuration for <DB_NAME> using LOCKLIST AUTOMATIC
db2set DB2_COMPATIBILITY_VECTOR=MYS
```

3. The user created in Step1 must then be granted the adequate roles and privileges on the SAP CC Core Database. Execute the following procedure:

```
db2cmd
db2
connect to <DB_NAME>
grant dbadm on database to <DB_USER>
connect reset
```

4. The data storage of your SAP CC Core Database must then be optimized by creating the following mandatory tablespaces:

Tablespace name	Description
CATALOG_DATA	Tables used to store the objects belonging to the catalog, and the operations audit
CATALOG_INDX	Indexes used for the objects belonging to the catalog, and the operations audit
SUBSCRIBER_DATA	Tables containing objects belonging to provider contracts and subscriptions
SUBSCRIBER_INDX	Indexes used for the objects belonging to provider contracts and subscriptions
ACCESS_DATA	Tables used to store chronologies of mappings between technical identifiers and corresponding contract or subscription
ACCESS_INDX	Indexes used for chronologies of mappings between technical identifiers and corresponding contract or subscription
PRERATING_DATA	Tables used to store prerating data
PRERATING_INDX	Indexes used for prerating data
COUNTER_DATA	Tables used to store the counters declared in subscriptions and provider contracts
COUNTER_INDX	Indexes used for counters
C_SNAPSHOT_DATA	Tables used to store snapshots of counter values for subscriptions and provider contracts.
C_SNAPSHOT_INDX	Indexes used for snapshots of counter values.

Tablespace name	Description
SESSION_DATA	Tables used to store session-based charging events and charging session histories for failover purposes
SESSION_INDX	Indexes used for session-based charging events and charging session histories for failover purposes
OBJECT_CHANGE_DATA	Tables used to store data related to the object change log for audited operations
OBJECT_CHANGE_INDX	Indexes used for object change log for audited operations
ALLOWANCE_DATA	Tables used to store allowances
ALLOWANCE_INDX	Indexes used for allowances
MONITORING_DATA	Tables used to store the objects used for monitoring purpose, such as data files metadata
MONITORING_INDX	Indexes used for the objects used for monitoring purpose, such as data files metadata

To create the mandatory tablespaces, execute the following procedure:

```
db2cmd
db2
connect to <DB_NAME>
```

And then, for each tablespace listed in the above table:

```
create tablespace <TABLESPACE_NAME> MANAGED BY DATABASE USING (file
'<FILE_LOCATION><TABLESPACE_NAME>' 256000) EXTENTSIZE 2 DROPPED TABLE
RECOVERY OFF AUTORESIZE YES
```

Example

For an IBM DB2 RDBMS installed on a Microsoft Windows host, you can use the following statement to create the CATALOG_DATA tablespace:

```
create tablespace CATALOG_DATA MANAGED BY DATABASE USING (file
'C:\DB2\tablespaces\CATALOG_DATA' 256000) EXTENTSIZE 2 DROPPED TABLE RECOVERY
OFF AUTORESIZE YES
```

7.1.15 Installing a Core Server on a Monohost Landscape

Description

The installation of a Core Server system on a mono-host landscape is performed using the SAP installer. This procedure explains you how to configure the different steps of the installation scenario handled by SAP installer in order to install:

- A stand-alone landscape that is not integrated with any third-party system

- A landscape that is integrated with SAP Convergent Invoicing
- A landscape that is integrated with SAP Business Suite 4 SAP HANA Cloud Public Edition

Preliminary Notes

For more information about license keys, refer to SAP Notes 197623 and 94998.

Prerequisites

- The installation materials of SAP Convergent Charging 2020 must be downloaded and available on the host. Refer to the Downloading the Installation Materials [page 92] procedure if necessary
- On Linux and UNIX operating systems hosts, a C shell (csh) must be installed and available. Refer to the Installing C Shell (csh) [page 101] procedure if necessary
- On Linux and UNIX operating systems hosts, you must install some required Linux runtime packages in order to use SAPinst 749 or higher (SWPM 1.0 SP19 or higher) on RHEL 6, SLES 11 or OL 6. For further information, refer to SAP Note 2195019
- On Microsoft Windows hosts, the “Microsoft Visual C++ 2013 runtime” libraries must be installed and available. For further information, refer to SAP Note 2676219
- A SID[48] must be chosen for the Core Server system. Refer to the Choosing System IDs [page 91] procedure if necessary
- The SAP CC Working Directory must be created on the host. Refer to the Creating Required Directories [page 99] procedure if necessary
- A database must be prepared and available. Depending of your landscape, refer if necessary to one of the following procedures:
  - Preparing the Oracle Core Database [page 106]
  - Preparing the SQL Server Core Database [page 109]
  - Preparing the SAP ASE Core Database [page 113]
  - Preparing the SAP HANA Core Database [page 118]
  - Preparing the IBM DB2 Core Database (w/o pureScale Feature) [page 120]
- If your Core Database is running under an Oracle RDBMS[49], the dedicated JDBC[50] driver must be downloaded and available on the host. Refer to the Downloading the Oracle JDBC Driver [page 97] procedure if necessary
- Port numbers required to contact SAP CC must be available. In case these ports are already used by another application, enter a new value at installation time included in the range of port numbers from 1024 to 49151.

[48] SAP System Identifier
[49] Relational Database Management System
[50] Java Database Connectivity

> → Tip
>
> Port numbers can be modified later on: refer to the Modifying the Ports of the Core Server System and Modifying the Ports of the BART Server System sections of the SAP CC 2020 Tuning Guide documentation.

Procedure

As this procedure relies on the SAP installer, you first need to run it using the Launching the SAP installer [page 190] dedicated procedure.

Once opened, the SL Common GUI tool displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. On the *Welcome* screen:
   - Click *SAP Convergent Charging* > *Install* > *Standard System Installation* > *Core Server - First Steps*
   - Click the *Next* button
2. **Parameter Settings** screen
   - *Parameter Mode*: Select *Custom*
   - Click the *Next* button
   - (Microsoft Windows only) You will be prompted to log off in order to let SAP installer grant you the authorizations that are required to perform the installation. Click the *OK* button to log off, and reconnect to the host. SAP installer automatically restarts to continue the installation from there
3. **General SAP System Parameters** screen
   - *SAP System ID (SAPSID)*: Fill with the chosen SID
   - Click the *Next* button
4. **Master Password** screen
   - *Password for All Users*: Fill with a password respecting the SAP password policy
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
5. (Microsoft Windows only) **Windows Domain** screen
   - *Domain Model*: Select *Local Domain*
   - Click the *Next* button
6. **Operating System Users** screen
   - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
   - *Confirm*: Re-type the chosen password
   - *Password of SAP Service User*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
7. **SAP Convergent Charging Host License** screen
   - Tick the *Generate a temporary license* checkbox

- Click the *Next* button

8. **Database Type** screen
   - *Database Type*: Select the adequate RDBMS
   - Click the *Next* button
9. **SAP HANA**, **Sybase ASE**, **Oracle**, **Oracle RAC**, **MS SQL Server**, **DB2**, or **DB2 pureScale** screen
   - **SAP HANA** screen

     > ⚠ Caution
     >
     > Do not install the Core Server on the same host than the SAP HANA database.
     >
     > Refer to the SAP Note 2770629 for more information.

     (Database Connection)
     - *Primary Host*: Fill with the IP address of the machine hosting the primary node
     - *Secondary Host*: Fill with the IP address of the machine hosting the secondary node
     - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
     - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
     - *Mode*: Select the mode that corresponds to the installation scenario of your SAP HANA database
       - *Single Container Mode*
         - *Port*: Fill with the relevant port of the system database of your SAP HANA system. For further information about the default ports of SAP products, refer to the dedicated section on SAP Help Portal: https://help.sap.com/viewer/ports
       - *Multiple Containers Mode*
         - *Tenant Name*: Fill with the name of the tenant database to connect to
         - *Port*: Fill with the relevant port of the tenant database of your SAP HANA system you want to connect to. Typical assignment rule for port number are:
           - 3`<INSTANCE_NUMBER>`15 for a single-container system
           - Between 3`<INSTANCE_NUMBER>`41 and 3`<INSTANCE_NUMBER>`98 for a multitenant system
             For further information about the value of such port, refer to the SAP Note 2365930 or to the section dedicated to the default ports on SAP Help Portal: https://help.sap.com/viewer/ports
   - **Sybase ASE** screen
     (Primary Database Connection)
     - *Host*: Fill with the IP address of the machine hosting the database
     - *Port*: Fill with the port specified during the installation of the database
     - *Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
     - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
     - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

(Backup Database Connection)

Tick the *Use backup database connection* checkbox if you want to use a backup database for high availability purpose. If you tick this checkbox, specify the following information:

- *Host*: Fill with the IP address of the machine hosting the backup database
- *Port*: Fill with the port used to connect to the backup database

- **Oracle** screen

(JDBC Driver)

- *JDBC Driver JAR Archive*: Click the *Browse* button and select the JAR archive you downloaded using the dedicated procedure [page 97]

(Database Authentication)

- *Host*: Fill with the IP address of the machine hosting the database
- *Port*: Fill with the port specified during the installation of the database
- *Instance Id*: Fill with the unique name of the database instance within which the content of the Core Database will be installed

> **Note**
>
> For an Oracle database, you can use the following command to fill the *Instance Id* field:
>
> ```
> sqlplus / as sysdba
> SELECT sys_context('USERENV', 'DB_NAME') FROM DUAL;
> ```

- *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
- *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

- **Oracle RAC** screen

(JDBC Driver)

- *JDBC Driver JAR Archive*: Click the *Browse* button and select the JAR archive you downloaded using the dedicated procedure [page 97]

(Database Authentication)

- *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
- *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

(Database Instances)

Click the *Add* button to add instances according to your needs, each instance containing the following information:

- *Host*: Fill with the IP address of the machine hosting the database instance
- *Port*: Fill with the port specified during the installation of the database instance
- *Instance Id*: Fill with the unique name of the database instance

- **MS SQL Server** screen

(Database Connection)

- *Host*: Fill with the IP address of the machine hosting the database

- *Port*: Fill with the port specified during the installation of the database
- *Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
- *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
- *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

- **DB2** screen
  (Database Connection)
  - *Host*: Fill with the IP address of the machine hosting the database
  - *Port*: Fill with the port specified during the installation of the database
  - *Database Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
  - *Schema Name*: Fill with the name of the database schema you want to create within the Core Database
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
  - *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary
- **DB2 pureScale** screen
  (Database Parameters)
  - *Database Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
  - *Schema Name*: Fill with the name of the database schema you want to create within the Core Database
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

  (Database Instances)
  Click the *Add* button to add instances according to your needs, each instance containing the following information:
  - *Host*: Fill with the IP address of the machine hosting the database instance
  - *Port*: Fill with the port specified during the installation of the database instance
- Click the *Next* button

10. **Tablespaces / Filegroups** screen
    - Ensure that each table or index is associated to the correct filegroup or tablespace created during the preparation of your Core Database. When necessary, modify the associations accordingly
    - Click the *Next* button
11. **SAP Convergent Charging Working Directory** screen

- *Working Directory*: Click the *Browse* button and select the directory you specified during the creation of the required directories [page 99]
- Click the *Next* button

12. **CA Introscope Java Agent** screen
    - *Use monitoring with CA Introscope*: Untick the checkbox
    - Click the *Next* button
13. **System Landscape Directory** screen
    - *Configure System Landscape Directory*: Untick the checkbox
    - Click the *Next* button
14. **SAP Convergent Charging Security** screen
    - *Security*: Select the *Disabled* option
    - Click the *Next* button
15. **SAP Convergent Charging Administrator User** screen
    - *Password*: Fill with a password respecting the SAP password policy
    - *Confirm*: Re-type the chosen password
    - Click the *Next* button
16. **SAP Convergent Charging Tax** screen
    - Only tick the *VAT* taxation framework
    - Click the *Next* button
17. **SAP Convergent Charging Integration Scenario** screen
    - *Scenario*: Select the scenario that corresponds to your landscape:
      - If you select the *Stand-alone* integration scenario, no dedicated intermediate screen needs to be filled
      - If you select the *Billing and Invoicing in SAP Convergent Invoicing*, *Billing, Invoicing, and Storage of Consumption Data in SAP Convergent Invoicing* or *Billing and Invoicing in SAP S/4HANA Cloud for Contract Accounting and Invoicing* integration scenario, use the following recommendations to fill these dedicated intermediate screens:
        - **JCo Connection Configuration** screen
          - Click the *Add* button to add the adequate JCo destination(s)
          - *Name*: Fill with the name of the JCo destination

            > ⚠ Caution
            >
            > Ensure that the JCo destination name does not contain spaces and/or special characters (especially slash and backslash).

          - *SAP Application*: Select the relevant type of targeted SAP Application
          - *Application Server Host*: Specify the full name of the machine hosting the SAP Application
          - *SAP System Number*: Specify the relevant SAP system number
          - *SAP Client*: Specify the relevant SAP client number
          - *Logon User*: Specify the name of the SAP user you want to use for connecting to the SAP Application
          - *Logon Password*: Specify the password of the previously specified user
          - *Install SAP Cryptographic Library*: Untick the checkbox
          - Click the *Next* button
        - **Customer Management Area and ERP Reference System** screen

- *Default Customer Management Area*: Specify a name for the default CMA
- *JCo Destination for ERP*: Select the JCo destination to use for the default CMA
- *JCo Destination for CRM*: Select the JCo destination to use for the default CMA
- *JCo Destination*: Select the JCo destination to use for connecting to the ERP reference system
- Click the *Next* button

- **SAP S/4HANA Cloud Public Edition Web Service Configuration** screen
  (WS Destinations)
  - Click the *Add* button to add the adequate WS destination(s)
  - *Name*: Fill with the name of the WS destination
  - *URL*: Fill with the URL of the targeted "Convergent Invoicing Billable Items – Create – Request (Single)" web service, configured in a Communication Arrangement within SAP S/4HANA Cloud Public Edition
  - *Login*: Specify the name of the SAP user you want to use for connecting to the web service
  - *Password*: Specify the password of the previously specified user
  - *Certificate*: In case the targeted web service uses HTTPS[51], click the *Browse* button to select the certificate to use

  (HTTPS Proxy Server)
  - *Use a proxy server*: Tick the checkbox if you want to use and configure an HTTPS proxy server
  - *Host*: Fill with the name or IP address (IPv4 or IPv6) of the HTTPS proxy server used for sending requests to the web service
  - *Port*: Fill with the port of the HTTPS proxy server used for sending requests to the web service
  - *Login*: Fill with the name of the technical user used to authenticate to the HTTPS proxy server
  - *Password*: Fill with the password of the specified technical user, that respects the SAP password policy

  (Connection)
  - *Check connection*: Tick the checkbox if you want SAP installer to verify the connection settings
- Click the *Next* button

18. **SAP Convergent Charging Instances** screen
    - Click the *Add* button to add instances to your landscape according to your needs. Each type of added instance will be configured within dedicated screens described afterwards.
    - Click the *Next* button
19. **SAP Convergent Charging Dispatcher Instance** screen
    - `<SERVICE>` *Port*: Keep the default configuration, or set specific values for each configurable port
    - `<SERVICE>` *Host*: Specify the full name of the machine hosting the dispatcher instance. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify one of these IP addresses
    - Click the *Next* button
20. **SAP Convergent Charging Updater Instance** screen

[51] HyperText Transfer Protocol Secured

- `<SERVICE>` *Port*: Keep the default configuration, or set specific values for each configurable port
- `<SERVICE>` *Host*: Specify the full name of the machine hosting the updater instance. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify one of these IP addresses
- Click the *Next* button

21. **Start Instances after Installation** screen
    - *Start SAP Convergent Charging instances after installation*: Tick the checkbox to automatically start the instances at the end of the installation process

      > ⓘ Note
      >
      > The SAP Management Console is deployed during the installation process and gives you the possibility to manage each instance of your landscape

    - Click the *Next* button
22. **Prerequisites Checker** screen
    - Click the *Next* button
23. **Host Agent and SAPOSCOL Log Directory** screen
    - Click the *Next* button
24. (Microsoft Windows only) **Windows Domain for Host Agent** screen
    - *Domain Model*: Select the *Local Domain* option
    - Click the *Next* button
25. **Unpack Archives** screen
    - Click the *Next* button
26. **Parameter Summary** screen
    - Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button
    - If all your settings are correctly taken into account, click the *Next* button to start the installation
27. **Task Progress** screen
    - This screen contains the different steps of your installation scenario, whose advancement can be followed

7.1.16 Installing a Core Server on a Multihost Landscape

Description

The installation and deployment of a Core Server system on a multi-hosts landscape is performed using the SAP installer. This procedure explains you how to configure the different steps of the installation scenario handled by SAP installer in order to install:

- A "stand-alone" SAP system landscape that is not integrated with any third-party system

- A landscape that is integrated with SAP Convergent Invoicing in SAP S/4HANA or SAP ERP
- A landscape that is integrated with SAP S/4HANA Cloud Public Edition

Preliminary Notes

For more information about license keys, refer to SAP Notes 197623 and 94998.

Prerequisites

- The installation materials of SAP Convergent Charging 2020 must be downloaded and available on the host. Refer to the Downloading the Installation Materials [page 92] procedure if necessary
- On Linux and UNIX operating systems hosts, a C shell script must be installed and available. Refer to the Installing C Shell (csh) [page 101] procedure if necessary
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2013 runtime" libraries must be installed and available. For further information, refer to SAP Note 2676219
- A SID[52] must be chosen for the Core Server system. Refer to the Choosing System IDs [page 91] procedure if necessary
- The adequate directories must be created. Refer to the Creating Required Directories [page 99] procedure if necessary
- A database must be prepared and available. Depending of your landscape, refer if necessary to one of the following procedures:
  - Preparing the Oracle Core Database [page 106]
  - Preparing the SQL Server Core Database [page 109]
  - Preparing the SAP ASE Core Database [page 113]
  - Preparing the SAP HANA Core Database [page 118]
  - Preparing the IBM DB2 Core Database (w/o pureScale Feature) [page 120]
- If your Core Database is running under an Oracle RDBMS[53], the dedicated JDBC[54] driver must be downloaded and available on the host. Refer to the Downloading the Oracle JDBC Driver [page 97] procedure if necessary
- Port numbers required to contact SAP CC must be available. In case these ports are already used by another application, enter a new value at installation time included in the range of port numbers from 1024 to 49151.

> → Tip
>
> Port numbers can be modified later on: refer to the Modifying the Ports of the Core Server System and Modifying the Ports of the BART Server System sections of the SAP CC 2020 Tuning Guide documentation.

[52] SAP System Identifier
[53] Relational Database Management System
[54] Java Database Connectivity

Procedure

As this procedure relies on the SAP installer for SAP CC, you first need to run it using the Launching the SAP installer [page 190] dedicated procedure.

Once opened, the SL Common GUI tool displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. On the *Welcome* screen:
   - Click *SAP Convergent Charging* > *Install* > *Standard System Installation* > *Core Server - First Steps*
   - Click the *Next* button
2. **Parameter Settings** screen
   - *Parameter Mode*: Select *Custom*
   - Click the *Next* button
3. **General SAP System Parameters** screen
   - *SAP System ID (SAPSID)*: Fill with the chosen SID
   - Click the *Next* button
4. **Master Password** screen
   - *Password for All Users*: Fill with a password respecting the SAP password policy
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
5. (Linux and UNIX only) **SAP System Administrator** screen (related to the SAP CC system)
   - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
   - *User ID*: *left empty*
   - *Group ID of sapsys*: *left empty*
   - Click the *Next* button
6. **SAP Convergent Charging Host License**screen
   - Tick the *Generate a temporary license* checkbox
   - Click the *Next* button
7. **Database Type** screen
   - *Database Type*: Select the adequate RDBMS
   - Click the *Next* button
8. **SAP HANA**, **Sybase ASE**, **Oracle**, **Oracle RAC**, **MS SQL Server**, **DB2**, or **DB2 pureScale** screen
   - **SAP HANA** screen
     (Database Connection)
     - *Primary Host*: Fill with the IP address of the machine hosting the primary node
     - *Secondary Host*: Fill with the IP address of the machine hosting the secondary node
     - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
     - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
     - *Mode*: Select the mode that corresponds to the installation scenario of your SAP HANA database
       - *Single Container Mode*

        - *Port*: Fill with the relevant port of the system database of your SAP HANA system. For further information about the default ports of SAP products, refer to the dedicated section on SAP Help Portal: https://help.sap.com/viewer/ports
    - *Multiple Containers Mode*
        - *Tenant Name*: Fill with the name of the tenant database to connect to
        - *Port*: Fill with the relevant port of the tenant database of your SAP HANA system you want to connect to. Typical assignment rule for port number are:
            - 3`<INSTANCE_NUMBER>`15 for a single-container system
            - Between 3`<INSTANCE_NUMBER>`41 and 3`<INSTANCE_NUMBER>`98 for a multitenant system
              For further information about the value of such port, refer to the SAP Note 2365930 or to the section dedicated to the default ports on SAP Help Portal: https://help.sap.com/viewer/ports
- **Sybase ASE** screen
  (Primary Database Connection)
    - *Host*: Fill with the IP address of the machine hosting the database
    - *Port*: Fill with the port specified during the installation of the database
    - *Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
    - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
    - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
    - *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

  (Backup Database Connection)
  Tick the *Use backup database connection* checkbox if you want to use a backup database pour high availability purpose. If you tick this checkbox, specify the following information:
    - *Host*: Fill with the IP address of the machine hosting the backup database
    - *Port*: Fill with the port used to connect to the backup database
- **Oracle** screen
  (JDBC Driver)
    - *JDBC Driver JAR Archive*: Click the *Browse* button and select the JAR archive you downloaded using the dedicated procedure [page 97]

  (Database Authentication)
    - *Host*: Fill with the IP address of the machine hosting the database
    - *Port*: Fill with the port specified during the installation of the database
    - *Instance Id*: Fill with the unique name of the database instance within which the content of the Core Database will be installed

      > **Note**
      >
      > For an Oracle database, you can use the following command to fill the *Instance Id* field:
      >
      > ```
      > sqlplus / as sysdba
      > SELECT sys_context('USERENV', 'DB_NAME') FROM DUAL;
      > ```

- *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
- *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

- **Oracle RAC** screen
  (JDBC Driver)
  - *JDBC Driver JAR Archive*: Click the *Browse* button and select the JAR archive you downloaded using the dedicated procedure [page 97]

  (Database Authentication)
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

  (Database Instances)
  Click the *Add* button to add instances according to your needs, each instance containing the following information:
  - *Host*: Fill with the IP address of the machine hosting the database instance
  - *Port*: Fill with the port specified during the installation of the database instance
  - *Instance Id*: Fill with the unique name of the database instance
- **MS SQL Server** screen
  (Database Connection)
  - *Host*: Fill with the IP address of the machine hosting the database
  - *Port*: Fill with the port specified during the installation of the database
  - *Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database
  - *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary
- **DB2** screen
  (Database Connection)
  - *Host*: Fill with the IP address of the machine hosting the database
  - *Port*: Fill with the port specified during the installation of the database
  - *Database Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
  - *Schema Name*: Fill with the name of the database schema you want to create within the Core Database
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

- **DB2 pureScale** screen
  (Database Parameters)
  - *Database Name*: Fill with the unique name of the database instance within which the content of the Core Database will be installed
  - *Schema Name*: Fill with the name of the database schema you want to create within the Core Database
  - *User*: Fill with the name of the Core Database administrator you specified during the preparation of your Core Database
  - *Password*: Fill with the password of the Core Database administrator you specified during the preparation of your Core Database

  (Database Instances)
  Click the *Add* button to add instances according to your needs, each instance containing the following information:
  - *Host*: Fill with the IP address of the machine hosting the database instance
  - *Port*: Fill with the port specified during the installation of the database instance
- Click the *Next* button

9. **Tablespaces / Filegroups** screen
   - Ensure that each table or index is associated to the correct filegroup or tablespace created during the preparation of your Core Database. When necessary, modify the associations accordingly
   - Click the *Next* button
10. **SAP Convergent Charging Working Directory** screen
    - *Working Directory*: Click the *Browse* button and select the directory you specified during the creation of the required directories [page 99]
    - Click the *Next* button
11. **CA Introscope Java Agent** screen
    - *Use monitoring with CA Introscope*: Untick the checkbox
    - Click the *Next* button
12. **System Landscape Directory** screen
    - *Configure System Landscape Directory*: Untick the checkbox
    - Click the *Next* button
13. **SAP Convergent Charging Security** screen
    - *Security*: Select the *Disabled* option
    - Click the *Next* button
14. **SAP Convergent Charging Administrator User** screen
    - *Password*: Fill with a password respecting the SAP password policy
    - *Confirm*: Re-type the chosen password
    - Click the *Next* button
15. **SAP Convergent Charging Tax** screen
    - Only tick the *VAT* taxation framework
    - Click the *Next* button
16. **SAP Convergent Charging Integration Scenario** screen
    - *Scenario*: Select the scenario that corresponds to your landscape:

- If you select the *Stand-alone* integration scenario, no dedicated intermediate screen needs to be filled
- If you select the *Billing and Invoicing in SAP Convergent Invoicing*, *Billing, Invoicing, and Storage of Consumption Data in SAP Convergent Invoicing* or *Billing and Invoicing in SAP S/4HANA Cloud for Contract Accounting and Invoicing* integration scenario, use the following recommendations to fill these dedicated intermediate screens:
  - **JCo Connection Configuration** screen
    - Click the *Add* button to add the adequate JCo[55] destination(s)
    - *Name*: Fill with the name of the JCo destination

      > ⚠ Caution
      >
      > Ensure that the JCo destination name does not contain spaces and/or special characters (especially slash and backslash).

    - *SAP Application*: Select the relevant type of targeted SAP Application
    - *Application Server Host*: Specify the full name of the machine hosting the SAP Application
    - *SAP System Number*: Specify the relevant SAP system number
    - *SAP Client*: Specify the relevant SAP client number
    - *Logon User*: Specify the name of the SAP user you want to use for connecting to the SAP Application
    - *Logon Password*: Specify the password of the previously specified user
    - *Install SAP Cryptographic Library*: Untick the checkbox
    - Click the *Next* button
  - **Customer Management Area and ERP Reference System** screen
    - *Default Customer Management Area*: Specify a name for the default CMA[56]
    - *JCo Destination for ERP*: Select the JCo destination to use for the default CMA
    - *JCo Destination for CRM*: Select the JCo destination to use for the default CMA
    - *JCo Destination*: Select the JCo destination to use for connecting to the ERP reference system
    - Click the *Next* button
  - **SAP S/4HANA Cloud Public Edition Web Service Configuration** screen
    (WS Destinations)
    - Click the *Add* button to add the adequate WS destination(s)
    - *Name*: Fill with the name of the WS destination
    - *URL*: Fill with the URL of the targeted "Convergent Invoicing Billable Items – Create – Request (Single)" web service, configured within a Communication Arrangement in SAP S/4HANA Cloud Public Edition
    - *Login*: Specify the name of the SAP user you want to use for connecting to the web service
    - *Password*: Specify the password of the previously specified user
    - *Certificate*: In case the targeted web service uses HTTPS[57], click the *Browse* button to select the certificate to use

    (HTTPS Proxy Server)

[55] Java Connector
[56] Customer Management Area
[57] HyperText Transfer Protocol Secured

- *Use a proxy server*: Tick the checkbox if you want to use and configure an HTTPS proxy server
- *Host*: Fill with the name or IP address (IPv4 or IPv6) of the HTTPS proxy server used for sending requests to the web service
- *Port*: Fill with the port of the HTTPS proxy server used for sending requests to the web service
- *Login*: Fill with the name of the technical user used to authenticate to the HTTPS proxy server
- *Password*: Fill with the password of the specified technical user, that respects the SAP password policy

(Connection)

- *Check connection*: Tick the checkbox if you want SAP installer to verify the connection settings to the HTTPS proxy server

- Click the *Next* button

17. **SAP Convergent Charging Instances** screen
    - Click the *Add* button to add the following instances to your landscape:
        - *Dispatcher*
        - *Guider*
    - Click the *Next* button
18. **SAP Convergent Charging Dispatcher Instance** screen
    - `<SERVICE>` *Port*: Keep the default configuration, or set specific values for each configurable port
    - `<SERVICE>` *Host*: Specify the full name of the machine hosting the dispatcher instance. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify one of these IP addresses
    - Click the *Next* button
19. **Start Instances after Installation** screen
    - *Start SAP Convergent Charging instances after installation*: Tick the checkbox to automatically start the instances at the end of the installation process

      > **Note**
      >
      > The SAP Management Console is deployed during the installation process and gives you the possibility to manage each instance of your landscape

    - Click the *Next* button
20. **Prerequisites Checker** screen
    - Click the *Next* button
21. (Linux and UNIX only) **SAP System Administrator** screen (global user for all SAP systems)
    - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
    - *User ID*: *left empty*
    - *Group ID of sapsys*: *left empty*
    - Click the *Next* button
22. **Unpack Archives** screen
    - Click the *Next* button
23. **Parameter Summary** screen

- Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button
- If all your settings are correctly taken into account, click the *Next* button to start the installation

24. **Task Progress** screen
    - This screen contains the different steps of your installation scenario, whose advancement can be followed

7.1.17 Testing a Landscape Basically

Description

Once an SAP CC Core Server system has been installed, you can use this procedure to perform some basic checks in your SAP system landscape relating to:

- The SAP Central Repository of the Global Host
- The Working Directory of each host
- The log files related to the installation process
- The status of the instances of your Core Server system

Preliminary Notes

- For more information about the Admin+ user interface, refer to its dedicated documentation: SAP CC 2020 Primary Help for Admin+.
- This procedure uses the following variables:
    - `<DRIVE>`, which corresponds to the drive of the Microsoft Windows host on which the system is installed
    - `<SYSTEM_ID>`, which corresponds to the identifier of your system
    - `<INSTANCE_NAME>`, which corresponds to the name of the concerned instance
    - `<SAPCC_SYSADM_USERNAME>`, which corresponds to the name of the administrator of your Core Server system (`<admin>` by default)
    - `<SAPCC_SYSADM_PASSWORD>`, which corresponds to the password of the administrator of your Core Server system, that you specified at installation time in the "SAP Convergent Charging *Administrator User* screen of the SAP installer

Checking the SAP Central Repository and the Working Directory

As described in the Document Definitions [page 12] section, the SAP Central Repository is used to centralize the configuration of each installed system within the following directory:

- `/sapmnt/<SYSTEM_ID>/SYS` for Linux and UNIX operating systems
- `<DRIVE>:\usr\sap\<SYSTEM_ID>\SYS` for a Microsoft Windows operating systems

You can browse this directory to get information about your installed system(s). In addition to the SAP Central Repository, the SAP installer creates a working directory for each deployed instance, in order to centralize elements such as configuration files, works logs, and so on. You can browse the following directories:

- `/usr/sap/<SYSTEM_ID>/<INSTANCE_NAME>` for Linux and UNIX operating systems
- `<DRIVE>:\usr\sap\<SYSTEM_ID>\<INSTANCE_NAME>` for a Microsoft Windows operating systems

Checking the installation logs

When an SAP CC system has been installed, you can check some log files to get detailed information about:

- The execution of the different installation steps performed by the SAP installer
- The status of the installed elements

The following table contains information about folders containing log files that you can get information from:

SAP installer installation steps	Depending whether your host is the Global Host of your system or not, you can browse the `FIRST` or the `ADDITIONAL` folders of the following directory:
	Linux and UNIX operating systems
	`/tmp/sapinst_instdir/CC2020/INSTALL/STANDARD/`
	Microsoft Windows operating systems
	`<DRIVE>:\Program Files\sapinst_instdir\CC2020\INSTALL\STANDARD\`

Instances status	The log files relating to a given instance are available within the following directories:
	Linux and UNIX operating systems
	`/usr/sap/<SYSTEM_ID>/<INSTANCE_NAME>/work`
	`/usr/sap/<SYSTEM_ID>/<INSTANCE_NAME>/work/log`
	Microsoft Windows operating systems
	`<DRIVE>:\usr\sap\<SYSTEM_ID>\<INSTANCE_NAME>\work`
	`<DRIVE>:\usr\sap\<SYSTEM_ID>\<INSTANCE_NAME>\work\log`

Checking the instances status using the Display System Status app of Cockpit

Checking the status of the different instances available within your landscape consists in launching [page 206] Cockpit and use the Display System Status app to display the status of the instances (that should be `Started` for alive instances).

Checking the instances status using Admin+

Checking the status of the different instances available within your landscape consists in the following operations:

- Launching [page 208] the Admin+ user interface
- Identifying as the administrator of the Core Server system and displaying the following table filled with the status of the different instances:

```
+----+---------+-----+---------+---------+-------------+-----------+-------+
|ID  |HOSTNAME |HTTP |EXTERNAL |INTERNAL |CURRENT DATE |START DATE |STATUS |
+----+---------+-----+---------+---------+-------------+-----------+-------+
:    :         :     :         :         :             :           :       :
+----+---------+-----+---------+---------+-------------+-----------+-------+
```

As this procedure relies on the Admin+ user interface, you first need to launch [page 208] it.

Once launched, execute the following command within Admin+ in order to display the status of the instances (that should be `ON` for alive instances):

```
list
```

7.1.18 Setting Up a New Host for a Landscape

Description

For many reasons such as business growth, performances improvement, and so on, it might be necessary to extend your landscape to add new system(s) or instance(s) of an existing system. This procedure explains you how to set up a new host in an existing SAP system landscape.

Procedure

To set up a new physical host, execute the following procedures:

- Checking Free Space [page 98], considering the installation materials and the system(s) to install
- Creating Required Directories [page 99], with the adequate permissions
- Installing C Shell (csh) [page 101], required by the SAP installer for Linux and UNIX operating systems
- Setting Up the System Encoding [page 100], because the SAP installer requires an UTF-8 character encoding for Linux and UNIX operating systems
- Setting Up the System Time Zone [page 102], to ensure that the time zone is correctly configured
- Setting Up the System Environment Variables [page 104]

7.1.19 Adding Core Server Instances in a Multi-Hosts Landscape

Description

For many reasons such as business growth, performances improvement, and so on, it might be necessary to extend your Core Server system. This procedure explains you how to add new instance(s) on a given host of an existing Core Server system, using the SAP installer.

> **Note**
>
> The new instances can be deployed:
>
> - On an existing host of your landscape, that already contains instances of a Core Server system
> - On a new host, that must be set up before launching the SAP installer. Refer to the Setting Up a New Host for a Landscape [page 141] procedure if necessary.

Preliminary Notes

For more information about license keys, refer to SAP Notes 197623 and 94998.

Prerequisites

The SAP Global Host [page 12] must be installed before executing this procedure. Refer to the Installing a Core Server on a Multihost Landscape [page 131] procedure if necessary.

A new host or an existing host is available for the new system instance. If necessary, see the Setting Up a New Host for a Landscape [page 141] procedure.

Procedure

As this procedure relies on the SAP installer, you first need to launch it using the Launching the SAP installer [page 190] dedicated procedure.

Once opened, the SAP installer displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. **Welcome** screen
   - Click *SAP Convergent Charging* > *Install* > *Standard System Installation* > *Core Server - Add Instances* >
   - Click the *Next* button
2. **Parameter Settings** screen
   - *Parameter Mode*: Select *Custom*
   - Click the *Next* button
3. **General SAP System Parameters** screen
   - *Profile Directory*: Filled by default with the location of the SAP system profile directory. Ensure that this location is correct
   - Click the *Next* button
4. **Master Password** screen
   - *Password for All Users*: Fill with a password respecting the SAP password policy
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
5. **SAP System Administrator** screen (related to the SAP CC system)
   - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
   - *User ID*: Fill with the identifier of the user that corresponds to the same user as for the Global Host

     > **Note**
     >
     > You can use the following command on a Linux or UNIX operating system host:
     >
     > ```
     > cat /etc/passwd | grep -i <SAP_System_ID>adm | cut -d':' -f3
     > ```

   - *Group ID of sapsys*: *left empty*
   - Click the *Next* button
6. **SAP Convergent Charging Host License** screen
   - Tick the *Generate a temporary license* checkbox
   - Click the *Next* button
7. **SAP Convergent Charging Instances** screen
   - Click the *Add* button to add the adequate instance(s) to your landscape
   - Click the *Next* button
8. **SAP Convergent Charging Updater Instance** screen
   - `<SERVICE>` *Port*: Keep the default configuration, or set specific values for each configurable port
   - `<SERVICE>` *Host*: Specify the full name of the machine hosting the dispatcher instance. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify one of these IP addresses

- Click the *Next* button

9. **Start Instances after Installation** screen
   - *Start SAP Convergent Charging instances after installation*: Tick the checkbox to automatically start the instances at the end of the installation process

     > **Note**
     >
     > The SAP Management Console is deployed during the installation process and gives you the possibility to manage each instance of your landscape

   - Click the *Next* button
10. **Prerequisites Checker** screen
    - Click the *Next* button
11. **SAP System Administrator** screen (global user for all SAP systems)
    - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
    - *User ID*: *left empty*
    - *Group ID of sapsys*: *left empty*
    - Click the *Next* button
12. **Unpack Archives** screen
    - Click the *Next* button
13. **Parameter Summary** screen
    - Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button
    - If all your settings are correctly taken into account, click the *Next* button to start the installation
14. **Task Progress** screen
    - This screen contains the different steps of your installation scenario, whose advancement can be followed

7.1.20 Starting and Stopping the Server Systems

Description

During the installation of an SAP CC system, the following applications are automatically deployed to manage your SAP Convergent Charging system landscape:

- SAP Management Console (SAP MC)
- SAP Microsoft Management Console (SAP MMC)
- SAPControl

The following table shows the availability of each application according to the operating system of your host:

Application	Application Type		OS Availability	
	GUI	Command-Line	Microsoft Windows	Linux and UNIX Operating Systems
SAP Management Console (SAP MC)	■		■	■
SAP Microsoft Management Console (SAP MMC)	■		■	
SAPControl		■	■	■

You can use this procedure to get information about the use of each application.

Preliminary Notes

- It is recommended to respect the following sequence when starting the SAP systems you installed within your landscape:
  1. Core Server system and its instances, respecting the following recommendations:

     → Recommendation

     - Start or restart one instance at a time: one after the other. Wait for an instance to be fully (re)started before (re)starting another one.
     - Always first (re)start the dispatcher instances.
     - SAP SE recommends this specific order for (re)starting:
       1. dispatcher
       2. updater
       3. guider
       4. rater
       5. bulkloader
  2. BART Server system
  3. Import/Export Connector application
- For more information about the SAP Management Console application, refer to its dedicated documentation: SAP Management Console
- For more information about the SAP Microsoft Management Console application, refer to its dedicated documentation: SAP Microsoft Management Console
- For more information about SAPControl, refer to its dedicated documentation: https://help.sap.com/viewer/ff18034f08af4d7bb33894c2047c3b71/latest/en-US/471d6feeff6e0d46e10000000a155369.html

Java Startup Framework

All the applications mentioned in this procedure rely on the Java Startup Framework, which provides centralized management of the Java server processes and gives the possibility to monitor their life cycle.

This framework handles the following **shutdown modes** that can be used to stop or restart an instance in an SAP Convergent Charging system:

- **Hard shutdown,** which consists in waiting for a given period during which the instance in the SAP is supposed to stop properly by itself. This period is defined via a timeout specified in the `shutdownTimeout` property of the `jstart.config` configuration file (set by default to 150 seconds but capped by the operating system to 180 seconds on Microsoft Windows and 300 on Linux and UNIX operating systems). When this timeout is reached, the Startup Framework stops the Java process.

  > ⚠ Caution
  >
  > SAP SE recommends that you avoid using this shutdown mode, as the timeout may be too small to let instances stop properly.

- **Soft shutdown with timeout,** which consists in waiting for a given period during which the instance in the SAP system is supposed to stop properly by itself. When this timeout is reached, the Startup Framework tries additional internal methods to stop the instance, which means that the instance may be definitively stopped only few minutes after the timeout has been reached
- **Soft shutdown without timeout,** which consists in waiting for the instance to stop, whatever the time it will take. In case of unexpected situation, the result of these shutdown operations cannot be ensured. When such a situation occurs, or when the instance takes too much time to stop, it is possible to stop the instance by executing a hard shutdown operation or by killing the process.

> ⓘ Note
>
> In an SAP Convergent Charging system, an instance can also stop due to the following reasons:
>
> - A signal has been sent by the operating system:
>   - `SIGKILL`, which corresponds to an immediate shutdown
>   - `SIGTERM`, which uses an internal timeout before shutting down the process
> - An individual user executed a dedicated command of a user interface (such as the `server stop` command of the Admin+ tool or the `sapcontrol Stop` command of SAPControl)
> - An applicative shutdown has been triggered, due to internal reasons such as bad startup configuration, invalid license, starting request refused by a dispatcher, conflict between dispatchers, and so on
> - An unrecoverable error occurred:
>   - `java.lang.VirtualMachineError`, coming from the Java Virtual Machine and leading to an immediate shutdown
>   - `java.lang.Error` and `java.lang.Throwable`, which both lead to a shutdown after an internal timeout

Using the SAP Management Console (SAP MC)

If you have installed a Java Runtime Environment (JRE, version 5.0 and higher) and if you use a web browser that is able to run Java applets, then you can launch the SAP Management Console (SAP MC) as a Java applet in your web browser using the following procedure.

> → Remember
>
> **If your web browser no longer supports the Java applet technology**, you can configure the SAP MC application to run locally on your PC.

For example, the error message `Your browser is completely ignoring the <APPLET> tag!` was displayed.

For more information about the existing procedures, see the section *Configuring SAP MC locally* in SAP Note 1014480 to complete your task.

To launch the SAP Management Console:

1. Open your web browser.
2. Call the following URL:
   `http://<HOST>:5<INSTANCE_NUMBER>13`
   Where:
   - `<HOST>` corresponds to the name or IP address of a host of your SAP system
   - `<INSTANCE_NUMBER>` corresponds to the number of the instance

   Example

   Considering an SAP CC system containing a host:
   - Whose hostname is **`SAPCCHOST1`** and IP address is 192.168.1.1
   - That contains an instance whose name is **`CCD02`**

   You can launch the SAP Management Console application using the following URLs: `http://192.168.1.1:50213` or `http://SAPCCHOST1:50213`
3. If your web browser displays a security warning message, configure the options in order to trust the Java applets and reopen the URL.

   Note

   If you want to connect over HTTPS, you have to use the `<sapstartsrv>` HTTPS port. The port number is composed as 5`<INSTANCE_NUMBER>`14.

   Before connecting over HTTPS, you first have to set up SSL correctly for the monitored SAP instances.

To manage an instance in an SAP CC system using the SAP Management Console application, use the following procedure:

1. Open the SAP Management Console application using the procedures described above.
2. Expand the *SAP Systems* node and expand the system which contains the instance you want to manage.
3. To manage instances, you must identify with the login information you provided at installation time. Click the instance you want to manage and fill the *Logon* screen with the information relating to the `<SID>adm` user.
4. Right-click the instance you want to manage, and select the action you want to perform:
   - *Start*, which directly starts the process without any intermediate action
   - *Stop*, which opens a dialog window that gives you the possibility to specify which kind of shutdown mode you want to use and tune the associated timeouts
   - *Restart*, which indeed corresponds to a *Stop* action followed by a *Start* action, and thus opens the related dialog window

Using the SAP Microsoft Management Console (SAP MMC)

Microsoft Windows Operating System

To launch the SAP Microsoft Management Console, connect to the host of your landscape that contains the SAP CC system you want to manage, and open the SAP Management Console program either from the shortcut available on the Windows desktop or from the *Microsoft Windows Start* menu.

To manage an instance in an SAP CC system using the SAP Microsoft Management Console application, use the following procedure:

1. Open the SAP Microsoft Management Console application using the procedures described above.
2. Expand the *SAP Systems* node and expand the system which contains the instance you want to manage.
3. To manage instances, you must identify with the login information you provided at installation time. Click the instance you want to manage and fill the *Logon* screen with the information relating to the `<SID>adm` user.
4. Right-click the instance you want to manage, and select the action you want to perform:
   - *Start*, which directly starts the process without any intermediate action
   - *Stop*, which opens a dialog window that gives you the possibility to specify which kind of shutdown mode you want to use and tune the associated timeouts
   - *Restart*, which indeed corresponds to a *Stop* action followed by a *Start* action, and thus opens the related dialog window

Using SAPControl

You can use SAPControl to start or stop an instance in the SAP Convergent Charging system from a command line.

You need to be logged on to the host of the SAP Convergent Charging system as the user `<sid>adm` where `<SID>` is the SAP System ID of your SAP CC system.

> **Note**
>
> `<sid>adm` user comes with direct access to SAPControl (SAPControl is available in the PATH of this user). The following table contains the list of installation locations for SAPControl :

OS	Location
Linux or UNIX operating systems	`/usr/sap/hostctrl/exe/sapcontrol`
Microsoft Windows	`%ProgramFiles%\SAP\hostctrl\exe\sapcontrol.exe`

The SAPControl command-line application has the following partial synopsis:

```
sapcontrol [-trace <file_name>]
           [-debug]
           [-user <user_id> <password>]
           [-host <hostname>]
           -nr <instance_number>
           -function <webmethod> [parameter list]
```

Where:

- The `-trace` option can be used to write SOAP request and/or response information into the specified file
- The `-debug` option can be used to write local trace to the `stderr` writing channel
- The `-user` option gives the possibility to execute the operation using a specific user, such as an SAP CC individual or service user granted the Administrator role
- The `-host` option gives the possibility to specify the host concerned by the operation
- The `-nr` parameter is used to specify the number of the instance concerned by the operation
- The `-function` parameter is used to specify the operation to perform on the target instance, such as:
  - **Start**, which directly starts the process
  - **Stop**, which corresponds to:
    - A **hard shutdown** when no timeout is specified
    - A **soft shutdown with timeout** when a timeout is specified
  - **Shutdown**, which corresponds to a **soft shutdown without timeout** (that SAP SE or an SAP affiliate company recommends you to avoid)
  - **RestartInstance**, which indeed corresponds to:
    - A **hard shutdown** when no timeout is specified
    - A **soft shutdown with timeout** when a timeout is specified

    followed by a **Start** action
  - And so on

> ### Note
>
> To get the whole synopsis of the SAPControl application (particularly the list of supported functions), consult its documentation: https://help.sap.com/viewer/ff18034f08af4d7bb33894c2047c3b71/7.31.24/en-US/471d6feeff6e0d46e10000000a155369.html

> ### Example
>
> Considering a landscape containing a host:
>
> - Whose hostname is **`SAPCCHOST1`** and IP address is 192.168.1.1
> - That contains a rater instance whose name is **`CCR04`** and belonging to the **`CCD`** Core Server system whose administrator username is **`CCDadm`** and password is **`CCDpwd`**
>
> You can use the following synopsis to execute a soft shutdown with a 10s timeout on this rater:
>
> ```
> sapcontrol -host 192.168.1.1 -user CCDadm CCDpwd -nr 4 -function Stop 10
> ```

Linux and UNIX Operating Systems

- **Starting an instance in the SAP Convergent Charging system**
  You can start an instance in the SAP Convergent Charging system by executing the following command from the command line (`<instance_number>` can be the number of any instance of the SAP Convergent Charging system with SAP system ID `<SID>`):

  ```
  sapcontrol -nr <instance_number> -function Start
  ```

For remote instances, the syntax is slightly different, because the `-host` and `-user` parameters also have to be specified:

```
sapcontrol -nr <instance_number> -host <remote_host> -user <sid>adm
<password> -function Start
```

- **Stopping an instance in the SAP Convergent Charging system**
  You can stop an instance in the SAP Convergent Charging system by executing the following command from the command line (`<instance_number>` can be the number of any instance of the SAP Convergent Charging system with SAP system ID `<SID>`):

  ```
  sapcontrol -nr <instance_number> -function Stop
  ```

  For remote instances, the syntax is slightly different, because the `-host` and `-user` parameters also have to be specified:

  ```
  sapcontrol -nr <instance_number> -host <remote_host> -user <sid>adm
  <password> -function Stop
  ```

Microsoft Windows Operating System

- **Starting an instance in the SAP Convergent Charging system**
  You can start an instance in the SAP Convergent Charging system by executing the following command from the command line (`<instance_number>` can be the number of any instance of the SAP Convergent Charging system with SAP system ID `<SID>`):

  ```
  sapcontrol.exe -nr <instance_number> -function Start
  ```

  For remote instances, the syntax is slightly different, because the `-host` and `-user` parameters also have to be specified:

  ```
  sapcontrol.exe -nr <instance_number> -host <remote_host> -user <sid>adm
  <password> -function Start
  ```

- **Stopping an instance in the SAP Convergent Charging system**
  You can stop an instance in the SAP Convergent Charging system by executing the following command from the command line (`<instance_number>` can be the number of any instance of the SAP Convergent Charging system with SAP system ID `<SID>`):

  ```
  sapcontrol.exe -nr <instance_number> -function Stop
  ```

  For remote instances, the syntax is slightly different, because the `-host` and `-user` parameters also have to be specified:

  ```
  sapcontrol.exe -nr <instance_number> -host <remote_host> -user <sid>adm
  <password> -function Stop
  ```

7.1.21 Creating a Database User/Account of the Core Database and Dedicated to Communications Between SAP CC Cockpit and the SAP CC Core Database

In your SAP CC system landscape, a database user/account is required for the communications between an SAP CC Cockpit web application and the SAP CC Core Database of the Core Server system.

> → Remember
>
> This database user (or database account) is required before the installation of the SAP CC Cockpit web application.
>
> SAP CC Cockpit is an optional component and may not be part of your installation scenario.

As a system database administrator (sys dba) or database administrator (dba) and depending on the RDBMS of the SAP CC Core Database, you create a database user/account.

This database user (or database account) must be granted a **read access** to the database views pointing on some database tables in the SAP CC Core Database.

The complete list of SAP CC database tables is:

- **CERT_STORE (as of SAP CC 2020 FPS 2)**
- FILE_METADATA (as of version 5.0 SP 1)
- FILE_METADATA_PATH (as of version 5.0 SP 1)
- METRIC_RECORD (as of version 5.0 SP 2)
- OBJECT_CHANGE_LOG (as of version 5.0 SP 2)
- OBJECT_SNAPSHOT (as of version 5.0 SP 2)
- USER_OPERATION (as of version 5.0 SP 2)

See also: Installing and Launching Cockpit [page 201]

Related Information

Database Users/Accounts

7.1.22 Preparing the BART Database

> → Remember
>
> The BART Database is required if a BART Server system is part of your SAP Convergent Charging (SAP CC) system landscape. See Installing a BART Server System in an Existing Landscape [page 167].
>
> Otherwise do not install this database.

Preparing the Oracle BART Database [page 152]

Preparing the SQL Server BART Database [page 154]

Preparing the SAP ASE BART Database [page 158]

Preparing the SAP HANA BART Database [page 162]

Preparing the IBM DB2 BART Database (w/o pureScale Feature) [page 164]

7.1.22.1 Preparing the Oracle BART Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from 1 (the Core Database) to 4 databases (the Core Database, the Session Database, the BART Database and the IEC Database). This procedure explains you how to prepare your BART Database running under an Oracle RDBMS[58] (Standard or Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Oracle, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/oracle
  - http://scn.sap.com/docs/DOC-7888

Prerequisites

An instance of an Oracle RDBMS must be installed and available in your landscape. This database instance must use the UTF-8 character encoding to work in conjunction with the different SAP CC software components, and the *Oracle SQL*plus* tool must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC BART Database.

[58] Relational Database Management System

Procedure

The preparation of your SAP CC BART Database consists in:

- Setting up some initialization parameters
- Creating a user account and granting the adequate roles and privileges to this user
- Creating the different tablespaces

1. SAP Convergent Charging requires to modify some initialization parameters of the Oracle database, located in the `init.ora` file which contains the persistent parameters used during database startup. You can use the following procedure to:
   - Set the `_optim_peek_user_binds` parameter to **FALSE** in order to disable the bind peeking feature, which is not compatible with SAP CC
   - Increase the maximum number of processes and sessions by setting the `processes` and `sessions` parameters to 300 multiplied by the number of databases installed on this Oracle instance, i.e. 1 (as only the BART Database is installed on this instance)

   To correctly set up the initialization parameters of your BART Database, execute the following procedure:

   ```
   sqlplus / as sysdba
   alter system set "_optim_peek_user_binds"=false scope=spfile;
   alter system set processes=300 scope=spfile;
   alter system set sessions=300 scope=spfile;
   shutdown immediate
   startup
   ```

2. An Oracle user must be created and granted the adequate roles and privileges to administrate the BART Database. Execute the following procedure by replacing `<DB_USER>` by the name of the BART Database administrator and `<DB_PASSWORD>` by its password:

   ```
   sqlplus / as sysdba
   define dbuser="<DB_USER>"
   define dbpwd="<DB_PASSWORD>"
   create user &dbuser identified by &dbpwd;
   grant CONNECT, RESOURCE TO &dbuser;
   grant EXECUTE_CATALOG_ROLE to &dbuser;
   grant EXP_FULL_DATABASE, IMP_FULL_DATABASE to &dbuser;
   grant CREATE PROCEDURE to &dbuser ;
   grant CREATE SESSION to &dbuser ;
   grant CREATE TABLESPACE, DROP TABLESPACE to &dbuser ;
   grant ALTER TABLESPACE to &dbuser ;
   grant CREATE ANY DIRECTORY to &dbuser ;
   grant DROP ANY DIRECTORY to &dbuser ;
   grant EXECUTE ON dbms_lock TO &dbuser ;
   ```

3. The data storage of your SAP CC BART Database must be optimized by creating the following mandatory tablespaces, whose size depends on the edition of your RDBMS (Standard or Enterprise edition):

Tablespace name	Description	Minimal Size (MB)	
		SE	EE
BART_DATA	Tables used to store the BART data which are not related to CDRs	50	50
BART_INDX	Indexes used for the BART data which are not related to CDRs	10	10

Tablespace name	Description	Minimal Size (MB)	
		SE	EE
CDR_DATA	Tables containing data related to CDRs	100	2000
CDR_INDX	Indexes used for the data related to CDRs	600	10000

To create the mandatory tablespaces, execute the following procedure by replacing:

- `<tablespace_name>` by the name of the tablespace to create
- `<file_location>` by the full path of the file that will contain the tablespace, located in the oradata directory
- `<size>` by the recommended value associated to the tablespace, listed in the above table and using the *AUTO_EXTEND* option

```
sqlplus / as sysdba
```

And then, for each tablespace listed in the above table:

```
CREATE BIGFILE TABLESPACE <tablespace_name> DATAFILE '<file_location>' SIZE
<size> AUTOEXTEND ON MAXSIZE UNLIMITED UNIFORM SIZE 1M LOGGING EXTENT MANAGEMENT
LOCAL SEGMENT SPACE MANAGEMENT AUTO;
```

Followed by this command for Oracle 12c databases:

```
ALTER USER <dbuser> QUOTA UNLIMITED ON <tablespace_name>;
```

Example

For an Oracle 12c Standard Edition RDBMS installed on a Linux or UNIX opertaing system host, you can use the following statement:

- To create the **BART_DATA** tablespace
- To give the **sapccbart** user the possibility to access this tablespace

```
CREATE BIGFILE TABLESPACE BART_DATA DATAFILE '/oradata/vol1/BART_DATA.dbf'
SIZE 100M AUTOEXTEND ON MAXSIZE UNLIMITED UNIFORM SIZE 1M LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
ALTER USER sapccbart QUOTA UNLIMITED ON BART_DATA;
```

7.1.22.2 Preparing the SQL Server BART Database

Description

According to your business requirements, your SAP Convergent Charging landscape can contain from one (the Core Database) to four databases (the Core Database, the Session Database, the BART Database and the IEC

Database). This procedure explains you how to prepare your BART Database running under a Microsoft SQL Server RDBMS[59] (Standard or Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on the SAP Community:
  - http://scn.sap.com/community/sqlserver
  - http://scn.sap.com/docs/DOC-8286

Prerequisites

An instance of a SQL Server RDBMS must be installed and available in your landscape. The *SQL Server Management Studio* application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC BART Database.

Procedure

The preparation of your SAP CC BART Database consists in the following operations:

- Creating a SQL Server user account and granting the adequate roles on the adequate server to this user
- Creating the BART Database with the user as the owner
- Creating the different filegroups within this database

1. An SQL Server user must be created for your SAP CC BART Database, and granted the adequate roles and privileges. Execute the following procedure to create this SAP CC user:
   - Open the *SQL Server Management Studio* application
   - The *Connect to Server* dialog window opens. Use the following settings:
     - *Server Name*: Select *(local)*
     - *Authentication*: Select *Windows Authentication*, in order to use your operating system user
     - Click the *Connect* button
   - Open the *View / Object Explorer* menu to display the *Object Explorer* and expand the *(local)* server
   - Right-click the *Security* element of the tree and open the *New / Login...* menu to display the *Login* screen
   - Fill the *General* tab with the following information:
     - *Login name*: Fill with the name of the BART Database administrator

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- Select the *SQL Server authentication* option
- *Password*: Fill with a password respecting the SAP password policy
- *Confirm password*: Re-type the chosen password
- *User must change password at next login*: Untick the checkbox
- *Default database*: Select the *master* database

- In the *Server Roles* tab, select the following roles:
  - *dbcreator*
  - *diskadmin*
  - *serveradmin*
- Click the *OK* button
- Right-click the *(local)* server and click *Disconnect*

> → Tip
>
> The SQL Server authentication method can be replaced by the Windows authentication method:
>
> 1. In the *General* tab, enter the following information:
>   - *Login name*: Fill with the `<DOMAIN>`\SAPService`<SID>` login
>   - Select the *Windows authentication* option
>   - Ensure that the Default Database is the master one
> 2. Change the launch configuration:
>   - Edit the `bart.config` file in the folder:
>
>   `<drive>:\usr\sap\<SID>\CAB<XX>\config\bart.config`
>   - Add the string property `IntegratedSecurity=true` to the value of the `db.SQLHELPER_JDBC_URI` system parameter to enable the Windows authenticatio login method
>
>   > ✤ Example
>   >
>   > db.SQLHELPER_JDBC_URI=jdbc\:sqlserver\://`<dbhost>`\:`<dbport>`;Databasename\=`<dbname>`;IntegratedSecurity\=true
>   - (Optional) Remove the values of the following parameters:
>
>   `db.SQLHELPER_LOGIN`
>
>   `db.SQLHELPER_PASSWORD`
> 3. Restart all instances

2. Once the SQL Server user dedicated to the BART Database has been created, it is necessary to reconnect to the server in order to create the BART Database. Execute the following procedure:
   - Open the *File / Connect Object Explorer...* menu or click the *Connect Object Explorer* button to re-open the *Connect to Server* dialog window. Use the following settings:
     - *Server Name*: Specify the full name of the machine hosting the database
     - *Authentication*: Select *SQL Server authentication*
     - *Login*: Fill with the previously created user
     - *Password*: Fill with the password of the previously created user
     - Click the *Connect* button
   - Right-click the *Databases* element of the tree and open the *New Database...* menu to display the *New Database* screen

- Fill the *General* tab with the following information:
  - *Database name*: Fill with the name of the BART Database
  - *Owner*: Fill with the name of the previously created user
- Fill the *Options* tab with the following information:
  - *Collation*: Collation defines the way strings are managed in SQL Server databases. The BART Database must be case-sensitive (CS) and compatible with ASCI Unicode (AS). Thus select the element that corresponds to the appropriate language, ending by the *CS_AS* string (e.g. *Latin1_General_CS_AS*)
  - *Recovery model*: Select *Full*
  - *Compatibility level*: Ensure that *SQL Server 2008 (100)* is selected
  - Click the *OK* button

3. The data storage of your SAP CC BART Database must then be optimized by creating the following mandatory filegroups:

Tablespace name	Description
BART_DATA	Tables used to store the BART data which are not related to CDRs
BART_INDX	Indexes used for the BART data which are not related to CDRs
CDR_DATA	Tables containing data related to CDRs
CDR_INDX	Indexes used for the data related to CDRs

To create the mandatory filegroups, execute the following procedure by replacing `<DATABASE_NAME>` by the name of the BART Database `<FILEGROUP_NAME>` by the name of the filegroup to create, and `<FILE_LOCATION>` by the full path of the file that will contain the filegroup directory:

- Right-click your server and open the *New Query* menu
- For each filegroup listed in the above table, type the following query and execute it:

```
ALTER DATABASE "<DATABASE_NAME>" ADD FILEGROUP <FILEGROUP_NAME>;
ALTER DATABASE "<DATABASE_NAME>" ADD FILE
(
NAME = <FILEGROUP_NAME>,
FILENAME = '<FILE_LOCATION>',
SIZE = 1MB,
MAXSIZE = 1000MB,
FILEGROWTH = 5MB
)
TO FILEGROUP <FILEGROUP_NAME>;
```

Example

To create the **BART_DATA** filegroup in the **SAPCCBART** BART Database of an SQL Server RDBMS, you can use the following query:

```
ALTER DATABASE "SAPCCBART" ADD FILEGROUP BART_DATA;
ALTER DATABASE "SAPCCBART" ADD FILE
(
NAME = BART_DATA,
FILENAME ='c:\tmp\BART_DATA.ndf',
SIZE = 1MB,
MAXSIZE = 1000MB,
FILEGROWTH = 5MB
)
TO FILEGROUP BART_DATA;
```

7.1.22.3 Preparing the SAP ASE BART Database

Description

This procedure explains you how to prepare your BART Database running under an SAP Adaptive Server Enterprise (SAP ASE) RDBMS.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on the SAP Community:
  - http://infocenter.sybase.com/
  - http://scn.sap.com/community/developer-center/oltp-db
  - http://scn.sap.com/docs/DOC-34995
  - http://scn.sap.com/docs/DOC-34996

Prerequisites

An instance of an SAP ASE RDBMS must be installed and available in your landscape. This instance must be:

- Correctly licensed in order to fit specific needs such as data partitioning, communication securing, system high-availability, and so on
- Installed with the following configuration:
  - *Application Type*: Mixed (OLTP/DSS)
  - *Page Size*: 8k. Note that this value cannot be modified after the installation. The modification of the value leads to a warning message informing that the newly specified page size must be similar between databases used for import purposes
  - *Default Character Set*: utf-8 : Unicode 3.1 for UTF-8 Character Set
  - *Default Sort Order*: binary (id=25) or binaryalt (id=99)

In addition, the *Interactive SQL* tool must be installed to execute the different steps of this procedure. For further information about this tool, refer afterwards.

To get information about these prerequisites relating to the SAP CC BART Database, refer to your System Administrator.

> ⚠ Caution
>
> Free Developer Edition or Small Business Edition of SAP ASE are not supported by SAP CC. **You must install a licensed copy of the SAP software product** *SAP Adaptive Server Enterprise*.

Using the Interactive SQL tool

Interactive SQL is a command-line tool provided by SAP ASE to execute SQL statements:

- Interactively, to execute individual commands
- In a batch mode, to execute script files that contain multiple commands

To use the interactive mode, you have to:

- Launch isql using the following synopsis, providing the username and password specified at installation time:

  ```
  isql -U <USERNAME> -P <PASSWORD>
  ```

- Type the command to execute, followed by the `go` statement:

  ```
  <command to execute>
  go
  ```

To use the batch mode, you have to:

- Create a text file containing the list of commands to execute, each command being followed by the `go` statement:

  ```
  <command to execute>
  go
  <command to execute>
  go
  <command to execute>
  go
  and so on
  ```

- Launch isql using the following synopsis, providing the username and password specified at installation time, and specifying the previously created script file:

  ```
  isql -U <username> -P <password> -i <script file>
  ```

> ⓘ Note
>
> - Whatever the execution mode is, you can use the `quit` command to exit the isql tool
> - For further information about the isql tool, refer to the dedicated documentation.

Procedure

The preparation of your SAP CC BART Database consists in the following operations:

- Setting up some parameters of the SAP Adaptive Server Enterprise (SAP ASE) system
- Creating the devices for data and logs, and the database relying on these devices
- Setting up some database options
- Creating a user account and granting the adequate roles and privileges to this user
- Creating the different segments

1. Some parameters of the SAP Adaptive Server Enterprise (SAP ASE) system must be modified in order to fit specific needs of SAP Convergent Charging. Use the following list of commands to modify the configuration of the SAP ASE system accordingly:

```
use master
GO
sp_configure "number of open objects", 10000
GO
sp_configure "number of open indexes", 2000
GO
sp_configure "max network packet size", 16384
GO
sp_configure "default network packet size", 16384
GO
sp_configure "optimization goal", 0, 'allrows_mix'
GO
sp_configure "disable varbinary truncation", 1
GO
sp_configure "number of user connections", 200
GO
sp_configure "number of locks", 1000000
GO
sp_configure "deadlock checking period", 800
GO
sp_configure "lock hashtable size", 16384
GO
sp_configure "row lock promotion HWM", 214748364
GO
sp_configure "row lock promotion LWM", 214748364
GO
sp_configure "SQL batch capture", 1
GO
sp_configure "kernel resource memory", 12000
GO
sp_configure "enable monitoring", 1
GO
sp_configure "sql text pipe active", 1
GO
sp_configure "sql text pipe max messages", 2000
GO
sp_configure "plan text pipe active", 1
GO
sp_configure "plan text pipe max messages", 2000
GO
sp_configure "statement pipe active", 1
GO
sp_configure "statement pipe max messages", 2000
GO
sp_configure "errorlog pipe active", 1
GO
sp_configure "errorlog pipe max messages", 2000
GO
sp_configure "deadlock pipe active", 1
```

```
GO
sp_configure "deadlock pipe max messages", 2000
GO
sp_configure "lock timeout pipe max messages", 2000
GO
sp_configure "lock timeout pipe active", 1
GO
sp_configure "wait event timing", 1
GO
sp_configure "process wait events", 1
GO
sp_configure "object lockwait timing", 1
GO
sp_configure "statement statistics active", 1
GO
sp_configure "per object statistics active", 1
GO
sp_configure "max SQL text monitored", 512
GO
sp_configure "enable stmt cache monitoring", 1
GO
sp_configure "aux scan descriptors", 1000
GO
sp_configure "select for update", 1
GO
```

If you use the database partitioning function for your database, execute the following commands:

```
sp_configure "enable semantic partitioning", 1
GO
sp_configure "number of open partitions", 100000
GO
```

If you use secured communications with your database, execute the following command:

```
sp_configure "enable SSL", 1
GO
```

2. To create the devices for data and logs, and then the database that relies on these devices, use the following list of commands considering that:
   - <DB_NAME> corresponds to the name of your database
   - <DEVICE_PATH> corresponds to the path of the device
   - <DEVICE_SIZE> corresponds to the amount of space to allocate to the device, made up with a quantity followed by one of these supported unit specifiers:
     - k or K (kilobytes)
     - m or M (megabytes)
     - g or G (gigabytes
     - t or T (terabytes)

```
disk init name="<DB_NAME>_data", physname="<DEVICE_PATH>\<DB_NAME>_data",
size=<DEVICE_SIZE>, dsync = true
GO
disk init name="<DB_NAME>_log", physname="<DEVICE_PATH>\<DB_NAME>_log",
size=<DEVICE_SIZE>, dsync = true
GO
create database <DB_NAME> on <DB_NAME>_data=2000000 log on
<DB_NAME>_log=2000000
GO
```

3. Some options of the previously created database must be modified. Use the following list of commands, considering that <DB_NAME> corresponds to the name of your database:

```
sp_dboption <DB_NAME>,"ddl in tran", true
GO
sp_dboption <DB_NAME>, "select into", true
GO
```

4. A dedicated user must be created to administrate the BART Database, and must be declared as the owner of the BART Database. Use the following list of commands considering that:
   - <DB_USER> corresponds to the name of the BART Database administrator
   - <DB_PASSWORD> corresponds to the password of the BART Database administrator, that respects the SAP password policy
   - <DB_NAME> corresponds to the name of the BART Database

```
create login <DB_USER> with password <DB_PASSWORD>
GO
use <DB_NAME>
GO
sp_cangedbowner <DB_USER>
GO
```

5. To optimize the storage of your data, you need to create mandatory segments within your database. Use the following list of commands, considering that <DB_NAME> corresponds to the name of your BART Database:

```
use <DB_NAME>
GO
sp_addsegment 'BART_DATA', '<DB_NAME>', '<DB_NAME>_data'
GO
sp_addsegment 'BART_INDX', '<DB_NAME>', '<DB_NAME>_data'
GO
sp_addsegment 'CDR_DATA', '<DB_NAME>', '<DB_NAME>_data'
GO
sp_addsegment 'CDR_INDX', '<DB_NAME>', '<DB_NAME>_data'
GO
```

7.1.22.4 Preparing the SAP HANA BART Database

Description

This procedure explains you how to prepare your BART Database running under an SAP HANA in-memory database system using a single container or a multiple containers mode.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of SAP HANA, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/developer-center/hana
  - http://scn.sap.com/docs/DOC-53955

For further information about data types, operators, functions, statements, and so on, refer to the SAP HANA SQL and System Views Reference documentation available on SAP Help Portal:https://help.sap.com

Prerequisites

An SAP HANA database system must be installed and available in your landscape. The SAP HANA Studio application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC BART Database.

Procedure

The preparation of your SAP CC BART Database consists in the following operations:

- Declaring your system within the SAP HANA Administration Console of the *SAP HANA Studio* application
- Creating a user account and granting the adequate roles and privileges to this user on this system
- Disabling the expiration of the user's password

1. An SAP HANA system must be created to administrate your SAP CC BART Database. Execute the following procedure to create this system:
   - Open the *SAP HANA Studio* application.
   - Open the *Window* > *Perspective* > *Open Perspective* > *SAP HANA Administration Console* menu to display the perspective dedicated to the management of SAP HANA systems.
   - Open the *Window* > *Show View* > *Systems* menu to display the view containing the list of SAP HANA systems.
   - Click the *Add System* icon to open the *System* dialog window.
   - Use the following settings to fill the *Specify System* screen:
     - *Host Name*: Fill with the master host name.
     - *Instance Number*: Fill with the instance number of your SAP HANA system.
     - *Mode*: Select *Single Container* or *Multiple containers*, according to your needs.
     - *Description*: Fill with a text that describes your system.
     - Click the *Next* button.
   - The *Connection Properties* screen opens. Select the *Authentication by database user* option and use the following settings:

- *User Name*: Fill with the name of the SAP HANA system administrator.
- *Password*: Fill with a password following the SAP password policy.
- Click the *Finish* button to log on the newly declared SAP HANA system.

2. Once you are connected to the relevant SAP HANA system, it is necessary to create a user account and grant the adequate roles and privileges to this user in order to administrate the BART Database. Execute the following procedure to create this user:
   - Expand the *Security* folder, and right-click the *Users* element to display the context menu corresponds to the name of your database.
   - Right-click the *Users* element of the tree and open the *New User* menu to display the screen dedicated to the creation of new users.
   - Fill the *User* tab with the following information:
     - *User Name*: Fill with the name of the BART Database administrator.
     - Select the *Password* option in the *Authentication* fieldset.
     - *Password*: Fill with a password following the SAP password policy.
     - *Confirm password*: Re-type the chosen password.
   - In the *Granted Roles* subtab, click the + button to add the following roles to the user:
     - *CONTENT_ADMIN*
     - *MODELING*
   - In the *System Privileges* subtab, click the + button to add the following privileges to the user:
     - *EXPORT*
     - *IMPORT*
   - Click the *Deploy* button to finalize the creation of the user.
3. Once the user has been created and deployed on your SAP HANA system, it is necessary to disable the expiration of its password. Execute the following procedure:
   - Right-click the previously created SAP HANA system and open the *Open SQL Console* menu to display the screen dedicated to the execution of SQL statements.
   - Type the following SQL statement within the editor, considering that `<USERNAME>` corresponds to the name of your previously created user. For further information, refer to the documentation of the `ALTER USER` command available in the SAP HANA documentation:

     ```
     ALTER USER <USERNAME> DISABLE PASSWORD LIFETIME;
     ```

   - Click the *Execute* button to execute the SQL statement.

7.1.22.5 Preparing the IBM DB2 BART Database (w/o pureScale Feature)

Description

This procedure explains you how to prepare your BART Database running under an IBM DB2 RDBMS[60] without the pureScale feature.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on the SAP Community:
  - http://scn.sap.com/community/db2-for-linux-unix-windows
  - http://scn.sap.com/docs/DOC-8211

This procedure uses the following variables:

- `<DB_GROUP>`, which corresponds to the name of the group containing the DB2 administrators, specified at DB2 installation time
- `<DB_NAME>`, which corresponds to the name of the BART Database
- `<DB_USER>`, which corresponds to the name of the BART Database administrator, which corresponds to a user granted adequate roles and privileges
- `<DB_PASSWORD>`, which corresponds to the password of the BART Database administrator, following the SAP password policy but without any expiration date
- `<TABLESPACE_NAME>`, which corresponds to the name of the tablespace to create
- `<FILE_LOCATION>`, which corresponds to the full path of the directory that will contain the tablespaces

Prerequisites

An instance of an IBM DB2 database must be installed and available in your landscape. The *IBM DB2 db2cmd* tool must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC BART Database.

Procedure

The preparation of your SAP CC BART Database consists in the following operations:

- Creating a user account and adding this user to the relevant group
- Creating the BART Database and setting up some configuration parameters
- Granting the newly created user the adequate roles and privileges
- Creating the different tablespaces within this database

1. SAP Convergent Charging requires to create a dedicated user for your SAP CCBART Database, and add this user to the group containing the DB2 administrators. Execute the following procedure:
   - For a Microsoft Windows operating system, execute the following commands into a command line prompt:

     ```
     net user <DB_USER> "<DB_PASSWORD>" /ADD /PASSWORDCHG:NO /EXPIRES:never
     ```

[60] Relational Database Management System

```
wmic UserAccount where Name="<DB_USER>" set PasswordExpires=False
net localgroup <DB_GROUP> <DB_USER> /add
```

- For Linux and UNIX operating systems, execute the following commands in order to retrieve the identifier of the group containing the DB2 administrators, create the user within the adequate group and set a password for this user:

```
cat /etc/group | grep -i <DB_GROUP> | cut -d':' -f3
useradd -g <GROUP_ID> <DB_USER>
passwd <DB_USER>
```

2. SAP Convergent Charging requires to modify some configuration parameters when creating the DB2 database. You can use the following procedure to:
   - Create the database
   - Set the default page size of the database to 16K
   - Specify an amount of storage for recovery log files by pre-allocating a number of 50 primary log files
   - Set the amount of the database heap memory to use as a buffer for recording logs before writing these records to disk to 1024 4KB pages
   - Set the size of each primary and secondary log file to 16K 4KB pages. This size limits the number of records that can be written in these files before they become full and require the creation of new ones
   - Specify the percentage of changed pages at which the asynchronous page cleaners will be started (if they are not currently active)
   - Activate the dynamically distribution of the available memory resources between the memory consumers
   - Activate the automatic sampling rate determination, based on table size
   - Set the default prefetch size of tablespaces to 960 pages
   - Set an automatic percentage for the lock list held by an application that must be filled before performing lock escalation
   - Set an automatic amount of 4K pages to store the lock list

   To create the SAP CC BART Database, execute the following procedure:

```
db2cmd
db2
create database <DB_NAME> pagesize 16384
update database configuration for <DB_NAME> using LOGPRIMARY 50
update database configuration for <DB_NAME> using DECFLT_ROUNDING
ROUND_CEILING
update database configuration for <DB_NAME> using LOGBUFSZ 1024
update database configuration for <DB_NAME> using LOGFILSIZ 16384
update database configuration for <DB_NAME> using CHNGPGS_THRES 40
update database configuration for <DB_NAME> using SELF_TUNING_MEM ON
update database configuration for <DB_NAME> using AUTO_SAMPLING ON
update database configuration for <DB_NAME> using DFT_PREFETCH_SZ 960
update database configuration for <DB_NAME> using MAXLOCKS AUTOMATIC
update database configuration for <DB_NAME> using LOCKLIST AUTOMATIC
```

3. The user created in Step1 must then be granted the adequate roles and privileges on the SAP CC BART Database. Execute the following procedure:

```
db2cmd
db2
connect to <DB_NAME>
grant dbadm on database to <DB_USER>
connect reset
```

4. The data storage of your SAP CC BART Database must then be optimized by creating the following mandatory tablespaces:

Tablespace name	Description
BART_DATA	Tables used to store the BART data which are not related to CDRs
BART_INDX	Indexes used for the BART data which are not related to CDRs
CDR_DATA	Tables containing data related to CDRs
CDR_INDX	Indexes used for the data related to CDRs

To create the mandatory tablespaces, execute the following procedure:

```
db2cmd
db2
connect to <DB_NAME>
```

And then, for each tablespace listed in the above table:

```
create tablespace <TABLESPACE_NAME> MANAGED BY DATABASE USING (file
'<FILE_LOCATION><TABLESPACE_NAME>' 256000) EXTENTSIZE 2 DROPPED TABLE
RECOVERY OFF AUTORESIZE YES
```

Example

For an IBM DB2 RDBMS installed on a Microsoft Windows host, you can use the following statement to create the **BART_DATA** tablespace:

```
create tablespace BART_DATA MANAGED BY DATABASE USING (file
'C:\DB2\tablespaces\BART_DATA' 256000) EXTENTSIZE 2 DROPPED TABLE RECOVERY
OFF AUTORESIZE YES
```

7.1.23 Installing a BART Server System in an Existing Landscape

→ Remember

In an integrated *SAP Solution* scenario and as of SAP Convergent Charging 4.0, the BART Server component is replaced with SAP Convergent Invoicing and its *Consumption Item Management* feature. SAP Convergent Invoicing is already part of your SAP system landscape. A transition guide is available and enables planning the transition.

Description

The optional BART Server component is a supplementary SAP CC system that is dedicated to business operations on chargeable items such as acquisition, consolidation, charging or rerating, all performed using a batch execution mode. The chargeable items represent some service consumptions (or product uses).

> **Note**
>
> The BART Server system can be deployed:
>
> - On an existing host in your SAP system landscape, that already contains other systems.
> - On a new host in your SAP system landscape, that is dedicated to the BART Server system.

This procedure explains you how to configure the different steps of the installation of the BART Server system handled by SAP installer, on a dedicated host.

Prerequisites

- A Core Server system must be available in your SAP CC system landscape.
  - According to your installation scenario, refer to the Installing a Core Server on a Monohost Landscape [page 123] or the Installing a Core Server on a Multihost Landscape [page 131] procedure if necessary.
  - In this Core Server system, a dispatcher instance must be available, and at least 1 instance must be connected to this dispatcher instance.
- A dedicated host must be set up and available in your SAP system landscape. Refer to the Setting Up a New Host for a Landscape [page 141] procedure if necessary.
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2017 runtime" libraries must be installed and available. For further information, refer to SAP Note 3143948.
- A database must be prepared and available. Depending of your landscape, refer if necessary to one of the following procedures:
  - Preparing the Oracle Core Database [page 106]
  - Preparing the SQL Server Core Database [page 109]
  - Preparing the SAP ASE Core Database [page 113]
  - Preparing the SAP HANA Core Database [page 118]
  - Preparing the IBM DB2 Core Database (w/o pureScale Feature) [page 120]
- If your BART Database is running under an Oracle RDBMS[61], the dedicated JDBC[62] driver must be downloaded and available on the host. Refer to the Downloading the Oracle JDBC Driver [page 97] procedure if necessary.
- A SID[63] must be chosen for the BART Server system. Refer to the Choosing System IDs [page 91] procedure if necessary.

Procedure

The installation of the BART Server system consists in the following operations:

- Creating an SAP CC service user and granting the adequate roles and privileges to this user. To avoid runtime problems such as password expiration or user locking, the password management policy does not apply to this service user.

[61] Relational Database Management System
[62] Java Database Connectivity
[63] SAP System Identifier

- Launching the SAP installer to install the system.

As described in the SAP CC 2020 Security Guide documentation, you must create a service user for the BART Server system and grant the *Process Manager* and *Message Charging Client* roles to this user. Execute the following procedure to create this service user:

1. Launch [page 193] the Core Tool user interface.
2. Connect to the host that contains the Core Server system for communicating with your BART Server system. To connect to the host, use an SAP CC user that is granted the adequate authorizations for creating new users
3. Create and set up a new user with the following information:
   - *Log on*: Fill with the name of the BART Server service user you want to create.
   - *Password*: Fill with a password respecting the SAP password policy.
   - *Security Profile*: Select *Service User*.
   - *Roles*: Tick the *Process Manager* and *Message Charging Client* roles.
   - Save your modifications.

Once the adequate service user has been created, you can use the Launching the SAP installer [page 190] dedicated procedure to launch the installation of the BART Server system. Once opened, the SL Common GUI tool displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. On the *Welcome* screen:
   - Click *SAP Convergent Charging* > *Install* > *Standard System Installation* > *BART Server*
   - Click the *Next* button
2. **Parameter Settings** screen
   - *Parameter Mode*: Select *Custom*
   - Click the *Next* button
   - (Microsoft Windows only) You will be prompted to log off in order to let SAP installer grant you the authorizations that are required to perform the installation. Click the *OK* button to log off, and reconnect to the host. SAP installer automatically restarts to continue the installation from there
3. **General SAP System Parameters** screen
   - *SAP System ID (SAPSID)*: Fill with the chosen SID for the BART Server
   - *Destination Drive or SAP Mount Directory*: Keep the default configuration
   - Click the *Next* button
4. **Master Password** screen
   - *Password for All Users*: Fill with a password respecting the SAP password policy
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
5. (Microsoft Windows only) **Microsoft Windows Domain** screen
   - *Domain Model*: Select *Local Domain*
   - Click the *Next* button
6. **Operating System Users** screen
   - **Linux and UNIX Operating Systems**
     - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
     - *Confirm*: Re-type the chosen password

- *User ID*: *left empty*
- *Group ID of sapsys*: *left empty*
- Click the *Next* button

- **Microsoft Windows Operating System**
  - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
  - *Confirm*: Re-type the chosen password
  - *Password of SAP System Service User*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
  - *Confirm*: Re-type the chosen password
  - Click the *Next* button

7. **Prerequisites Checker** screen
   - Click the *Next* button
8. **Database Type** screen
   - *Database Type*: Select the adequate RDBMS
   - Click the *Next* button
9. **SAP HANA**, **Sybase ASE**, **Oracle**, **MS SQL Server** or **DB2** screen
   - **SAP HANA** screen
     (Database Connection)
     - *Primary Host*: Fill with the IP address of the machine hosting the primary node
     - *Secondary Host*: Fill with the IP address of the machine hosting the secondary node
     - *User*: Fill with the name of the BART Database administrator you specified during the preparation of your BART Database
     - *Password*: Fill with the password of the BART Database administrator you specified during the preparation of your BART Database
     - *Mode*: Select the mode that corresponds to the installation scenario of your SAP HANA database
       - *Single Container Mode*
         - *Port*: Fill with the SQL[64] port for `indexserver` service of the system database of your SAP HANA system. For further information about the default ports of SAP products, refer to the dedicated section on SAP Help Portal: https://help.sap.com/viewer/ports
       - *Multiple Containers Mode*
         - *Tenant Name*: Fill with the name of the tenant database to connect to
         - *Port*: Fill with the relevant port of the tenant database of your SAP HANA system you want to connect to. For further information about the value of such port, refer to the SAP Note 2365930 or to the section dedicated to the default ports on SAP Help Portal: https://help.sap.com/viewer/ports
   - **Sybase ASE** screen
     (Primary Database Connection)
     - *Host*: Fill with the IP address of the machine hosting the database
     - *Port*: Fill with the port specified during the installation of the database
     - *Name*: Fill with the unique name of the database instance within which the content of the BART Database will be installed

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- *User*: Fill with the name of the BART Database administrator you specified during the preparation of your BART Database
- *Password*: Fill with the password of the BART Database administrator you specified during the preparation of your BART Database
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary

(Backup Database Connection)

Tick the *Use backup database connection* checkbox if you want to use a backup database pour high availability purpose. If you tick this checkbox, specify the following information:

- *Host*: Fill with the IP address of the machine hosting the backup database
- *Port*: Fill with the port used to connect to the backup database

- **Oracle** screen

(JDBC Driver)

- *JDBC Driver JAR Archive*: Click the *Browse* button and select the JAR archive you downloaded using the dedicated procedure [page 97]

(Database Authentication)

- *Host*: Fill with the IP address of the machine hosting the database.
- *Port*: Fill with the port specified during the installation of the database.
- *Instance Id*: Fill with the unique name of the database instance within which the content of the BART Database will be installed.

  > **Note**
  >
  > For an Oracle database, you can use the following command to fill the *Instance Id* field:
  >
  > ```
  > sqlplus / as sysdba
  > SELECT sys_context('USERENV', 'DB_NAME') FROM DUAL;
  > ```

- *User*: Fill with the name of the BART Database administrator you specified during the preparation of your BART Database.
- *Password*: Fill with the password of the BART Database administrator you specified during the preparation of your BART Database.
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary.

- **MS SQL Server** screen

(Database Connection)

- *Host*: Fill with the IP address of the machine hosting the database.
- *Port*: Fill with the port specified during the installation of the database.
- *Name*: Fill with the unique name of the database instance within which the content of the BART Database will be installed.
- *User*: Fill with the name of the BART Database administrator you specified during the preparation of your BART Database.
- *Password*: Fill with the password of the BART Database administrator you specified during the preparation of your BART Database.
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary.

- **DB2** screen

(Database Connection)

- *Host*: Fill with the IP address of the machine hosting the database.
- *Port*: Fill with the port specified during the installation of the database.
- *Database Name*: Fill with the unique name of the database instance within which the content of the BART Database will be installed.
- *Schema Name*: Fill with the name of the database schema you want to create within the BART Database.
- *User*: Fill with the name of the BART Database administrator you specified during the preparation of your BART Database.
- *Password*: Fill with the password of the BART Database administrator you specified during the preparation of your BART Database.
- *Use Partitioning*: Tick the checkbox if your RDBMS supports the partitioning feature. Refer to your database administrator if necessary.

- Click the *Next* button.

10. **Tablespaces / Filegroups / Segments** screen
    - Ensure that each table or index is associated to the correct filegroup, tablespace or segment created during the preparation of your BART Database. When necessary, modify the associations accordingly.
    - Click the *Next* button.
11. **CA Introscope Java Agent** screen
    - *Use monitoring with CA Introscope*: Untick the checkbox.
    - Click the *Next* button.
12. **System Landscape Directory** screen
    - *Configure System Landscape Directory*: Untick the checkbox.
    - Click the *Next* button.
13. **SAP Convergent Charging Security** screen
    - *Security*: Select the *Disabled* option.
    - Click the *Next* button.
14. **BART Server Settings** screen BART Database
    - `<SERVICE>` *Port*: Keep the default configuration, or set specific values for each configurable port.
    - *Message Host*: Specify the full name of the machine hosting the BART Server. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify one of these IP addresses.
    - Click the *Next* button.
15. **SAP Convergent Charging Core Server Connection Security** screen
    - *Security*: Select the *None* option.
    - Click the *Next* button.
16. **SAP Convergent Charging Core Server Connection** screen
    - Click the *Add* button to add a new line in the table containing the list of dispatchers available within your landscape.
    - Click the *Dispatcher HTTP Host* column of the newly inserted row, and fill with the IP address of an available dispatcher.
    - Click the *Dispatcher HTTP Port* column of the newly inserted row, and fill with the port number of this available dispatcher.
    - *Login*: Fill with the name of the service user for the BART Server, created previously.
    - *Password*: Fill with the password of the service user for the BART Server.

- Click the *Next* button.

17. **Start Instances after Installation** screen
    - *Start SAP Convergent Charging instances after installation*: Tick the checkbox to automatically start the instances at the end of the installation process.

      > ⓘ Note
      >
      > The SAP Management Console is deployed during the installation process and gives you the possibility to manage each instance of your landscape.

    - Click the *Next* button.
18. **Unpack Archives** screen
    - Click the *Next* button.
19. **Parameter Summary** screen
    - Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button.
    - If all your settings are correctly taken into account, click the *Next* button to start the installation.
20. **Task Progress** screen
    - This screen contains the different steps of your installation scenario, whose advancement can be followed.

7.1.24 Installing a Diameter Server in an Existing Landscape

> ⚠ Caution
>
> From SAP Convergent Charging SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

Description

The Diameter Server component represents a message translator which is used to convert credit-control messages coming from a Diameter client application into charging messages that the Core Server system is able to handle. It behaves as an online mediation system providing credit-control capabilities such as real-time services control, prepaid balances management, customers notification, sessions termination, and so on.

> ⓘ Note
>
> The Diameter Server can be deployed:
>
> - On an existing host of your landscape, that already contains other systems
> - On a new host within your landscape, that is dedicated to the Diameter Server system

This procedure explains you how to configure the different steps of the installation of the Diameter Server handled by SAP installer, on a dedicated host.

Prerequisites

- A Core Server system must be available within your landscape. According to your installation scenario, refer to the Installing a Core Server on a Monohost Landscape [page 123] or the Installing a Core Server on a Multihost Landscape [page 131] procedure if necessary. A dispatcher of this Core Server system must be available, and at least 1 instance must be connected to this dispatcher
- A dedicated host must be set up and available within your landscape. Refer to the Setting Up a New Host for a Landscape [page 141] procedure if necessary
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2013 runtime" libraries must be installed and available. For further information, refer to SAP Note 2676219
- A SID must be chosen for the Diameter Server system. Refer to the Choosing System IDs [page 91] procedure if necessary

Procedure

The installation of the Diameter Server system consists in the following operations:

- Creating an SAP CC service user and granting the adequate roles and privileges to this user. To avoid runtime problems such as password expiration or user locking, the password management policy does not apply to this service user
- Launching the SAP installer to install the system

As described in the SAP CC 2020 Security Guide documentation, you must create a service user for the Diameter Server system and grant the *Message Charging Client* role to this user. Execute the following procedure to create this service user:

1. Launch [page 193] the Core Tool
2. Connect to the host that contains the Core Server system for communicating with your Diameter Server system. To connect to the host, use an SAP CC user that is granted the adequate authorizations for creating new users
3. Create and set up a new user with the following information:
   - *Log on*: Fill with the name of the Diameter Server service user you want to create
   - *Password*: Fill with a password respecting the SAP password policy
   - *Security Profile*: Select *Service User*
   - *Roles*: Tick the *Message Charging Client* role
   - Save your modifications

Once the adequate service user has been created, you can use the Launching the SAP installer [page 190] dedicated procedure to launch the installation of the Diameter Server system. Once opened, the SL Common GUI tool displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. On the *Welcome* screen:
   - Click *SAP Convergent Charging* > *Install* > *Standard System Installation* > *Diameter Server*
   - Click the *Next* button
2. **Parameter Settings** screen
   - *Parameter Mode*: Select *Custom*

- Click the *Next* button
- (Microsoft Windows only) You will be prompted to log off in order to let SAP installer grant you the authorizations that are required to perform the installation. Click the *OK* button to log off, and reconnect to the host. SAP installer automatically restarts to continue the installation from there

3. **General SAP System Parameters** screen
   - *SAP System ID (SAPSID)*: Fill with the chosen SID for the Diameter Server
   - Click the *Next* button
4. **Master Password** screen
   - *Password for All Users*: Fill with a password respecting the SAP password policy
   - *Confirm*: Re-type the chosen password
   - Click the *Next* button
5. (Microsoft Windows only) **Microsoft Windows Domain** screen
   - *Domain Model*: Select *Local Domain*
   - Click the *Next* button
6. **Operating System Users** screen
   - **Linux and UNIX Operating Systems**
     - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
     - *Confirm*: Re-type the chosen password
     - *User ID*: *left empty*
     - *Group ID of sapsys*: *left empty*
     - Click the *Next* button
   - **Microsoft Windows Operating System**
     - *Password of SAP System Administrator*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
     - *Confirm*: Re-type the chosen password
     - *Password of SAP System Service User*: Filled by default with the password you specified in the **Master Password** screen, you can specify a different password
     - *Confirm*: Re-type the chosen password
     - Click the *Next* button
7. **Prerequisites Checker** screen
   - Click the *Next* button
8. **OpenBlox License** screen
   - *License Key*: Fill with your license key provided by the Traffix Systems company
   - Click the *Next* button
9. **CA Introscope Java Agent** screen
   - *Use monitoring with CA Introscope*: Untick the checkbox
   - Click the *Next* button
10. **System Landscape Directory** screen
    - *Configure System Landscape Directory*: Untick the checkbox
    - Click the *Next* button
11. **SAP Convergent Charging Security screen**
    - *Security*: Select the *Disabled* option

- Click the *Next* button

12. **Diameter Server Settings** screen
    - *Diameter Port*: Keep the default configuration, or set a specific value
    - *Diameter Host*: Specify the full name of the machine hosting the Diameter Server. If this host has several network interfaces and thus different IP addresses, tick the *Bind on a specific interface* checkbox and specify oneof these IP addresses
    - Click the *Next* button
13. **SAP Convergent Charging Core Server Connection Security** screen
    - *Security*: Select the *None* option
    - Click the *Next* button
14. **SAP Convergent Charging Core Server Connection** screen
    - Click the *Add* button to add a new line in the table containing the list of dispatchers available within your landscape
    - Click the *Dispatcher Message Host* column of the newly inserted row, and fill with the IP address of an available dispatcher
    - Click the *Dispatcher Message Port* column of the newly inserted row, and fill with the port number of this available dispatcher
    - *Login*: Fill with the name of the service user for the Diameter Server, created previously
    - *Password*: Fill with the password of the service user for the Diameter Server
    - Click the *Next* button
15. **Start Instances after Installation** screen
    - *Start SAP Convergent Charging instances after installation*: Tick the checkbox to automatically start the instances at the end of the installation process

      > **Note**
      >
      > The SAP Management Console is deployed during the installation process and gives you the possibility to manage each instance of your landscape

    - Click the *Next* button
16. **Unpack Archives** screen
    - Click the *Next* button
17. **Parameter Summary** screen
    - Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button
    - If all your settings are correctly taken into account, click the *Next* button to start the installation
18. **Task Progress** screen
    - This screen contains the different steps of your installation scenario, whose advancement can be followed

7.1.25 Preparing the Oracle Session Database

Description

This procedure explains you how to prepare your Session Database running under an Oracle RDBMS[65] (Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Oracle, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/oracle
  - http://scn.sap.com/docs/DOC-7888

Prerequisites

An instance of an Oracle RDBMS must be installed and available in your landscape. This database instance must use the UTF-8 character encoding to work in conjunction with the different SAP CC software components, and the *Oracle SQL*plus* tool must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC Session Database.

Procedure

The preparation of your SAP CC Session Database consists in the following operations:

- Setting up some initialization parameters
- Creating a user account and granting the adequate roles and privileges to this user
- Creating the different tablespaces
- Executing the SQL script to create the structure of the database

As a DBA (database administrator), you complete the following tasks:

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1. SAP Convergent Charging requires to modify some initialization parameters of the Oracle database, located in the `init.ora` file which contains the persistent parameters used during database startup. You can use the following procedure to:
   - Set the `_optim_peek_user_binds` parameter to **FALSE** in order to disable the bind peeking feature, which is not compatible with SAP CC
   - Increase the maximum number of processes and sessions by setting the `processes` and `sessions` parameters to 300 multiplied by the number of databases installed on this Oracle instance, i.e. 1 (as only the Session Database is installed on this instance)

   To correctly set up the initialization parameters of your Session Database, execute the following procedure:

```
sqlplus / as sysdba
alter system set "_optim_peek_user_binds"=false scope=spfile;
alter system set processes=300 scope=spfile;
alter system set sessions=300 scope=spfile;
shutdown immediate
startup
```

2. An Oracle user must be created and granted the adequate roles and privileges to administrate the Session Database. Execute the following procedure by replacing <DB_USER> by the login of the Session Database administrator and <DB_PASSWORD> by its password:

```
sqlplus / as sysdba
define dbuser="<DB_USER>";
define dbpwd="<DB_PASSWORD>";
create user &dbuser identified by &dbpwd;
grant CONNECT, RESOURCE TO &dbuser;
```

3. The data storage of your SAP CC Session Database, must be optimized by creating the following mandatory tablespaces:

Tablespace name	Description	Size (MB)
SESSION_RATING_DATA	Tables used to store the data related to the rating sessions	7000
SESSION_RATING_INDX	Indexes used for the data related to the rating sessions	500

   To create the mandatory tablespaces, execute the following procedure by replacing:
   - <TABLESPACE_NAME> by the name of the tablespace to create
   - <FILE_LOCATION> by the full path of the file that will contain the tablespace, located in the oradata directory
   - <SIZE> by the recommended value associated to the tablespace, listed in the above table and using the AUTO_EXTEND option

```
sqlplus / as sysdba
```

   And then, for each tablespace listed in the above table:

```
CREATE BIGFILE TABLESPACE <TABLESPACE_NAME> DATAFILE '<FILE_LOCATION>' SIZE
<SIZE> AUTOEXTEND ON MAXSIZE UNLIMITED UNIFORM SIZE 1M LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
```

   Followed by this command for Oracle 12c databases:

```
ALTER USER <DB_USER> QUOTA UNLIMITED ON <TABLESPACE_NAME>;
```

Example

For an Oracle 12c Enterprise Edition RDBMS installed on a Linux or UNIX operating system host, you can use the following statement:

- To create the **SESSION_RATING_DATA** tablespace
- To give the **sapccsession** user the possibility to access this tablespace

```
CREATE BIGFILE TABLESPACE SESSION_RATING_DATA DATAFILE '/oradata/vol1/
SESSION_RATING_DATA.dbf' SIZE 7000M AUTOEXTEND ON MAXSIZE UNLIMITED
UNIFORM SIZE 1M LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT
AUTO;
ALTER USER sapccsession QUOTA UNLIMITED ON SESSION_RATING_DATA;
```

4. To create the structure of the SAP CC , execute the following procedure:
   - Open the following folder of the SAP Central Repository directory (refer to the Document Definitions [page 12] section for further information):

     ```
     exe/uc/<OS>/CC_CORE_SERVER/session_sql/
     ```

   - Locate the `oracle_enterprise_sessiondb_create.sql` file that corresponds to the SQL script to use for creating the structure of the Session Database
   - Execute the following command, replacing <DB_USER> by the login and <DB_PASSWORD> by the password of the previously created user:

     ```
     sqlplus <DB_USER>/<DB_PASSWORD> oracle_enterprise_sessiondb_create.sql
     ```

7.1.26 Preparing the IEC Database

→ Remember

The IEC Database is required if an import/export connector (IEC) is part of your SAP Convergent Charging (SAP CC) system landscape. See Installing an IEC (Import/Export Connector) in an Existing Landscape [page 187].

Otherwise do not install this database.

Preparing the Oracle IEC Database [page 180]

Preparing the SQL Server IEC Database [page 182]

Preparing the SAP HANA IEC Database [page 184]

7.1.26.1 Preparing the Oracle IEC Database

Description

This procedure explains you how to prepare your IEC Database running under an Oracle RDBMS[66] (Standard or Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Oracle, refer to the following pages available on SAP Community:
  - http://scn.sap.com/community/oracle
  - http://scn.sap.com/docs/DOC-7888

Prerequisites

An instance of an Oracle RDBMS must be installed and available in your landscape. This database instance must use the UTF-8 character encoding to work in conjunction with the different SAP CC software components, and the *Oracle SQL*plus* tool must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC IEC Database.

Procedure

The preparation of your SAP CC IEC Database consists in the following operations:

- Setting up some initialization parameters
- Creating a user account and granting the adequate roles and privileges to this user
- Executing the SQL script to create the structure of the database

As a DBA (database administrator), you complete the following tasks:

1. SAP Convergent Charging requires to modify some initialization parameters of the Oracle database, located in the `init.ora` file which contains the persistent parameters used during database startup. You can use the following procedure to:

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- Set the `_optim_peek_user_binds` parameter to **FALSE** in order to disable the bind peeking feature, which is not compatible with SAP CC
- Increase the maximum number of processes and sessions by setting the `processes` and `sessions` parameters to 300 multiplied by the number of databases installed on this Oracle instance, i.e. 1 (as only the IEC Database is installed on this instance)

To correctly set up the initialization parameters of your IEC Database, execute the following procedure:

```
sqlplus / as sysdba
alter system set "_optim_peek_user_binds"=false scope=spfile;
alter system set processes=300 scope=spfile;
alter system set sessions=300 scope=spfile;
shutdown immediate
startup
```

2. An Oracle user must be created and granted the adequate roles and privileges to administrate the IEC Database. Execute the following procedure by replacing `<dbuser>` by the login of the IEC Database administrator and `<dbpwd>` by its password:

```
sqlplus / as sysdba
define dbuser="<dbuser>";
define dbpwd="<dbpwd>";
create user &dbuser identified by &dbpwd;
grant CONNECT, RESOURCE TO &dbuser;
grant EXECUTE_CATALOG_ROLE to &dbuser;
grant EXP_FULL_DATABASE, IMP_FULL_DATABASE to &dbuser;
grant CREATE PROCEDURE to &dbuser;
grant CREATE SESSION to &dbuser;
grant CREATE TABLESPACE, DROP TABLESPACE to &dbuser;
grant ALTER TABLESPACE to &dbuser;
grant CREATE ANY DIRECTORY to &dbuser;
grant DROP ANY DIRECTORY to &dbuser;
grant EXECUTE ON dbms_lock TO &dbuser
```

3. To create the structure of the SAP CC IEC Database, execute the following procedure:
   - Open the following folder of the installation materials:`/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND`
   - Uncompress the iec.zip archive using the following command:

     ```
     %SAPCC_JAVA_HOME%/bin/jar.exe xvf iec.zip -C iec
     ```

   - Open the newly created `iec/iec_sql` directory, and locate the `oracle_iecdb_create.sql` file that corresponds to the SQL script to use for creating the structure of the IEC database
   - Execute the following command, replacing `<dbuser>` by the login and `<dbpwd>` by the password of the previously created user:

     ```
     sqlplus <dbuser>/<dbpwd> oracle_iecdb_create.sql
     ```

7.1.26.2 Preparing the SQL Server IEC Database

Description

This procedure explains you how to prepare your IEC Database running under a Microsoft SQL Server RDBMS[67] (Standard or Enterprise edition).

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on the SAP Community:
  - http://scn.sap.com/community/sqlserver
  - http://scn.sap.com/docs/DOC-8286

Prerequisites

An instance of a SQL Server RDBMS must be installed and available in your landscape. The *SQL Server Management Studio* application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC IEC Database.

Procedure

The preparation of your SAP CC IEC Database consists in the following operations:

- Creating a SQL Server user account and granting the adequate roles on the adequate server to this user
- Creating the IEC Database with the user as the owner
- Executing the SQL script to create the structure of the database

1. An SQL Server user must be created for your SAP CC IEC Database, and granted the adequate roles and privileges. Execute the following procedure to create this SAP CC user:
   - Open the *SQL Server Management Studio* application.
   - The *Connect to Server* dialog window opens. Use the following settings:
     - *Server Name*: Select *(local)*.

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    - *Authentication*: Select *Windows Authentication*, in order to use your operating system user.
    - Click the *Connect* button.
- Open the *View* > *Object Explorer* menu to display the *Object Explorer* and expand the *(local)* server
- Right-click the *Security* element of the tree and open the *New / Login...* menu to display the **Login** screen
- Fill the *General* tab with the following information:
    - *Login name*: Fill with the name of the IEC Database administrator
    - Select the *SQL Server authentication* option
    - *Password*: Fill with a password following the SAP password policy
    - *Confirm password*: Re-type the chosen password
    - *User must change password at next login*: Untick the checkbox
    - *Default database*: Select the *master* database
- In the *Server Roles* tab, select the following roles:
    - *dbcreator*
    - *diskadmin*
    - *serveradmin*
- Click the *OK* button
- Right-click the *(local)* server and click *Disconnect*

2. Once the SQL Server user dedicated to the IEC Database has been created, it is necessary to reconnect to the server in order to create the IEC Database. Execute the following procedure:
    - Open the *File* > *Connect Object Explorer...* menu or click the *Connect Object Explorer* button to re-open the *Connect to Server* dialog window. Use the following settings:
        - *Server Name*: Specify the full name of the machine hosting the database
        - *Authentication*: Select *SQL Server authentication*
        - *Login*: Fill with the previously created user
        - *Password*: Fill with the password of the previously created user
        - Click the *Connect* button
    - Right-click the *Databases* element of the tree and open the *New Database...* menu to display the **New Database** screen
    - Fill the *General* tab with the following information:
        - *Database name*: Fill with the name of the IEC Database
        - *Owner*: Fill with the name of the previously created user
    - Fill the *Options* tab with the following information:
        - *Collation*: Collation defines the way strings are managed in SQL Server databases. The IEC Database must be case-sensitive (CS) and compatible with ASCI Unicode (AS). Thus select the element that corresponds to the appropriate language, ending by the *CS_AS* string (e.g. *Latin1_General_CS_AS*)
        - *Recovery model*: Select *Full*
        - *Compatibility level*: Ensure that *SQL Server 2008 (100)* is selected
        - Click the *OK* button
3. The SQL Server database must be in read committed snapshot mode.
   Execute the command:

```
ALTER DATABASE "<DB NAME>" SET READ_COMMITTED_SNAPSHOT ON;
```

To ensure that all users reading data will get access to a snapshot and not be blocked by concurrent transactions, SAP Convergent Charging requires that you enable this Microsoft SQL Server options.

4. To create the structure of the SAP CC IEC Database, execute the following procedure:
   - Open the following folder of the installation materials: `/DATA_UNITS/CC50_TOOLS_CONTENT_UC_OSIND`
   - Uncompress the iec.zip archive using the following command:

     ```
     %SAPCC_JAVA_HOME%/bin/jar.exe xvf iec.zip -C iec
     ```

   - Right-click the previously created database, and open the *New Query* menu to display the SQLQuery screen.
   - Open the *Edit / Insert File As Text...* menu and browse the uncompressed `iec/iec_sql` directory to select the `sqlserver_iecdb_create.sql` file that corresponds to the SQL script to use for creating the structure of the IEC Database.
   - Open the *Query* > *Execute* menu to create the structure of the database.

7.1.26.3 Preparing the SAP HANA IEC Database

Description

This procedure explains you how to prepare your IEC Database running under an SAP HANA in-memory database system using a single container or a multiple containers mode.

Preliminary Notes

- In a development landscape, you can install all the SAP CC databases on the same physical host, but using different schemas. Nevertheless, SAP SE highly recommends that you install the different databases on separated hosts, in order to ease administration tasks and improve performances
- For further information about the installation of Microsoft SQL Server, refer to the following pages available on the SAP Community:
  - http://scn.sap.com/community/developer-center/hana
  - http://scn.sap.com/docs/DOC-53955

For further information about data types, operators, functions, statements, and so on, refer to the SAP HANA SQL and System Views Reference documentation available on SAP Help Portal:https://help.sap.com

Prerequisites

An SAP HANA database system must be installed and available in your landscape. The SAP HANA Studio application must be installed to execute this procedure. Refer to your System Administrator to get information about these prerequisites relating to the SAP CC IEC Database.

Procedure

The preparation of your SAP CC IEC Database consists in the following operations:

- Declaring your system within the SAP HANA Administration Console of the *SAP HANA Studio* application
- Creating a user account and granting the adequate roles and privileges to this user on this system
- Disabling the expiration of the user's password
- Executing the SQL script to create the structure of the database

1. An SAP HANA system must be created to administrate your SAP CC IEC Database. Execute the following procedure to create this system:
   - Open the *SAP HANA Studio* application
   - Open the *Window* > *Perspective* > *Open Perspective* > *SAP HANA Administration Console* menu to display the perspective dedicated to the management of SAP HANA systems
   - Open the *Window* > *Show View* > *Systems* menu to display the view containing the list of SAP HANA systems
   - Click the *Add System* icon to open the *System* dialog window
   - Use the following settings to fill the *Specify System* screen
     - *Host Name*: Fill with the master host name
     - *Instance Number*: Fill with the instance number of your SAP HANA system
     - *Mode*: Select *Single Container* or *Multiple containers*, according to your needs
     - *Description*: Fill with a text that describes your system
     - Click the *Next* button
   - The **Connection Properties** screen opens. Select the *Authentication by database user* option and use the following settings:
     - *User Name*: Fill with the name of the SAP HANA system administrator
     - *Password*: Fill with a password following the SAP password policy
     - Click the *Finish* button to log on the newly declared SAP HANA system
2. Once you are connected to the relevant SAP HANA system, it is necessary to create a user account and grant the adequate roles and privileges to this user in order to administrate the IEC Database. Execute the following procedure to create this user:
   - Expand the *Security* folder, and right-click the *Users* element to display the context menu corresponds to the name of your database
   - Right-click the *Users* element of the tree and open the "New User" menu to display the screen dedicated to the creation of new users
   - Fill the *User* tab with the following information:
     - *User Name*: Fill with the name of the IEC Database administrator

        - Select the *Password* option in the *Authentication* fieldset
        - *Password*: Fill with a password following the SAP password policy
        - *Confirm password*: Re-type the chosen password
    - In the *Granted Roles* subtab, click the + button to add the following roles to the user:
        - *CONTENT_ADMIN*
        - *MODELING*
    - In the *System Privileges* subtab, click the + button to add the following privileges to the user:
        - *EXPORT*
        - *IMPORT*
    - Click the *Deploy* button to finalize the creation of the user
3. Once the user has been created and deployed on your SAP HANA system, it is necessary to disable the expiration of its password. Execute the following procedure:
    - Right-click the previously created SAP HANA system and open the *Open SQL Console* menu to display the screen dedicated to the execution of SQL statements
    - Type the following SQL statement within the editor, considering that `<USERNAME>` corresponds to the name of your previously created user. For further information, refer to the documentation of the `ALTER USER` command available in the SAP HANA documentation:

      ```
      ALTER USER <USERNAME> DISABLE PASSWORD LIFETIME;
      ```
    - Click the *Execute* button to execute the SQL statement
4. To create the structure of the SAP CC IEC Database, execute the following procedure:
    - Open the following folder of the installation materials:
      `/DATA_UNITS/CC50_TOOLS_CONTENT_UC_OSIND`
    - Uncompress the `iec.zip` archive using the following command:

      ```
      %SAPCC_JAVA_HOME%/bin/jar.exe xvf iec.zip -C iec
      ```
    - Right-click the previously created SAP HANA system and click the *Add System with Different User...* menu to authenticate as the IEC Database administrator
    - Select the *Authentication by database user* option, and specify the following information:
        - *User Name*: Fill with the name of the IEC Database administrator, specified in Step 2
        - *Password*: Fill with the password of the IEC Database administrator, specified in Step 2
    - Click the *Finish* button to create the new system
    - Right-click the newly created SAP HANA system and click the *Open SQL Console* menu to display the screen dedicated to the execution of SQL statements
    - Right-click the opened SQL Console and click the *Open File...* menu to open the file dialog window
    - Browse the uncompressed `iec/iec_sql` directory to select the `hdb_iecdb_create.sql`file that corresponds to the SQL script to use for creating the structure of the IEC Database
    - Click the *Execute* button to execute the SQL statement

7.1.27 Installing an IEC (Import/Export Connector) in an Existing Landscape

Description

The IEC (Import/Export Connector) is an SAP CC system that is used to connect the SAP CC Core Server system to external systems such as back office, legacy applications, and so on, and perform operations such as data transfer, modification or consolidation. This procedure explains you how to install and start the IEC.

> **Note**
>
> The IEC can be deployed:
>
> - On an existing host of your landscape, that already contains other systems
> - On a new host within your landscape, that is dedicated to the IEC

> **Caution**
>
> The IEC must not be deployed in a system landscape used of production purpose. You can only deploy the IEC on a host of a landscape that is used for development or test purposes.

Prerequisites

- A Core Server system must be available within your landscape. According to your installation scenario, refer to the Installing a Core Server on a Monohost Landscape [page 123] or the Installing a Core Server on a Multihost Landscape [page 131] procedure if necessary
- A dedicated host must be set up and available within your landscape. Refer to the Setting Up a New Host for a Landscape [page 141] procedure if necessary
- If you plan to use the *Export to Database* or *Import XCI from Database* components within your scenario, the IEC Database must be prepared and available. Depending of your landscape, refer to the Preparing the SAP HANA IEC Database [page 184], Preparing the Oracle IEC Database [page 180] or the Preparing the SQL Server IEC Database [page 182] procedure if necessary
- If your IEC Database is running under an Oracle RDBMS[68], the dedicated JDBC[69] driver must be downloaded and available on the host. Refer to the Downloading the Oracle JDBC Driver [page 97] procedure if necessary

[68] Relational Database Management System
[69] Java Database Connectivity

Installing IEC

The IEC is delivered within a dedicated archive that must be uncompressed using the following procedure:

- Open the following folder of the installation materials: `/DATA_UNITS/CC50_TOOLS_CONTENT_UC_OSIND`
- Uncompress the iec.zip archive using the following command:

```
%SAPCC_JAVA_HOME%/bin/jar.exe xvf iec.zip -C iec
```

If your landscape contains Oracle databases, copy the JAR file of the downloaded Oracle JDBC driver into the `jars` subfolder of the newly created `iec` folder.

Launching IEC in a standalone mode

To launch the IEC in a standalone mode, execute the following procedure:

- Open the previously uncompressed `iec/bin` directory, and locate the script file that corresponds to your operating system:
  - `iec.bat` for Microsoft Windows operating systems
  - `iec.sh` for Linux and UNIX operating systems, that needs to be:
    - Granted the executable right using the following command:

      ```
      chmod a+x iec.sh
      ```

    - Converted into the adequate format using the following command:

      ```
      dos2unix iec.sh
      ```

- Execute the launch script, using the following synopsis:

```
iec <scenario_dir>
    -login <user login>
    -password <visible user password>
    -uri <dispatchers list>
    [-rootDir <log folder>]
    [-scheduled]
```

  Where:
  - The `scenario_dir` parameter represents the path of the scenario file to execute
  - The `-login` parameter is used to specify the service user to connect with (that must be granted the Connector Administrator role)
  - The `-password` parameter gives the possibility to specify the password of the provided user
  - The `-uri` parameter is used to specify a comma-separated list of URIs relating to dispatcher instances, each URI using the following syntax: `<PROT>://<HOST_ADDRESS>:<MSG_TCP_PORT>`, where:
    - `<PROT>` corresponds to the tcp (or tcps) communication channel
    - `<HOST_ADDRESS>` corresponds to the name or IP address (IPv4 or IPv6) of the machine hosting the concerned dispatcher
    - `<MSG_TCP_PORT>` is the port number used by the dispatcher to communicate via the Message over TCP communication channel

- The `-rootDir` parameter is an optional parameter that can be used to specify a directory (`<log folder>`) to use for saving log files. When this parameter is not set, the working directory of the user is used by default
- The `-scheduled` parameter is an optional parameter that can be used to enable the scheduler configured within the scenario to execute. When this parameter is not set, the scenario is executed only once

Launching IEC in a remote mode

To launch the IEC in a remote mode, execute the following procedure:

- Open the previously uncompressed `iec/bin` directory, and locate the script file that corresponds to your operating system:
  - `iec.remote.bat` for Microsoft Windows operating systems
  - `iec.remote.sh` for Linux and UNIX operating systems, that needs to be:
    - Granted the executable right using the following command:

      ```
      chmod a+x iec_remote.sh
      ```

    - Converted into the adequate format using the following command:

      ```
      dos2unix iec_remote.sh
      ```

- Execute the launch script, using the following synopsis:

  ```
  iec_remote -uri <dispatchers list>
             [-repositoryDir <repository folder>]
             [-port <listening port>
             [-logDir <log folder>]
  ```

  Where:
  - The `-uri` parameter is used to specify a comma-separated list of URIs relating to dispatcher instances, each URI using the following syntax: `<PROT>://<HOST_ADDRESS>:<MSG_TCP_PORT>`, where:
    - `<PROT>` corresponds to the tcp (or tcps) communication channel
    - `<HOST_ADDRESS>` corresponds to the name or IP address (IPv4 or IPv6) of the machine hosting the concerned dispatcher
    - `<MSG_TCP_PORT>` is the port number used by the dispatcher to communicate via the Message over TCP communication channel
  - The `-repositoryDir` parameter is an optional parameter that can be used to specify a folder containing scenario files. When this parameter is not set, the `iec-repository` parent folder is used by default
  - The `-port` parameter is an optional parameter that can be used to specify a port for listening to connections from the CAT Tool user interface. When this parameter is not set, the 9002 port number is used by default
  - The `-logDir` parameter is an optional parameter that can be used to specify a directory (`<log folder>`) to use for saving log files. When this parameter is not set, the `<repository folder>/log` subfolder is used by default

7.1.28 Launching the SAP installer

Description

The installation of the different SAP Convergent Charging systems relies on a web-based GUI[70] named **SL Common GUI** and its back-end application SAPinst Framework. You can use this procedure to launch the SAP installer and perform your SAP CC installation within the SL Common GUI.

Preliminary Notes

For more information about the SAP installer and the SL Common GUI, refer to SAP Notes 2393060 and 2480194.

Prerequisites

- The SAP installer is available within the installation materials of SAP Convergent Charging 2020. Refer to the Downloading the Installation Materials [page 92] procedure if necessary.
- Check the supported and recommended web browsers [page 191] for SL Common GUI.

Launching the SAP installer

Linux and UNIX Operating Systems

1. Open the `/DATA_UNITS/CC2020_IM_<OS>` folder of the installation materials of SAP Convergent Charging 2020, where `<OS>` corresponds to the version of your Linux or UNIX operating system.
2. Execute the following command to update the access rights of the SAP installer launch script and make it executable:

   ```
   chmod +x sapinst
   ```

3. Open `./sapinst` to start the SAP installer and wait for incoming connections from the SL Common GUI.

Microsoft Windows Operating System

1. Open the `/DATA_UNITS/CC2020_IM_WINDOWS_X86_64` folder of the installation materials of SAP Convergent Charging 2020.
2. Open `sapinst.exe` to start the SAP installer and wait for incoming connections from the SL Common GUI.

[70] Graphical User Interface

3. If a *User Account Control* dialog window opens, click the *Yes* button to grant the SAP installer the required administrator access right.

> **Note**
>
> Once started, the SAP installer listens to the following defaults ports:
>
> - **4237**, used for HTTPS[71] connections with the SL Common GUI. If this port cannot be used, you can assign a free port number by launching the SAP installer with the following command line parameter: SAPINST_HTTPS_PORT=<FREE_PORT_NUMBER>
> - **4239**, used for managing the feedback evaluation form displayed at the end of the installation process. If this port cannot be used, you can assign a free port number by executing the SAP installer with the following command line parameter: SAPINST_HTTP_PORT=<FREE_PORT_NUMBER>

Using the SL Common GUI

To use the SL Common GUI, make sure that you use the latest version of one of the following supported browsers:

- Google Chrome (recommended)
- Mozilla Firefox
- Microsoft Edge
- Microsoft Internet Explorer 11 or higher

1. If you have a supported browser installed on the host where you run the `sapinst` executable, the SL Common GUI application starts automatically a browser.
2. Alternatively you can open a supported web browser on any device and run the URL from there. Then, open the URL[72] of the SL Common GUI, that is available at the bottom of the shell from which you launched the SAP installer, in the following information section:

```
...
*********************************************************************
Open your browser and paste the following URL address to access the GUI
https://[<hostname>]:4237/sapinst/docs/index.html
Logon users: [<users>]
*********************************************************************
...
```

> **Note**
>
> If you copy the URL of the SL Common GUI application manually in the browser window, make sure that you open a new Web browser window in private browsing mode (Internet Explorer), incognito mode (Chrome) or private browsing mode (Firefox). This is to prevent Web browser plugins and settings from interfering with the SL Common GUI.

[71] HyperText Transfer Protocol Secured
[72] Uniform Resource Locator

Troubleshooting

If you run into any problems with your personalized version of the SAP installer for SAP Convergent Charging 2020, see the topic Troubleshooting in our primary help.

7.1.29 Installing and Launching Core Tool

Description

The SAP CC Core Tool is the main user interface of an SAP Convergent Charging system landscape. This desktop application provides a Java-based GUI[73] that simplifies the management of a large number of processes or operations available in SAP CC, from implementation to production phases. You can use this procedure to launch the Core Tool from any host.

Preliminary Notes

For more information about the Core Tool user interface, refer to its dedicated documentation.

Prerequisites

- An SAP Convergent Charging Core Server system, with at least one dispatcher instance and one updater instance up and running, is needed to work with Core Tool. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary
- The SAP JVM 8.1 must be installed on the host, and the SAPCC_JAVA_HOME environment variable must be set to the path of the JVM[74]. Refer to the Downloading the SAP JVM [page 94] and Setting Up the System Environment Variables [page 104] procedures if necessary
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2013 runtime" libraries must be installed and available. For further information, refer to SAP Note 2676219
- On Mac OS X, SAP is not providing with SAP JVM, you can download and use the current version of Java for OS X directly from https://java.com

Getting Core Tool [page 193]

Starting Core Tool [page 193]

[73] Graphical User Interface
[74] Java Virtual Machine

7.1.29.1 Getting Core Tool

The Core Tool user interface is available within a ZIP archive located:

- In the Installation materials of SAP Convergent Charging 2020
- On SAP ONE Support Launchpad

If you want to get the Core Tool from the installation materials, take it from the following folder: `/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND`

If you want to download the latest version compatible with your Core Server version, execute the following procedure:

> → Remember
>
> Your Core Tool version must always match your Core Server version.

1. Go to SAP ONE Support Launchpad at the following address: https://launchpad.support.sap.com/
2. Click the *Software Downloads* launch tile of the *System Operations and Maintenance* section
3. Click the *Support Packages and Patches* section
4. Expand the *By Alphabetical Index (A-Z)* section
5. Click the *C* letter
6. Click the *SAP Convergent Charging* element of the list
7. Click the *SAP Convergent Charging 2020* element of the list
8. Click the *CORE TOOL 2020* element of the list to open the *Downloads* section
9. Choose the item matching your SAP Convergent Charging 2020 Core Server system Support Package and Patch level: `CCDESKTL<SP_LEVEL>P_<PATCH_LEVEL>-XXXXXXXX.ZIP`
   Where:
   - `<SP_LEVEL>` is the SAP Convergent Charging 2020 Core Server Support Package level
   - `<PATCH_LEVEL>` is the SAP Convergent Charging 2020 Core Server Patch level

7.1.29.2 Starting Core Tool

Launching the SAP CC Core Tool user interface consists in the following operations:

- Checking the prerequisite related to the SAPCC_JAVA_HOME environment variable
- Unzipping the retrieved archive that contains the Core Tool
- Executing the launch script

Linux and UNIX Operating Systems

- Execute the following command to check the existence of the SAPCC_JAVA_HOME environment variable:

```
echo $SAPCC_JAVA_HOME
```

If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure

- Uncompress the `core_tool.zip` archive
- Open the newly created `core_tool/bin` directory
- Execute the following command to update the access rights of the Core Tool launch script and make it executable:

```
chmod +x core_tool.sh
```

- Open `./core_tool.sh` and log in:
  - With credentials relating to an SAP CC user granted the adequate role within SAP CC
  - Using the name and port of a dispatcher instance that will distribute the login request to the adequate updater that will interact with the Core Tool

Microsoft Windows Operating System

- Execute the following command into a command line prompt in order to check the existence of the SAPCC_JAVA_HOME environment variable:

```
echo %SAPCC_JAVA_HOME%
```

If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure

- Uncompress the `core_tool.zip` archive
- Open the newly created `core_tool/bin` directory
- Open `core_tool.bat` and log in:
  - With credentials relating to an SAP CC user granted the adequate role within SAP CC
  - Using the name and port of a dispatcher instance that will distribute the login request to the adequate updater that will interact with the Core Tool

Mac OS X

- On Mac OS X, there is no need to set SAPCC_JAVA_HOME environment variable
- Uncompress the `core_tool.zip` archive
- Open the newly created `core_tool/bin` directory
- Execute the following command to update the access rights of the Core Tool launch script and make it executable:

```
chmod +x CoreTool.app/Contents/MacOS/core_tool
```

- Launch the `CoreTool.app` application from the `bin` directory to start the Core Tool user interface and log in:
  - with credentials relating to an SAP CC user granted the adequate role within SAP CC
  - using the name and port of a dispatcher instance that will distribute the login request to the adequate updater that will interact with the Core Tool

> **Note**
>
> Once installed, you can use the following procedure to modify the configuration file of the Core Tool in order to specify the location of the dedicated online helps:
>
> - Open the `core_tool/config` directory
> - Copy the `core_tool.config.sk` to a new `core_tool.config` file
> - Edit the `core_tool.config` file and uncomment the last 3 lines that contain the different links related to online helps

7.1.30 Installing and Launching BART Tool

Description

The SAP CC BART Tool is a user interface of an SAP Convergent Charging system landscape which is dedicated to the supervision of a BART Server system. This desktop application provides a Java-based GUI[75] that can be used to monitor the BART Server system and ease operations such as CDR acquisition, jobs administration, and so on. You can use this procedure to launch the BART Tool from any host.

Preliminary Notes

For more information about the BART Tool user interface, refer to its dedicated documentation.

Prerequisites

- A BART Server system must be available within your landscape. Refer to the Installing a BART Server System in an Existing Landscape [page 167] procedure if necessary
- The SAP JVM 8.1 must be installed on the host, and the SAPCC_JAVA_HOME environment variable must be set to the path of the JVM. Refer to the Downloading the SAP JVM [page 94] and Setting Up the System Environment Variables [page 104] appendices if necessary
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2013 runtime" libraries must be installed and available. For further information, refer to SAP Note 2676219
- On Mac OS X, SAP is not providing with SAP JVM, you can download and use the current version of Java for OS X directly from https://java.com

Getting BART Tool [page 196]

Starting BART Tool [page 196]

[75] Graphical User Interface

7.1.30.1 Getting BART Tool

The BART Tool user interface is available within a ZIP archive located:

- In the Installation materials of SAP Convergent Charging 2020
- On SAP ONE Support Launchpad

If you want to get the BART Tool from the installation materials, take it from the following folder: `/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND` If you want to download the latest version, execute the following procedure:

1. Go to SAP ONE Support Launchpad at the following address: https://launchpad.support.sap.com/
2. Click the *Software Downloads* launch tile of the *System Operations and Maintenance* section
3. Click the *Support Packages and Patches* section
4. Expand the *By Alphabetical Index (A-Z)* section
5. Click the *C* letter
6. Click the "*SAP Convergent Charging* element of the list
7. Click the *SAP Convergent Charging 2020* element of the list

   > **Note**
   >
   > Previous versions are also available but this procedure only concerns the 2020 release

8. Click the *BART TOOL 2020* element of the list to open the *Downloads* section
9. Choose the item matching your SAP Convergent Charging 2020 Core Server system Support Package and Patch level: `CCBARTTL<SP_LEVEL>P_<PATCH_LEVEL>-XXXXXXXX.ZIP`
   Where:
   - `<SP_LEVEL>` is the SAP Convergent Charging 2020 Core Server Support Package level
   - `<PATCH_LEVEL>` is the SAP Convergent Charging 2020 Core Server Patch level

7.1.30.2 Starting BART Tool

Launching the SAP CC BART Tool user interface consists in the following operations:

- Checking the prerequisite related to the SAPCC_JAVA_HOME environment variable
- Unzipping the archive that contains the BART Tool
- Modifying the default configuration to communicate with the BART Server system
- Executing the launch script

Linux and UNIX Operating Systems

- Execute the following command to check the existence of the SAPCC_JAVA_HOME environment variable:

  ```
  echo $SAPCC_JAVA_HOME
  ```

  If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure

- Uncompress the `bart_tool.zip` archive using the following command:

```
unzip bart_tool.zip -d bart_tool
```

- Open the newly created `bart_tool/config` directory, and copy the `bart_tool.config.sk` to a new `bart_tool.config` file
- Edit the `bart_tool.config` file and modify the `BARTServerURL` parameter with the URL[76] of the BART Server system you want to interact with, using the following syntax: `http(s)://<BART_SERVER_HOST>:<BART_SERVER_PORT>/`, where `<BART_SERVER_HOST>` corresponds to the name or IP address (IPv4 or IPv6) of a host of your landscape where a BART Server is installed and is running
- Open the `bart_tool/bin` directory
- Execute the following command to update the access rights of the BART Tool launch script and make it executable:

```
chmod +x bart_tool.sh
```

- Open `./bart_tool.sh` and log in with an SAP CC user granted the **"Batch Rating Administrator"** role within SAP CC

Microsoft Windows Operating System

- Execute the following command into a command line prompt in order to check the existence of the SAPCC_JAVA_HOME environment variable:

```
echo %SAPCC_JAVA_HOME%
```

  If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure
- Uncompress the `bart_tool.zip` archive using the following command:

```
%SAPCC_JAVA_HOME%\bin\jar.exe xvf bart_tool.zip -C bart_tool
```

- Open the newly created `bart_tool/config` directory, and copy the `bart_tool.config.sk` to a new `bart_tool.config` file
- Edit the `bart_tool.config` file and modify the `BARTServerURL` parameter with the URL of the BART Server system you want to interact with, using the following syntax: `http(s)://<BART_SERVER_HOST>:<BART_SERVER_PORT>/`, where `<BART_SERVER_HOST>` corresponds to the name or IP address (IPv4 or IPv6) of a host of your landscape where a BART Server is installed and is running
- Open the `bart_tool/bin` directory
- Open `bart_tool.bat` and log in with an SAP CC user granted the role within SAP CC

Mac OS X

- On Mac OS X, there is no need to set SAPCC_JAVA_HOME environment variable
- Uncompress the `bart_tool.zip` archive using the following command:

```
unzip bart_tool.zip -d bart_tool
```

- Open the newly created `bart_tool/config` directory, and copy the `bart_tool.config.sk` to a new `bart_tool.config` file

[76] Uniform Resource Locator

- Edit the `bart_tool.config` file and modify the `BARTServerURL` parameter with the URL2 of the BART Server system you want to interact with, using the following syntax: `http(s)://<BART_SERVER_HOST>:<BART_SERVER_PORT>/`, where `<BART_SERVER_HOST>` corresponds to the name or IP address (IPv4 or IPv6) of a host of your landscape where a BART Server is installed and is running
- Open the `bart_tool/bin` directory
- Execute the following command to update the access rights of the BART Tool launch script and make it executable:

```
chmod +x BARTTool.app/Contents/MacOS/bart_tool
```

- Launch the `BARTTool.app` application from the `bin` directory to start the BART Tool user interface and log in with an SAP CC user granted the **"Batch Rating Administrator"** role within SAP CC

> **Note**
>
> Once installed, you can use the following procedure to modify the configuration file of the BART Tool in order to specify the location of the dedicated online helps:
>
> - Open the `bart_tool.config` file located in the `bart_tool/config` directory
> - Uncomment the 2 lines that contain the different links related to online helps

7.1.31 Installing and Launching CAT Tool

Description

The SAP CC CAT Tool is a user interface of an SAP Convergent Charging system landscape which is dedicated to the definition and management of scenarios executed within an Import/Export Connector Server system. This desktop application provides a Java-based GUI[77] that can be used to perform operations such as data transfers between SAP CC and external systems, data modification, Java classes execution, and so on. You can use this procedure to launch the CAT Tool from any host.

Preliminary Notes

For more information about the CAT Tool user interface, refer to its dedicated documentation.

[77] Graphical User Interface

Prerequisites

- An IEC system must be available within your landscape. Refer to the Installing an IEC (Import/Export Connector) in an Existing Landscape [page 187] procedure if necessary. According to the components used within your scenarios, other systems might be available within your landscape. Refer to the adequate procedures if necessary
- The SAP JVM 8.1 must be installed on the host, and the SAPCC_JAVA_HOME environment variable must be set to the path of the JVM[78]. Refer to the Downloading the SAP JVM [page 94] and Setting Up the System Environment Variables [page 104] appendices if necessary
- On Microsoft Windows hosts, the "Microsoft Visual C++ 2013 runtime" libraries must be installed and available. For further information, refer to SAP Note 2676219.
- On Mac OS X, SAP is not providing with SAP JVM, you can download and use the current version of Java for OS X directly from https://java.com

Getting CAT Tool [page 199]

Starting CAT Tool [page 200]

7.1.31.1 Getting CAT Tool

The CAT Tool user interface is available within a ZIP archive located:

- In the Installation materials of SAP Convergent Charging 2020
- On SAP ONE Support Launchpad

If you want to get the CAT Tool from the installation materials, take it from the following folder: `/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND`

If you want to download the latest version, execute the following procedure:

1. Go to SAP ONE Support Launchpad at the following address: https://launchpad.support.sap.com/
2. Click the *Software Downloads* launch tile of the *System Operations and Maintenance* section
3. Click the *Support Packages and Patches* section
4. Expand the *By Alphabetical Index (A-Z)* section
5. Click the *C* letter
6. Click the *SAP Convergent Charging* element of the list
7. Click the *SAP Convergent Charging 2020* element of the list

> **Note**
>
> Previous versions are also available but this procedure only concerns the 2020 release

8. Click the *CAT TOOL 2020* element of the list to open the *Downloads* section
9. Choose the item matching your SAP Convergent Charging 2020 Core Server system Support Package and Patch level: `CCCAT<SP_LEVEL>P_<PATCH_LEVEL>-XXXXXXXX.ZIP`
   Where:
   - `<SP_LEVEL>` is the SAP Convergent Charging 2020 Core Server Support Package level

[78] Java Virtual Machine

- <PATCH_LEVEL> is the SAP Convergent Charging 2020 Core Server Patch level

7.1.31.2 Starting CAT Tool

Launching the SAP CC CAT Tool user interface consists in the following operations:

- Checking the prerequisite related to the SAPCC_JAVA_HOME environment variable
- Unzipping the archive that contains the CAT Tool
- Initializing the configuration file from the provided default one
- Executing the launch script

Linux and UNIX Operating Systems

1. Execute the following command to check the existence of the SAPCC_JAVA_HOME environment variable:

   ```
   echo $SAPCC_JAVA_HOME
   ```

   If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure.
2. Uncompress the `cat_tool.zip` archive using the following command:

   ```
   unzip cat_tool.zip -d cat_tool
   ```
3. Open the newly created `cat_tool/config` directory, and copy the `cat_tool.config.sk` to a new `cat_tool.config` file.
4. Open the `cat_tool/bin` directory.
5. Execute the following command to update the access rights of the CAT Tool launch script and make it executable:

   ```
   chmod +x cat_tool.sh
   ```
6. Open `./cat_tool.sh` and log in with an SAP CC user granted the **"Connector Administrator"** role within SAP CC.

Microsoft Windows Operating System

1. Execute the following command into a command line prompt in order to check the existence of the SAPCC_JAVA_HOME environment variable:

   ```
   echo %SAPCC_JAVA_HOME%
   ```

   If you get an empty result, it means that the variable is not set. To set this variable, refer to the Setting Up the System Environment Variables [page 104] procedure.
2. Uncompress the `cat_tool.zip` archive using the following command:

   ```
   %SAPCC_JAVA_HOME%\bin\jar.exe xvf cat_tool.zip -C cat_tool
   ```
3. Open the newly created `cat_tool/config` directory, and copy the `cat_tool.config.sk` to a new `cat_tool.config` file.
4. Open the `cat_tool/bin` directory.
5. Open `cat_tool.bat` and log in with an SAP CC user granted the **"Connector Administrator"** role within SAP CC.

Mac OS X

1. On Mac OS X, there is no need to set SAPCC_JAVA_HOME environment variable.
2. Uncompress the `cat_tool.zip` archive using the following command:

```
unzip cat_tool.zip -d cat_tool
```

3. Open the newly created `cat_tool/bin` directory.
4. Execute the following command to update the access rights of the CAT Tool launch script and make it executable:

```
chmod +x CATTool.app/Contents/MacOS/cat_tool
```

5. Launch the `CATTool.app` application from the `bin` directory to start the CAT Tool user interface and log in with an SAP CC user granted the **"Connector Administrator"** role within SAP CC.

> **Note**
>
> Once installed, you can use the following procedure to modify the configuration file of the CAT Tool in order to specify the location of the dedicated online helps:
>
> - Open the `cat_tool.config` file located in the `cat_tool/config` directory.
> - Uncomment the 2 lines that contain the different links related to online helps.

7.1.32 Installing and Launching Cockpit

Description

SAP CC Cockpit aka **Convergent Charging Cockpit** is a web-based user interface that provides a few apps that are dedicated to the administration and supervision of an SAP CC Core Server system or to the pricing configuration/administration for the service or product to monetize. Deployed and run within a Java Web Server (Apache Tomcat server for example), the Cockpit web application simplifies and empowers administrators' and pricing specialists' activities and tasks.

Preliminary Notes

Microsoft Windows Operating System

The Cockpit application uses a configuration file that you must set up in the home directory of the operating system user of the Apache Tomcat server. When Tomcat is installed as a Windows service, the OS[79] user of Tomcat might be a special Windows user such as `LocalSystem` or `LocalService`. The home directory of these particular OS users is a special folder located within the Microsoft Windows administration folder.

[79] Operating System

If you install Tomcat as a Windows service, SAP recommends that you configure this Windows service to run the Tomcat process as a dedicated local OS user created for that purpose. This local user has a regular home directory that you can use to configure the different Cockpit applications.

For more information, refer to the Apache Tomcat server documentation.

Prerequisites

- The SAP CC Core Server system you want to administer with the Cockpit application must be available in your SAP system landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary, and the Product Availability Matrix (PAM) here: support.sap.com/pam.
- You have identified the SAP system ID (SID) of the targeted Core Server system and the list of HTTP URLs of the dispatcher instances running in the system you want to connect to.
- You have identified the RDBMS that hosts the SAP CC Core Database associated to the targeted Core Server system. A DBA has created a database user or account in this RDBMS for the SAP CC Cockpit web application to deploy now. See Creating a Database User/Account of the Core Database and Dedicated to Communications Between SAP CC Cockpit and the SAP CC Core Database [page 151].
- **A Tomcat server using SAP JVM 8 must be available.** For further information, use the Tomcat documentation and recommendations.
  - In terms of security requirements, the installation must comply with the recommendations of the SAP CC 2020 Security Guide documentation. As shown on the architecture schema available in the SAP CC Software Components [page 21] section, the Tomcat server must also be able to communicate with:
    - The dispatcher and updater instances running in the Core Server system

      > ⓘ Note
      >
      > Applications belonging to the groups:
      > - Pricing Data Structure
      > - Pricing Definition
      > - Pricing Administration
      >
      > of Cockpit need a running updater to work.

    - The Core Database
  - In terms of sizing requirements, SAP SE recommends that you increase the size of the Tomcat heap memory (-Xmx option of the JVM) to 6GB in order to manage large objects within Cockpit.

    > → Recommendation
    >
    > 1GB is enough if no large objects.

    To be able to upload large files, a large mapping table for example, reconfigure each deployed Cockpit web application. Consider the `http.multipart.max-file-size` setting and refer to the *Data Files Management Settings* section in the Configuring Cockpit topic.
  - The time zone used by the Tomcat server must be the same as the system time zone of the connected Core Server system. Refer to the Setting Up the System Time Zone [page 102] procedure to ensure that the time zone is correctly configured.

- You install and maintain the same version for both the Core Server system and the Cockpit application.

Oracle RDBMS

If your landscape contains one or multiple Oracle databases, you need to execute the following procedure in order to update your Apache Tomcat server with the relevant Oracle JDBC[80] driver:

1. Execute the Downloading the Oracle JDBC Driver [page 97] procedure to download the relevant JAR archive.
2. Copy the downloaded JAR file into the `lib` subfolder that is available in the root directory of the Apache Tomcat server.

1. Installing the Oracle JDBC Driver in an Apache Tomcat Server System [page 203]
2. Getting Cockpit [page 204]
3. Installing Cockpit [page 204]
4. Starting and Stopping the Cockpit Back-End Application [page 206]
5. Launching Cockpit in a Web Browser [page 206]

7.1.32.1 Installing the Oracle JDBC Driver in an Apache Tomcat Server System

Procedure

To pepare your Apache Tomcat Server system:

- If your SAP CC system landscape contains one or multiple Oracle databases, you need to execute the following procedure in order to update your Apache Tomcat server with the relevant Oracle JDBC[81] driver:
  1. Complete the Downloading the Oracle JDBC Driver [page 97] procedure to download the relevant JAR file.
  2. Copy the downloaded JAR file to the `lib` subfolder that is available in the root directory of the Apache Tomcat Server system.

Parent topic: Installing and Launching Cockpit [page 201]

Next: Getting Cockpit [page 204]

80 Java Database Connectivity
81 Java Database Connectivity

7.1.32.2 Getting Cockpit

The SAP CC Cockpit application is delivered within a ZIP archive. The `ZIP` archive file is available:

- In the Installation materials of SAP Convergent Charging 2020 (in this case, the archive is named `cockpit.zip`)
- On SAP ONE Support Launchpad on SAP Support Portal

If you want to get the Cockpit software from the installation materials, take it from the following folder: `/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND`

If you want to download the latest version, execute the following procedure:

1. Go to SAP ONE Support Launchpad at the following address: https://launchpad.support.sap.com/
2. Click the *Software Downloads* launch tile of the *System Operations and Maintenance* section
3. Click the *Support Packages and Patches* section
4. Expand the *By Alphabetical Index (A-Z)* section
5. Click the *C* letter
6. Click the *SAP Convergent Charging* element of the list
7. Click the *SAP Convergent Charging 2020* element of the list

   > **Note**
   >
   > Previous versions are also available but this procedure only concerns the 2020 release

8. Click the *CC COCKPIT 2020* element of the list to open the *Downloads* section
9. Choose the item matching your SAP Convergent Charging 2020 Core Server system Support Package and Patch level: `CCCOCKPIT<SP_LEVEL>P_<PATCH_LEVEL>-XXXXXXXX.ZIP`
   Where:
   - `<SP_LEVEL>` is the SAP Convergent Charging 2020 Core Server Support Package level
   - `<PATCH_LEVEL>` is the SAP Convergent Charging 2020 Core Server Patch level

Parent topic: Installing and Launching Cockpit [page 201]

Previous: Installing the Oracle JDBC Driver in an Apache Tomcat Server System [page 203]

Next: Installing Cockpit [page 204]

7.1.32.3 Installing Cockpit

The previously retrieved `ZIP` archive file contains a `cccockpit.war` file that represents the SAP CC Cockpit application. According to the configuration of the targeted Tomcat Server, this war file:

- Can be automatically deployed if the *Automatic Application Deployment* feature of Tomcat is **enabled** (which is the case for fresh installations of Tomcat servers)
- Must be specified as a dedicated Context within the configuration file of the Tomcat server if the *Automatic Application Deployment* feature of Tomcat is **disabled**

To install the SAP CC Cockpit application, execute the following procedure:

1. Uncompress the `ZIP` archive, that includes the `cccockpit.war` file
2. Rename this `WAR` file if you want to customize the installation. The name of the file determines the URI that is used to access the Web application and the location of the Web application in the Tomcat Server.

   > ### Example
   >
   > - `cccockpit_<SID>.war`, where `<SID>` corresponds to the identifier of the targeted Core Server system
   > - `mycccockpit.war`

3. Copy the template file (delivered in the ZIP file) named `sap_cc_cockpit_[WEB_APP_NAME].properties.sk` to initialize a configuration file for the Cockpit application to deploy
4. Suppress the `.sk` extension of the copied file
5. Rename this `sap_cc_cockpit_[WEB_APP_NAME].properties` file to `sap_cc_cockpit_<war_file_name>.properties`, where `<war_file_name>` corresponds to the name of the deployed `WAR` file.

   > ### Example
   >
   > Considering the second example in step **2**, the file name becomes: `sap_cc_cockpit_mycccockpit.properties`, where **`mycccockpit`** is the name of the `WAR` file

6. Open the newly created configuration file and modify the different parameters that are either mandatory or that necessary to fit your specific needs.

   > ### → Recommendation
   >
   > For further information and recommendations, refer to the Configuring Cockpit section available in the SAP CC 2020 Tuning Guide documentation.

   > ### ⚠ Caution
   >
   > Passwords stored in the configuration file of the Cockpit are not encrypted. For security reasons, it is highly recommended to encrypt them. To do so, follow the steps below:
   >
   > 1. Locate the `.passphrase` file in the `\usr\sap\<SID>\<INSTANCE_DIRECTORY>\config\` directory, where:
   >   - `<SID>` is the SAP Convergent Charging 2020 System ID
   >   - `<INSTANCE_DIRECTORY>` is the directory of the instance of your choice (CC`<Instance name first letter (D for Dispatcher, U for Updater...)><Number of the instance with two digits>`). For example: CCD03
   > 2. Copy this file in the folder containing the configuration file
   > 3. Rename the freshly copied `.passphrase` file to `.sap_cc_cockpit_<WEB_APP_NAME>`
   > 4. Restart the Cockpit

7. Save your modifications.
8. Move this configuration file to the home directory of the OS user running the Tomcat Server as explained in the preliminary note. Enable the **read** access right to this file for this user account
9. If the *Automatic Application Deployment* feature of Tomcat is enabled, move the `WAR` file to webapps folder of the Tomcat Server. The Cockpit application will be automatically deployed by Tomcat while running.

If the *Automatic Application Deployment* feature of Tomcat is **disabled**, set up a dedicated Context in the configuration file of the Tomcat server. For further information about the creation of a Tomcat context, refer to its dedicated documentation.

10. Verify that the deployment is done by:
    - Launching the Web application in your favorite web browser. For further information, refer to the Launching Cockpit in a Web Browser [page 206] dedicated procedure afterwards
    - Searching within the Tomcat log files the following log entry: `Deployment of web application archive..\webapps\<war_file_name> has finished` , where `<war_file_name>` corresponds to the name of the deployed `WAR` file

When the installation process is completed, you can provide your Cockpit users with the relevant URLs of the deployed user interfaces.

Parent topic: Installing and Launching Cockpit [page 201]

Previous: Getting Cockpit [page 204]

Next: Starting and Stopping the Cockpit Back-End Application [page 206]

7.1.32.4 Starting and Stopping the Cockpit Back-End Application

There are several ways of stopping (and starting) a Web application in your server. For example, use the Tomcat Manager Web UI (`/manager`) or its commands to shut down the deployed Web application.

Monitor the startup and detect errors.

See Troubleshooting Your Cockpit Installation [page 297].

Parent topic: Installing and Launching Cockpit [page 201]

Previous: Installing Cockpit [page 204]

Next: Launching Cockpit in a Web Browser [page 206]

7.1.32.5 Launching Cockpit in a Web Browser

To launch an SAP CC Cockpit user interface to administer a Core Server system:

1. Open a web browser.
2. Navigate to the deployed Web application by entering the URL that corresponds to the SAP CC Cockpit application you want to run, using the following syntax: `http(s)://<IP_OR_DNS_ADDRESS>:<PORT_NB>/<CCCOCKPIT_URI>/`

Where:

- `<IP_OR_DNS_ADDRESS>` corresponds to the IP address or DNS name of the Tomcat Server host.
- `<PORT_NB>` corresponds to the dedicated communication port.
- `<CCCOCKPIT_URI>` corresponds to the path to the Web application in the Tomcat Server or to the "Context Path" of the Cockpit application.

3. A sign-in window appears.
4. As an SAP CC user granted the adequate roles and privileges, log on to the SAP Convergent Charging Cockpit application to access the available apps.

> → Tip
>
> For more information about the Cockpit user interface, refer to its dedicated documentation: SAP CC 2020 Primary Help for Cockpit.

Parent topic: Installing and Launching Cockpit [page 201]

Previous: Starting and Stopping the Cockpit Back-End Application [page 206]

Related Information

Launching Cockpit

7.1.33 Installing and Launching Admin+

Description

The SAP CC Admin+ user interface is dedicated to the administration operations relating to a Core Server system, such as getting and setting parameters' values, resetting data caches, managing batch groups, and so on. You can use this procedure to launch Admin+ from any host.

Preliminary Notes

For more information about the Admin+ user interface, refer to its dedicated documentation.

Prerequisites

A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary

Getting Admin+ [page 208]

Starting Admin+ [page 208]

7.1.33.1 Getting Admin+

The Admin+ user interface is available:

- In the `script` folder of each installed dispatcher
- In the `bin` folder of the Core Tool user interface. For further information, refer to the Installing and Launching Core Tool [page 192] procedure if necessary

7.1.33.2 Starting Admin+

To launch Admin+, execute the following procedure considering that:

- `<SYSTEM_ID>` corresponds to the identifier of your Core Server system
- `<DISPATCHER_INSTANCE_NUMBER>` corresponds to the SAP_INSTANCE_NUMBER (00-97) of the dispatcher instance
- `<DISPATCHER_HOST_ADDRESS>` corresponds to the name or IP address (IPv4 or IPv6) of a host of your landscape where a dispatcher is installed and is running
- `<DISPATCHER_PORT>` corresponds to the port used by the targeted dispatcher
- `<CORE_TOOL_INSTALLATION_PATH>` corresponds to the location of the Core Tool user interface

Linux and UNIX Operating Systems

Connect to **a host of your landscape where a dispatcher is installed**, and execute the following commands:

```
cd /usr/sap/<SYSTEM_ID>/CCD<DISPATCHER_INSTANCE_NUMBER>/script
./admin+.sh
```

Or connect to **a host where a Core Tool is installed**, and execute the following commands:

```
cd <CORE_TOOL_INSTALLATION_PATH>/bin
./admin+.sh <DISPATCHER_HOST_ADDRESS> <DISPATCHER_PORT>
```

Microsoft Windows Operating System

Connect to **a host of your landscape where a dispatcher is installed and is running**, and execute the following commands into a command line prompt:

```
cd C:\usr\sap\<SYSTEM_ID>\CCD<DISPATCHER_INSTANCE_NUMBER>\script
admin+
```

Or connect to **a host where a Core Tool is installed**, and execute the following commands into a command line prompt:

```
cd <CORE_TOOL_INSTALLATION_PATH>/bin
admin+ <DISPATCHER_HOST_ADDRESS> <DISPATCHER_PORT>
```

7.1.34 Installing and Launching BART+

Description

The SAP CC BART+ user interface is dedicated to the administration operations relating to a BART Server system, such as getting and setting parameters' values, monitoring the batch acquisition sessions, and so on. You can use this procedure to launch BART+ from any host.

Preliminary Notes

For more information about the BART+ user interface, refer to its dedicated documentation.

Prerequisites

A BART Server system must be available within your landscape. Refer to the Installing a BART Server System in an Existing Landscape [page 167] procedure if necessary.

Getting BART+ [page 209]

Starting BART+ [page 210]

7.1.34.1 Getting BART+

The BART+ user interface is available:

- In the `script` folder of each installed BART Server system
- In the `bin` folder of the BART Tool user interface. For further information, refer to the Installing and Launching BART Tool [page 195] procedure if necessary

7.1.34.2 Starting BART+

Starting

To launch BART+, execute the following procedure considering that:

- <SYSTEM_ID> corresponds to the identifier of your BART Server system
- <BART_INSTANCE_NUMBER> corresponds to the number of the BART Server instance
- <BART_SERVER_HOST_ADDRESS> corresponds to the name or IP address (IPv4 or IPv6) of a host of your landscape where a BART Server system is installed and is running
- <BART_SERVER_PORT> corresponds to the port used by the targeted BART Server
- <BART_TOOL_INSTALLATION_PATH> corresponds to the location of the BART Tool user interface

Linux and UNIX Operating Systems

Connect to **a host of your landscape where a BART Server is installed and is running**, and execute the following commands:

```
cd /usr/sap/<SYSTEM_ID>/CAB<BART_INSTANCE_NUMBER>/script
./bart+.sh
```

Or connect to **a host where a BART Tool is installed**, and execute the following commands:

```
cd <BART_TOOL_INSTALLATION_PATH>/bin
./bart+ <BART_SERVER_HOST_ADDRESS> <BART_SERVER_PORT>
```

Microsoft Windows Operating System

Connect to **a host of your landscape where a BART Server is installed and is running**, and execute the following commands into a command line prompt:

```
cd C:\usr\sap\<SYSTEM_ID>\CAB<BART_SERVER_INSTANCE_NUMBER>\script
bart+
```

Or connect to **a host where a BART Tool is installed**, and execute the following commands into a command line prompt:

```
cd <BART_TOOL_INSTALLATION_PATH>/bin
bart+ <BART_SERVER_HOST_ADDRESS> <BART_SERVER_PORT>
```

7.1.35 Installing and Launching Setup Tool

Description

The SAP CC Setup Tool is a user interface of an SAP Convergent Charging system landscape which is dedicated to configuration and maintenance operations of a Core Server system. You can use this procedure to launch the Setup Tool from any host.

Preliminary Notes

For more information about the Setup Tool user interface, refer to its dedicated documentation.

Prerequisites

A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary.

Getting Setup Tool [page 211]

Starting Setup Tool [page 211]

7.1.35.1 Getting Setup Tool

The Setup Tool user interface is available in the `script` folder of each installed dispatcher.

7.1.35.2 Starting Setup Tool

The Setup Tool user interface is a command-line application that has the following partial synopsis:

```
setup <command> [<arguments>]
```

Where:

- The `command` parameter is used to specify the operation to perform, such as:
  - certentry list/import/export/... that are used to manage client certificates for secured landscapes

- keyentry list/(un)link/import/export/... that are used to manage key entries for secured landscapes
- cif delete/import/export, that are used to manage the CIF configuration
- instancemap delete/import/export, that are used to manage the instance map
- license import/export/..., that are used to manage the licensing of your landscape
- And so on

> **Note**
>
> To get the whole synopsis of the Setup Tool user interface, consult its documentation.

To launch Setup Tool, execute the following procedure considering that:

- `<SYSTEM_ID>` corresponds to the identifier of your Core Server system
- `<DISPATCHER_INSTANCE_NUMBER>` corresponds to the SAP_INSTANCE_NUMBER (00-97) of the dispatcher instance
- `<SYNOPSIS>` corresponds to the commands and options to execute

Linux and UNIX Operating Systems

Connect to **a host of your landscape where a dispatcher is installed**, and execute the following commands:

```
cd /usr/sap/<SYSTEM_ID>/CCD<DISPATCHER_INSTANCE_NUMBER>/script
./setup.sh <SYNOPSIS>
```

Microsoft Windows Operating System

Connect to **a host of your landscape where a dispatcher is installed and is running**, and execute the following commands into a command line prompt:

```
cd C:\usr\sap\<SYSTEM_ID>\CCD<DISPATCHER_INSTANCE_NUMBER>\script
setup <SYNOPSIS>
```

7.1.36 Installing and Launching BART Setup Tool

Description

The SAP CC BART Setup Tool is a user interface of an SAP Convergent Charging system landscape which is dedicated to the management of security settings for securing the communication channels used by the BART Server system. You can use this procedure to launch the BART Setup Tool from any host.

Preliminary Notes

For more information about the BART Setup Tool user interface, refer to its dedicated documentation.

Prerequisites

A BART Server system must be available within your landscape. Refer to the procedure Installing a BART Server System in an Existing Landscape [page 167] if necessary.

Getting BART Setup Tool [page 213]

Starting BART Setup Tool [page 213]

7.1.36.1 Getting BART Setup Tool

The BART Setup Tool user interface is available in the `script` folder of each installed BART Server.

7.1.36.2 Starting BART Setup Tool

> ⚠ Caution
>
> Before running the BART Setup Tool user interface, ensure that at least one dispatcher instance is up and running.

The BART Setup Tool user interface is a command-line application that has the following partial synopsis:

```
setup <command> [<arguments>]
[-login=<user login>]
[-password=<visible user password>]
```

Where:

- The `command` parameter is used to specify the operation to perform, such as:
  - certentry list/import/export/... that are used to manage client certificates for secured landscapes
  - keyentry list/(un)link/import/export/... that are used to manage key entries for secured landscapes
- The `-login` optional parameter is used to specify the individual user to connect with (that must be granted the Administrator role, except for the `help` and `about` commands). If you do not use the `-login` option, you will be prompted to enter this username
- The `-password` optional parameter gives the possibility to specify the password of the provided user. Please note that the specified password is displayed on your screen, and thus visible. If you do not use the `-password` option, you will be prompted to enter the password, which is a safer method to authenticate

> ⓘ Note
>
> To get the whole synopsis of the BART Setup Tool user interface, consult its documentation.

To launch BART Setup Tool, execute the following procedure considering that:

- `<SYSTEM_ID>` corresponds to the identifier of your BART Server system
- `<BART_INSTANCE_NUMBER>` corresponds to the SAP instance number of the BART Server instance

- <SYNOPSIS> corresponds to the commands and options to execute

Linux and UNIX Operating Systems

Connect to **a host of your landscape where a BART Server is installed**, and execute the following commands:

```
cd /usr/sap/<SYSTEM_ID>/CAB<BART_INSTANCE_NUMBER>/script
./setup.sh <SYNOPSIS>
```

Microsoft Windows Operating System

Connect to **a host of your landscape where a BART Server is installed**, and execute the following commands into a command line prompt:

```
cd C:\usr\sap\<SYSTEM_ID>\CAB<BART_INSTANCE_NUMBER>\script
setup <SYNOPSIS>
```

7.1.37 Installing and Launching Config Tool

Description

The SAP CC Config Tool is a user interface of an SAP Convergent Charging system landscape which is dedicated to configuration operations of a Core Server system. You can use this procedure to launch the Config Tool from any host.

Preliminary Notes

For more information about the Config Tool user interface, refer to its dedicated documentation.

Prerequisites

A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary.

Getting Config Tool [page 215]

Starting Config Tool [page 215]

7.1.37.1 Getting Config Tool

The Config Tool user interface is available in the `script` folder of each installed dispatcher.

7.1.37.2 Starting Config Tool

The Config Tool user interface is a command-line application that has the following partial synopsis:

```
config <command> [<arguments>]
[-login=<user login>]
[-password=<visible user password>]
```

Where:

- The `-login` optional parameter is used to specify the individual user to connect with (that must be granted the Administrator role, except for the `help` and `about` commands). If you do not use the `-login` option, you will be prompted to enter this username
- The `-password` optional parameter gives the possibility to specify the password of the provided user. Please note that the specified password is displayed on your screen, and thus visible. If you do not use the `-password` option, you will be prompted to enter the password, which is a safer method to authenticate
- The `command` parameter is used to specify the operation to perform

> **Note**
>
> To get the whole synopsis of the Config Tool user interface, consult its documentation.

To launch Config Tool, execute the following procedure considering that:

- `<SYSTEM_ID>` corresponds to the identifier of your Core Server system
- `<DISPATCHER_INSTANCE_NUMBER>` corresponds to the SAP_INSTANCE_NUMBER (00-97) of the dispatcher instance
- `<SYNOPSIS>` corresponds to the commands and options to execute

Linux and UNIX Operating Systems

Connect to **a host of your landscape where a dispatcher is installed**, and execute the following commands:

```
cd /usr/sap/<SYSTEM_ID>/CCD<DISPATCHER_INSTANCE_NUMBER>/script
./config.sh <SYNOPSIS>
```

Microsoft Windows Operating System

Connect to **a host of your landscape where a dispatcher is installed and is running**, and execute the following commands into a command line prompt:

```
cd C:\usr\sap\<SYSTEM_ID>\CCD<DISPATCHER_INSTANCE_NUMBER>\script
config <SYNOPSIS>
```

7.1.38 Securing an SAP CC System Landscape

Description

The communication between the different systems of an SAP Convergent Charging landscape relies on different communication channels that can be secured to fit security requirements. You can consider this procedure as an entry point for implementing a secured landscape, within which:

- Communications channels used by clients and servers can be secured using either a "dual" or a "oneway" mode
- SAP CC server systems can be configured to specify a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS

This procedure contains:

- A common procedure that you can use to generate the securing materials required within the different procedures relating to each SAP CC server system
- Cross-references to procedures dedicated to the securing of:
  - The Core Server system
  - The BART Server system
  - The Diameter Server system
  - The Cockpit application and its Tomcat Server system

Preliminary Notes

- For more information about the different communication channels, refer to the SAP CC 2020 Security Guide documentation
- For more information about securing SAP Convergent Charging, refer to the guided answer How to Secure Communications in and with SAP Convergent Charging?
- The procedure used to secure your landscape can be adapted to fit specific security needs. In case you need to adapt the securing procedure and implement another securing policy for your landscape, SAP SE highly recommends that you to refer to your technology consultants and security experts
- This procedure contains commands which refer to command-line applications:
  - Delivered with the operating system
  - Provided within the SAP CC solution
  - Available within the materials downloaded in the previous installation steps (such as *keytool*, available in the SAP JVM [page 94])
  - Available for download on dedicated websites, where we invite you to get the latest version (such as the openSSL tool, available at the following location: http://www.openssl.org/)

Common procedure for generating the securing materials

See Common Procedure for Generating the Securing Materials [page 219].

Securing communications with the Core Server system

Securing the Core Server system consists in the following operations:

- Specifying a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS
- Securing the communications channels relating to the Core Server system

1. To specify a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS, you must modify the TLS_PROTOCOLS and TLS_CIPHER_SUITES parameters. Execute one of the following procedures to modify these system parameters:
   - Launch [page 206] the Cockpit and use the Manage SAP CC System Parameters app to edit the adequate parameters
   - - Launch [page 208] the Admin+ user interface
     - Execute the following commands within Admin+ in order to identify as the administrator of the Core Server system and modify the adequate parameters:

       ```
       user <SAPCC_SYSADM_USERNAME> <SAPCC_SYSADM_PASSWORD>
       set TLS_PROTOCOLS <TLS_PROTOCOLS> persistent <TARGET_INSTANCES>
       set TLS_CIPHER_SUITES <TLS_TLS_CIPHER_SUITES> persistent
       <TARGET_INSTANCES>
       ```

       Where:
       - `<SAPCC_SYSADM_USERNAME>` and `<SAPCC_SYSADM_PASSWORD>` correspond to the name and password of the user granted with the Administrator role
       - `<TLS_PROTOCOLS>` corresponds to the list of protocols to enable
       - `<TLS_CIPHER_SUITES>` corresponds to the list of cipher suites to enable
       - `<TARGET_INSTANCES>` corresponds to the concerned instances

> ### Note
>
> For further information about the `TLS_PROTOCOLS` and `TLS_CIPHER_SUITES` parameters, refer to the SAP CC 2020 System Parameter Reference documentation.

2. To secure the communication channels relating to the Core Server system using a "dual" or a "oneway" mode, refer to the Securing Communications with the Core Server System [page 221] dedicated procedure

Securing communications with the BART Server system

Securing the BART Server system consists in the following operations:

- Specifying a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS

- Securing the communications channels relating to the BART Server system

1. To specify a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS, you must modify the configuration file of the BART Server system. Execute the following procedure to modify these system parameters:
   - Open the `bart.config` file
   - Declare the protocols and cipher suites to enable by modifying the following parameters accordingly:

     ```
     tls.protocols=<TLS_PROTOCOLS>
     tls.cipher_suites=<TLS_CIPHER_SUITES>
     ```

     Where:
     - `<TLS_PROTOCOLS>` corresponds to the list of protocols to enable
     - `<TLS_CIPHER_SUITES>` corresponds to the list of cipher suites to enable
2. To secure the communication channels relating to the BART Server system using a "dual" or a "oneway" mode, refer to the Securing Communications with the BART Server System [page 235] dedicated procedure

Securing communications with the Diameter Server system

> ⚠ Caution
>
> From SAP Convergent Charging SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

Securing the Diameter Server system consists in the following operations:

- Specifying a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS
- Securing the communications channels relating to the Diameter Server system

1. To specify a list of protocols and cipher suites that can be enabled for encrypted communications using SSL/TLS, you must modify the configuration file of the Diameter Server system. Execute the following procedure to modify these system parameters:
   - Open the `serverConfig.xml` file
   - Declare the protocols and cipher suites to enable by modifying the following parameters accordingly:

     ```
     <entry key="TLSEnabledProtocols" value="<TLS_PROTOCOLS>"/>
     <entry key="TLSEnabledCipherSuites" value="<TLS_CIPHER_SUITES>"/>
     ```

     Where:
     - `<TLS_PROTOCOLS>` corresponds to the list of protocols to enable
     - `<TLS_CIPHER_SUITES>` corresponds to the list of cipher suites to enable
2. To secure the communication channels relating to the Diameter Server system using a "dual" or a "oneway" mode, refer to the Securing Communications with the Diameter Server System [page 242] dedicated procedure

Securing communications with the Cockpit application and its Tomcat Server system

Securing the Cockpit application and its Apache Tomcat Server consists in:

- Securing the communications between the Web browsers and the Tomcat Server system that runs the Cockpit application
- Securing the communications between the Tomcat Server system and the Core Server system as an external client application

To secure the communication channels using a "dual" or a "oneway" mode, refer to the Securing Communications with Cockpit and Tomcat Server [page 246] dedicated procedure.

7.1.38.1 Common Procedure for Generating the Securing Materials

The securing procedure relies on certificates and private keys, which are used to secure the different communication channels used by the different SAP CC systems. Execute the following procedure to create:

- A private key
- A X.509 v3 digital certificate, named `certificate.der` and DER-encoded
- A keystore stored in a PKCS#12 archive file named `keystore.p12` and containing the created private certificate and private key

1. **Generating a private key**
   To generate a private key named `key.pem`, use the following command:

   ```
   openssl genrsa -out key.pem <KEY_SIZE>
   ```

   Where the `<KEY_SIZE>` parameter represents the size (in bits) of the private key you want to generate. Possible values are: 1024,2048 or 4096
2. **Creating a X.509 v3 digital certificate**

   > ⓘ Note
   >
   > As far as certificates related to client applications are concerned, you can use X.509 v1 certificates, which do not require any configuration file. In this case, skip the `-config` argument of the `openssl` command below.

   The creation of a X.509 v3 digital certificate is based upon a configuration file taken into account by the openssl command. This configuration file contains a property named `subjectAltName`, which gives the possibility to provide an exhaustive list of hostnames and/or IP addresses (IPv4 or IPv6) used to identify the host to trust and connect to. This property uses the following format:
   IP:{$localHostAddress},DNS:{$localHostName},DNS:{$localCanonicalHostName}
   You can use a generic domain name (e.g. DNS:*.mydomain.com) in order to easily add possible new hosts, but SAP SE highly recommends you to specify a list of IP addresses (IPv4 or IPv6) and/or hostnames (e.g. IP:192.168.0.1, IP: fe80::192.168.0.2, DNS:host1).

To create a configuration file named `openssl.config`, use the following text content (adapting the `SubjectAltName` property to your own configuration):

```
[ req ]
distinguished_name = req_distinguished_name
x509_extensions = v3_req
[ req_distinguished_name ]
[ v3_req ]
subjectAltName = DNS:*.mydomain.com
```

> ⚠ Caution
>
> When a condensed IPv6 address is used in the `subjectAltName` property, the `openSSL` tool automatically records the complete address. This complete address corresponds to the address which must be provided when connecting to a given service secured with this certificate.

In addition, SAP SE also recommends you to use a pretty name when filling the common name, in order to distinguish each generated certificate. This common name contains information related to the trusting authority, i.e. the host to trust and connect to.
To generate the `certificate.pem` certificate, type the following command:

```
openssl req -new -x509 -key key.pem -out certificate.pem -days 365 -config
openssl.config -subj "/CN=<PRETTY_NAME>"
```

3. **Creating the keystore used to store the certificate and its private key**
   To create a keystore named **keystore.p12**, secured with a passphrase and containing the previously created private key and certificate, type the following command:

   ```
   openssl pkcs12 -export -in certificate.pem -inkey key.pem -out keystore.p12
   -passout pass:<KEYSTORE_PASSPHRASE>
   ```

4. **Deleting the private key**
   For security reasons, delete the previously created private key using the following command:
   - For Linux and UNIX operating systems:

     ```
     rm key.pem
     ```

   - For Microsoft Windows operating systems:

     ```
     del key.pem
     ```

5. **Modifying the certificate's format from PEM to DER**
   To convert the certificate from a PEM format to a DER one, type the following command:

   ```
   openssl x509 -in certificate.pem -out certificate.der -outform der
   ```

7.1.39 Securing Communications with the Core Server System

Description

The communications relating to the Core Server system mostly concern the Core Server instances, which:

- Communicate together during internal operations
- Can be contacted by client applications such as SAP CC user interfaces or other systems (SAP CC systems and/or third-party systems)
- Are connected with the Core Database

Internal communications between the instances rely on the Packets over TCP/IP procedure. Refer to it when necessary [82] communication channel. The communication channels which are used for communications with client applications depend on the concerned instance. The following table summarizes these interactions:

	(primary) Dispatcher Instance	(active) Updater Instance	guider/Rater Instances	Bulkloader Instances	Core Database
Core Server					
(primary) dispatcher	-	Packets over TCP/IP	Packets over TCP/IP	Packets over TCP/IP	JDBC[83]
(secondary) dispatcher	Packets over TCP/IP	-	-	-	JDBC
(active) updater	Packets over TCP/IP + XML[84] over HTTP[85]	SOAP[86] over HTTP[87]	-	-	JDBC
guider/rater	Packets over TCP/IP	-	-	-	JDBC
bulkloader	Packets over TCP/IP	-	-	-	JDBC
Client applications					
Core Tool	XML over HTTP[88]	XML over HTTP	-	-	-
Setup Tool	-	-	-	-	JDBC
Config Tool	-	-	-	-	JDBC
Admin+	XML over HTTP	-	-	-	-

82 Transmission Control Protocol/Internet Protocol
83 Java Database Connectivity
84 eXtended Markup Language
85 When using the Catalog Transport feature of SAP Convergent Charging
86 Simple Object Access Protocol
87 When using the Catalog Transport feature of SAP Convergent Charging
88 HyperText Transfer Protocol

	(primary) Dispatcher Instance	(active) Updater Instance	guider/Rater Instances	Bulkloader Instances	Core Database
Cockpit	XML over HTTP	XML over HTTP	-	-	-
HTTP Client	XML over HTTP	XML over HTTP	-	-	-
Third-party systems					
Client application	Packets over TCP/IP + SOAP over HTTP + XML over HTTP	XML over HTTP	-	RFC[89] procedure. over TCP/IP	-
Mediation	Packets over TCP/IP + SOAP over HTTP + XML over HTTP	-	-	-	-
SAP CI	SOAP over HTTP + RFC over TCP/IP	-	-	RFC over TCP/IP	-
SAP CRM	-	SOAP over HTTP	-	-	-
SAP SLD	XML over HTTP	-	-	-	-
SAP SMD	XML over HTTP	-	-	-	-
SAP CC CTS+ plugin	XML over HTTP	SOAP over HTTP	-	-	-
Identity Provisioning Service (IPS)	REST[90] over HTTP				

The securing of communications performed on the Packets over TCP/IP, XML over HTTP, SOAP over HTTP and REST over HTTP communications channels relies on dedicated properties located in the static instance map. Internal communications between instances can be secured using the `InternalSecure` property of the static instance map. The securing policy of external communications relies on different properties, according to the concerned communication channel. The following table displays the correspondence between communication channels and related securing properties, with the possible values for each concerned property:

Communication channel	Property	dispatcher			updater		
		off	one-way	dual	off	one-way	dual
Packets over TCP/IP	`InternalSecure`	■	□	■	□	□	□
	`ExternalSecure`	■	■	■	□	□	□
XML over HTTP	`HCISecure`	■	■	■	■	■	■
SOAP over HTTP	`WsSecure`	■	■	■	■	■	■

89 Remote Function Call
90 REpresentational State Transfer

Communication channel	Property	dispatcher			updater		
		off	one-way	dual	off	one-way	dual
REST over HTTP	`WsSecure`	■	■	■	■	■	■

■ *Available,* ☐ *Not Available*

The securing of communications performed on the RFC over TCP/IP communication channel and on the JDBC protocol relies on parameters set in associated parameters files or command-line arguments. For further information about these protocols, refer to the SAP CC 2020 Security Guide documentation.

The securing procedure consists in securing both the Core Server system and its client applications, which corresponds to the "dual" mode. The operations described afterwards give you the possibility to:

- Secure the internal communications between the different instances, which consists in:
  - Generating a certificate related to the Core Server
  - Importing this certificate into the keystore of the Core Server
  - Linking this certificate to all the SAP CC instances, for the “internal” communication channel
- Secure the communications between the Core Server instances and a client application, considered as external communications and consisting in:
  - Generating a certificate related to the Core Server
  - Generating a certificate related to the client application
  - Importing the Core Server certificate into the keystore of the Core Server
  - Linking the Core Server certificate to all the concerned instances, for the adequate communication channels
  - Importing the Core Server certificate into the truststore of the JVM[91] used by the client application, so that this application can trust the Core Server
  - Updating the keystore of the client application, by declaring the client certificate towards the application
  - Importing the client certificate into the truststore of Core Server, so that the Core Server system can trust the client application
  - Linking the client certificate to all the concerned instances, for all the communication channels
- Secure the communications with the Core Database, that rely on the JDBC[92] protocol
- Secure the communications between the Core Server and other systems such as SAP CI or SAP CRM, that rely on the RFC over TCP/IP communication channel
- Enable this defined securing policy, which consists in:
  - Modifying the instance map
  - Restarting the Core Server, which finally runs in a dual authentication mode

Preliminary Notes

- Some of the procedures described afterwards refer to the Common procedure for generating the securing materials [page 217] that is available in the Securing an SAP CC System Landscape [page 216].

[91] Java Virtual Machine
[92] Java Database Connectivity

- If your client application is not a Java application, you must adapt some of the following procedures to fit the technology of your application.

Prerequisites

- The `setup` command specified in the following operations refers to the SAP CC 2020 Setup Tool user interface. Refer to the Installing and Launching Setup Tool [page 211] procedure if necessary.
- The `config` command specified in the following operations refers to the SAP CC 2020 Primary Help for Config Tool user interface. Refer to the Installing and Launching Config Tool [page 214] procedure if necessary.

Securing the Core Server internal communications

Considering a keystore named `keystore.int.p12`, generated using the common procedure and containing a private certificate with its private key, use the following procedure to:

- Import the certificate and its private key into the keystore of Core Server
- Configure all the instances of the Core Server system so that they use this private key when communicating one with another on the “internal” communication channel

1. **Generating a certificate and its private key for the Core Server system**
   Use the common procedure to generate a keystore named `keystore.int.p12` and containing a certificate with its private key.
2. **Importing the certificate and its private key into the keystore of the Core Server system**
   To import the content of the `keystore.int.p12` keystore into the SAP CC keystore under the `<InternalAlias>` alias, type the following command:

   ```
   setup keyentry import_p12 <InternalAlias> keystore.int.p12
   ```
3. **Linking the imported certificate**
   To specify that the Core Server system must use the imported certificate for internal communications between the different instances, type the following command:

   ```
   setup keyentry link <InternalAlias> all internal
   ```

Securing the Core Server external communications

To secure the communications between the Core Server system and a client application, execute the following operations in order to:

- Use the common procedure in order to generate a keystore:
  - Related to the Core Server
  - Named `keystore.ext.p12`

    - Containing a `certificate.ext.der` certificate and its private key
- Use the common procedure in order to generate a keystore:
    - Related to the client application
    - Named `keystore.cli.p12`
    - Containing a `certificate.cli.der` certificate and its private key
- Import the content of the `keystore.ext.p12` keystore into the keystore of the Core Server, using the `<ExternalAlias>` alias
- Link the `<ExternalAlias>` certificate to all the instances, for the "http", "ws" and "external" communication channels
- Import the `certificate.ext.der` certificate of the Core Server into the truststore of the client application
- Enable the `keystore.cli.p12` keystore on the client application side, by declaring this keystore on the client application side
- Import the `certificate.cli.der` certificate into the truststore of the Core Server, using the `<ClientAlias>` alias
- Link the `<ClientAlias>` certificate to all the instances, for all the communication channels

1. **Generating a certificate and its private key for the Core Server**
   Use the common procedure to generate a keystore named `keystore.ext.p12` and containing a certificate with its private key.
2. **Generating a certificate related to a client application**
   Use the common procedure to generate a keystore named `keystore.cli.p12` and containing a certificate with its private key.

   > **Note**
   >
   > It is necessary to remember the passphrase of the keystore file in order to execute the step 6 described below.

3. **Importing the certificate of Core Server (and its private key) into the keystore of the Core Server system**
   To import the content of the `keystore.ext.p12` keystore into the keystore of the Core Server, under the `<ExternalAlias>` alias, type the following command:

   ```
   setup keyentry import_p12 <ExternalAlias> keystore.ext.p12
   ```

   > **Note**
   >
   > You will be prompted to type the passphrase of the keystore file which contains the certificate and private key to import.

4. **Linking the imported certificate**
   To specify that the Core Server must use the imported certificate for external communications relying on the "http", "ws" and "external" communication channels, type the following command:

   ```
   setup keyentry link <ExternalAlias> all "http,ws,external"
   ```

5. **Importing the certificate of Core Server into the truststore of the client application**
   To give the possibility for a client application to trust the Core Server, it is necessary to import the certificate of the Core Server into the truststore used by this client application. To import the

`certificate.ext.der` certificate into a Java keystore named `keystore.cli.jks`, type the following command:

```
keytool -importcert -file certificate.ext.der -keystore keystore.cli.jks
-noprompt
```

> **Note**
>
> You will be prompted to type a passphrase for the Java keystore file which will contain the certificate to import. It is necessary to remember this passphrase in order to execute the following command.

The client application must then be configured to use the above created keystore as a truststore. To make the client application use the `keystore.cli.jks` keystore as its default truststore, add the following options to the command-line of the application:

```
-Djavax.net.ssl.trustStore=<Path_of_the_Java_keystore>
-Djavax.net.ssl.trustStorePassword=<Passphrase_of_the_Java_keystore>
```

If your client application is not a Java application, you must adapt this step to fit the technology of your application.

6. **Enabling the client keystore on the client application side**
   The client application must take into account its certificate in order to use it when communicating in a secured mode. To enable this certificate, it is necessary to declare the related keystore in the command-line of the application. To declare the `keystore.cli.p12` client keystore, add the following options to the command-line of the application:

   ```
   -Djavax.net.ssl.keyStoreType=pkcs12
   -Djavax.net.ssl.keyStore=<Path_of_the_client_keystore>
   -Djavax.net.ssl.keyStorePassword=<Passphrase_of_the_client_keystore>
   ```

7. **Importing the certificate of the client keystore into the truststore of the Core Server**
   To import the certificate of the `keystore.cli.p12` keystore into the SAP CC truststore under the `<ClientAlias>` alias, type the following command:

   ```
   setup certentry import <ClientAlias> certificate.cli.der
   ```

8. **Linking the imported certificate**
   To specify that the Core Server must use the imported certificate when communicating with a client application, whatever the communication channel is, type the following command:

   ```
   setup certentry link <ClientAlias> all all
   ```

Securing the Core Server JDBC connections

To secure the communications between the Core Server and the Core Database, it is necessary to execute the following operations:

- Execute the securing procedures related to your database provider
- Update the configuration of the Core Server to modify the connection URL with the adequate parameters

1. **Executing the securing procedures related to your database provider**

Each database provider delivers specific procedures to encrypt communications with their databases management system. Refer to the documentation of your database provider to create the different materials secure the communications with the Core Database.

2. **Updating the configuration of the Core Server**
   The Core Server supports the following parameters and associated values to secure the communications relying on the JDBC protocol:

Database Provider	Parameter	Value
SAP HANA	`encrypt`	True
	`validateCertificate`	False[93]
SAP ASE	`ENABLE_SSL`	True
Oracle	`N/A`	N/A
Microsoft SQL Server	`encrypt`	True
IBM DB2	`trustServerCertificate`	False

To secure the communications relying on the JDBC protocol, it is necessary to update the following parameters, both in the `boot.config` file (located in the `boot` directory of the SAP Profile Folder) and in the Core Database:

- `SQLHELPER_JDBC_URIx`, where x can be empty when no HA is required, or a value between 1 and 12 for the Oracle and DB2 database providers (depending on the HA configuration)
- `SQLHELPER_JDBC_PROPERTIES`, which corresponds to a CSV string used to specify the additional parameters to use when connecting to the database

To update the `SQLHELPER_JDBC_URIx` parameter into the `boot.config` file, open the configuration file and declare the adequate parameters to secure the communications with the Core Database, e.g.:

Database Provider	Value
SAP HANA	**`jdbc:sap:<server>:<port>`**
SAP ASE	**`jdbc:sybase:Tds:<server>:<port>/<database_name>`**
Oracle	**`jdbc:oracle:thin:@//<server>:<port>/<ServiceName>`** **`jdbc:oracle:thin:@<server>:<port>:<SID>`**
Microsoft SQL Server	**`jdbc:sqlserver://<server>:<port>;DatabaseName=<database_name>`**
IBM DB2	**`jdbc:db2://<server>:<port>/<instance_name>:currentSchema=<schema_name>;`**

[93] When this value is set to **`true`**, it is necessary to specify additional parameters to the `SQLHELPER_JDBC_PROPERTIES` parameter, depending on the securing mode. See below for further information about these parameters.

To update the `SQLHELPER_JDBC_PROPERTIES` parameter into the `boot.config` file, open it in a text editor and declare the adequate parameters and values corresponding to the database provider, e.g.:

Database Provider	Value
SAP HANA	**`encrypt=true,validateCertificate=false`** [94]
SAP ASE	**`ENABLE_SSL=true`**
IBM DB2	**`sslConnection=true`**

To update these parameters into the Core Database, first export the existing system configuration into an XML file, using the following command:

```
config configuration export parameters.xml all
```

Then, modify the values of the above modified parameters located in the XML file according to the database provider's format, and finally update the system configuration using the using the following command:

```
config configuration import parameters.xml
```

Securing the RFC over TCP/IP communications

To secure the communications with other systems such as SAP CI, SAP ERP/FI-CA or SAP CRM, which all rely on the RFC over TCP/IP communication channel, it is necessary to execute the following operations:

- Download and install the SAP Cryptographic Library
- Setup a Personal Security Environment (PSE) for SNC, that contains information related to the SAP CC public key which includes its private key, its certificate and the list of trusted certificates
- Create the credentials that are required at runtime by SAP CC for accessing its PSE
- Exchange public key certificates between SAP CC and the Application Server
- Update the Core Server system configuration related to JCo[95] settings
- Maintain System ACL[96] and Extended User ACL on the SAP Netweaver Application Server

1. **Downloading and installing the SAP Cryptographic Library**
   To download and install the SAP Cryptographic Library, refer to the Downloading the SAP Cryptographic Library [page 96] dedicated procedure.
2. **Setting up a Personal Security Environment (PSE) for SNC**
   To create the SNC PSE for SAP Convergent Charging, execute the following command:

   ```
   sapgenpse4cc get_pse -p <PSE_NAME> -x <PIN> <DISTINGUISHED_NAME>
   ```

[94] When the value of the `validateCertificate` parameter is set to **`true`**, it is necessary to specify the following additional parameters:
- `trustStore`, `trustStoreType` and `trustStorePassword` when using a "oneway" authentication mode
- `trustStore`, `trustStoreType`, `trustStorePassword`, `keyStore`, `keyStoreType` and `keyStorePassword` when using a "dual" authentication mode

[95] Java Connector

[96] Access Control List

Where:

- <PSE_NAME> corresponds to the name of the PSE
- <PIN> corresponds to a PIN code
- <DISTINGUISHED_NAME> corresponds to the absolute name of SAP Convergent Charging, made up with the following elements:
  - CN=<COMMON_NAME>
  - OU=<ORGANIZATIONAL_UNIT >
  - O=<ORGANIZATION>
  - C=<COUNTRY>

> ### Example
>
> The following command line creates a SAP_CC.pse file that is protected with the **ccpin** PIN code:
>
> ```
> sapgenpse4cc get_pse -p SAP_CC.pse -x ccpin “CN=SAPCC, O=MyCompany, C=US”
> ```
>
> When using this PSE, SAP Convergent Charging has the following distinguished name: CN=**SAPCC**, O=**MyCompany**, C=**US**

3. **Creating credentials**
   The credentials required by SAP CC are located in the /usr/sap/<SID>/sys/profile/sec/cred_v2 file, that must only be accessible by the OS user running the Core Server system. To create these credentials, execute the following command:

   ```
   sapgenpse4cc seclogin -p <PSE_NAME> -x <PIN_CODE> -O <USER_ID>
   ```

   Where:

   - <PSE_NAME> corresponds to the name of the PSE created in step 2
   - <PIN_CODE> corresponds to the PIN code provided at PSE creation time in step 2
   - <USER_ID> corresponds to name of the concerned user, including the domain name in case of Microsoft Windows operating system

   > ### Example
   >
   > The following command line creates credentials for the <SID>adm user so that this user can access the previously created SAP_CC.pse file:
   >
   > **Linux Operating Systems**:
   >
   > ```
   > sapgenpse4cc seclogin -p SAP_CC.pse -x ccpin -O <SID>adm
   > ```
   >
   > **Microsoft Windows Operating Systems**:
   >
   > ```
   > sapgenpse4cc seclogin -p SAP_CC.pse -x ccpin -O SAPService<SID>
   > ```
   >
   > Where <SID> corresponds to the system identifier of the concerned Core Server system.

4. **Exchanging certificates**
   To communicate using SNC, SAP CC and the SAP Netweaver Application Server must identify and trust each other using their respective certificates available in the relevant PSEs (which means the PSE of Application Server is available). To exchange certificates between SAP CC and the Application Server, it is necessary to:

- Export the certificate of SAP CC, using the following command:

  ```
  sapgenpse4cc export_own_cert -o <OUTPUT_FILE> -p <PSE_NAME> -x <PIN_CODE>
  ```

  Where:

  - `<OUTPUT_FILE>` corresponds to the absolute path of the file containing the exported certificate
  - `<PSE_NAME>` corresponds to the name of the PSE related to SAP CC
  - `<PIN_CODE>` corresponds to the PIN code provided during the creation of the PSE related to SAP CC
- Import the certificate of SAP CC into the SNC PSE of the Application Server. If the Application Server is an SAP Web Application Server with release 6.20 or later, you can use the Trust Manager ("STRUST" transaction) to import the certificate. Otherwise, use the following command:

  ```
  sapgenpse4cc maintain_pk -a <CERT_FILE> -p <PSE_NAME> -x <PIN_CODE>
  ```

  Where:

  - `<CERT_FILE>` corresponds to the absolute path of the file containing the previously exported certificate
  - `<PSE_NAME>` corresponds to the name of the PSE related to SAP CC
  - `<PIN_CODE>` corresponds to the PIN code provided during the creation of the PSE related to SAP CC
- Export the certificate of the Application Server. If the Application Server is an SAP Web Application Server with release 6.20 or later, you can use the Trust Manager ("STRUST" transaction) to export the certificate. Otherwise, use the `export_own_cert` command of the `sapgenpse` tool installed on the Application Server
- Import the certificate of the Application Server in the SNC PSE of SAP CC.

5. **Updating JCo settings of the Core Server system**

   To modify the configuration of the Core Server related to JCo settings, it is necessary to:

   - Export the SAP CC configuration of the relevant SAP system.
     - If the concerned system is the SAP ERP/FI-CA

       ```
       setup sapci exportConfiguration
       ```

     - If the concerned system is the SAP CRM system, execute the following command: system, execute the following command:

       ```
       setup sapci jcoDestination export crm <OUTPUT_FILE>
       ```

       Where `<OUTPUT_FILE>` corresponds to the absolute path of the file containing the exported settings
   - Edit the exported file and modify the values of the following parameters:
     - `jco.client.snc_mode`
     - `jco.client.snc_lib`
     - `jco.client.snc_qop`
     - `jco.client.snc_myname`
     - `jco.client.snc_partnername`
     - `jco.client.snc_sso`

     For further information about these parameters, refer to the Configuration Settings for JCo section of the SAP CC 2020 Tuning Guide documentation.

- Import the modified configuration.
  - If the concerned system is the SAP ERP/FI-CA system, execute the following command:

    ```
    setup sapci importConfiguration <INPUT_FILE>
    ```

    Where:
    - `<LOGIN>` corresponds to an SAP CC **individual** user granted the adequate role
    - `<PASSWORD>` corresponds to the password of the specified user
    - `<INPUT_FILE>` corresponds to the absolute path of the file containing the modified settings
  - If the concerned system is the SAP CRM system, execute the following command:

    ```
    setup sapci jcoDestination import crm <INPUT_FILE>
    ```

    Where `<INPUT_FILE>` corresponds to the absolute path of the file containing the modified settings
- Restart all dispatcher and bulkloader instances

6. **Maintaining system ACL and extended user ACL on the SAP Netweaver Application Server**
   In addition to being able to identify its communication partner using the SNC layer, SAP NetWeaver uses system Access Control Lists (ACLs) to make sure that it is communicating with the correct components. To maintain the SNC system ACL (table SNCSYSACL, view VSNCSYSACL, type=E), execute the following procedure:
   - Use the "SM30" transaction used for table maintenance operations
     - Enter an asterisk (*) as a wildcard in the *User* field
     - Use the sequence number if you have multiple entries with the same user
     - Enter SAP Convergent Charging's SNC name in the *SNC name* field. Make sure you enter the server's SNC name and not the Distinguished Name. The server's SNC name is the Distinguished Name prefixed with "p:"
   - Save your modifications
   - Use the "SM30" transaction used for table maintenance operations
   - Confirm the warning
   - Save your modifications

Enabling the securing policy

To activate the securing policy which has been configured during the previous operations, it is necessary to execute the following operations:

- Update the instance map according to the defined securing policy
- Restart the Core Server instances, in order to get a secured system

1. **Modifying the instance map**
   To modify the instance map, you need to:
   - Export the instance map into a CSV file named `instanceMap.csv`, using the following command:

     ```
     setup instancemap export instanceMap.csv
     ```

- Open this file with a text editor, and modify the values of the `InternalSecure`, `ExternalSecure`, `HCISecure` and `WsSecure` properties with the "dual" value, according to the possibilities described above
- (Re)import the modified instance map in the system, using the following command:

```
setup instancemap import -clean instanceMap.csv
```

- The `BOOT_DISPATCHER_LIST` parameter located in the boot.config file is not compatible with secured landscapes. Open the `boot.config` file and remove this parameter when exists

2. **Restarting the Core Server instances**
   To restart the Core Server instances, refer to the Starting and Stopping the Server Systems [page 144] procedure.

Securing the communications with the SAP SLD system

To communicate with a secured version of the SAP SLD system, it is necessary to execute the following operations:

- Update the configuration of the Core Server regarding the connection to the SLD system
- Import the certificate of the SAP SLD system into the truststore of the JVM used by the Core Server, so that the Core Server can trust the SAP SLD system
- Restart the Core Server instances, in order to apply the new configuration

1. **Updating the configuration of the Core Server**
   To configure the access to an SAP SLD system, refer to the "System settings for the System Landscape Directory" procedure described in the SAP Convergent Charging SAP CC 2020 Tuning Guide documentation.
2. **Importing the SAP SLD certificate into the truststore of the JVM used by the Core Server**
   To give the possibility to the Core Server to trust the SAP SLD system, it is necessary to import the certificate of the SAP SLD system into the truststore of the JVM used by the Core Server.
   Once you have retrieved the SLD certificate (X.509 v3 digital certificate, DER-encoded), you need to import it into the truststore of the JVM used by the Core Server, using the following command:

```
keytool -importcert -file certificate.sld.der -keystore keystore.core.jks
-noprompt
```

> **Note**
>
> You will be prompted to type a passphrase for the Java keystore file which will contain the certificate to import. It is necessary to remember this passphrase in order to execute the following command.

   The Core Server must then be configured to use the above updated keystore as a truststore. To do this, open the `jstart.config` file of each dispatcher instance, located in the `jstart` directory of the SAP Profile Folder.
   Then, update the `dispatcher-x.javaParameters` configuration parameter with the following information:

```
-Djavax.net.ssl.trustStore=<Path_of_the_Core_Server_Java_keystore>
-Djavax.net.ssl.trustStorePassword=<Passphrase_of_the_keystore>
```

3. **Restarting the Core Server instances**
   To restart the Core Server instances, refer to the Starting and Stopping the Server Systems [page 144] procedure.

Securing communications with the SAP CC CTS+ plugin

To communicate with a Core Server secured with dual secure mode, it is necessary to execute the following technical operations:

- Update the configuration of the SAP CC CTS+ plugin to trust the Core Server
- Use the common procedure in order to generate a keystore:
  - Related to the SAP CC CTS+ plugin
  - Named `keystore.ctsplugin.p12`
  - Containing a `certificate.ctsplugin.der` certificate and its private key
- Update the configuration of the SAP CC CTS+ plugin to enable its keystore
- Import the certificate of the SAP CC CTS+ plugin into the truststore of the Core Server, so that the Core Server can trust the SAP CC CTS+ plugin
- Update SAP CTS system to enable secure connection

1. **Configuring the SAP CC CTS+ plugin to trust the Core Server**
   To give the possibility for the SAP CC CTS+ plugin to trust an SAP CC Core Server, it is necessary to import the certificate of the SAP CC Core Server into the Java truststore used by the SAP CC CTS+ plugin.
   To import the certificate into the Java truststore, you need to use the keytool utility delivered within the SAP JVM:

   ```
   <SAPJVM_PATH>/bin/keytool -import -alias <alias> -file <server_certificate>
   -keystore <truststore_path> -storepass <truststore_passphrase>
   ```

   Where:
   - `<server_certificate>` is the certificate of the SAP CC Core Server
   - `<truststore_path>` is the path of the Java truststore that will be used by the SAP CC CTS+ plugin
   - `<trustore_passphrase>` is the passphrase of this Java truststore

   > **Note**
   >
   > You will be prompted to type a passphrase for the Java truststore file which will contain the certificate to import. It is necessary to remember this passphrase as it will be required to configure the SAP CC CTS+ plugin.

   To make the SAP CC CTS+ plugin use the Java truststore you just created, uncomment the following lines in the SAP CC CTS+ plugin script file corresponding to your system and provide the missing information between angle brackets "<...>":

   ```
   -Djavax.net.ssl.trustStore=<Path_of_the_SAP_CC_CTS_truststore>
   -Djavax.net.ssl.trustStorePassword=<Passphrase_of_SAP_CC_CTS_truststore>
   ```

   Where:
   - `<Path_of_the_SAP_CC_CTS_truststore>` is the path of the previously created Java truststore

- `<Passphrase_of_SAP_CC_CTS_truststore>` is the passphrase of this Java truststore

2. **Generating a certificate related to a client application**
   Use the common procedure [page 217] to generate a keystore named `keystore.ctsplugin.p12` and containing a certificate with its private key.
3. **Enabling the keystore of the SAP CC CTS+ plugin**
   The SAP CC CTS+ plugin must take into account its own keystore in order to use it when communicating with the Core Server in a dual secured mode. To enable this keystore, it is necessary to declare it in the SAP CC CTS+ plugin configuration. To declare the `keystore.ctsplugin.p12` client keystore, uncomment the following lines in the SAP CC CTS+ plugin script file corresponding to your system and provide the missing information between angle brackets "<...>":

   ```
   -Djavax.net.ssl.keyStoreType=pkcs12
   -Djavax.net.ssl.keyStore=<Path_of_the_SAPCC_CTS_keystore>
   -Djavax.net.ssl.keyStorePassword=<Passphrase_of_SAPCC_CTS_keystore>
   ```

   Where:
   - `<Path_of_the_SAP_CC_CTS_keystore>` is the path to the `keystore.ctsplugin.p12` keystore
   - `<Passphrase_of_SAP_CC_CTS_keystore>` is the passphrase of this keystore
4. **Importing the content of the SAP CC CTS+ plugin keystore into the truststore of the Core Server**
   To import the content of the `keystore.ctsplugin.p12` keystore into the SAP CC truststore under the `<CTSAlias>` alias, type the following command:

   ```
   setup certentry import <CTSAlias> certificate.ctsplugin.der
   ```

   > **Note**
   >
   > The certificate must be imported in each SAP CC system that the SAP CC CTS+ plugin has to connect with.

5. **Linking the imported certificate**
   To specify that the Core Server must use the imported certificate when communicating with the SAP CC CTS+ plugin through the "transport" APIs[97], type the following command:

   ```
   setup certentry link <CTSAlias> all "http,ws"
   ```

6. **Updating SAP CTS system to enable secure connection**
   Refer to the Configuring the HTTP Connection section of the SAP CTS system documentation to get information about the configuration of transport landscapes.

Securing the communications with the OAuth 2.0 Authorization Server

As the Cockpit user interface and the SAP CC Core Server system must validate the access and ID tokens, they need a public key provided by the authorization server (AS) system.

This public key, hold within a certificate, must be imported into the Core Database of SAP Convergent Charging.

The technical operations consist in:

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- Saving the certificate of the authorization server, containing the public key, to a `*.CER`, or a `*.PEM` file.
- Importing the certificate into the Core Database.

As an SAP CC system administrator, you complete the following tasks:

1. **Register the certificate of the Authorization Server (AS) system.**
   Retrieve the certificate containing the public key from the authorization server (AS) system (most of time by requesting a URL given by the authorization provider) and store it in a PEM-format (X.509 certificate with base64 encryption) into a `*.PEM` or `*.CER` file (not `*.DER` which is reserved for a binary storage).

   > ⓘ Note
   >
   > Consult the official documentation of the authorization server or refer to its system administrator.

2. **Import the AS certificate into the Core Database.**
   The import of the AS certificate requires a dedicated and reserved alias: `oauth2`
   To import the certificate into the SAP CC Core Database, use the Setup Tool CLI in an operating system (OS) console. See Executing Setup Tool.
   Execute the following `certentry import` command (considering a `file.PEM` file at the first step):

   ```
   setup.bat certentry import oauth2 file.PEM
   ```

   ```
   ./setup.h certentry import oauth2 file.PEM
   ```

   > → Remember
   >
   > `oauth2` is the dedicated and reserved alias name that must be used to import the AS certificate.

> → Tip
>
> Refer to the Common Procedure for Generating the Securing Materials [page 219] section of the current documentation if you need to generate your own security certificates instead of using the native AS certificates.

7.1.40 Securing Communications with the BART Server System

Description

The communications with the BART Server system rely on:

- The Packets over TCP/IP communication channel to communicate with:
  - An offline mediation system
  - A dispatcher instance of the Core Server

- The XML over HTTP communication channel to communicate with:
  - Client applications such as BART Tool
  - An updater instance of the Core Server
  - The SLD system, at launch time
- The JDBC[98] protocol for interactions between the BART Setup Tool and the BART Database

The securing procedure consists in securing both the BART Server and its client applications. This procedure requires the use of the BART Setup Tool console application, which needs to communicate with the previously secured Core Server system for authentication purposes, and which must thus be first secured.

Once the BART Setup Tool has been secured, the following operations give you the possibility to:

- Secure the external communications of the BART Server, which concern:
  - The communications with the Core Server
  - The communications with a client application
- Secure the communications with the BART Database, relying on the JDBC protocol
- Enable this defined securing policy, which consists in:
  - Modifying the configuration of the BART Server
  - Restarting the BART Server, which finally runs in a secured mode

Preliminary Notes

- Some of the procedures described afterwards refer to the Common procedure for generating the securing materials [page 217] that is available in the Securing an SAP CC System Landscape [page 216] procedure. Refer to it when necessary
- If your client application is not a java application, you must adapt some of the following procedures to fit the technology of your application

Prerequisites

The `setup` command specified in the following operations refers to the BART Setup Tool user interface. Refer to the Installing and Launching BART Setup Tool [page 212] procedure if necessary

Securing the BART Setup Tool console application

To secure the BART Setup Tool user interface, refer to the 5 to 8 steps of the Securing the Core Server external communications [page 224] section.

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Securing the BART Server external communications

External communications related to the BART Server system both concern:

- The Core Server system, with whom the BART Server system communicates
- Client applications, which can interact with BART Server to perform operations such as chargeable items batch charging or recharging

As far as communications with the Core Server system are concerned, execute the following operations:

- Use the `certificate.ext.der` certificate generated into the Securing the Core Server external communications [page 224] procedure, in order to give BART Server the possibility to trust Core Server:
  - Import this certificate into the truststore of the BART Server system, using the "CoreClientAlias" alias
  - Link this "CoreClientAlias" certificate to the Core Server system, for all the communication channels
- Use the `keystore.cli.p12` keystore generated into the Securing the Core Server external communications [page 224] procedure, so that the BART Server system can be trusted by the Core Server system (as BART Server can also be considered as a Java-based client application):
  - Import this keystore into the keystore of the BART Server system, using the "CoreClientAlias" alias
  - Link this "CoreClientAlias" certificate to the Core Server system, for all the communication channels
- Update the URLs of the dispatcher instances declared in the configuration of the BART Server system

1. **Trusting the Core Server**
   To trust the Core Server, use the certificate generated into the Securing the Core Server external communications [page 221] procedure and located in the `keystore.ext.p12` keystore.
   Import the `certificate.ext.der` certificate into the truststore of the BART Server under the "CoreClientAlias" alias, using the following command:

   ```
   setup certentry import <CoreClientAlias> certificate.ext.der
   ```

   Specify that the BART Server must use the imported certificate for external communications, whatever the communication channel is, using the following command:

   ```
   setup certentry link <CoreClientAlias> dispatcher all
   ```

2. **Communicate with the Core Server as a trusted client application**
   To be considered by the Core Server as a trusted application, use the `keystore.cli.p12` keystore generated into the Securing the Core Server external communications [page 224] procedure.
   Import the content of this keystore into the keystore of the BART Server, under the "CoreClientAlias" alias, using the following command:

   ```
   setup keyentry import_p12 <CoreClientAlias> keystore.cli.p12
   ```

   > ⓘ Note
   >
   > You will be prompted to type the passphrase of the keystore file which contains the certificate and private key to import.

   Specify that the BART Server must use the imported certificate when communicating with the Core Server, whatever the used communication channel is, using the following command:

   ```
   setup keyentry link <CoreClientAlias> dispatcher all
   ```

3. **Updating the configuration of the BART Server**

Update the `CC.DISPATCHER.BOOT.HTTP.URL.LIST` parameter located in the bart.config file, in order to specify the list of dispatchers to contact in a secured mode, e.g.: `https://<host>:<port>`

To secure the communications between the BART Server and a client application, it is necessary to execute the following operations:

- Use the common procedure in order to generate a keystore:
  - Related to the BART Server
  - Named `keystore.bart.p12`
  - Containing a `certificate.bart.der` certificate and its private key
- Use the common procedure in order to generate a keystore:
  - Related to the client application
  - Named `keystore.bart.cli.p12`
  - Containing a `certificate.bart.cli.der` certificate and its private key
- Import the content of the `keystore.bart.p12` keystore into the keystore of the BART Server, using the "ExternalAlias" alias
- Link the "ExternalAlias" certificate to the BART Server, for all the communication channels
- Import the `certificate.bart.der` certificate of the BART Server into the truststore of the client application
- Enable the `keystore.bart.cli.p12` keystore on the client application side, by declaring this keystore in the Java command-line
- Import the `certificate.bart.cli.der` certificate into the truststore of the BART Server, using the "ClientAlias" alias
- Link the "ClientAlias" certificate to the BART Server, for all the communication channels

> **Note**
>
> If your client application is not a java application, you must adapt some of the following operations to fit the technology of your application.

1. **Generating a certificate and its private key for the BART Server**
   Use the common procedure to generate a keystore named `keystore.bart.p12` and containing a certificate with its private key.
2. **Generating a certificate and its private key for a client application**
   Use the common procedure to generate a keystore named `keystore.bart.cli.p12` and containing a `certificate.bart.cli.der` certificate with its private key.

   > **Note**
   >
   > It is necessary to remember the passphrase of the keystore file in order to execute the step 5 described below.
3. **Importing the BART Server certificate into the keystore of the BART Server**
   To import the content of the `keystore.bart.p12` keystore into the keystore of the BART Server, under the "ExternalAlias" alias, type the following command:

   ```
   setup keyentry import_p12 <ExternalAlias> keystore.bart.p12
   ```

> **Note**
>
> You will be prompted to type the passphrase of the keystore file which contains the certificate and private key to import.

4. **Linking the imported certificate**
   To specify that the BART Server must use the imported certificate for external communications, whatever the communication channel is, type the following command:

```
setup keyentry link <ExternalAlias> bart all
```

5. **Importing the BART Server certificate into the truststore of the client application**
   To give the possibility for a client application to trust the BART Server, it is necessary to import the certificate of the BART Server into the truststore used by this client application.
   To import the `certificate.bart.der` certificate into a java keystore named `keystore.cli.jks`, type the following command:

```
keytool -importcert -file certificate.bart.der -keystore keystore.cli.jks
-noprompt
```

> **Note**
>
> You will be prompted to type a passphrase for the java keystore file which will contain the certificate to import. It is necessary to remember this passphrase in order to execute the following command.

   The client application must then be configured to use the above created keystore as a truststore. To make the client application use the `keystore.cli.jks` keystore as its default truststore, add the following options to the command-line of the application:

```
-Djavax.net.ssl.trustStore=<Path_of_the_java_keystore>
-Djavax.net.ssl.trustStorePassword=<Passphrase_of_the_java_keystore>
```

6. **Enabling the `keystore.bart.cli.p12` keystore on the client application side**
   The client application must take into account its own certificate in order to use it when communicating in a secured mode with the BART Server. To enable this certificate, it is necessary to declare the related keystore in the command-line of the application. To declare the `keystore.bart.cli.p12` client keystore, add the following options to the command-line of the client application:

```
-Djavax.net.ssl.keyStoreType=pkcs12
-Djavax.net.ssl.keyStore=<Path_of_the_client_keystore>
-Djavax.net.ssl.keyStorePassword=<Passphrase_of_the_client_keystore>
```

7. **Importing the client application's certificate into the truststore of the BART Server**
   To import the `certificate.bart.cli.der` client certificate into the truststore of the BART Server under the "ClientAlias" alias, type the following command:

```
setup certentry import <ClientAlias> certificate.bart.cli.der
```

8. **Linking the imported certificate**
   To specify that the BART Server must use the imported certificate when communicating with a client application, whatever the communication channel is, type the following command:

```
setup certentry link <ClientAlias> bart all
```

Securing the BART Server JDBC connections

To secure the communications with the BART Database, it is necessary to execute the following operations:

- Execute the securing procedures related to your database provider
- Update the configuration of the BART Server to modify the connection URL with the adequate parameters

1. **Executing the securing procedures related to your database provider**
   Each database provider delivers specific procedures to encrypt communications with their databases management system. Refer to the documentation of your database provider to create the different materials secure the communications with the BART Database.
2. **Updating the configuration of the BART Server**
   The BART Server supports the following parameters and associated values to secure the communications relying on the JDBC protocol:

Database Provider	Parameter	Value
SAP HANA	`encrypt`	True
	`validateCertificate`	False [99]
SAP ASE	`ENABLE_SSL`	True
Oracle	`N/A`	N/A
Microsoft SQL Server	`encrypt`	True
	`trustServerCertificate`	False
IBM DB2	`sslConnection`	True

To secure the communications relying on the JDBC protocol, it is necessary to update the following parameters in the `bart.config` file (located in the `/usr/sap/<SYSTEM_ID>/<INSTANCE_NAME>/config` directory):

- `db.SQLHELPER_JDBC_URIx`, where x can be empty when no HA is required, or a value between 1 and 12 for the Oracle and DB2 database providers (depending on the HA configuration)
- `db.SQLHELPER_JDBC_PROPERTIES`, which corresponds to a CSV string used to specify the additional parameters to use when connecting to the database

To update the `db.SQLHELPER_JDBC_URIx` parameter into the `bart.config` file, open the configuration file and declare the adequate parameters to secure the communications with the BART Server, e.g.:

Database Provider	Value
SAP HANA	`jdbc:sap:<server>:<port>`
SAP ASE	`jdbc:sybase:Tds:<server>:<port>/<database_name>`

[99] When this value is set to `true`, it is necessary to specify additional parameters to the `SQLHELPER_JDBC_PROPERTIES` parameter, depending on the securing mode. See below for further information about these parameters.

Database Provider	Value
Oracle	**jdbc:oracle:thin:@//<server>:<port>/<ServiceName>** **jdbc:oracle:thin:@<server>:<port>:<SID>**
Microsoft SQL Server	**jdbc:sqlserver://<server>:<port>;DatabaseName=<database_name>**
IBM DB2	**jdbc:db2://<server>:<port>/<instance_name>:currentSchema=<schema_name>;**

To update the `SQLHELPER_JDBC_PROPERTIES` parameter into the `bart.config` file, open it in a text editor and declare the adequate parameters and values corresponding to the database provider, e.g.:

Database Provider	Value
SAP HANA	**encrypt=true,validateCertificate=false** [100]
SAP ASE	**ENABLE_SSL=true**
IBM DB2	**sslConnection=true**

Enabling the securing policy

To activate the securing policy which has been configured during the previous operations, it is necessary to execute the following operations:

- Update the configuration of the BART Server according to the defined securing policy
- Restart the BART Server, in order to get a secured system

1. **Updating the configuration of the BART Server**
   Update the `BART.HTTP.SECURE` and `BART.COLLECTOR.SECURE` parameters located in the bart.config file, in order to declare the secured version, e.g.:

   ```
   bart.http.secure=true
   bart.http.secure.client_auth=true
   bart.collector.secure=true
   bart.collector.secure.client_auth=true
   ```

2. **Restarting the BART Server**
   To restart the BART Server, refer to the Starting and Stopping the Server Systems [page 144] procedure.

[100] When the value of the `validateCertificate` parameter is set to **`true`**, it is necessary to specify the following additional parameters:
- `trustStore`, `trustStoreType` and `trustStorePassword` when using a "oneway" authentication mode
- `trustStore`, `trustStoreType`, `trustStorePassword`, `keyStore`, `keyStoreType` and `keyStorePassword` when using a "dual" authentication mode

Securing the communications with the SAP SLD system

To communicate with a secured version of the SAP SLD system, it is necessary to execute the following operations:

- Update the configuration of the BART Server regarding the connection to the SLD system
- Import the certificate of the SAP SLD system into the truststore of the JVM used by the BART Server
- Restart the BART Server, in order to apply the new configuration

1. **Updating the configuration of the BART Server**
   Update the following parameters located in the `bart.config file`, in order to configure the access to a secured SLD system:

   ```
   bart.sld.url=<URL_of_the_secured_SLD_system>
   bart.sld.user=<Username_used_to_connect_to_the_SLD_system>
   bart.sld.password=<Password_used_to_connect_to_the_SLD_system>
   ```

2. **Importing the SAP SLD certificate into the truststore of the JVM used by the BART Server**
   To give the possibility to the BART Server to trust the SAP SLD system, it is necessary to import the certificate of the SAP SLD system into the truststore of the JVM used by the BART Server.
   Once you retrieved the SLD certificate (X.509 v3 digital certificate, DER encoded) you need to import it into the truststore of the JVM used by the BART Server, using the following command:

   ```
   keytool -importcert -file certificate.sld.der -keystore keystore.bart.jks
   -noprompt
   ```

   > **Note**
   >
   > You will be prompted to type a passphrase for the java keystore file which will contain the certificate to import. It is necessary to remember this passphrase in order to execute the following command.

   The BART Server must then be configured to use the above updated keystore as a truststore. To do this, open the `jstart.config` file from the server's profile directory, located in the following folder:
   `<DRIVE>:\usr\sap\<SID>\SYS\profile\jstart\<SID>_<INSTANCE_NAME>_<HOST>`
   Then, update the `BART.javaParameters` configuration parameter with the following information:

   ```
   -Djavax.net.ssl.trustStore=<Path_of_the_BART_Server_java_keystore>
   -Djavax.net.ssl.trustStorePassword=<Passphrase_of_the_keystore>
   ```

3. **Restarting the BART Server**
   To restart the BART Server, refer to the Starting and Stopping the Server Systems [page 144] procedure.

7.1.41 Securing Communications with the Diameter Server System

> **Caution**
>
> From SAP Convergent Charging SP 4, there is no further planned enhancements to the Diameter Server component. For further information, refer to SAP Note 2792189.

Description

The communications with the Diameter Server system rely on:

- The Packets over TCP/IP communication channel to communicate with a dispatcher instance of the Core Server
- The XML over HTTP communication channel to communicate with the SAP SLD system, at launch time
- The Diameter protocol for interactions with a network element

The securing procedure consists in:

- Configuring the Diameter Server as a client application of the Core Server
- Activating (when not already activated during the installation made with SAP installer) the secured version of the Diameter Stack in order to secure the communications with network elements
- Enabling this securing policy, which consists in:
  - Modifying the configuration file of the Diameter Server
  - Restarting the Diameter Server, which finally runs in a secured mode

Preliminary Notes

- Some of the procedures described afterwards refer to the Common procedure for generating the securing materials [page 217] that is available in the Securing an SAP CC System Landscape [page 216] procedure. Refer to it when necessary
- If your client application is not a java application, you must adapt some of the following procedures to fit the technology of your application

Securing the communications between the Diameter Server and the Core Server

To configure the Diameter Server as a java client application, refer to the 5 to 8 steps of the Securing the Core Server external communications [page 221] section.

Securing the communications with the network elements

To communicate with a network element using the Diameter protocol, the Diameter Server uses the OpenBlox Java Diameter Stack provided by the Traffix Systems company. To use the secured version of this Diameter Stack, the configuration of the Diameter Server must contain the following information:

- A valid license key
- An adequate routing table used to handle the proxiable requests
- The TLS information used to secure the communication channel used by the network elements

1. **Valid license key**
   Check (or specify) the `LicenseKey` parameter located in the `serverConfig.xml` file, to ensure that it contains a valid license key which must be used by the Diameter Server:

```
<entry key="LicenseKeys">
  <properties>
    <entry key="LicenseKey" value="<Valid_License_key>"/>
  </properties>
</entry>
```

> **Note**
>
> To get a valid license key, contact the Traffix Systems company, or refer to your technology consultants.

2. **Routing table**
   Check (or specify) that the `RoutingTable` parameter located in the `serverConfig.xml` file refers to a realm-based routing table that the Diameter Server must use:

```
<entry key="RoutingTable">
  <properties>
    <entry key="RealmName" value="<Realm_name>"/>
    <entry key="RealmApplicationId" value="4"/>
    <entry key="RealmLocalAction" value="LOCAL"/>
  </properties>
</entry>
```

3. **TLS information**
   To communicate in a secured mode with a network element using the Diameter protocol, it is necessary to:
   - Generate a keystore:
     - Related to the Diameter Server
     - Named `keystore.diameter.p12`
     - Containing a `certificate.diameter.der` certificate and its private key
   - Update the configuration of the Diameter Server, in order to configure the necessary TLS parameters

   1. To generate a keystore related to the Diameter Server, use the common procedure.

      > **Note**
      >
      > You will be prompted to type a passphrase for the java keystore file which will contain the certificate and its private key. It is necessary to remember this passphrase in order to update the configuration of the Diameter Server, as described below.

   2. Once this keystore is created, update the `serverConfig.xml` file, in order to declare the TLS[101] parameters which must be used by the Diameter Server:

```
<entry key="InbandSecurityIds">
  <properties>
    <entry key="SecurityId" value="TLS"/>
  </properties>
</entry>
<entry key="TLSKeyStoreFile" value="<Path_of_the_server_keystore>"/>
<entry key="TLSTrustStoreFile" value="<Path_of_the_server_keystore>"/>
<entry key="TLSKeyStorePassword"
value="<Passphrase_of_the_server_keystore>"/>
<entry key="TLSTrustStorePassword"
value="<Passphrase_of_the_server_keystore>"/>
```

[101] Transport Layer Security

Enabling the securing policy

To activate the securing policy which has been configured during the previous operations, it is necessary to execute the following operations:

- Update the configuration of the Diameter Server according to the defined securing policy
- Restart the Diameter Server, in order to get a fully secured system

1. **Updating the configuration of the Diameter Server**
   Update the OCSSecured parameter located in the `serverConfig.xml` file, in order to declare the secured version:

   ```
   <entry key="OCSSecured" value="true"/>
   ```

2. **Restarting the Diameter Server**
   To restart the Diameter Server, refer to the Starting and Stopping the Server Systems [page 144] procedure.

Securing the communications between the Diameter Server and the SAP SLD system

To communicate with a secured version of the SAP SLD system, it is necessary to execute the following operations:

- Update the configuration of the Diameter Server regarding the connection to the SLD system
- Import the certificate of the SAP SLD system into the truststore of the JVM used by the Diameter Server, so that the Diameter Server can trust the SAP SLD system
- Restart the Diameter Server, in order to apply the new configuration

1. **Updating the configuration of the Diameter Server**
   Update the following parameters located in the `serverConfig.xml`, in order to configure the access to a secured SLD system:

   ```
   <entry key="SLDURL" value="<URL_of_the_secured_SLD_system>"/>
   <entry key="SLDUser" value="=<Username_used_to_connect_to_the_SLD_system>"/>
   <entry key="SLDPassword"
   value="<Password_used_to_connect_to_the_SLD_system>"/>
   ```

2. **Importing the SAP SLD certificate into the truststore of the JVM used by the Diameter Server**
   To give the possibility to the Diameter Server to trust the SAP SLD system, it is necessary to import the certificate of the SAP SLD system into the truststore of the JVM used by the Diameter Server.
   Once you retrieved the SLD certificate (X.509 v3 digital certificate, DER encoded) you need to import it into the truststore of the JVM used by the Diameter Server, using the following command:

   ```
   keytool -importcert -file certificate.sld.der -keystore keystore.diameter.jks
   -noprompt
   ```

   > **Note**
   >
   > You will be prompted to type a passphrase for the java keystore file which will contain the certificate to import. It is necessary to remember this passphrase in order to execute the following command.

The Diameter Server must then be configured to use the above updated keystore as a truststore. To do this, open the `jstart.config` file from the server's profile directory, located in the following folder: `<DRIVE>:\usr\sap\<SID>\SYS\profile\jstart\<SID>_<INSTANCE_NAME>_<HOST>`
Then, update the `Diameter.javaParameters` configuration parameter with the following information:

```
-Djavax.net.ssl.trustStore=<Path_of_the_Diameter_Server_java_keystore>
-Djavax.net.ssl.trustStorePassword=<Passphrase_of_the_keystore>
```

3. **Restarting the Diameter Server**
   To restart the Diameter Server, refer to the Starting and Stopping the Server Systems [page 144] procedure.

7.1.42 Securing Communications with Cockpit and Tomcat Server

Description

Cockpit is a Web application deployed on Apache Tomcat Server. It includes two main components:

- A front-end component that corresponds to the user interface visible in the browser of the SAP power users
- A back-end component that is deployed on the Tomcat Server and communicates with the Core Server system

The communications with Cockpit rely on:

- The XML[102] over HTTP[103] communication channel, that is used by the back-end component of Cockpit to communicate with the dispatcher and updater instances of the Core Server system
- The HTTP or HTTPS[104] protocol, that is used for interactions between the browsers and the front-end component of Cockpit

SAP SE recommends that you configure your Tomcat Server with SSL[105]/TLS[106] to interact both with the user's browser and with the Core Server system.

To secure your SAP CC landscape, SAP SE recommends that you dedicate a Tomcat Server to each Cockpit application. If you need to secure several Cockpit applications, you need to install and secure several Tomcat Server systems. If you mix Cockpit applications that communicate with secured Core Server systems and non secured systems, you can deploy the non secured Cockpit applications on the same Apache Tomcat Server.

The securing procedure consists in:

- Securing the communications between the Apache Tomcat Server and the Core Server system by configuring the JVM[107] of the Apache Tomcat Server as a client application of the Core Server system

102 eXtended Markup Language
103 HyperText Transfer Protocol
104 HyperText Transfer Protocol Secured
105 Secure Socket Layer
106 Transport Layer Security

- Securing the communications between web browsers and the Cockpit application by enabling SSL/TLS in the Apache Tomcat Server

As a prerequisite, the Core Server system must already been secured as detailed in the Securing the Core Server external communications [page 224] procedure.

Preliminary Notes

Some of the procedures described afterwards refer to the Common procedure for generating the securing materials [page 217] that is available in the Securing an SAP CC System Landscape [page 216] procedure. Refer to it when necessary.

Prerequisites

- The Core Server system is already secured as detailed in the Securing the Core Server external communications [page 224] procedure
- You have generated a `keystore.ext.p12` keystore file. The keystore relates to the secured Core Server system. It includes the `certificate.ext.der` certificate and corresponding private key of the Core Server system
- You have imported the content of the `keystore.ext.p12` keystore into the keystore of the Core Server system

Securing the Apache Tomcat Server

To implement your security policy, we recommend that you configure the Tomcat Server system to support the SSL/TLS technologies and important secured functions. Refer to the Security Aspects of Apache Tomcat Server section of the SAP CC 2020 Security Guide documentation for more information about security recommendations such as:

- Implementing SSL
- Cross-Origin Resource Sharing (CORS)
- Cross-Site Request Forgery (CFRS)
- HTTP Header management
- Cookies management

[107] Java Virtual Machine

Securing the communications between the Cockpit and the Core Server system

To secure the communications between the Cockpit application and the Core Server system, you secure the JVM of the Apache Tomcat Server that runs the Web application.

To communicate with a secured version of the Core Server system in dual mode, it is necessary to execute the following operations:

- Use the common procedure to generate a keystore:
  - Related to the JVM of the Apache Tomcat Server that runs the Cockpit application
  - Named `tomcat.keystore.p12`
  - Containing a `tomcat.certificate.der` certificate and corresponding private key
- Import the `certificate.ext.der` certificate of Core Server into the truststore of the Tomcat JVM
- Enable the `tomcat.keystore.p12` keystore on the Tomcat JVM side, by declaring this keystore on the Tomcat JVM side
- Import the `tomcat.certificate.der` certificate into the truststore of the Core Server system, using the `<ClientAlias>` alias
- Link the `<ClientAlias>` certificate to all the instances, for all the communication channels
- Update the configuration of the Cockpit application to use secured communications
- Restart the Tomcat Server to apply the new configuration

1. **Generating a certificate related to the Tomcat JVM as a client application of the Core Server system**
   Use the common procedure to generate a keystore named `tomcat.keystore.p12` and containing a `tomcat.certificate.der` certificate with its private key.

   > **Note**
   >
   > It is necessary to remember the passphrase of the keystore file in order to execute the steps described below.

2. **Importing the certificate of the Core Server into the truststore of the JVM of the Apache Tomcat Server that runs the Cockpit application**
   To give the possibility for the Tomcat JVM to trust the Core Server system, it is necessary to import the certificate of the Core Server into the truststore used by the Tomcat JVM that runs the Cockpit application. Use the `certificate.ext.der` certificate generated into the Securing the Core Server external communications [page 224] procedure.
   - To import the `certificate.ext.der` certificate into a Java truststore named `tomcat.truststore`, use the keytool utility delivered with the SAP JVM:

     ```
     <SAPJVM_PATH>/bin/keytool -importcert -file certificate.ext.der -keystore
     tomcat.trustore -noprompt
     ```

     > **Note**
     >
     > - keytool is a key and certificate management utility available in the bin directory of the SAP JVM
     > - You will be prompted to type a passphrase for the Java truststore file that will contain the certificate to import. It is necessary to remember this passphrase to execute the following command

- The Tomcat JVM must then be configured to use the above created Java truststore. To make the Apache Tomcat Server use the `tomcat.truststore` as its default truststore, you must specify the following additional options for the java command that starts Tomcat:
  - `-Djavax.net.ssl.trustStore=<tomcat.truststore path>`
  - `-Djavax.net.ssl.trustStorePassword=<tomcat.truststore passphrase>`

  > **Note**
  >
  > Specification of additional options for the java command that starts Tomcat is frequently done using the CATALINA_OPTS environment variable. To define this kind of environment variable, the recommendation from Apache Tomcat is to use the `setenv` script.
  >
  > Note that all the environment variables described in the `setenv` script are used only if you use the standard scripts to launch Tomcat. For example, if you have installed Tomcat as a service on Microsoft Windows, the service wrapper launches Java directly and does not use the `setenv` script file.
  >
  > For more information on how to define the Java options of the Apache Tomcat Server, please refer to the Apache Tomcat documentation.

3. **Enabling the client keystore on the Apache Tomcat Server side**
   The Tomcat JVM that runs the Cockpit application must consider its `tomcat.certificate.der` certificate to use it when communicating in a secured mode with the Core Server system.
   To declare the related `tomcat.keystore.p12` client keystore in the Apache Tomcat Server that runs the Cockpit web application, you must specify the following additional options for the java command that starts Tomcat:
   - `-Djavax.net.ssl.keyStoreType=pkcs12`
   - `-Djavax.net.ssl.keyStore=<tomcat.keystore.p12 path>`
   - `-Djavax.net.ssl.keyStorePassword=<tomcat.keystore.p12 passphrase>`

   > **Note**
   >
   > Specification of additional options for the java command that starts Tomcat is frequently done using the CATALINA_OPTS environment variable. To define this kind of environment variable, the recommendation from Apache Tomcat is to use the `setenv` script.
   >
   > Note that all the environment variables described in the `setenv` script are used only if you use the standard scripts to launch Tomcat. For example, if you have installed Tomcat as a service on Microsoft Windows, the service wrapper launches Java directly and does not use the `setenv` script file.
   >
   > For more information on how to define the Java options of the Apache Tomcat Server, please refer to the Apache Tomcat documentation.

4. **Importing the content of the client keystore into the truststore of the Core Server system**
   To be considered by the Core Server system as a trusted client application, import the certificate of the Tomcat JVM to the truststore of the Core Server system.
   To import the content of the Tomcatr certificate (`tomcat.certificate.der`) into the SAP CC truststore under the `<CockpitAlias>` alias, use the `certentry import` command of SAP CC Setup Tool:

- For UNIX or Linux operating systems

```
./setup.sh certentry import <CockpitAlias> tomcat.certificate.der
```

- For Microsoft Windows operating systems

```
setup.bat certentry import <CockpitAlias> tomcat.certificate.der
```

5. **Linking the imported certificate**
   To specify that the Core Server system must use the imported certificate when communicating with the Cockpit web application, use the `certentry import` command of SAP CC Setup Tool:
   - For UNIX or Linux operating systems

```
./setup.sh certentry link <CockpitAlias> all all
```

   - For Microsoft Windows operating systems

```
setup.bat certentry link <CockpitAlias> all all
```

6. **Updating the configuration of the Cockpit application on Tomcat Server**
   In the configuration file of Cockpit, change the `hci.http.url` application parameter. The URLs must be based on the HTTPS protocol. Refer to the Installing and Launching Cockpit [page 201] procedure for more information on how and where to maintain the configuration settings for your installed Cockpit applications.
7. **Restarting the Tomcat Server to apply the new configuration**
   Restart the Apache Tomcat Server to take into account:
   - The additional options for the java command that starts Tomcat
   - The updated configuration of the Cockpit application

Securing the communications between Web browsers and the Cockpit web application

To communicate with a Web browser, the Apache Tomcat Server that runs your Cockpit application must use the SSL/TLS technologies. To implement SSL, the Tomcat Server must have a certificate. To avoid warnings in your Web browsers, the certificate needs to be signed by your trusted certificate authority (CA).

Depending on the version of your installed Apache Tomcat Server, use its product documentation to perform the following steps:

- If needed, create a keystore to store the certificate
- Create a local Certificate Signing Request (CSR) and send it to your certificate authority
- Import the certificate into your keystore
- Activate the SSL support in Tomcat by uncommenting and completing the appropriate SSL connector in the `server.xml` file

> **Note**
>
> To facilitate the user experience of Cockpit users, you can use the redirect mechanism of Tomcat to redirect the HTTP port (8080 by default) to the HTTPS port (8443 by default). It enables you to manage simple URLs without the port number precision.

Verify the technical implementation

To verify the technical implementation, launch the Cockpit application and log on to the user interface. For example, open your browser and use the following URL to check that there is no issue: `https://<TOMCAT_HOST>:<HTTPS_PORT>/<WEB_APP_NAME>`

7.1.43 Securing Communications with Admin+

Related Information

Installing and Launching Admin+ [page 207]

7.1.43.1 Configuring SAP CC Admin+ to Trust SAP CC Core Server

Configure the Admin+ User Interface to Trust the Core Server System

To give the possibility for the SAP CC Admin+ user interface (as a console application) to trust an SAP CC Core Server system, it is necessary to import the certificate of the Core Server system into the Java truststore used by the Admin+ console application.

To import the certificate into the Java truststore, you need to use the keytool utility delivered with the JVM:

```
<JVM_PATH>/bin/keytool -import -alias <alias> -file <server_certificate>
-keystore <truststore_path> -storepass <truststore_passphrase>
```

Where:

- `<server_certificate>` the certificate of the SAP CC Core Server system
- `<truststore_path>` the path to the Java truststore that will be used by the SAP CC Admin+ user interface
- `<truststore_passphrase>` the passphrase of this Java truststore

> **Note**
>
> *keytool* is a key and certificate management utility available in the `bin` directory of the JVM.

> ⚠ Caution
>
> You will be prompted to type a passphrase for the Java truststore file which will contain the certificate to import.
>
> It is necessary to remember this passphrase as it will be required to configure the Admin+ user interface.

To make the Admin+ user interface uses the Java truststore that you just created, modify the Admin+ launch script (`admin+.sh` or `admin+.bat`) as follow:

1. Open the Admin+ script (`admin+.sh` or `admin+.bat`).
2. Locate the Java command that runs `com.highdeal.admin.plus.AdminPlus`.
3. Add the following two options to this Java command:

   ```
   -Djavax.net.ssl.trustStore=<truststore_path>
   -Djavax.net.ssl.trustStorePassword=<truststore_passphrase>
   ```

   Where:
   - `<truststore_path>` is the path to the previously created Java truststore
   - `<truststore_passphrase>` is the passphrase of the previously created Java truststore
4. Save your modifications.
5. Test the configuration. See Checking That the One-Way Secure Mode Is Properly Set for Admin+ [page 252].

7.1.43.2 Checking That the One-Way Secure Mode Is Properly Set for Admin+

Check that the one-way secure mode is properly set for the Admin+ user interface

Now that Admin+ is configured to trust the SAP CC Core Server system, it is time to test that one-way secure mode is correctly set.

1. Start the Admin+ user interface using the secure HCI URL of a dispatcher instance of the secured SAP CC Core Server system.
   This URL is based on Hypertext Transfer Protocol Secure (HTTPS) for encrypted communications.
   - Microsoft Windows operating systems:

     ```
     admin+.bat https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
     ```
   - Linux and UNIX operating systems:

     ```
     ./admin+.sh https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
     ```

   Where:

- `<Dispatcher_HCI_Host>` is the HCI host (address or IP) of a dispatcher instance of the secure SAP CC Core Server system
- `<Dispatcher_HCI_Port>` is the HCI port of a dispatcher instance of the secure SAP CC Core Server system
- **`https`** is the required scheme for secured encrypted communications

> **Note**
>
> Make sure that the URL is the expected one by checking that the Admin+ user interface displays the following message in its console:
>
> ```
> The URL identifying the distant SAP CC system is: https://
> <Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>/
> ```

2. Log on using the `user` (or `login`) command to check that Admin+ is able to communicate with the Core Server system.

> **Note**
>
> If the following error is returned from the login attempt, there might be an issue with the secure configuration:
>
> ```
> Error: A communication failure occurred on
> <Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
> ```
>
> In this case, you may want to troubleshoot your configuration.

> **→ Recommendation**
>
> Once the one-way mode is properly set up, you may want to improve security by enhancing the secure mode configuration from one-way to dual.

7.1.43.3 Securing Communications with Admin+ in Dual Mode

Create the SAP CC Admin+ Private Key, Certificate, and the Corresponding Keystore [page 253]

Configure Admin+ to Declare the Keystore [page 254]

Import the Admin+ Certificate into the SAP CC Core Server System [page 254]

Link the Imported Admin+ Certificate to the http Service of the Updater Instances [page 255]

Check That the Dual Secure Mode Is Properly Set for Admin+ [page 255]

7.1.43.3.1 Create the SAP CC Admin+ Private Key, Certificate, and the Corresponding Keystore

For an SAP CC Admin+ user interface to be trusted by an SAP CC Core Server system, you will need to create a keystore containing the SAP CC Admin+ private key and certificate.

For every Admin+ user interface to secure, create a keystore that contains the SAP CC Admin+ private key and certificate.

Apply the procedure: Create the SAP CC Client Private Key, Certificate, and the Corresponding Keystore

7.1.43.3.2 Configure Admin+ to Declare the Keystore

To communicate with an SAP CC Core Server system in a dual secured mode, the Admin+ user interface must take into account its own keystore.

To declare the keystore in the Admin+ configuration, modify the Admin+ launch script (`admin+.sh` or `admin+.bat`) as follow:

1. Open the Admin+ script (`admin+.sh` or `admin+.bat`).
2. Locate the Java command that runs `com.highdeal.admin.plus.AdminPlus`.
3. Add the following three options to this Java command:

   ```
   -Djavax.net.ssl.keyStoreType=pkcs12
   -Djavax.net.ssl.keyStore=<keystore_path>
   -Djavax.net.ssl.keyStorePassword=<keystore_passphrase>
   ```

   Where:

   - `<keystore_path>` is the path to the Admin+ keystore
   - `<keystore_passphrase>` is the passphrase of the Admin+ keystore
4. Save your modifications.

Next

Next, you will need to:

1. Import the Admin+ certificate into the SAP CC Core Server system. See Import the SAP CC Client Certificate into the SAP CC Core Server System.
2. Link the imported Admin+ certificate to the `http` service of the dispatcher instances. See Link the Imported SAP CC Client Certificate to the Targeted SAP CC Core Server Services.

7.1.43.3.3 Import the Admin+ Certificate into the SAP CC Core Server System

To give the possibility for an SAP CC Core Server system to trust an SAP CC Admin+ user interface as a Client application, it is necessary to import the certificate of the SAP CC Admin+ user interface into this SAP CC Core Server system.

For all the Admin+ user interfaces to secure, import their certificates into the SAP Core Server system.

As an SAP CC system administrator:

1. For every Admin+ user interface, import the Admin+ certificate into the SAP CC Core Server system.
   Apply the procedure: Import the SAP CC Client Certificate into the SAP CC Core Server System

7.1.43.3.4 Link the Imported Admin+ Certificate to the `http` Service of the Updater Instances

For an SAP CC Core Server system to trust an SAP CC Admin+ user interface, it is necessary to link the imported certificate of the SAP CC Admin+ user interface to the relevant communication service of the SAP CC Core Server service. For working with SAP CC Admin+, the service to link is HCI (technical code: `http`) and the link is required for the dispatcher instances that compose the system.

As an SAP CC system administrator:

1. For every Admin+ user interface, link the imported Admin+ certificate to the `http` service of the dispatcher instances.
   Apply the procedure: Link the Imported SAP CC Client Certificate to the Targeted SAP CC Core Server Services

7.1.43.3.5 Check That the Dual Secure Mode Is Properly Set for Admin+

Now that the secure configurations of SAP CC Admin+ and SAP CC Core Server have been improved, it is time to test that dual secure mode is correctly set.

1. Start the Admin+ user interface using the secure HCI URL of a dispatcher instance of the secure SAP CC Core Server system.
   - Microsoft Windows operating systems:

     ```
     admin+.bat https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
     ```

   - Linux and UNIX operating systems:

     ```
     ./admin+.sh https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
     ```

   Where:
   - **`<Dispatcher_HCI_Host>`** is the HCI host (address or IP) of a dispatcher instance of the secure SAP CC Core Server system
   - `<Dispatcher_HCI_Port>` is the HCI port of a dispatcher instance of the secure SAP CC Core Server system
   - **`https`** is the required scheme for secured encrypted communications

   > **Note**
   >
   > Make sure that the URL is the expected one by checking that the Admin+ user interface displays the following message in its console:

```
The URL identifying the distant SAP CC system is: https://
<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>/
```

2. Log on using the `user` (or `login`) command to check that Admin+ is able to communicate with the Core Server system.

> **Note**
>
> If the following error is returned from the login attempt, there might be an issue with the secure configuration:
>
> ```
> Error: A communication failure occurred on
> <Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
> ```
>
> In this case, you may want to troubleshoot your configuration.

7.1.44 Securing Communications with Core Tool

Related Information

Installing and Launching Core Tool [page 192]

7.1.44.1 Configuring SAP CC Core Tool to Trust SAP CC Core Server

Configure the SAP CC Core Tool User Interface to Trust the Core Server System

To give the possibility for the SAP CC Core Tool user interface (as a desktop application) to trust an SAP CC Core Server system, it is necessary to import the certificate of the Core Server system into the Java truststore used by the Core Tool desktop application.

To import the certificate into the Java truststore, you need to use the keytool utility delivered with the JVM:

```
<JVM_PATH>/bin/keytool -import -alias <alias> -file <server_certificate>
-keystore <truststore_path> -storepass <truststore_passphrase>
```

Where:

- `<server_certificate>` the certificate of the SAP CC Core Server system
- `<truststore_path>` the path to the Java truststore that will be used by the SAP CC Core Tool user interface
- `<truststore_passphrase>` the passphrase of this Java truststore

> **Note**
>
> *keytool* is a key and certificate management utility available in the `bin` directory of the JVM.

> **Caution**
>
> You will be prompted to type a passphrase for the Java truststore file which will contain the certificate to import.
>
> It is necessary to remember this passphrase as it will be required to configure the Core Tool user interface.

To make the Core Tool user interface uses the Java truststore that you just created, modify the Core Tool launch script (`core_tool.sh` or `core_tool.bat`) as follow:

1. Open the Core Tool script (`core_tool.sh` or `core_tool.bat`).
2. Locate the Java command that runs `com.highdeal.uilauncher.PnRUILauncher`.
3. Add the following two options to this Java command:

   ```
   -Djavax.net.ssl.trustStore=<truststore_path>
   -Djavax.net.ssl.trustStorePassword=<truststore_passphrase>
   ```

   Where:
   - `<truststore_path>` is the path to the previously created Java truststore
   - `<truststore_passphrase>` is the passphrase of the previously created Java truststore
4. Save your modifications.
5. Test the configuration. See Checking That the One-Way Secure Mode Is Properly Set for Core Tool [page 257].

7.1.44.2 Checking That the One-Way Secure Mode Is Properly Set for Core Tool

Check That the One-Way Secure Mode is Properly Set for the Core Tool User Interface

Now that SAP CC Core Tool is configured to trust the SAP CC Core Server system, it is time to test that one-way secure mode is correctly set.

1. Start every Core Tool user interface to check:

- Microsoft Windows operating systems:

```
core_tool.bat
```

- Linux and UNIX operating systems:

```
./core_tool.sh
```

2. In the *Connection* dialog box, type the name of an individual user in the *Login* field.
3. Type the individual user password in the *Password* field.
4. Type the secure HCI URL of a dispatcher instance of the secured SAP CC Core Server system in the *Server URL* field:

```
https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
```

   Where:

   - `<Dispatcher_HCI_Host>` is the HCI host (address or IP) of a dispatcher instance of the secure SAP CC Core Server system.
   - `<Dispatcher_HCI_Port>` is the HCI port of a dispatcher instance of the secure SAP CC Core Server system.
   - `https` is the required scheme for secured encrypted communications

5. Click *OK* to check that Core Tool is able to communicate with the SAP CC Core Server system.

> ⓘ Note
>
> If the following error window is displayed while trying to log on, there might be an issue with the secure configuration:
>
> *An error occurred during communication with the server.*
>
> In this case, you may want to troubleshoot your configuration.

> → Recommendation
>
> Once the one-way mode is properly set up, you may want to improve security by enhancing the secure mode configuration from one-way to dual.

7.1.44.3 Securing Communications with Core Tool in Dual Mode

Create the SAP CC Core Tool Private Key, Certificate, and the Corresponding Keystore [page 259]

Configure Core Tool to Declare the Keystore [page 259]

Import the Core Tool Certificate into the SAP CC Core Server System [page 260]

Link the Imported Core Tool Certificate to the http Service of the Dispatcher and Updater Instances [page 260]

Check That the Dual Secure Mode Is Properly Set for Core Tool [page 260]

7.1.44.3.1 Create the SAP CC Core Tool Private Key, Certificate, and the Corresponding Keystore

For an SAP CC Core Tool user interface to be trusted by an SAP CC Core Server system, you will need to create a keystore containing the SAP CC Core Tool private key and certificate.

For every Core Tool user interface to secure, create a keystore that contains the SAP CC Core Tool private key and certificate.

Apply the procedure: Create the SAP CC Client Private Key, Certificate, and the Corresponding Keystore

7.1.44.3.2 Configure Core Tool to Declare the Keystore

To communicate with an SAP CC Core Server system in a dual secured mode, the Core Tool user interface must take into account its own keystore.

To declare the keystore in the Core Tool configuration, modify the Core Tool launch script (`core_tool.sh` or `core_tool.bat`) as follow:

1. Open the Core Tool script (core_tool.sh or core_tool.bat).
2. Locate the Java command that runs `com.highdeal.uilauncher.PnRUILauncher`.
3. Add the following three options to this Java command:

   ```
   -Djavax.net.ssl.keyStoreType=pkcs12
   -Djavax.net.ssl.keyStore=<keystore_path>
   -Djavax.net.ssl.keyStorePassword=<keystore_passphrase>
   ```

   Where:
   - `<keystore_path>` is the path to the Core Tool keystore
   - <keystore_passphrase> is the passphrase of the Core Tool keystore
4. Save your modifications.

Next

Next, you will need to:

1. Import the Core Tool certificate into the SAP CC Core Server system. See Import the SAP CC Client Certificate into the SAP CC Core Server System.
2. Link the imported Core Tool certificate to the `http` service of the dispatcher and updater instances. See Link the Imported SAP CC Client Certificate to the Targeted SAP CC Core Server Services.

7.1.44.3.3 Import the Core Tool Certificate into the SAP CC Core Server System

For all the Core Tool user interfaces to secure, import their certificates into the SAP Core Server system.

1. Import the Core Tool certificate into the SAP CC Core Server system.
   Apply the procedure Import the SAP CC Client Certificate into the SAP CC Core Server System.

7.1.44.3.4 Link the Imported Core Tool Certificate to the `http` Service of the Dispatcher and Updater Instances

1. Link the imported Core Tool certificate to the `http` service of the dispatcher and updater instances.
   Apply the procedure Link the Imported SAP CC Client Certificate to the Targeted SAP CC Core Server Services.

7.1.44.3.5 Check That the Dual Secure Mode Is Properly Set for Core Tool

Now that the secure configurations of SAP CC Core Tool and SAP CC Core Server have been improved, it is time to test that dual secure mode is correctly set.

1. Start every Core Tool user interface to check:
   - Microsoft Windows operating systems:

     ```
     core_tool.bat
     ```

   - Linux and UNIX operating systems:

     ```
     ./core_tool.sh
     ```

2. In the *Connection* dialog box, type the name of an individual user in the *Login* field.
3. Type the individual user password in the *Password* field.
4. Type the secure HCI URL of a dispatcher instance of the secure SAP CC Core Server system in the *Server URL* field:

   ```
   https://<Dispatcher_HCI_Host>:<Dispatcher_HCI_Port>
   ```

   Where:
   - **`<Dispatcher_HCI_Host>`** is the HCI host (address or IP) of a dispatcher instance of the secure SAP CC Core Server system
   - <Dispatcher_HCI_Port> is the HCI port of a dispatcher instance of the secure SAP CC Core Server system
   - `https` is the required scheme for secured encrypted communications

5. Click *OK* to check that SAP CC Core Tool is able to communicate with the SAP CC Core Server system.

> **Note**
>
> If the following error window is displayed while trying to log on, there might be an issue with the secure configuration:
>
> *An error occurred during communication with the server.*
>
> In this case, you may want to troubleshoot your configuration.

7.1.45 Installing a Permanent License

Description

During the execution of your installation scenario, a temporary license has been installed for your SAP Convergent Charging system, that is only valid for a limited period of time. You can use this procedure remove this time limitation by installing a permanent license.

Preliminary Notes

For more information about license keys, refer to SAP Notes 197623 and 94998.

Prerequisites

- An SAP Convergent Charging 2020 system must be installed and available before executing this procedure. Execute an installation scenario if necessary
- The `setup` command specified in the following operations refers to the Setup Tool user interface. Refer to the Installing and Launching Setup Tool [page 211] procedure if necessary

Procedure

The installation of an SAP permanent license consists in the following operations:

- Applying for a permanent license
- Installing the permanent license within your system landscape
- Restarting the instances to take the permanent license into account

1. To obtain a permanent license key, open the "Keys, Systems & Installations" section of the SAP Support Portal available at http://support.sap.com/licensekey. You need to provide the hardware key of each host of your SAP Convergent Charging landscape. The different hardware keys are available in the `hardware_keys.txt` file that is available in the SAP Central Repository [page 13] directory
2. Copy the license file to a host of your landscape where a dispatcher is installed
3. Connect to this host and launch the Setup Tool user interface (refer to the Installing and Launching Setup Tool [page 211] procedure if necessary). Execute the following command as the administrator of your SAP CC system landscape, considering that `<LICENSE_FILE_PATH>` corresponds to the absolute path of the previously transferred file containing the permanent license:

   ```
   setup license import <LICENSE_FILE_PATH>
   ```

4. Once the permanent license has been successfully installed within your SAP CC system, you need to restart the instances to take the new license into account. To restart the system, refer to the Starting and Stopping the Server Systems [page 144] procedure
5. In the `LOG` of an instance, check the license has been applied and the new expiration date by looking for the two entries below:

   ```
   <SID>, hardware key "<HARDWARE_KEY>", product
   "<PRODUCT_NAME>"#
   [...]
   License will expire on 2021-04-11T00:00:00#
   ```

7.1.46 Removing System(s) or Instance(s) from a Given Host

Description

This procedure explains you how to remove one or multiple systems or instances from a given host of an existing landscape, using the SAP installer.

Procedure

As this procedure relies on the SAP installer, you first need to launch it using the Launching the SAP installer [page 190] dedicated procedure.

Once opened, the installer displays a succession of screens that give you the possibility to configure your installation scenario. Use the following recommendations to fill the different screens:

1. On the *Welcome* screen:
   - Click *SAP Convergent Charging* > *Software Life-Cycle Options* > *Uninstall* > *Uninstall - SAP Systems or Single Instances*
   - Click the *Next* button to set up the uninstallation.
2. *General SAP System Parameter* screen

- Tick the *Profiles Available* checkbox when not ticked by default
- *Profile Directory*: Filled by default with the location of the SAP system profile directory. Ensure that this location is correct
- Click the *Next* button

3. *Uninstall Instances* screen
   - Untick the *Uninstall all instances from this host and delete the SAP CC system* checkbox to unlock the table and select a list of instances to remove.

     > **Note**
     >
     > When ticked, this option uninstalls all parts of the SAP system on this host: all instances, the default profile, and the directories of the SAP system. Select this option only when uninstalling instances from the SAP Global Host, after all the other hosts have been uninstalled.

4. *Parameter Summary* screen
   - Carefully check that all your settings have been taken into account before starting the installation. In case you need to correct one or multiple settings, tick the corresponding checkboxes and click the *Revise* button
   - If all your settings are correctly taken into account, click the *Next* button to uninstall the selected instances from your host
5. *Task Progress* screen
   - This screen contains the different steps of your installation scenario, whose advancement can be followed

7.1.47 Integrating SAP CC with a SCIM-Based System

Description

This procedure explains you how to configure your SAP CC Core Server system in order to communicate with SCIM-compliant systems or applications for user management operations in an external system. As of SAP CC 5.0 SP 6, SAP Convergent Charging provides some SCIM services that are based on the SCIM 2.0 protocol and available in the REST Web Services.

Preliminary Notes

- As of SAP CC 5.0 SP 6, SAP Convergent Charging provides a new feature based on the System for Cross-domain Identity Management (SCIM) protocol to enable implementing user management in external systems. For more information, refer to the security guide.
- Secured communications are required using the TLS/SSL protocol.

Prerequisites

- An SAP CC Core Server system must be available and running in your SAP system landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary.
- A SCIM-based system is available and implements the SCIM API of SAP CC. See SAP CC SCIM API.
- A dedicated SAP CC user is available in your SAP CC system landscape.
- To work with SAP CC, the SCIM-based system connects to the Core Server system using the REST Web Services of SAP CC.

Procedure

The integration of an SAP Convergent Charging system landscape with a SCIM-based system consists in the following operations:

- Securing the landscape
- Creating dedicated users
- Enabling the feature in SAP CC
- Restarting the system

As an SAP CC system administrator, complete the following procedure:

1. Secure the communications with the Core Server system. The communications on the REST over HTTP communication channel must be secured.
   See Securing Communications with the Core Server System [page 221].
2. Using Core Tool, configure a new SAP CC user.
3. Using Config Tool or Admin+, set up the SCIM_MANAGEMENT_ENABLED system parameter to true.
4. Restart the Core Server system.

7.1.48 Integrating SAP CC with CA Application Performance Management

Description

This procedure explains you how to configure your SAP CC systems in order to communicate with CA Application Performance Management (CA Application Performance Management) for monitoring purpose.

Preliminary Notes

- For more information about CA Application Performance Management, refer to SAP Note 1453216.
- This procedure uses the following variables:
  - `<SID>`, which corresponds to the SAP system identifier (SID, aka SAPSID) of your SAP CC system (Core Server or BART Server) to be monitorer
  - `<SAP_INSTANCE_NAME>`, which corresponds to the name of the concerned instance forming the SAP CC system to monitor
    It is made with a predefined 3-letter pattern combined with the SAP instance number (00-97) that was set by the SAP installer tool. The number depends on the installed host
    For example, CCD`<nn>` or CCR01, or CCU01, CCG01, CCB01

Prerequisites

- This procedure relies on the use of the Introscope Agent, that is automatically deployed when installing an instance of the Core Server system. Check that the following folder contains the `Agent.jar` file:
  - `/usr/sap/<SID>/<SAP_INSTANCE_NAME>/exe/wily` for Linux and UNIX operating systems
  - `\usr\sap\<SID>\<SAP_INSTANCE_NAME>\exe\wily` for a Microsoft Windows operating system
- To collect data using the Introscope Agent, an Introscope Enterprise Manager system must be available within your overall SAP system landscape. This element is usually installed on the host used for SAP Solution Manager.

Procedure

The integration of an SAP Convergent Charging system with CA Application Performance Management (formerly known as CA Wily Introscope) consists in the following operations:

- For each instance that is supposed to be monitored:
  - The Introscope Agent must be configured accordingly
  - The Java configuration of the instance must be updated to take the Introscope Agent into account
  - The Introscope Agent must be activated
- The instances must be restarted
- The result of the integration must be checked

For each instance of your SAP CC system that must be monitored via Introscope:

1. Configure the Introscope Agent, by retrieving the `IntroscopeAgent.profile` file located in the following subfolder of each instance working directory:
   - `/usr/sap/<SID>/<SAP_INSTANCE_NAME>/exe/wily/core/config` for Linux and UNIX operating systems
   - `\usr\sap\<SID>\<SAP_INSTANCE_NAME>\exe\wily\core\config` for a Microsoft Windows operating system
2. Copy this `IntroscopeAgent.profile` file into the following directory:

- `/usr/sap/<SID>/<SAP_INSTANCE_NAME>/config` for Linux and UNIX operating systems
- `\usr\sap\<SID>\<SAP_INSTANCE_NAME>\config` for a Microsoft Windows operating system

3. Edit the copied file and modify the following, uncommenting them when necessary:

Property name	Property value
`introscope.autoprobe.logfile`	`/usr/sap/<SID>/<SAP_INSTANCE_NAME>/work/introscope/IntroscopeAutoProbe.log`
`introscope.autoprobe.directivesFile`	`/usr/sap/<SID>/<SAP_INSTANCE_NAME>/exe/wily/core/config/sap_typical.jar,<INSTANCE_WKG_DIRECTORY>/exe/wily/core/config/sap_srm_crm.jar`
`log4j.appender.logfile.File`	`/usr/sap/<SID>/<SAP_INSTANCE_NAME>/work/introscope/IntroscopeAgent.log`
`introscope.agent.enterprisemanager.transport.tcp.host.DEFAULT`	`<Enterprise Manager hostname or IP address>`
`introscope.agent.enterprisemanager.transport.tcp.port.DEFAULT`	`<Enterprise Manager port>`
`introscope.agent.customProcessName`	**`SAP Convergent Charging`**
`introscope.agent.defaultProcessName`	**`SAP Convergent Charging`**
`introscope.agent.extensions.directory`	`/usr/sap/<SID>/<SAP_INSTANCE_NAME>/exe/wily/core/ext`
`introscope.agent.jmx.enable`	`true`
`com.sap.introscope.agent.autojmx.enable`	`false`
`introscope.agent.jmx.globalfilter`	**`com.highdeal:*`**
`introscope.agent.jmx.name.primarykeys`	**`type,name`**
`introscope.agent.jmx.name.filter`	**`com.highdeal`**
`introscope.agent.hostName`	`<Instance hostname (in lower case)>`

4. Save your changes.

Introscope Agent must now be activated, according to the following procedure:

1. Edit the `/usr/sap/<SID>/SYS/profile/jstart/<SID>_<SAP_INSTANCE_NAME>_<HOSTNAME>jstart.config` file and add the following options to the `javaParameters` property:

```
-javaagent:/usr/sap/<SID>/<SAP_INSTANCE_NAME>/exe/wily/Agent.jar
```

```
-Dcom.wily.introscope.agentProfile=/usr/sap/<SID>/<SAP_INSTANCE_NAME>/config/
IntroscopeAgent.profile/
-Dcom.wily.introscope.agent.agentName=<AGENT_NAME>
```

Where <AGENT_NAME> represents a unique name used by every instance using Introscope, and whose value depends on the concerned system:

- **<AGENT_NAME>=<SID>_<INSTANCE_TYPE>#<INSTANCE_NUMBER>_SCCCoreServer** for the Core Server system (for example: **CCT_dispatcher#1_SCCCoreServer**)
- **<AGENT_NAME>=<SID>_BART_SCCServer** for the BART Server system

2. Save your changes.
3. Once the Java configuration of the concerned instances has been successfully updated, you need to restart these instances to take this new configuration into account. Refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.
4. Once the instances successfully restarted, you can check the result of this integration procedure:
   - By checking the content of the following log file: `/usr/sap/<SID>/<SAP_INSTANCE_NAME>/work/introscope/IntroscopeAgent.<AGENT_NAME>.log`
   - By using an Introscope Workstation available in your overall landscape, and finding your system within the list of connected agents.

7.1.49 Integrating SAP CC with SAP CTS+

Description

This procedure explains you how to deploy and configure the SAP CC SAP CTS+ plugin, that is necessary to integrate your SAP CC Core Server system with the SAP enhanced Change and Transport System (SAP CTS+).

Preliminary Notes

- For more information about the enhanced Change and Transport System, refer to the following page available on the SAP Community Network: http://scn.sap.com/docs/DOC-8576
- This procedure only contains information about the SAP CTS+ plugin, and does not contain information about the configuration of the enhanced Change and Transport System. To get such information, refer to SAP Note 2156128

> → Remember
>
> The SAP CTS+ plugin contacts SAP CC via:
>
> 1. the HCI[108] port (9100 by default) of the dispatcher first (only to retrieve the URL[109] of the updater)

[108] HTTP Communication Interface
[109] Uniform Resource Locator

2. the WS[110] port (9080 by default) of the updater secondly (to run the actual transport)

Prerequisites

- A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary
- The SAP JVM 8.1 must be installed as standalone on the host where the SAP CC CTS+ plugin is planned to be deployed. This SAP JVM 8.1 must be accessible and executable by the OS user under which the AS JAVA (SAP CC SAP CTS+ plugin `deploy_CC` script) runs. The SAPCC_JAVA_HOME environment variable must be set to the path of the JVM. Refer to the Downloading the SAP JVM [page 94] and Setting Up the System Environment Variables [page 104] procedures if necessary

Getting the SAP CC SAP CTS+ plugin

The SAP CC SAP CTS+ plugin is available within a ZIP archive located:

- In the Installation materials of SAP Convergent Charging 2020
- On SAP ONE Support Launchpad

If you want to get the SAP CC CTS+ plugin from the installation materials, take it from the following folder: `/DATA_UNITS/CC2020_TOOLS_CONTENT_UC_OSIND`

If you want to download the latest version, execute the following procedure:

1. Go to SAP ONE Support Launchpad at the following address: https://launchpad.support.sap.com/
2. Click the *Software Downloads* launch tile of the *System Operations and Maintenance* section
3. Click the *Support Packages and Patches* section
4. Expand the *By Alphabetical Index (A-Z)* section
5. Click the *C* letter
6. Click the "*SAP Convergent Charging* element of the list
7. Click the **SAP Convergent Charging 2020** element of the list

> **Note**
>
> Previous versions are also available but this procedure only concerns the 2020 release

8. Click the **CTS+ PLUGIN 2020** element of the list to open the *Downloads* section
9. Choose the item matching the level of the Support Package and Patch of your SAP Convergent Charging 2020 Core Server system: `CCCTSPLUGIN<SP_LEVEL>_<PATCH_LEVEL>-XXXXXXXX.ZIP`
   Where:
   - `<SP_LEVEL>` is the SAP Convergent Charging 2020 Core Server Support Package level
   - `<PATCH_LEVEL>` is the SAP Convergent Charging 2020 Core Server Patch level

[110] Web Services

Deploying the SAP CC CTS+ plugin

Once downloaded, use the following procedure to deploy the SAP CC SAP CTS+ plugin on the targeted SAP CTS system:

Linux and UNIX Operating Systems

1. Copy the SAP CC SAP CTS+ plugin ZIP archive on the targeted SAP CTS system. Refer to Getting the SAP CC CTS+ plugin [page 268] procedure if necessary.
2. Create an `IS-CC` folder within the `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts` directory (the path will be `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts/IS-CC`).
   If this folder already exists, execute the following operations:
   - Back-up the `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts/IS-CC/bin` directory, that contains scripts files whose configuration might have been previously modified
   - Remove all elements of the `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts/IS-CC` folder
3. Uncompress the SAP CC SAP CTS+ plugin ZIP archive using the following command:

   ```
   unzip <SAP_CC_CTS_PLUGIN_ZIP_ARCHIVE> -d /usr/sap/<CTS_SYSTEM_SID>/global/
   CtsScripts/IS-CC
   ```

4. Ensure security settings according to your needs and security policy:
   - The `deploy_CC.sh` script should have "execute" permissions for the user under which the AS JAVA runs.
   - The `CtsScripts` directory should have restricted permissions for other users not involved into the deployment and setup process.
5. Test if the script can be successfully executed (by the proper user), for example in the "CONNECT" mode
6. Apply SAP Note 2792507 to enable the script-based deployment of the SAP enhanced Change and Transport System (SAP CTS+)

Microsoft Windows Operating System

1. Copy the SAP CC SAP CTS+ plugin ZIP archive on the targeted SAP CTS system. Refer to Getting the SAP CC CTS+ plugin [page 268] procedure if necessary.
2. Create an `IS-CC` folder within the `C:\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts` directory (the path will be `C:\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts\IS-CC`).
   If this folder already exists, execute the following operations:
   - Back-up the `C:\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts\IS-CC\bin` directory, that contains scripts files whose configuration might have been previously modified
   - Remove all elements of the `C:\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts\IS-CC` folder
3. Uncompress the SAP CC SAP CTS+ plugin ZIP archive using the following command:

   ```
   %SAPCC_JAVA_HOME%\bin\jar.exe xvf <SAP_CC_CTS_PLUGIN_ZIP_ARCHIVE> -C IS-CC
   ```

4. Ensure security settings according to your needs and security policy:
   - The `deploy_CC.bat` script should have "execute" permissions for the user under which the AS JAVA runs.
   - The `CtsScripts` directory should have restricted permissions for other users not involved into the deployment and setup process.
5. Test if the script can be successfully executed (by the proper user), for example in the "CONNECT" mode
6. Apply SAP Note 2792507 to enable the script-based deployment of the SAP enhanced Change and Transport System (SAP CTS+)

Configuring the SAP CC CTS+ plugin

The SAP CC SAP CTS+ plugin is delivered as a script file named:

- `deploy_CC.bat` for Microsoft Windows operating systems
- `deploy_CC.sh` for Linux and UNIX operating systems

To fit your environment, you can use the following procedure to modify the parameters used within the script file:

- `SAPCC_JAVA_HOME`, that corresponds to the home directory of the standalone SAP JVM 8.1
- `LIBRARY_PATH`, that corresponds to the folder containing the different libraries used by the plugin
- `LOG_SEVERITY`, that corresponds to the level of log messages received from the Core Server system, whose supported value can be:
  - `debug`, which is used to display all available information
  - `info`, which is used to display only relevant information
  - `warning`, which is used to display warnings and error messages
  - `error`, which is used to display only error messages

Linux and UNIX Operating Systems

- Open the `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts/IS-CC/bin/deploy_CC.sh` script file
- Uncomment and replace the `<JAVA_PATH>` string by the relevant value in the following line:

```
#SAPCC_JAVA_HOME=<JAVA_PATH>
```

- The default value of the `LIBRARY_PATH` parameter corresponds to the `/usr/sap/<CTS_SYSTEM_SID>/global/CtsScripts/IS-CC/jars` directory. In case you need to install these libraries in a different folder, modify the following line accordingly:

```
LIBRARY_PATH=../jars/
```

- The default value of the `LOG_SEVERITY` parameter corresponds to the info level. In case you need to modify this log level, modify the following line accordingly:

```
LOG_SEVERITY=info
```

- Save your modifications and exit your text editor

Microsoft Windows Operating System

- Open the `\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts\IS-CC\bin\deploy_CC.bat` script file
- Uncomment and replace the `<JAVA_PATH>` string by the relevant value in the following line:

```
::SET SAPCC_JAVA_HOME=<JAVA_PATH>
```

- The default value of the `LIBRARY_PATH` parameter corresponds to the `\usr\sap\<CTS_SYSTEM_SID>\global\CtsScripts\IS-CC\jars` directory. In case you need to install these libraries in a different folder, modify the following line accordingly:

```
SET LIBRARY_PATH=..\jars\
```

- The default value of the `LOG_SEVERITY` parameter corresponds to the info level. In case you need to modify this log level, modify the following line accordingly:

  ```
  SET LOG_SEVERITY=info
  ```

- Save your modifications and exit your text editor

7.1.50 Integrating SAP CC with SAP Convergent Invoicing

Description

This procedure explains you how to configure an SAP CC stand-alone Core Server system in order to communicate with SAP Convergent Invoicing in the SAP ERP/FI-CA or SAP S/4HANA system for billing and invoicing purposes.

Preliminary Notes

- In case you have just installed your SAP CC Core Server system using a stand-alone scenario instead of an integrated one, SAP SE recommends that you reinstall your system instead of using this procedure, in order to avoid any possible manual error
- This procedure requires to restart your landscape. SAP SE thus recommends that you prepare and test this integration procedure within a validation landscape before executing it on your production landscape itself
- This procedure can be used to integrate with SAP Convergent Invoicing and generate:
  - Charged items only (cit)
  - Charged items and consumptions items (cit_ci)

  According to your business, replace the `<MODE>` argument in the procedure with the relevant value.

Prerequisites

- A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary
- This procedure uses the Setup Tool, Config Tool and Core Tool user interfaces. For further information about these user interfaces, refer to the following procedures:
  - Installing and Launching Setup Tool [page 211]
  - Installing and Launching Config Tool [page 214]
  - Installing and Launching Core Tool [page 192]

Procedure

The integration of an SAP Convergent Charging landscape with SAP Convergent Invoicing consists in the following operations:

- Installing bulkloader instances
- Loading relevant information related to SAP Convergent Invoicing (in SAP ERP/FI-CA or SAP S/4HANA) in the Core Database:
  - JCo destination
  - Default mappings
  - CIF configuration
  - System parameters
- Restarting the updater and dispatcher instances
- Configuring the required master data:
  - Default CMA
  - External currencies
- Restarting the rater instances
- Starting the bulkloader instances

1. **Installing the bulkloader instances**
   To work in conjunction with SAP Convergent Invoicing, each rater of your landscape must be associated to a dedicated bulkloader:
   - Deployed on the same host as the rater
   - Sharing the same instance number

   To install the relevant bulkloaders within your landscape, execute the Adding Core Server Instances in a Multi-Hosts Landscape [page 142] procedure on each host where a rater is currently deployed, taking into consideration the previously mentioned requirements and without restarting the instances at the end of the installation process.
2. **Loading information related to SAP Convergent Invoicing in the Core Database**
   Once bulkloaders have been successfully installed, it is necessary to load the following information in the Core Database:
   - The JCo destination used to connect to the relevant SAP ERP/FI-CA or SAP S/4HANA system that includes SAP CI
   - The default mappings
   - The configuration of the CIF
   - Some additional system parameters
   - The configuration of the RIF

   1. To create the relevant JCo destination, execute the following operations:
      - Connect to the host of your landscape where a dispatcher is installed
      - Copy the `jco.destination.sk` file available in the `config/ci` folder of the dispatcher instance:
        - `/usr/sap/<SID>/CCDxx/exe/CC_CORE_SERVER/config/ci/jco.destination.sk` for Linux and UNIX operating systems
        - `<drive>:\usr\sap\<SID>\CCDxx\CC_CORE_SERVER\config\ci\jco.destination.sk` for a Microsoft Windows operating system

        in a new `jco.destination` file
      - Edit this newly created file and modify the following parameters to fit the targeted SAP CI system:

`jco.client.ashost`
`jco.client.sysnr`
`jco.client.client`
`jco.client.user`
`jco.client.passwd`

> → Remember
>
> For a **load-balanced** SAP ERP/FI-CA or SAP S/4HANA system, note that:
> - `jco.client.ashost`
> - `jco.client.sysnr`
>
> parameters must be respectively replaced at least by:
> - `jco.client.mshost`
> - `jco.client.msserv` or `jco.client.r3name`
>
> Find more information about load-balancing connections in the dedicated section of the table in the Configuration Settings for JCo chapter of the SAP CC 2020 Tuning Guide documentation.

- Execute the following command of the Setup Tool user interface to import these parameters in the Core Database:

  jcodestination import
- Execute the following command of the Setup Tool user interface to link this JCo destination to the targeted SAP CI system:

  erpreferencesystem setJCoDestination

2. To load the relevant default mappings related to charged items, execute the following operations:
   - Copy the `sapcc_ci_additionalFields.xml.sk` file available in the `config/standalone` folder of the dispatcher instance in a `new sapcc_ci_additionalFields.xml` file
   - Execute the following command of the Setup Tool user interface (using the `chargeableItem` type) to import these parameters in the Core Database:

     sapci importDefaultMapping
   - Copy the `sapci_cit_defaultMapping.xml.sk` file available in the `config/sapci` folder of the dispatcher instance in a new `sapci_cit_defaultMapping.xml` file
   - Execute the following command of the Setup Tool user interface (using the `chargedItem` type) to import these parameters in the Core Database:

     sapci importDefaultMapping
3. In case you want to use consumption items, it is necessary to load the relevant default mappings related to charged items and consumption items by executing the following operations:
   - Copy the `sapci_ci_defaultMapping.xml.sk` file available in the `config/sapci/cit_ci` folder of the dispatcher instance in a new `sapci_ci_defaultMapping.xml` file
   - Execute the following command of the Setup Tool user interface (using the `chargeableItem` type) to import these parameters in the Core Database: `sapci importDefaultMapping`

> ⚠ Caution
>
> If you use the account split feature implemented by SOM and the ERP, refer to the Executing a Charging Plan for Consumption Item Mapping section of the SAP CC 2020 Tuning Guide documentation.

4. To configure the generation of data files during the execution of the charging and bulkloading processes, execute the following operations:
   - Copy the `cif.rater.config.xml.sk` file available in the `config/sapci/<MODE>` folder of the dispatcher instance in a new `cif.rater.config.xml` file
   - In case you need to fit specific needs, modify the newly created `cif.rater.config.xml` file
   - Execute the following command of the Setup Tool user interface (using the `rater` type of targeted instance) to import these parameters in the Core Database: `cif import`
   - Copy the `cif.bulkloader.config.xml.sk` file available in the `config/sapci/<MODE>` folder of the dispatcher instance in a new `cif.bulkloader.config.xml` file
   - In case you need to fit specific needs, modify the newly created `cif.bulkloader.config.xml` file
   - Execute the following command of the Setup Tool user interface (using the `bulkLoader` type of targeted instance) to import these parameters in the Core Database: `cif import`

> ⓘ Note
>
> For further information about the configuration of the CIF, refer to the *Data Files Generation* section of the SAP CC 2020 Tuning Guide documentation.

5. To load the configuration parameters related to SAP Convergent Invoicing, execute the following operations:
   - Copy the `parameters.xml.sk` file available in the `config/sapci/<MODE>` folder of the dispatcher instance in a new `parameters.xml` file
   - Execute the following command of the Config Tool user interface to load this set of configuration parameters in the Core Database: `configuration import`
6. To configure the use of the rerating feature, execute the following operations:
   - Copy the `rif.rater.config.xml.sk` file available in the `config/sapci/<MODE>` folder of the dispatcher instance in a new `rif.rater.config.xml` file
   - Execute the following command of the Setup Tool user interface to import these parameters in the Core Database: `rif import`

3. **Restarting the updater and dispatcher instances**
   Once the Core Database has been updated with the relevant information, it is necessary to restart the updater and dispatcher instances to update their cached structures with the imported parameters that are required by the Core Tool in the next step. For further information, refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.
4. **Configuring required master data**
   Once the Core Database has been updated with the relevant information, it is necessary to execute the following operations within the Core Tool user interface:
   - Creation of the default CMA
   - Synchronization of the currencies defined in SAP Convergent Invoicing

   1. To work in conjunction with SAP ERP/FI-CA and SAP CRM systems, it is necessary to create a Customer Management Area used for connection purposes. To create this CMA, launch [page 193] the Core Tool user interface and click the *Home* > *Tools* > *Customer Management Areas* menu

2. As some differences may exist in terms of currencies defined in the 2 systems, it is necessary to synchronize SAP CC and SAP CI. To synchronize the currencies, launch [page 193] the Core Tool user interface and click the *Home* > *Tools* > *Currencies* > *Synchronize* menu

> **Note**
>
> For further information about customer management areas and currencies synchronization, refer to the dedicated sections available in the SAP Convergent Charging Core Tool online help documentation.

5. **Restarting the rater instances and starting the bulkloader instances**
   Once the rater and bulkloader instances have been successfully configured, you need to restart them in order to take this new configuration into account. SAP SE recommends that you first restart the raters, then start the bulkloaders. For further information, refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.

Once the instances of your SAP Convergent Charging landscape have been successfully restarted, you can use the Core Tool to set up your master data such as relevant billable item and/or consumption item mappings:

- On Existing Charged/chargeable Item Classes
- On New Charged/chargeable Item Classes

7.1.51 Integrating SAP CC with SAP S/4HANA Cloud Public Edition

Description

This procedure explains you how to configure a "stand-alone" SAP CC Core Server system in order to communicate with SAP S/4HANA Cloud Public Edition for billing and invoicing purposes.

> **Caution**
>
> This procedure does not explain how to integrate an SAP CC Core Server 2020 system with SAP S/4HANA Cloud Public Edition when this Core Server system is already integrated with SAP Convergent Invoicing in an SAP S/4HANA system.
>
> For such a situation, SAP SE recommends that you contact your SAP Support Team and consider a dedicated migration project.

Preliminary Notes

- In case you have just installed your SAP CC Core Server system using a stand-alone scenario instead of an integrated one, SAP SE recommends that you reinstall your system instead of using this procedure, in order to avoid any possible manual error.

- This procedure requires to restart your landscape. SAP SE thus recommends that you prepare and test this integration procedure within a validation landscape before executing it on your production landscape itself.
- This procedure can be used to integrate with SAP Business Suite 4 SAP HANA Cloud Public Edition and generate billable items.

Prerequisites

- A Core Server system must be available within your landscape. Refer to the Installing a Core Server on a Monohost Landscape [page 123] or to the Installing a Core Server on a Multihost Landscape [page 131] procedures if necessary
- This procedure uses the Setup Tool, Config Tool and Core Tool user interfaces. For further information about these user interfaces, refer to the following procedures:
  - Installing and Launching Setup Tool [page 211]
  - Installing and Launching Config Tool [page 214]
  - Installing and Launching Core Tool [page 192]

Procedure

The integration of an SAP Convergent Charging landscape with SAP S/4HANA Cloud Public Edition consists in the following operations:

- Deleting all the objects that relate to the Prepaid Accounts Management feature
- Updating the configuration of the concerned instances
- Deactivating the Prepaid Accounts Management and the Chargeable Items Recharging features
- Configuring the WS[111] destinations used to send billable items to SAP S/4HANA Cloud Public Edition
- Restarting dispatcher and updater instances
- Creating CMAs[112] within the Core Tool user interface
- Restarting rater and bulkloader instances
- Creating billable item mappings within the Core Tool user interface

1. **Deleting objects related to the Prepaid Accounts Management feature**
   As the Prepaid Accounts Management feature is not supported when working in conjunction with SAP S/4HANA Cloud Public Edition, it is necessary to delete all the SAP CC objects that refer to this feature, including:
   - The prepaid accounts defined within subscriber accounts
   - The refill item classes
   - The refill record classes
   - The refill logic objects
   - The refill plans
   - The prepaid charging references used within charges

[111] Web Services
[112] Customer Management Area

- The "Switching Comparator" and "Default Reference" components used within charges
- The prepaid account assignments used within charge plans
- The properties related to prepaid accounts used within charged item classes
- The properties related to prepaid accounts used within charged item mappings into both charge plans and offers

2. **Updating instances configuration**
   Once the different objects related to the Prepaid Accounts Management feature have been successfully deleted, it is necessary to load the following information in the Core Database:
   - The configuration parameters related to SAP S/4HANA Cloud Public Edition
   - The default mappings
   - The configuration of the CIF[113]

   1. To load the configuration parameters related to SAP S/4HANA Cloud Public Edition, execute the following operations:
      - Copy the `parameters.xml.sk` file available in the `config/s4hc` folder of the dispatcher instance in a new `parameters.xml` file
      - Execute the following command of the Config Tool user interface to load this set of configuration parameters in the Core Database: `config configuration import parameters.xml`
   2. To generate billable items, it is necessary to load the relevant default mappings by executing the following operations:
      - Copy the `s4hc_cit_defaultMapping.xml.sk` file available in the `config/s4hc` folder of the dispatcher instance in a new `s4hc_cit_defaultMapping.xml` file
      - Execute the following command of the Setup Tool user interface to import these parameters in the Core Database:

        ```
        setup sapci importDefaultMapping chargedItem s4hc_cit_defaultMapping.xml
        ```
   3. To configure the generation of data files during the execution of the charging and bulkloading processes, execute the following operations:
      - Copy the `cif.rater.config.xml.sk` file available in the `config/s4hc` folder of the dispatcher instance in a new `cif.rater.config.xml` file
      - In case you need to fit specific needs, modify the newly created `cif.rater.config.xml` file
      - Execute the following command of the Setup Tool user interface to import these parameters in the Core Database:

        ```
        setup cif import rater cif.rater.config.xml
        ```
      - Copy the `cif.bulkloader.config.xml.sk` file available in the `config/s4hc` folder of the dispatcher instance in a new `cif.bulkloader.config.xml` file
      - In case you need to fit specific needs, modify the newly created cif.bulkloader.config.xml file
      - Execute the following command of the Setup Tool user interface to import these parameters in the Core Database:

        ```
        setup cif import bulkLoader cif.bulkloader.config.xml
        ```

[113] Charging output Integration Framework

> **Note**
>
> For further information about the configuration of the CIF, refer to the *Data Files Generation* section of the SAP CC 2020 Tuning Guide documentation.

3. **Deactivating features**
   Once the configuration of raters and bulkloaders has been successfully updated, it is necessary to deactivate the Prepaid Accounts Management and Chargeable Items Recharging features.
   1. To deactivate the Prepaid Accounts Management feature, execute the following command of the Setup Tool user interface:

      ```
      setup migration deactivatePrepaid
      ```

   2. To deactivate the Chargeable Items Recharging feature, execute the following command of the Setup Tool user interface:

      ```
      setup migration deactivateRecharging
      ```

4. **Configuring WS destinations**
   Once the Prepaid Accounts Management and Chargeable Items Recharging features have been successfully deactivated, it is necessary to configure the different WS destinations that will be used to send billable items to SAP S/4HANA Cloud Public Edition. For each WS destination to create, execute the following operations:
   - Copy the `ws.destination.sk` file available in the `config/s4hc` folder of the dispatcher instance in a new `ws.destination_<DEST_NAME>` file, where <DEST_NAME> corresponds to the name of the WS destination to create
   - Modify the newly created `ws.destination_<DEST_NAME>` file to fit the targeted web service
   - Execute the following command of the Setup Tool user interface to import these WS destinations in the Core Database:

     ```
     setup wsdestination import <DEST_NAME> ws.destination_<DEST_NAME>
     ```

   - In case you need to import certificates, for example when the web services use HTTPS with a self-signed certificate, execute the following command within Setup Tool for each concerned certificate:

     ```
     setup certentry import <ALIAS> <CERT_FILE>
     ```

     , where:
     - <ALIAS> corresponds to the alias of the certificate
     - <CERT_FILE> corresponds to the absolute path of the certificate to import

5. **Restarting instances**
   Once the relevant WS destinations have been successfully created, it is necessary to restart the following instances:
   - dispatchers
   - updaters

   For further information, refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.
6. **Creating CMAs**
   Once dispatchers and updaters have correctly restarted, it is necessary to create the Customer Management Areas for connection purpose to the previously configured WS destinations. To create these

CMAs, launch [page 193] the Core Tool user interface and click the *Tools* > *Customer Management Areas* > *Edit* menu.

7. **Restarting instances**
   Once the relevant objects have been successfully created within the Core Tool user interface, it is necessary to restart the following instances:
   - raters
   - bulkloaders

   For further information, refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.
8. **Creating billable item mappings**
   Once raters and bulkloaders have correctly restarted, it is necessary to create the different billable item mappings that correspond to the charged item classes defined in SAP CC, to be able to send billable items to SAP S/4HANA Cloud Public Edition. To create these billable item mappings, launch [page 193] the Core Tool user interface and click the *File* > *New* > *Billable Item Mapping* menu.

7.1.52 Copying an SAP CC Core Server

> → Remember
>
> The Session Database is used only for SAP CC's session-based charging feature, and doesn't exist at all for SAP HANA. Just ignore any step about the Session Database in this procedure if not relevant.

Description

For multiple reasons such as development integration tests, system tuning, performance tests, backup operations, and so on, it might be necessary to copy a Core Server system from one landscape to another, an operation that consists in:

- Preparing the destination system
- Backing-up the tables of the destination Core Database and Session Database, according to the copy scenario
- Backing-up the source Core Database and Session Database
- Transferring the adequate data from the source Core Database and Session Database to the destination ones
- Restoring the adequate data within the destination Core Database and Session Database
- Restarting the destination system

You can use this procedure as **guidelines** for copying a Core Server system (considered as the source system) from:

- A production landscape to another landscape that is used for any purpose
- A non-production landscape to another non-production landscape

> ⚠ Caution
>
> This procedure requires to shut down both the source and the destination Core Server systems. This operation can impact your overall landscape and thus your business in case the source system is part of a production landscape. SAP SE thus recommends you to be very careful when executing this procedure.

Prerequisites

- A Core Server system must be available within your landscape. This Core Server system will be considered as the source system, whose version must correspond to the destination system

  > ⚠ Caution
  >
  > A version is composed of the:
  > - Major/minor release level (for example 4.1)
  > - Support Package (SP) level (for example SP 3)
  > - Patch level (for example 2)

- Both the source and the destination Core Databases must have the same software version
- If the Session Database is used, both the source and the destination Session Databases must have the same software version
- Both Core Server systems used as source and destination must be installed on physical hosts:
  - Compatible in terms of hardware sizing to ensure that the instances of the destination Core Server can start after this procedure
  - Running under the same operating system
  - Where concerned third-party systems (such as SAP Convergent Invoicing, CA Application Performance Management, and so on) are already installed and available, running under the same versions

    > ⓘ Note
    >
    > If the source Core Server system is integrated with SAP Convergent Invoicing, it is necessary to use a similar configuration for both the source and the destination systems to ensure that:
    > - The same currencies are defined in both the source and the destination SAP ERP/FI-CA systems
    > - The same "SAP Convergent Charging Integrated SAP Solution Scenario" is used ("Billing and Invoicing in SAP Convergent Invoicing" or "Billing, Invoicing, and Storage of Consumption Data in SAP Convergent Invoicing")
    > - The same Consumption Item Classes and Billable Item Classes are defined
    > - The names of the used JCo destinations are the same
    > - The lists of available Customer Management Area are the same
    > - The targeted SAP systems of the used JCo destinations must be the same if your copy scenario concerns a copy of a Core Server system from a production to another production landscape, or different otherwise
    >
    > Moreover, when using this procedure to copy a Core Server system from a production to another production landscape, ensure that:

- The same transport destinations are defined in both the source and the destination systems
- All pending data files containing charged items, chargeable items or notifications are:
  - Processed in the source system
  - Deleted from the destination

Preparing the destination system

The first step of this procedure consists in preparing the Core Server system that is considered as the destination system, by referring to the Installing SAP Convergent Charging [page 21] section of this documentation and respecting the prerequisites.

Once this destination system has been prepared, SAP SE highly recommends that you back up the configuration of this system, using the Config Tool and Setup Tool user interfaces. This backup will be used at the end of this procedure to ensure that this configuration has been successfully restored on the destination system after the data transfer from the source system. For further information about the Config Tool and Setup Tool user interfaces, refer to the following procedures:

- Installing and Launching Config Tool [page 214]
- Installing and Launching Setup Tool [page 211]

Backing-up the destination databases

The second step of this procedure consists in backing-up the tables of the destination Core and Session Databases, which means:

- Shutting down the destination Core Server
- Exporting all the tables of the Core Database in case of failure during the copy procedure
- Exporting all the tables of the Session Database in case of failure during the copy procedure
- Exporting a set of tables of the Core Database that relate to the configuration of the destination Core Server, whose content will be restored later

1. To ensure data consistency and avoid any locking problem during backup operations, it is necessary to shut down all the instances of the destination Core Server. Refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary
2. As data within the destination Core Database will be overwritten in the next steps, it might be necessary to restore the content of the destination Core Database in case of failure during the execution of the copy procedure. Refer to your DBA[114] for further information about backup procedures and associated commands that must be performed to create this safety backup of all the tables of this database
3. As data within the destination Session Database will be overwritten in the next steps, it might be necessary to restore the content of the destination Session Database in case of failure during the execution of the copy procedure. Refer to your DBA for further information about backup procedures and associated commands that must be performed to create this safety backup of all the tables of this database

[114] DataBase Administrator

4. In addition to the previously created safety backup, it is necessary to create a dedicated backup containing the following set of tables of the Core Database that relate to the configuration of the destination Core Server. For further information about backup procedures and associated commands, refer to your DBA.
   - **CERT_STORE** table, that contains the certificates of allowed clients when using SSL/TLS
   - **CONFIGURATION** table, that contains some specific configuration settings of SAP CC
   - **CUSTOMER_MANAGEMENT_AREA** table, that contains a set of customer management areas defined in the system landscape
   - **INSTANCE_CERT_STORE** table, that is used to link a service to a client certificate when using SSL/TLS
   - **INSTANCE_CONFIGURATION** table, that contains all the system parameters set for each type of instance
   - **INSTANCE_KEY_STORE** table, that is used to link a service to a key store entry when using SSL/TLS
   - **INSTANCE_MAP** table, that contains the map of available instances of the Core Server system
   - **KEY_STORE** table, that contains the private keys and their associated certificates when using SSL/TLS
   - **LICENSE** table, that contains information about the SAP Convergent Charging license
   - **VERSION** table, that contains the history of database changes performed on an installed version of SAP CC

   And if your copy scenario concerns a copy of a Core Server system from a production to a non-production landscape, also backup the following tables of the Core Database:
   - **HCI_USER** table, that contains information about the authorized users of SAP CC
   - **USER_FAILED_AUTHENTICATION** table, that contains the list of failed authentication attempts for the different users since their last successful authentication
   - **USER_PASSWORD** table, that contains the history of passwords of the SAP CC users
   - **USER_RIGHT** table, that contains the authorization roles of each SAP CC user

Backing-up the source databases

The third step of this procedure consists in making consistent backups of the tables of the source databases. To ensure data consistency and avoid any locking problem during backup operations, it is necessary to shut down all the instances of the destination Core Server, referring if necessary to the Starting and Stopping the Server Systems [page 144] procedure. Then, according if necessary to the recommendations of your DBA, create the consistent backup of:

- The tables of the Core Database
- The tables of the Session Database, in case your copy scenario concerns a copy of a Core Server system between 2 production landscapes

Transferring adequate data from the source databases

The fourth step of this procedure consists in transferring the content of the consistent backups from the source databases to the destination databases. This operation consists in:

- Emptying the tables of the destination Core Database
- Emptying the tables of the destination Session Database
- Importing the previously created consistent backup related to the Core Database
- Importing the previously created consistent backup related to the Session Database, in case:
  - Your copy scenario concerns a copy of a Core Server system between 2 production landscapes
  - Your Core Server system uses the Session Database

> ⚠ Caution
>
> To prevent replication errors, each replicated value must stay in its original column.
>
> As a general precaution, in your replication tool, you should enable options to explicitly reference the columns by their name, during the export and the import.
>
> Refer to SAP Note 3057000 for further information.

Restoring the destination Core Database

The fifth step of this procedure consists in managing the content of some tables of the destination Core Database to restore an adequate configuration of the destination Core Server. According to your copy scenario, the following table provides you with the necessary information to decide to:

- **Keep** the content imported from the source database
- **Empty** the content imported from the source database
- **Overwrite** the content imported from the source database with the configuration data backed-up in step 2, after having emptied the concerned table

	Action			
Table	**Keep**	**Empty**	**Over-write**	**Notes**
Core Database				
LICENSE	☐	☐	■	
VERSION	☐	☐	■	
CONFIGURATION	☐	☐	■	
CUSTOMER_MANAGE-MENT_AREA	☐	☐	■	
INSTANCE_CONFIGURA-TION	☐	☐	■	
INSTANCE_MAP	☐	☐	■	
KEY_STORE	☐	☐	■	
INSTANCE_KEY_STORE	☐	☐	■	
CERT_STORE	☐	☐	■	
INSTANCE_CERT_STORE	☐	☐	■	

Table	Action			Notes
	Keep	Empty	Over-write	
CHARGED_ITEM_FILES	☐	■	☐	This table should be empty due to prerequisites
WS_JOB	☐	■	☐	
HCI_USER	■	☐	■	**Overwrite** if your copy scenario concerns a copy of a Core Server system from a production to a non-production landscape. **Keep** otherwise.
USER_RIGHT	■	☐	■	
USER_PASSWORD	■	☐	■	
USER_FAILED_AUTHENTI-CATION	■	☐	■	
RATING_SESSION_@NUM-BER@	■	■	☐	**Empty** if your copy scenario concerns a copy of a Core Server system from a production to a non-production landscape. **Keep** otherwise. **Note** @NUMBER@ representing the partition number, the RATING_SESSION_@NUMBER@ tables represents a possible set of 480 tables to restore.
RENEW_RESERVA-TION_NOTIFICATION	■	■	☐	
SPENDING_STATUS_MONI-TORING	■	■	☐	
SPENDING_STATUS_RE-PORT	■	■	☐	
CHANGE_LIST	■	■	☐	
CHAN_OBJECT	■	■	☐	
CHAN_OBJECT_SNAP-SHOT	■	■	☐	
TRANSPORT_REQUEST	■	■	☐	
TRAN_DESTINATION	■	■	☐	

■*: Supported action,* ☐*: Unsupported action*

Note

If your RDBMS[115] provides statistics, you can re-compute them on the destination Core Database after the import.

Restarting the Core Server systems

The last step of this procedure consists in:

- Creating a new backup of the configuration of the destination system, used to compare with the configuration backed-up in Step 1 in order to ensure that this configuration has been successfully restored (in Step 4) and avoid situations such as:
  - Bad instance configuration (including database connection parameters)

[115] Relational Database Management System

- Bad securing information
- Bad JCo destinations
- And so on

- Starting the different instances of both the source and the destination Core Servers. Refer to the Starting and Stopping the Server Systems [page 144] procedure if necessary.

> ⚠ Caution
>
> For security and confidentiality reasons, SAP SE highly recommends that you delete the backed-up content once you ensured that the copy procedure successfully ended.

7.1.53 Configuring the Required SAP CC Users and Accounts

As part of your post-installation operations, you create the SAP CC users who are relevant and required for your implementation scenario.

Depending on your corporate security policy and business requirements, you create the SAP CC user accounts that are relevant for your implementation scenario of SAP Convergent Charging.

During the post-installation phase, you create the necessary user accounts for your SAP CC users:

- System administrators who manage the SAP CC systems in an SAP system landscape
- Application administrators who manage the business configurations at the system level
- User administrators who manage the SAP CC users and accounts in an SAP system landscape

> ⚠ Caution
>
> In a production SAP CC system landscape, delete the super administrator user freshly created by the SAP CC Installer tool.

See Also

Deleting the Super Administrator and Emergency User [page 286]

More Information

- User Administration and Authentication
  - Required Users

7.1.54 Deleting the Super Administrator and Emergency User

> → Recommendation
>
> During the post-installation phase, it is highly recommended that you create all the necessary users in SAP Convergent Charging **and then delete the super administrator user**.

As an administrator of SAP CC, use the Core Tool user interface to delete the super administrator user created by the SAP CC Installer tool. See Deleting a User.

7.2 Update and Maintenance

One-Off Maintenance

SAP has adapted our major release strategy to better fit your adoption cycle. To ensure that you benefit from new developments and innovations while minimizing the impact on your core operational systems, SAP uses support packages (SPs) to speed up the delivery of new functions.

With SAP support packages, you can install and activate new functions depending on your business needs and requirements without having to perform a system upgrade.

You can selectively implement the new functions and activate the software upon business demand. As a result, you can isolate the impact of software updates and make new functions available faster due to shortened test cycles.

Regular Maintenance

Regular maintenance is key to keep your systems efficient.

- Planning the Software Maintenance [page 287]
- Testing and Validating the Technical Implementation [page 287]
- Determining the Minimal Production Landscape [page 287]
- Preparing the Core Server Systems [page 288]
- Preparing the Back-End Database Systems [page 288]
- Updating the Server Systems [page 288]
- Updating the Client Applications Based on the Java Libraries [page 291]
- Updating the SAP CC GUIs and the IEC Application [page 292]
- Updating the Cockpit Applications [page 292]

Updating BillSoft EZTax Technology Monthly [page 293]

Post Processing [page 294]

7.2.1 Planning the Software Maintenance

You decide when to start the update. If you have special requirements regarding the availability of your SAP system, adjust the process and the length of the upgrade to fit the expected amount of downtime or restricted services in high availability (HA) environment.

To schedule the software maintenance, prefer periods with low business traffic when possible in your environment. Otherwise, consider off-peak times to avoid peak traffic periods. Estimate the level of the business traffic flow.

In a high availability (HA) environment, consider periods with limited business traffic. In a non HA environment, you stop the SAP systems.

During the maintenance update, various actions are performed in the system. The runtime of these actions depends on various factors and is therefore specific to each SAP system.

Test and validate the update, by applying the maintenance operations to a preproduction system.

7.2.2 Testing and Validating the Technical Implementation

Test and validate the maintenance update (support package, patch level) before to be able to update the production system. First, update your preproduction, validation, or quality assurance systems.

7.2.3 Determining the Minimal Production Landscape

In high-availability (HA) environment, you determine for each type of instances in the running SAP CC Core Server system, the count of instances that must run together to handle the business traffic that is expected according to your estimation. You determine the minimal production landscape.

Use the preproduction systems (or validation systems). Simulate the business traffic and stop some instances.

> ⚠ Caution
>
> If you cannot define this minimal production landscape, do restart the instances one-by-one after the maintenance update with the SAP installer. Refer to the Starting and Stopping the Server Systems [page 144] section of this documentation.

7.2.4 Preparing the Core Server Systems

Increase the SQLHELPER_CONNECTION_COUNT Parameter Values

If you update from SP 0 to SP 1 minimum, you must increase the runtime and persistent values of the SQLHELPER_CONNECTION_COUNT system parameter in all the rater instances that compose your SAP CC Core Server systems.

As of SP 1, SAP Convergent Charging supports the new system parameter RATING_SESSION_SERVICE_THREAD_COUNT. Before launching the SAP installer, make sure that the running system to update is prepared adequately.

7.2.5 Preparing the Back-End Database Systems

Before updating your SAP CC system landscape, be sure to back up the existing data to be able to roll back your maintenance operations.

According to your database technology, you must determine and create the following tablespaces or filegroups, **if you update from SP 0 to SP 1 minimum**:

- `MONITORING_DATA`
- `MONITORING_INDX`

These tablespaces or filegroups include data and database indexes related to the new technical feature that allows you to store metadata information about the data files generated by the Core Server system during business operations (charging, refilling, and activation). These data files are also known as "item files" in the SAP documentation.

After the software maintenance operations, your Core Database includes such data when you activate the feature. For further information, refer to:

- The FILE_METADATA_RECORDING_ENABLED system parameter
- The database preparation procedure that relates to your RDBMS in the Installation Procedures [page 88] section of this documentation

7.2.6 Updating the Server Systems

At the scheduled times, you operate the update of the SAP CC systems in your development, test, quality assurance, preproduction system landscapes to validate the implementation of the new support package or patch level.

Finally, you update the productive SAP CC systems in your production system landscape.

You perform the update of the server systems with the SAP installer on the SAP Global Host [page 12] of each SAP CC system.

The update consists in the following stages:

Stage	Core Server	BART Server	Diameter Server
Data update in the database	■	■	N/A
SAP CC 2020 software binaries update	■	■	■
Server system restart	■	■	■

→ Remember

You do not need to stop the server systems before the maintenance update with the SAP installer. After the update, you must manually restart [page 144] all the Core Server instances and other SAP CC server systems.

Identify the Hosts

To update a server system and an SAP CC system landscape, you will have to launch the SAP installer:

- On the machines hosting the BART Server system and the Diameter Server system
- On the SAP Global Host [page 12] of the Core Server system

You must identify these hosts before starting the update.

ⓘ Note

The SAP Global Host of a Core Server system is the first installed host of the SAP system. The SAP Central Instance is installed on this host and must be a dispatcher instance of the SAP CC system. To find the *global host* for the Core Server system:

- Go to the following directory on any machine hosting instances of the Core Server system:
  - `<DRIVE>:\usr\sap\<SID>\SYS\profile\` for Microsoft Windows
  - `/usr/sap/<SID>/SYS/profile/` for Linux and UNIX operating systems
- Open the profile file `DEFAULT.PFL`
- See the value of the variable `<SAPGLOBALHOST>`

Copy the Installation Materials

To update a server system, you will have to launch the SAP installer from the delivered installation materials:

- Go to the installation materials [page 92]
- Fully or partially copy the maintenance material to the host(s) of an SAP CC system

Launch the Installer

> ⚠ Caution
>
> If you update from SP 0 of the main release 5.0 to SP 1 minimum, make sure you prepared:
>
> - The Core Database with the newly created tablespaces or filegroups (refer to the Preparing the Back-End Database Systems [page 288] section of this documentation)
> - The previously installed Core Server system with the increased values of the SQLHELPER_CONNECTION_COUNT system parameter in all the rater instances (see the Preparing the Core Server Systems [page 288] section of this documentation)

To update an SAP CC server system in your SAP system landscape, you have to launch the SAP installer:

- On the machines hosting the SAP CC BART Server system and the Diameter Server system (if needed)
- On the global host for the Core Server system
- Execute the following procedure:
  1. Go to the `CC2020_IM_*` subdirectory in the installation folder where * is your OS[116]
  2. Start the installer by running the `sapinst.exe` program for Microsoft Windows operating systems or the sapinst script for other OSs
  3. Click *SAP Convergent Charging* > *Software Life-Cycle Options* > *Update* > *Update - SAP Systems*

> → Remember
>
> In a multihost environment, you do not need to launch the SAP installer on the other hosts of the SAP CC Core Server system to implement a support package. The instances located on other hosts are automatically updated when they restart.

The installer prompts you for the following parameters:

1. **Choose Option** screen:
   (Installation Mode)
   - *Typical*: the update is performed with default settings. You only have to respond to a small selection of prompts
   - *Custom*: you are prompted for all the parameters

   > ⓘ Note
   >
   > If you do not choose the *Custom* mode, you can still change later any parameter on the *Parameter Summary* screen.
2. **Define Parameters** screen:
   - *SAP System Identification*: specify the Profile Directory of the system to update
   - *SAP System Administrator Password*: enter the password of the SAP System Administrator (Microsoft Windows with remote hosts only)
   - *Database User*: enter the user and the password used to connect to the relevant Oracle database
   - *Database Update*: select the SQL[117] scripts to execute

[116] Operating System
[117] Structured Query Language

- *Update Archives*: specify the archives to be updated
- *Restart Instances*: choose if you want to automatically restart the running SAP CC instances at the end fo the update process.
  At the end of the upgrade process, each SAP CC instance needs a restart to make the update effective. As long as an instance is not restarted, it keeps on running in its previous version.

> → Recommendation
>
> The *Restart SAP Convergent Charging instances after installation* option is ticked by default, meaning that SAP CC instances will be automatically restarted by SAP installer at the end of the upgrade process.
>
> - In production (or preproduction) landscapes, SAP SE highly recommends to untick this option. This will allow you to manually manage the SAP CC instances restart sequence and thus keep control of the update.
> - In HA[118] environments, SAP SE also recommends to untick this option. This will allow you to check that all the updater instances are stopped.
> - Because the Diameter Server system is used for online charging, restart it at off-peak times to avoid service disruption.

> ⓘ Note
>
> In development or test landscapes, you can leave this option ticked

3. **Summary** screen: the installer displays the summary of the actions scheduled. You can modify any of the selected parameters by clicking on the appropriate checkboxes and then clicking on the *Revise* button.

The installer displays a final summary for the operations it will perform. When you are ready, click *Next* to start the next stage.

Manually Restart the Core Server Instances and Other SAP CC Systems

After the update of the SAP CC Core Server system, you must manually restart all its system instances in case of HA setup.

Refer to the Starting and Stopping the Server Systems [page 144] section of this documentation.

7.2.7 Updating the Client Applications Based on the Java Libraries

You must update all the external applications or systems that use the Java Libraries of SAP CC 2020.

[118] High Availability

7.2.8 Updating the SAP CC GUIs and the IEC Application

You must update all the graphical user interfaces and the IEC[119] application of SAP CC 2020.

To update the user interfaces and the IEC application, you have to:

1. Get the `ZIP` files. They are available on the installation materials in the directory `CC2020_TOOLS_CONTENT_UC_OSIND`
2. Unzip the appropriate file in the installation directory
3. Save copies of the old configuration files (`*.config`)
4. Rename (from `*.config.sk` to `*.config`) and customize the new configuration files accordingly
5. Download and install the SAP JVM 8.1 [page 94]

7.2.9 Updating the Cockpit Applications

By default, you must update all the installed Cockpit user interfaces of SAP CC 2020 with the latest delivered patch, FPS, or support package (SP).

Before to start this software maintenance operation, consult the Installing SAP Convergent Charging [page 21] section of this documentation for more information about the installation [page 88] procedures and recommendations.

To update the Web applications in your Tomcat Server systems, you perform the following procedure:

1. Verify the last-minute information in the latest SAP Notes delivered with your SAP Convergent Charging version (patch notes, SP notes, technical implementation notes).
2. Make sure there is no supplementary procedure that relates to the technical implementation of the SAP CC software.
3. Determine the current installation settings:
   - Basic installation and deployment in a Tomcat Server: Verify the app name and path in the Java Web server.
   - Advanced installation based on Tomcat Contexts: Verify the location and usage of the `WAR` file that was installed previously.
4. Plan the update processing and inform the SAP CC users about the software maintenance.
5. Get the `ZIP` file. It is available on the installation media in the directory `CC2020_TOOLS_CONTENT_UC_OSIND`. See the Downloading the Installation Materials [page 92] section for more information about the media and the zip files to use.
6. Unzip the file in the installation directory to extract the `WAR` file.
7. Verify that there are no users connected to the Web application.
8. Depending on your installation, deploy, rename and deploy, or manage the Tomcat Contexts and the `WAR` file.
9. Save copies of the old configuration files (`*.properties`).
10. Create the necessary configuration files by copying the newly delivered templates and customize the configuration of the corresponding Cockpit user interfaces.

[119] Import/Export Connector

11. In Tomcat, restart each updated Web application initially to verify that they can run and display the sign-in window.
12. Close and relaunch the browser.
13. Navigate to the valid URL and verify that Convergent Charging Cockpit can run and display the sign-in window.
14. Sign in.
15. As you have updated the Web application of the Cockpit user interface, you and other SAP CC users must clear the Web browser caches by using the CTRL + F5 combination of keys.

7.2.10 Updating BillSoft EZTax Technology Monthly

The monthly update of the BillSoft EZTax installation(s) is mandatory for the tax function for US telecommunications.

You have to update three categories of units:

- The BillSoft EZTax installations
- A central repository `CC_EZTAX` used by SAP Convergent Charging
- The relevant server systems of SAP CC (taxer instances of the Core Server)

At the original installation time, the SAP installer of SAP CC has created a specific central repository dedicated to the management of application data. It is named `CC_EZTAX` in the following procedures and may differ on your landscape.

Maintenance of BillSoft EZTax Software

You need to monthly update your BillSoft EZTax software and data.

It is installed on each machine that hosts one or several Taxer instances of the SAP CC Core Server.

Refer to the maintenance procedures from the BillSoft EZTax documentation.

Maintenance of CC_EZTAX Central Repository

You need to monthly update the `CC_EZTAX` central repository that includes EZTax software files and data required by SAP Convergent Charging software. It has been created at the installation time by the SAP installer of SAP CC.

This folder must contain the changed files:

- the `transervdesc.txt` file
- the subdirectories `db`, `jar` and `lib`:
  - `db/` contains EZTax database files
  - `jar/` contains the file `eztax.jar`

- `lib/` contains EZTax library files depending on the operating system: `*.dll` for Microsoft Windows or `*.so` for Linux and UNIX operating systems

Once you have updated the BillSoft EZTax installations, you update the `CC_EZTAX` central repository by copying necessary files and directories listed above.

> **Note**
>
> - Verify if the `CC_EZTAX` central repository is shared by several SAP CC Server Systems: taxer instances of the Core Server. These systems will be automatically updated when you restart them
> - Verify if the `CC_EZTAX` central repository is shared by another major version of SAP Convergent Charging. These systems will be automatically updated when you restart them

Maintenance of taxer Instances of the Core Server

You need to monthly update your BillSoft EZTax software files and application data. It has been installed on each machine that hosts one or several taxer instances of the Core Server system.

The `CC_EZTAX` central repository must be up-to-date.

1. Stop and restart sequentially all the taxer instances of the Core Server system. When restarting, each Taxer instance will update its local installation folder by copying all these files to the folder: `/config/eztax/...`
2. Use the Setup Tool user interface of SAP CC for updating the Core Database with the latest transaction/service types declared in the `transervdesc.txt` file present in the `CC_EZTAX` repository:
   - Use the `eztax importTSTypes` command and this file for inserting the new types in the database
3. Use the Admin+ user interface of SAP CC for updating the the relevant instances of the Core Server system:
   - Start Admin+ and log on as administrator
   - Use the `refresh_transerv_list` command for updating the updater instance(s)

> **Caution**
>
> Do never stop all the taxer instances at the same time. Prefer updating one instance after the other.

7.2.11 Post Processing

Verify the last-minute information in the relevant Technical Implementation Note document provided with each support package of SAP CC.

Post-Processing Tasks After a Core Server Update

1. As of SP 1, SAP recommends that you verify the values of the LS_TRC_DOMAIN system parameter in your running SAP CC Core Server systems. New tracing domains are available: DATA, PERF, and PPI. In a

production SAP CC system, set up the system parameter with all the existing domains. You can use the `reset` command of Admin+

2. In an integrated SAP Solution with SAP Convergent Invoicing (SAP CI) in SAP ERP, please verify that the `Event Date Shifting` function is enabled in the rater instances of your running SAP CC system. There is no basic system parameter to view but there is an advanced configuration record to check. In Setup Tool, export the CIF[120] configuration settings and check that the `eventDateShifter XML` nodes are present in the file. See the Data Files Generation (Item Files) section of the SAP CC 2020 Tuning Guide documentation for more information and user assistance

7.3 Upgrade Procedures

[120] Charging output Integration Framework

8 Troubleshooting

Information for addressing common problems that may arise during the installation, maintenance update, or upgrade project of SAP Convergent Charging (SAP CC in your SAP system landscapes.

> → Remember
>
> In a production SAP system landscape, contact your SAP Support Team.

8.1 Installation Troubleshooting

Determine the cause of installation problems and apply the recommended solution.

To determine the cause of an error, first look in the log files of the utilities being used, to identify the task each utility was performing when it failed. Then check the server error log.

Read tips and troubleshooting solutions related to the installation of this version of SAP Convergent Charging (SAP CC).

Troubleshooting Your Core Server Installation [page 296]
Determine the cause of installation problems for *SAP CC Core Server* and apply the recommended solution.

Troubleshooting Your Cockpit Installation [page 297]
Determine the cause of installation problems for *SAP CC Cockpit* and apply the recommended solution.

8.1.1 Troubleshooting Your Core Server Installation

Determine the cause of installation problems for *SAP CC Core Server* and apply the recommended solution.

Configuration changes can lead to new issues.

Tips and Troubleshooting Solutions (Core Server Installations) [page 297]
Read tips and troubleshooting solutions related to the installation of this version of SAP Convergent Charging Core Server system.

Related Information

Installing a Core Server on a Monohost Landscape [page 123]
Installing a Core Server on a Multihost Landscape [page 131]

8.1.1.1 Tips and Troubleshooting Solutions (Core Server Installations)

Read tips and troubleshooting solutions related to the installation of this version of SAP Convergent Charging Core Server system.

8.1.2 Troubleshooting Your Cockpit Installation

Determine the cause of installation problems for *SAP CC Cockpit* and apply the recommended solution.

Configuration changes can lead to new issues.

Tips and Troubleshooting Solutions (SAP CC Cockpit/Apache Server Installations) [page 297]
Read tips and troubleshooting solutions related to the installation of this version of SAP Convergent Charging Cockpit.

Related Information

Installing and Launching Cockpit [page 201]

8.1.2.1 Tips and Troubleshooting Solutions (SAP CC Cockpit/Apache Server Installations)

Read tips and troubleshooting solutions related to the installation of this version of SAP Convergent Charging Cockpit.

See also: Troubleshooting Problems of the User Interface

Fatal Error: "Invalid JDBC URI or related JDBC driver not found" [page 298]

Issue: Charging is not available [page 299]

8.1.2.1.1 Fatal Error: "Invalid JDBC URI or related JDBC driver not found"

Symptom

SAP CC Cockpit fails to start with the following error:

```
com.highdeal.sql.SQLManagerException: Invalid JDBC URI
(jdbc:oracle:thin:XXXXXXX:XXXX:XXX) or related JDBC driver not found
```

This error may occur when initially installing a new version of SAP CC higher than version 5.0 SP 4, when updating an SAP CC system landscape landscape up to version 5.0 SP 5 or SP 6, or when upgrading your landscape to the major release 2020 or to a higher release.

Environments

SAP CC 5.0 SP 4 and higher versions

Reason

As of SAP CC Cockpit 5.0 SP 4, the Oracle JDBC driver is not delivered within SAP CC. It is now necessary to download the Oracle JDBC driver manually.

Solution

You need to update your installed Apache Tomcat Server system with the relevant Oracle JDBC driver. See Installing the Oracle JDBC Driver in an Apache Tomcat Server System [page 203].

8.1.2.1.2 Issue: Charging is not available

Symptom

After the installation, the tile of the app Process a Chargeable Item is grayed out and displays the error message: `Charging is not available`

Solution

See Enabling and Displaying the App Process a Chargeable Item and Configuring Cockpit.

8.2 Uninstallation Troubleshooting

Determine the cause of uninstallation problems and apply the recommended solution.

8.3 Update Troubleshooting

Determine the cause of update (Feature Package Stack (FPS) or patch level) problems and apply the recommended solution.

8.4 Upgrade Troubleshooting

Determine the cause of upgrade problems and apply the recommended solution.

For troubleshooting the upgrade, you can use the logs from:

- SAP installer
- The SAP CC system being upgraded

8.4.1 Logs of SAP Installer

During the execution of the upgrade process, SAP installer writes detailed logs of all the actions it performs on the file system of the host where it is executed.

These log files are written in the following directory:

- Microsoft Windows operating systems: `<DRIVE>:\Program Files\sapinst_instdir\CC2020\SLM\UPDATE\`
- Linux and UNIX operating systems and other operating systems: `/tmp/sapinst_instdir/CC2020/SLM/UPDATE/`

This directory includes subdirectories that contain the log files and control files of previous executions of SAP installer. These subdirectories have a naming convention: `log` is used as prefix and the date/time is added to the name.

> **Example**
>
> The `UPDATE` directory includes a directory named: `log_24_May_2019_08_53_07`

Among the files contained in this directory, 2 files are particularly important. These files describe the actions performed by SAP installer during the upgrade process:

- `sapinst.log` (summary of actions)
- `sapinst_dev.log` (this log file includes more detailed information than `sapinst.log`)

The directory also contains an archive named `SUPPORT.SAR` that contains all the important log files for troubleshooting the issue.

8.4.2 Logs of SAP CC

The individual log files of the instances of the SAP CC system(s) can be used to troubleshoot issues encountered during the upgrade process.

Each instance of the system has a directory where its own log files are written:

- Microsoft Windows operating system: `<DRIVE>:\usr\sap\<SID>\<INSTANCE_NAME>\work\`
- Linux and UNIX operating systems and other operating systems: `/usr/sap/<SID>/<INSTANCE_NAME>/work/`

Refer to the SAP CC 2020 Administration Guide for more information.

8.4.3 Restoring the Software

All instances of the SAP Convergent Charging server system have a local copy of the software binary files. These copies are retrieved from the SAP Central Repository [page 13] of the SAP CC system when the instances start.

To restore the software binary files to the previous version, you need to restore first the files on the SAP Central Repository, and then restart all the instances of the SAP CC server system.

The SAP Central Repository is available in the SAP Global Host [page 12] or from the network:

OS[121] of the Host Machines	Local Access (from the SAP Global Host)	Network Access (from a host machine of the SAP system)
Microsoft Windows	`<DRIVE>:\usr\sap\<SID>\SYS\`	`\\<SAP_GLOBAL_HOST>\sapmnt\<SID>\SYS\`
Linux and UNIX operating systems and other operating systems	`/usr/sap/<SID>/SYS/`	`/sapmnt/<SID>/SYS/`

The directory in the SAP Central Repository containing the software files is located in the following folder:

- Microsoft Windows operating system: `<SAP_CENTRAL_REPOSITORY>\exe\uc\<OS_PLATFORM>\`
- Linux and UNIX operating systems or other operating systems: `/sapmnt/<SID>/exe/uc/<OS_PLATFORM>/`

Where: `<OS>` corresponds to the operating system where the system is installed:

Operating System	Directory Name
Microsoft Windows 64-bit	`ntamd64`
AIX	`rs6000_64`
HP-UX	`hpia64`
Linux and UNIX operating systems	`linuxx86_64`
Solaris	`sunx86_64`

This central repository contains the archives of the previous upgrade or update operations. It includes the following folders created by SAP installer:

- Subdirectories named `SOFT_UPG_FILES` that contain the current version of the software binary files
- Backup subdirectories named `<SOFT_UPG_ARCHIVED>.<ARCHIVE_DATE>` that contain the backup copy of the software binary files composing a previous version.

Different software units are handled. The possible subdirectory names (for `<SOFT_UPG_FILES>`) are:

- `CC_CORE_SERVER`
- `CC_BART_SERVER`
- `CC_DIAMETER_SERVER`
- `sapjvm_8`
- `wily`

Some subdirectories correspond to old updates (implementation of a support package) of former versions of SAP Convergent Charging

[121] Operating System

Core Server

To restore the software files to the last version before the upgrade:

- Remove the subdirectory `CC_CORE_SERVER`
- Locate the subdirectory `CC_CORE_SERVER.<timestamp>` with the highest timestamp value and rename it to `CC_CORE_SERVER`
- Restart all the instances of the SAP CC system you want to restore the version so that each instance retrieves a copy of the restored software files from the central repository.

Other Server Systems

To restore the software files to the last version before the upgrade:

- Remove the subdirectory `CC_<system>`
- Locate the subdirectory `CC_<system>.<timestamp>` with the highest timestamp value and rename it to `CC_<system>`
- Restart all the instances of the SAP CC system you want to restore the version so that each instance retrieves a copy of the restored software files from the central repository.

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